Unary primitive recursive functions in Z/EVES

Formalization of Section 6 of D. E. Severin, *Unary primitive recursive functions*, J. Symbolic Logic 73(4), 2008, at the level of closures (as in the Mizar version), and of the two open questions answered in the README.

Conventions: one-argument functions are the elements of $\mathbb{N} \to \mathbb{N}$. A function or operator X of the paper is introduced by an axiomatic box (`funX` or `opX`) whose defining equations are labelled rewrite rules. Every axiomatic box is followed by a consistency theorem `XEx`, which exhibits a witness (`XWit`, defined by an explicit term) satisfying the box; so no definition adds an axiom that could be inconsistent.

1 Building functions

Z/EVES has no rules showing that a λ-term is a total function; these four lemmas provide them. A relation that is single-valued and total on X is a total function.

theorem disabled rule RelPow $[X, Y]$
$X \leftrightarrow Y = \mathbb{P}(X \times Y)$

proof
use relDefinition$[X,\ Y]$;
prove;
■

theorem disabled rule PfunIntro $[X, Y]$
$f \in X \leftrightarrow Y \land (\forall x : X;\ y, z : Y \mid (x, y) \in f \land (x, z) \in f \bullet y = z) \Rightarrow (f \in X \pfun Y \Leftrightarrow \mathit{true})$

proof
prove;
use pfunDef$[X,\ Y][R\ :=\ f]$;
prove;
with enabled (*inPower*, *compDef*) *prove*;
with enabled (*invDef*, *inId*) *prove by reduce*;
instantiate $x__0\ ==\ y,\ y__0\ ==\ x,\ z__0\ ==\ z$;
prove;
■

theorem disabled rule DomTotal $[X, Y]$
$f \in X \leftrightarrow Y \land (\forall x : X \bullet \exists y : Y \bullet (x, y) \in f) \Rightarrow \mathrm{dom}\, f = X$

proof
apply extensionality;
prove;
with enabled (*inDom*) *prove*;
instantiate $x\ ==\ y$;
prove;
instantiate $y__2\ ==\ y__0$;
prove;
■

theorem disabled rule FunIntro $[X, Y]$
$f \in X \leftrightarrow Y \land (\forall x : X \bullet \exists y : Y \bullet (x, y) \in f) \land (\forall x : X;\ y, z : Y \mid (x, y) \in f \land (x, z) \in f \bullet y = z) \Rightarrow (f \in X \to Y \Leftrightarrow \mathit{true})$

proof
use pfunIsFun$[X,\ Y][A\ :=\ X,\ B\ :=\ Y,\ f\ :=\ f]$;
with enabled (*PfunIntro*, *DomTotal*) *prove*;
■

theorem rule FunIsPfun $[X, Y]$
$f \in X \to Y \Rightarrow (f \in X \nrightarrow Y \Leftrightarrow true)$

proof
use fun_type$[X,\ Y]$;
with enabled (*inPower*) *prove*;
■

Relational composition and iteration of one-argument functions (the witness for iteration $F^{\square(a)}$ uses the Toolkit's iteration F^n).

theorem rule FunIsRel $[X, Y]$
$f \in X \to Y \Rightarrow (f \in X \leftrightarrow Y \Leftrightarrow true)$

proof
use inPowerTransitive$[X := X \to Y,\ Y := X \nrightarrow Y,\ Z := X \leftrightarrow Y]$;
use fun_type$[X,\ Y]$;
use pfun_type$[X,\ Y]$;
prove;
■

theorem DomFunGen $[X, Y]$
$\forall f : X \to Y \bullet \operatorname{dom} f = X$

proof
prove;
■

theorem BigcapSub $[X]$
$\forall T : \mathbb{P}(\mathbb{P} X);\ Y : \mathbb{P} X \mid Y \in T \bullet \bigcap T \subseteq Y$

proof
prove;
■

theorem SubAntisym $[X]$
$\forall A, B : \mathbb{P} X \mid A \subseteq B \land B \subseteq A \bullet A = B$

proof
prove;
apply extensionality to predicate $A = B$;
with enabled (*subsetDef*, *inPower*) *prove*;
■

theorem InPair $[X]$
$\forall u, p, q : X \mid u \in \{p, q\} \bullet u = p \lor u = q$

proof
prove;
■

theorem rule CompRelFun
$\forall f, g : \mathbb{N} \to \mathbb{N} \bullet f \fatsemi [\mathbb{N}, \mathbb{N}, \mathbb{N}] g \in \mathbb{N} \to \mathbb{N}$

proof
prove;
with enabled (*easyDomComp*) *prove*;
■

theorem rule CompRelApp
$\forall f, g : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet (f \fatsemi [\mathbb{N}, \mathbb{N}, \mathbb{N}] g)\, x = g\,(f\, x)$

proof
prove;
■

$IterSet == \{n : \mathbb{N} \mid \forall F : \mathbb{N} \to \mathbb{N} \bullet iter[\mathbb{N}]\, n\, F \in \mathbb{N} \to \mathbb{N}\}$

proof [IterSet$domainCheck]
prove;
use applyInRanFun$[\mathbb{Z},\ \mathbb{P}\,(\mathbb{N} \times \mathbb{N}) \to \mathbb{P}\,(\mathbb{N} \times \mathbb{N})][f := iter[\mathbb{N}],\ a := n]$;
use DomFunGen$[\mathbb{P}\,(\mathbb{N} \times \mathbb{N}),\ \mathbb{P}\,(\mathbb{N} \times \mathbb{N})][f := iter[\mathbb{N}]\, n]$;
prove;
equality substitute;
prove;
■

theorem rule IterBase
$0 \in IterSet$

proof
invoke IterSet;
prove;
■

theorem rule IterStep
$\forall n : IterSet \bullet 1 + n \in IterSet$

proof
invoke IterSet;
prove;
use iterPositive$[\mathbb{N}][R__1 := F,\ n__0 := n]$;
instantiate $F__0 == F$;
with enabled (*easyDomComp*) *prove*;
■

theorem IterSub
$\mathbb{N} \subseteq IterSet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq IterSet$;
prove;
invoke IterSet;
prove;
■

theorem rule IterFun
$\forall F : \mathbb{N} \to \mathbb{N};\ n : \mathbb{N} \bullet iter[\mathbb{N}]\, n\, F \in \mathbb{N} \to \mathbb{N}$

proof
prove;
use IterSub;
with enabled (*inPower*) *prove*;
instantiate $e == n$;
invoke IterSet;
prove;
instantiate $F__1 == F$;
prove;
■

theorem IterSucR
$\forall F : \mathbb{N} \to \mathbb{N};\ k, a : \mathbb{N} \bullet iter[\mathbb{N}]\,(1 + k)\,F\,a = iter[\mathbb{N}]\,k\,F\,(F\,a)$

proof
prove;
use composePositiveIterates$[\mathbb{N}][n := 1,\ k := k,\ R := F,\ Y := \mathbb{N}]$;
use CompRelApp$[f := F,\ g := iter[\mathbb{N}]\,k\,F,\ x := a]$;
use IterFun$[F := F,\ n := k]$;
prove;
■

theorem rule IterInDom
$\forall F : \mathbb{N} \to \mathbb{N};\ n : \mathbb{N} \bullet F \in \mathrm{dom}(iter[\mathbb{N}]\,n)$

proof
prove;
use applyInRanFun$[\mathbb{Z},\ \mathbb{P}\,(\mathbb{N} \times \mathbb{N}) \to \mathbb{P}\,(\mathbb{N} \times \mathbb{N})][f := iter[\mathbb{N}],\ a := n]$;
use DomFunGen$[\mathbb{P}\,(\mathbb{N} \times \mathbb{N}),\ \mathbb{P}\,(\mathbb{N} \times \mathbb{N})][f := iter[\mathbb{N}]\,n]$;
prove;
■

theorem rule IterZero
$\forall F : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \bullet iter[\mathbb{N}]\,0\,F\,a = a$

proof
prove;
■

theorem rule IterSuc
$\forall F : \mathbb{N} \to \mathbb{N};\ n, a : \mathbb{N} \bullet iter[\mathbb{N}]\,(1 + n)\,F\,a = F\,(iter[\mathbb{N}]\,n\,F\,a)$

proof
prove;
use composePositiveIterates$[\mathbb{N}][n := n,\ k := 1,\ R := F,\ Y := \mathbb{N}]$;
use CompRelApp$[f := iter[\mathbb{N}]\,n\,F,\ g := F,\ x := a]$;
use IterFun$[F := F,\ n := n]$;
prove;
■

The domain of a one-argument function, and extensionality.

theorem rule DomUF
$\forall f : \mathbb{N} \to \mathbb{N} \bullet \mathrm{dom}\, f = \mathbb{N}$

proof
prove;
■

theorem rule UFAppNat
$\forall f : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet f\ x \in \mathbb{N}$

proof
prove;
use applyInRanFun[$\mathbb{N}$, $\mathbb{N}$][$f := f,\ a := x$];
prove;
■

theorem LeOrGt
$\forall a, b : \mathbb{Z} \bullet a \leq b \lor b < a$

proof
prove;
■

theorem UFNonneg
$\forall f : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet f\ x \geq 0$

proof
prove;
use applyInRanFun[$\mathbb{N}$, $\mathbb{N}$][$f := f,\ a := x$];
with disabled (*UFAppNat*) *prove*;
■

theorem rule NatAdd
$\forall a, b : \mathbb{N} \bullet a + b \in \mathbb{N}$

proof
prove;
■

theorem rule NatMul
$\forall a, b : \mathbb{N} \bullet a * b \in \mathbb{N}$

proof
prove;
■

theorem rule DomUOP
$\forall u : (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}) \bullet \mathrm{dom}\ u = \mathbb{N} \to \mathbb{N}$

proof
prove;
■

theorem rule DomBOP
$\forall b : (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}) \bullet \mathrm{dom}\ b = (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N})$

proof
prove;
■

theorem rule AppliesUOP
$\forall\, u : (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N});\ F : \mathbb{N} \to \mathbb{N} \bullet \mathit{uapplies\$toF}$

proof
prove;
■

theorem rule AppliesBOP
$\forall\, b : (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N});\ F, G : \mathbb{N} \to \mathbb{N} \bullet \mathit{bapplies\$to}(F, G)$

proof
prove;
■

theorem rule AppliesUF
$\forall\, f : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet \mathit{fapplies\$tox}$

proof
prove;
■

theorem rule UOPApp
$\forall\, u : (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N});\ F : \mathbb{N} \to \mathbb{N} \bullet u\, F \in \mathbb{N} \to \mathbb{N}$

proof
prove;
■

theorem rule BOPApp
$\forall\, b : (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N});\ F, G : \mathbb{N} \to \mathbb{N} \bullet b(F, G) \in \mathbb{N} \to \mathbb{N}$

proof
prove;
■

theorem rule NatUOPApp
$\forall\, b : \mathbb{N} \to ((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ a : \mathbb{N} \bullet b\, a \in (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
prove;
■

theorem disabled rule UFExt
$\forall\, F, G : \mathbb{N} \to \mathbb{N} \bullet F = G \Leftrightarrow (\forall\, x : \mathbb{N} \bullet F\, x = G\, x)$

proof
use funExtensionality$[\mathbb{N},\ \mathbb{N}][f\ :=\ F,\ g\ :=\ G]$;
prove;
■

2 Section 2: functions and operators

2.1 The identity and the constants

$$funI : \mathbb{N} \to \mathbb{N}$$

$\langle\!\langle$ rule IVal $\rangle\!\rangle$
$\forall x : \mathbb{N} \bullet funI\ x = x$

proof [funI$domainCheck]
prove;
■

$IWit == (\lambda x : \mathbb{N} \bullet x)$

theorem IWitFun
$IWit \in \mathbb{N} \to \mathbb{N}$

proof
invoke IWit;
apply FunIntro to predicate $(\lambda\ x : \mathbb{N} \bullet x) \in \mathbb{N} \to \mathbb{N}$;
prove by reduce;
with enabled (*RelPow*) *prove by reduce*;
■

theorem rule IWitVal
$\forall x : \mathbb{N} \bullet IWit\ x = x$

proof
prove;
use IWitFun;
use pairInFunction$[\mathbb{N}, \mathbb{N}][f := IWit, y := x]$;
invoke IWit;
prove by reduce;
■

theorem IEx
$\exists funI : \mathbb{N} \to \mathbb{N} \bullet \forall x : \mathbb{N} \bullet funI\ x = x$

proof
instantiate funI $==$ *IWit*;
use IWitFun;
prove;
■

$$cst : \mathbb{N} \to (\mathbb{N} \to \mathbb{N})$$

$\langle\!\langle$ rule CstVal $\rangle\!\rangle$
$\forall n, x : \mathbb{N} \bullet cst\ n\ x = n$

proof [cst$domainCheck]
prove;
■

$CstWit == (\lambda n : \mathbb{N} \bullet (\lambda x : \mathbb{N} \bullet n))$

theorem CstWitFun
$CstWit \in \mathbb{N} \to (\mathbb{N} \to \mathbb{N})$

proof
invoke CstWit;
apply FunIntro to predicate $(\lambda\ n : \mathbb{N} \bullet (\lambda\ x : \mathbb{N} \bullet n)) \in \mathbb{N} \to (\mathbb{N} \to \mathbb{N})$;
prove by reduce;
with enabled (*RelPow*) *prove by reduce*;
■

theorem CstWitApp
$\forall\, n : \mathbb{N} \bullet CstWit\ n = (\lambda\, x : \mathbb{N} \bullet n)$

proof
prove;
use CstWitFun;
use pairInFunction$[\mathbb{N},\ \mathbb{N} \to \mathbb{N}][f := CstWit,\ x := n,\ y := (\lambda\ x : \mathbb{N} \bullet n)]$;
invoke CstWit;
prove by reduce;
■

theorem rule CstWitVal
$\forall\, n, x : \mathbb{N} \bullet CstWit\ n\ x = n$

proof
prove;
use CstWitApp;
prove;
equality substitute CstWit n;
prove;
■

theorem CstEx
$\exists\, cst : \mathbb{N} \to (\mathbb{N} \to \mathbb{N}) \bullet \forall\, n, x : \mathbb{N} \bullet cst\ n\ x = n$

proof
instantiate cst == *CstWit*;
use CstWitFun;
prove;
■

2.2 Arithmetic functions

$funS : \mathbb{N} \to \mathbb{N}$

$\langle\!\langle$ rule SVal $\rangle\!\rangle$
$\forall\, x : \mathbb{N} \bullet funS\ x = x + 1$

proof [funS$domainCheck]
prove;
■

$SWit == (\lambda\, x : \mathbb{N} \bullet x + 1)$

theorem SWitFun
$SWit \in \mathbb{N} \to \mathbb{N}$

proof
invoke SWit;
apply FunIntro to predicate $(\lambda\ x : \mathbb{N} \bullet x + 1) \in \mathbb{N} \to \mathbb{N}$;
prove by reduce;
with enabled (*RelPow*) *prove by reduce*;
■

theorem rule SWitVal
$\forall x : \mathbb{N} \bullet SWit\ x = x + 1$

proof
prove;
use SWitFun;
use pairInFunction$[\mathbb{N}, \mathbb{N}][f := SWit, x := x, y := x + 1]$;
invoke SWit;
prove by reduce;
■

theorem SEx
$\exists funS : \mathbb{N} \to \mathbb{N} \bullet \forall x : \mathbb{N} \bullet funS\ x = x + 1$

proof
instantiate funS $==$ *SWit*;
use SWitFun;
prove;
■

$funD : \mathbb{N} \to \mathbb{N}$

$\langle\!\langle$ rule DVal $\rangle\!\rangle$
$\forall x : \mathbb{N} \bullet funD\ x = 2 * x$

proof [funD$domainCheck]
prove;
■

$DWit == (\lambda x : \mathbb{N} \bullet 2 * x)$

theorem DWitFun
$DWit \in \mathbb{N} \to \mathbb{N}$

proof
invoke DWit;
apply FunIntro to predicate $(\lambda\ x : \mathbb{N} \bullet 2 * x) \in \mathbb{N} \to \mathbb{N}$;
prove by reduce;
with enabled (*RelPow*) *prove by reduce*;
■

theorem rule DWitVal
$\forall x : \mathbb{N} \bullet DWit\ x = 2 * x$

proof
prove;
use DWitFun;
use pairInFunction$[\mathbb{N}, \mathbb{N}][f := DWit, x := x, y := 2 * x]$;
invoke DWit;
prove by reduce;
■

theorem DEx
$\exists\, funD : \mathbb{N} \to \mathbb{N} \bullet \forall\, x : \mathbb{N} \bullet funD\, x = 2 * x$

proof
instantiate funD == DWit;
use DWitFun;
prove;
■

$funSq : \mathbb{N} \to \mathbb{N}$

$\langle\!\langle$ rule SqVal $\rangle\!\rangle$
$\forall\, x : \mathbb{N} \bullet funSq\, x = x * x$

proof [funSq$domainCheck]
prove;
■

$SqWit == (\lambda\, x : \mathbb{N} \bullet x * x)$

theorem SqWitFun
$SqWit \in \mathbb{N} \to \mathbb{N}$

proof
invoke SqWit;
apply FunIntro to predicate $(\lambda\ x : \mathbb{N} \bullet x * x) \in \mathbb{N} \to \mathbb{N}$;
prove by reduce;
with enabled (*RelPow*) *prove by reduce*;
■

theorem rule SqWitVal
$\forall\, x : \mathbb{N} \bullet SqWit\, x = x * x$

proof
prove;
use SqWitFun;
use pairInFunction$[\mathbb{N},\ \mathbb{N}][f := SqWit,\ x := x,\ y := x * x]$;
invoke SqWit;
prove by reduce;
■

theorem SqEx
$\exists\, funSq : \mathbb{N} \to \mathbb{N} \bullet \forall\, x : \mathbb{N} \bullet funSq\, x = x * x$

proof
instantiate funSq == SqWit;
use SqWitFun;
prove;
■

$funHf : \mathbb{N} \to \mathbb{N}$

$\langle\!\langle$ rule HfVal $\rangle\!\rangle$
$\forall\, x : \mathbb{N} \bullet funHf\, x = x \operatorname{div} 2$

proof [funHf$domainCheck]
prove;
■

$HfWit == (\lambda\, x : \mathbb{N} \bullet x \;\mathsf{div}\; 2)$

proof [HfWit$domainCheck]
prove;
■

theorem HfWitFun
$HfWit \in \mathbb{N} \to \mathbb{N}$

proof
invoke HfWit;
apply FunIntro to predicate $(\lambda\ x : \mathbb{N} \bullet x\ \mathsf{div}\ 2) \in \mathbb{N} \to \mathbb{N}$;
prove by reduce;
with enabled (*RelPow*) *prove by reduce*;
■

theorem rule HfWitVal
$\forall\, x : \mathbb{N} \bullet HfWit\, x = x \;\mathsf{div}\; 2$

proof
prove;
use HfWitFun;
use pairInFunction$[\mathbb{N},\ \mathbb{N}][f := HfWit,\ x := x,\ y := x\ \mathsf{div}\ 2]$;
invoke HfWit;
prove by reduce;
■

theorem HfEx
$\exists\, funHf : \mathbb{N} \to \mathbb{N} \bullet \forall\, x : \mathbb{N} \bullet funHf\, x = x \;\mathsf{div}\; 2$

proof
instantiate funHf $==$ *HfWit*;
use HfWitFun;
prove;
■

$funO : \mathbb{N} \to \mathbb{N}$

$\langle\!\langle$ rule OVal $\rangle\!\rangle$
$\forall\, x : \mathbb{N} \bullet funO\, x = \mathbf{if}\ x = 0\ \mathbf{then}\ 1\ \mathbf{else}\ 0$

proof [funO$domainCheck]
prove;
■

$OWit == (\lambda\, x : \mathbb{N} \bullet 0) \oplus \{0 \mapsto 1\}$

theorem OWitFun
$OWit \in \mathbb{N} \to \mathbb{N}$

proof
invoke OWit;
prove by reduce;
■

theorem rule OWitVal
$\forall x : \mathbb{N} \bullet OWit\, x = \textbf{if } x = 0 \textbf{ then } 1 \textbf{ else } 0$

proof
invoke OWit;
prove by reduce;
■

theorem OEx
$\exists funO : \mathbb{N} \to \mathbb{N} \bullet \forall x : \mathbb{N} \bullet funO\, x = \textbf{if } x = 0 \textbf{ then } 1 \textbf{ else } 0$

proof
instantiate funO == OWit;
use OWitFun;
prove;
■

$funSgn : \mathbb{N} \to \mathbb{N}$

$\langle\!\langle$ rule SgnVal $\rangle\!\rangle$
$\forall x : \mathbb{N} \bullet funSgn\, x = \textbf{if } x = 0 \textbf{ then } 0 \textbf{ else } 1$

proof [funSgn$domainCheck]
prove;
■

$SgnWit == (\lambda x : \mathbb{N} \bullet 1) \oplus \{0 \mapsto 0\}$

theorem SgnWitFun
$SgnWit \in \mathbb{N} \to \mathbb{N}$

proof
invoke SgnWit;
prove by reduce;
■

theorem rule SgnWitVal
$\forall x : \mathbb{N} \bullet SgnWit\, x = \textbf{if } x = 0 \textbf{ then } 0 \textbf{ else } 1$

proof
invoke SgnWit;
prove by reduce;
■

theorem SgnEx
$\exists funSgn : \mathbb{N} \to \mathbb{N} \bullet \forall x : \mathbb{N} \bullet funSgn\, x = \textbf{if } x = 0 \textbf{ then } 0 \textbf{ else } 1$

proof
instantiate funSgn == SgnWit;
use SgnWitFun;
prove;
■

2.3 Integer square root and the excess over a square

$$\begin{array}{|l} \mathit{funRt} : \mathbb{N} \to \mathbb{N} \\ \hline \langle\!\langle\, \mathrm{RtSpec} \,\rangle\!\rangle \\ \forall\, x : \mathbb{N} \bullet \mathit{funRt}\, x * \mathit{funRt}\, x \leq x \land x < (\mathit{funRt}\, x + 1) * (\mathit{funRt}\, x + 1) \end{array}$$

proof [funRt$domainCheck]
prove;
■

theorem RtUnique
$\forall\, r, s, x : \mathbb{N} \mid r * r \leq x \land x < (r+1) * (r+1) \land s * s \leq x \land x < (s+1) * (s+1) \bullet r = s$

proof
prove;
split $r < s$;
cases;
use *timesMonotonic2*$[a := r + 1,\ b := r + 1,\ c := s,\ d := s]$;
prove;
next;
split $s < r$;
cases;
use *timesMonotonic2*$[a := s + 1,\ b := s + 1,\ c := r,\ d := r]$;
prove;
next;
prove;
next;
■

$\mathit{RtExSet} == \{x : \mathbb{N} \mid \exists\, r : \mathbb{N} \bullet r * r \leq x \land x < (r+1) * (r+1)\}$

theorem rule RtExBase
$0 \in \mathit{RtExSet}$

proof
invoke *RtExSet*;
prove;
instantiate $r == 0$;
prove;
■

theorem rule RtExStep
$\forall\, x : \mathit{RtExSet} \bullet 1 + x \in \mathit{RtExSet}$

proof
invoke *RtExSet*;
prove;
split $x + 1 < (r + 1) * (r + 1)$;
cases;
instantiate $r__0 == r$;
prove;
next;
instantiate $r__1 == r + 1$;
prove;
next;
■

theorem RtExSub
$\mathbb{N} \subseteq RtExSet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq RtExSet$;
prove;
invoke RtExSet;
prove;
■

theorem RtExists
$\forall x : \mathbb{N} \bullet \exists r : \mathbb{N} \bullet r * r \leq x \land x < (r+1) * (r+1)$

proof
prove;
use RtExSub;
with enabled (*inPower*) *prove*;
instantiate $e == x$;
invoke RtExSet;
prove;
■

$RtWit == \{x, r : \mathbb{N} \mid r * r \leq x \land x < (r+1) * (r+1) \bullet (x, r)\}$

theorem RtWitFun
$RtWit \in \mathbb{N} \to \mathbb{N}$

proof
invoke RtWit;
apply FunIntro to predicate $\{ x, r : \mathbb{N} \mid r * r \leq x \land x < (r+1) * (r+1) \bullet (x, r) \} \in \mathbb{N} \to \mathbb{N}$;
with enabled (*RelPow*) *prove by reduce*;
use RtExists$[x := x]$;
use RtUnique$[r := y, s := z, x := x__0]$;
prove;
instantiate $y__0 == r$;
prove;
■

theorem rule RtWitLe
$\forall x : \mathbb{N} \bullet RtWit\, x * RtWit\, x \leq x$

proof
prove;
use RtExists$[x := x]$;
use RtWitFun;
prove;
use pairInFunction$[\mathbb{N}, \mathbb{N}][f := RtWit, x := x, y := r]$;
invoke RtWit;
prove by reduce;
■

theorem rule RtWitLt
$\forall x : \mathbb{N} \bullet x < (1 + RtWit\, x) * (1 + RtWit\, x)$

proof
prove;
use RtExists[$x := x$];
use RtWitFun;
prove;
use pairInFunction[$\mathbb{N}$, $\mathbb{N}$][$f := RtWit$, $x := x$, $y := r$];
invoke RtWit;
prove by reduce;
■

theorem RtEx
$\exists\, funRt : \mathbb{N} \to \mathbb{N} \bullet \forall\, x : \mathbb{N} \bullet funRt\, x * funRt\, x \leq x \land x < (funRt\, x + 1) * (funRt\, x + 1)$

proof
instantiate funRt $==$ *RtWit*;
use RtWitFun;
prove;
■

$\mathrm{Rt}(x) = r$ whenever $r^2 \leq x < (r+1)^2$.

theorem RtI
$\forall\, r, x : \mathbb{N} \mid r * r \leq x \land x < (r+1) * (r+1) \bullet funRt\, x = r$

proof
prove;
use RtSpec[$x := x$];
use RtUnique[$r := funRt\, x$, $s := r$, $x := x$];
use UFNonneg[$f := funRt$, $x := x$];
prove;
■

$funE : \mathbb{N} \to \mathbb{N}$

$\langle\!\langle$ disabled rule EVal $\rangle\!\rangle$
$\forall\, x : \mathbb{N} \bullet funE\, x = x - funRt\, x * funRt\, x$

proof [funE$domainCheck]
prove;
■

$EWit == (\lambda\, x : \mathbb{N} \bullet x - funRt\, x * funRt\, x)$

proof [EWit$domainCheck]
prove;
■

theorem EWitRel
$(\lambda\, x : \mathbb{N} \bullet x - funRt\, x * funRt\, x) \in \mathbb{P}(\mathbb{N} \times \mathbb{N})$

proof
with enabled (*inPower*) *prove by reduce*;
use RtSpec[$x := x$];
prove;
■

theorem EWitFun
$EWit \in \mathbb{N} \to \mathbb{N}$

proof
invoke EWit;
use EWitRel;
apply FunIntro to predicate $(\lambda\ x : \mathbb{N} \bullet x - funRt\, x * funRt\, x) \in \mathbb{N} \to \mathbb{N}$;
with enabled (*RelPow*) *prove by reduce*;
use RtSpec$[x := x]$;
prove;
■

theorem rule EWitVal
$\forall x : \mathbb{N} \bullet EWit\, x = x - funRt\, x * funRt\, x$

proof
prove;
use EWitFun;
use pairInFunction$[\mathbb{N}, \mathbb{N}][f := EWit, x := x, y := x - funRt\, x * funRt\, x]$;
invoke EWit;
prove by reduce;
■

theorem EEx
$\exists funE : \mathbb{N} \to \mathbb{N} \bullet \forall x : \mathbb{N} \bullet funE\, x = x - funRt\, x * funRt\, x$

proof
instantiate funE == *EWit*;
use EWitFun;
prove;
■

2.4 Operators

$F^+(x) = F(x) + x.$

$opPlusI : (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

⟪ rule PlusIVal ⟫
$\forall F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet opPlusI\, F\, x = F\, x + x$

proof [opPlusI$domainCheck]
prove;
■

theorem rule PlusIInner
$\forall F : \mathbb{N} \to \mathbb{N} \bullet (\lambda x : \mathbb{N} \bullet F\, x + x) \in \mathbb{N} \to \mathbb{N}$

proof
apply FunIntro to predicate $(\lambda\ x : \mathbb{N} \bullet F\, x + x) \in \mathbb{N} \to \mathbb{N}$;
with enabled (*RelPow*, *inPower*) *prove by reduce*;
■

$PlusIWit == (\lambda F : \mathbb{N} \to \mathbb{N} \bullet (\lambda x : \mathbb{N} \bullet F\, x + x))$

proof [PlusIWit$domainCheck]
prove;
■

theorem PlusIWitFun
$PlusIWit \in (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
invoke PlusIWit;
apply FunIntro to predicate $(\lambda\ F : \mathbb{N} \to \mathbb{N} \bullet (\lambda\ x : \mathbb{N} \bullet F\,x + x)) \in (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$;
with enabled $(RelPow,\ inPower)$ *prove by reduce*;
■

theorem PlusIWitApp
$\forall F : \mathbb{N} \to \mathbb{N} \bullet PlusIWit\ F = (\lambda x : \mathbb{N} \bullet F\,x + x)$

proof
use PlusIWitFun;
use pairInFunction$[\mathbb{N} \to \mathbb{N},\ \mathbb{N} \to \mathbb{N}][f := PlusIWit,\ x := F,\ y := (\lambda\ x : \mathbb{N} \bullet F\,x + x)]$;
invoke PlusIWit;
prove by reduce;
■

theorem rule PlusIWitVal
$\forall F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet PlusIWit\ F\,x = F\,x + x$

proof
prove;
use PlusIWitApp$[F := F]$;
use pairInFunction$[\mathbb{N},\ \mathbb{N}][f := (\lambda\ x : \mathbb{N} \bullet F\,x + x),\ x := x,\ y := F\,x + x]$;
prove;
■

theorem PlusIEx
$\exists\, opPlusI : (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}) \bullet \forall F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet opPlusI\ F\,x = F\,x + x$

proof
instantiate opPlusI $==$ *PlusIWit*;
use PlusIWitFun;
prove;
■

Binary operators are functions on pairs of one-argument functions: $F + G$ and the composition FG.

$opPlus : (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

$\langle\!\langle$ rule PlusVal $\rangle\!\rangle$
$\forall F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet opPlus(F, G)\,x = F\,x + G\,x$

proof [opPlus$domainCheck]
prove;
■

theorem rule PlusInner
$\forall F, G : \mathbb{N} \to \mathbb{N} \bullet (\lambda x : \mathbb{N} \bullet F\,x + G\,x) \in \mathbb{N} \to \mathbb{N}$

proof
apply FunIntro to predicate $(\lambda\ x : \mathbb{N} \bullet F\,x + G\,x) \in \mathbb{N} \to \mathbb{N}$;
with enabled $(RelPow,\ inPower)$ *prove by reduce*;
■

$PlusWit == (\lambda\, F, G : \mathbb{N} \to \mathbb{N} \bullet (\lambda\, x : \mathbb{N} \bullet F\, x + G\, x))$

proof [PlusWit$domainCheck]
prove;
■

theorem PlusWitFun
$PlusWit \in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
invoke PlusWit;
apply FunIntro to predicate $(\lambda\ F,\ G : \mathbb{N} \to \mathbb{N} \bullet (\lambda\ x : \mathbb{N} \bullet F\, x + G\, x)) \in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$;
with enabled $(RelPow,\ inPower)$ *prove by reduce*;
instantiate $F == x.1,\ G == x.2$;
prove;
■

theorem PlusWitApp
$\forall\, F, G : \mathbb{N} \to \mathbb{N} \bullet PlusWit(F, G) = (\lambda\, x : \mathbb{N} \bullet F\, x + G\, x)$

proof
use PlusWitFun;
use pairInFunction$[(\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}), \mathbb{N} \to \mathbb{N}][f := PlusWit,\ x := (F,\ G),\ y := (\lambda\ x : \mathbb{N} \bullet F\, x + G\, x)]$;
invoke PlusWit;
prove by reduce;
■

theorem rule PlusWitVal
$\forall\, F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet PlusWit(F, G)\, x = F\, x + G\, x$

proof
prove;
use PlusWitApp$[F := F,\ G := G]$;
use pairInFunction$[\mathbb{N},\ \mathbb{N}][f := (\lambda\ x : \mathbb{N} \bullet F\, x + G\, x),\ x := x,\ y := F\, x + G\, x]$;
prove;
■

theorem PlusEx
$\exists\, opPlus : (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}) \bullet \forall\, F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet opPlus(F, G)\, x = F\, x + G\, x$

proof
instantiate $opPlus == PlusWit$;
use PlusWitFun;
prove;
■

$comp : (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

$\langle\!\langle$ rule CompVal $\rangle\!\rangle$
$\forall\, F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet comp(F, G)\, x = F\, (G\, x)$

proof [comp$domainCheck]
prove;
■

theorem rule CompInner
$\forall F, G : \mathbb{N} \to \mathbb{N} \bullet (\lambda x : \mathbb{N} \bullet F(G x)) \in \mathbb{N} \to \mathbb{N}$

proof
apply FunIntro to predicate $(\lambda\ x : \mathbb{N} \bullet F(G x)) \in \mathbb{N} \to \mathbb{N}$;
with enabled $(RelPow,\ inPower)$ *prove by reduce*;
■

$CompWit == (\lambda F, G : \mathbb{N} \to \mathbb{N} \bullet (\lambda x : \mathbb{N} \bullet F(G x)))$

proof [CompWit$domainCheck]
prove;
■

theorem CompWitFun
$CompWit \in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
invoke CompWit;
apply FunIntro to predicate $(\lambda\ F,\ G : \mathbb{N} \to \mathbb{N} \bullet (\lambda\ x : \mathbb{N} \bullet F(G x))) \in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$;
with enabled $(RelPow,\ inPower)$ *prove by reduce*;
instantiate $F == x.1,\ G == x.2$;
prove;
■

theorem CompWitApp
$\forall F, G : \mathbb{N} \to \mathbb{N} \bullet CompWit(F, G) = (\lambda x : \mathbb{N} \bullet F(G x))$

proof
use CompWitFun;
use pairInFunction$[(\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}), \mathbb{N} \to \mathbb{N}][f := CompWit,\ x := (F,\ G),\ y := (\lambda\ x : \mathbb{N} \bullet F(G x))]$;
invoke CompWit;
prove by reduce;
■

theorem rule CompWitVal
$\forall F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet CompWit(F, G)\, x = F(G x)$

proof
prove;
use CompWitApp$[F := F,\ G := G]$;
use pairInFunction$[\mathbb{N},\ \mathbb{N}][f := (\lambda\ x : \mathbb{N} \bullet F(G x)),\ x := x,\ y := F(G x)]$;
prove;
■

theorem CompEx
$\exists comp : (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}) \bullet \forall F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet comp(F, G)\, x = F(G x)$

proof
instantiate comp $==$ *CompWit*;
use CompWitFun;
prove;
■

Iteration from a: $F^{\Box(a)}(0) = a$ and $F^{\Box(a)}(x+1) = F(F^{\Box(a)}(x))$ (in the defining equations Z/EVES's normal form $1 + x$ is used for $x + 1$).

$opBoxA : \mathbb{N} \to ((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}))$

$\langle\!\langle$ rule BoxA0 $\rangle\!\rangle$
$\forall\, a : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N} \bullet opBoxA\ a\ F\ 0 = a$
$\langle\!\langle$ rule BoxASuc $\rangle\!\rangle$
$\forall\, a : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet opBoxA\ a\ F\ (1 + x) = F\ (opBoxA\ a\ F\ x)$

proof [opBoxA$domainCheck]
prove;
■

theorem rule BoxAInner
$\forall\, a : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N} \bullet (\lambda\, n : \mathbb{N} \bullet iter[\mathbb{N}]\ n\ F\ a) \in \mathbb{N} \to \mathbb{N}$

proof
apply FunIntro to predicate $(\lambda\ n : \mathbb{N} \bullet iter[\mathbb{N}]\ n\ F\ a) \in \mathbb{N} \to \mathbb{N}$;
with enabled $(RelPow,\ inPower)$ *prove by reduce*;
■

theorem rule BoxAMid
$\forall\, a : \mathbb{N} \bullet (\lambda\, F : \mathbb{N} \to \mathbb{N} \bullet (\lambda\, n : \mathbb{N} \bullet iter[\mathbb{N}]\ n\ F\ a)) \in (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
apply FunIntro to predicate $(\lambda\ F : \mathbb{N} \to \mathbb{N} \bullet (\lambda\ n : \mathbb{N} \bullet iter[\mathbb{N}]\ n\ F\ a)) \in (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$;
with enabled $(RelPow,\ inPower)$ *prove by reduce*;
■

$BoxAWit == (\lambda\, a : \mathbb{N} \bullet (\lambda\, F : \mathbb{N} \to \mathbb{N} \bullet (\lambda\, n : \mathbb{N} \bullet iter[\mathbb{N}]\ n\ F\ a)))$

proof [BoxAWit$domainCheck]
prove;
■

theorem BoxAWitFun
$BoxAWit \in \mathbb{N} \to ((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}))$

proof
invoke BoxAWit;
apply FunIntro to predicate $(\lambda\ a : \mathbb{N} \bullet (\lambda\ F : \mathbb{N} \to \mathbb{N} \bullet (\lambda\ n : \mathbb{N} \bullet iter[\mathbb{N}]\ n\ F\ a))) \in \mathbb{N} \to ((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}))$;
with enabled $(RelPow,\ inPower)$ *prove by reduce*;
■

theorem BoxAWitApp
$\forall\, a : \mathbb{N} \bullet BoxAWit\ a = (\lambda\, F : \mathbb{N} \to \mathbb{N} \bullet (\lambda\, n : \mathbb{N} \bullet iter[\mathbb{N}]\ n\ F\ a))$

proof
use BoxAWitFun;
use pairInFunction$[\mathbb{N},\ (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})][f := BoxAWit,\ x := a,\ y := (\lambda\ F : \mathbb{N} \to \mathbb{N} \bullet (\lambda\ n : \mathbb{N} \bullet iter[\mathbb{N}]\ n\ F\ a))]$;
invoke BoxAWit;
prove by reduce;
■

theorem BoxAWitApp2
$\forall\, a : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N} \bullet \mathit{BoxAWit}\ a\ F = (\lambda\, n : \mathbb{N} \bullet \mathit{iter}[\mathbb{N}]\ n\ F\ a)$

proof
prove;
use *BoxAWitApp*$[a := a]$;
use *pairInFunction*$[\mathbb{N} \to \mathbb{N},\ \mathbb{N} \to \mathbb{N}][f := (\lambda\ F : \mathbb{N} \to \mathbb{N} \bullet (\lambda\ n : \mathbb{N} \bullet \mathit{iter}[\mathbb{N}]\ n\ F\ a)),\ x := F,\ y := (\lambda\ n : \mathbb{N} \bullet \mathit{iter}[\mathbb{N}]\ n\ F\ a)]$;
prove;
■

theorem rule BoxAWitVal
$\forall\, a : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N};\ n : \mathbb{N} \bullet \mathit{BoxAWit}\ a\ F\ n = \mathit{iter}[\mathbb{N}]\ n\ F\ a$

proof
prove;
use *BoxAWitApp2*$[a := a,\ F := F]$;
use *pairInFunction*$[\mathbb{N},\ \mathbb{N}][f := (\lambda\ n : \mathbb{N} \bullet \mathit{iter}[\mathbb{N}]\ n\ F\ a),\ x := n,\ y := \mathit{iter}[\mathbb{N}]\ n\ F\ a]$;
prove;
■

theorem BoxAEx
$\exists\, \mathit{opBoxA} : \mathbb{N} \to ((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet (\forall\, a : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N} \bullet \mathit{opBoxA}\ a\ F\ 0 = a) \land (\forall\, a : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet \mathit{opBoxA}\ a\ F\ (1 + x) = F\ (\mathit{opBoxA}\ a\ F\ x))$

proof
instantiate *opBoxA* == *BoxAWit*;
use *BoxAWitFun*;
prove;
■

Pure iteration $F^{\Box}$, i.e. iteration from 0 (its witness is $F^{\Box(0)}$).

$\mathit{opBox} : (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

$\langle\!\langle$ rule Box0 $\rangle\!\rangle$
$\forall\, F : \mathbb{N} \to \mathbb{N} \bullet \mathit{opBox}\ F\ 0 = 0$
$\langle\!\langle$ rule BoxSuc $\rangle\!\rangle$
$\forall\, F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet \mathit{opBox}\ F\ (1 + x) = F\ (\mathit{opBox}\ F\ x)$

proof [opBox$domainCheck]
prove;
■

theorem BoxEx
$\exists\, \mathit{opBox} : (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}) \bullet (\forall\, F : \mathbb{N} \to \mathbb{N} \bullet \mathit{opBox}\ F\ 0 = 0) \land (\forall\, F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet \mathit{opBox}\ F\ (1 + x) = F\ (\mathit{opBox}\ F\ x))$

proof
instantiate *opBox* == *opBoxA* 0;
prove;
■

2.5 Bases and closures

A basis is given by a set I of initial functions, a set U of unary operators and a set B of binary operators; its closure is the least set containing I and closed under U and B, i.e. the intersection of all such sets. First, the sets closed under U and those closed under B.

$UClosed : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{P}(\mathbb{N} \to \mathbb{N}))$

⟪ disabled rule UClosedDef ⟫
$\forall\, U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet UClosed\ U = \{X : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \mid \forall\, u : U;\ F : X \bullet u\, F \in X\}$

proof [UClosed$domainCheck]
prove;
■

$UClosedWit == (\lambda\, U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \{X : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \mid \forall\, u : U;\ F : X \bullet u\, F \in X\})$

proof [UClosedWit$domainCheck]
prove;
■

theorem UClosedWitFun
$UClosedWit \in \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{P}(\mathbb{N} \to \mathbb{N}))$

proof
invoke UClosedWit;
apply FunIntro to predicate $(\lambda\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \{ X : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \mid \forall\ u : U;\ F : X \bullet u\, F \in X \}) \in \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{P}(\mathbb{N} \to \mathbb{N}))$;
with enabled (*RelPow*, *inPower*) *prove by reduce*;
■

theorem disabled rule UClosedWitVal
$\forall\, U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet UClosedWit\ U = \{X : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \mid \forall\, u : U;\ F : X \bullet u\, F \in X\}$

proof
use UClosedWitFun;
use pairInFunction$[\mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})), \mathbb{P}(\mathbb{P}(\mathbb{N} \to \mathbb{N}))][f := UClosedWit,\ x := U,\ y := \{ X : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \mid \forall\ u : U;\ F : X \bullet u\, F \in X \}]$;
invoke UClosedWit;
prove by reduce;
■

theorem UClosedEx
$\exists\, UClosed : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{P}(\mathbb{N} \to \mathbb{N})) \bullet \forall\, U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet UClosed\ U = \{X : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \mid \forall\, u : U;\ F : X \bullet u\, F \in X\}$

proof
instantiate UClosed $==$ *UClosedWit*;
use UClosedWitFun;
with enabled (*UClosedWitVal*) *prove*;
■

$BClosed : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{P}(\mathbb{N} \to \mathbb{N}))$

⟪ disabled rule BClosedDef ⟫
$\forall\, B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet BClosed\ B = \{X : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \mid \forall\, b : B;\ F, G : X \bullet b(F, G) \in X\}$

proof [BClosed$domainCheck]
prove;
■

$BClosedWit == (\lambda\, B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \{X : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \mid \forall\, b : B;\ F, G : X \bullet b(F, G) \in X\})$

proof [BClosedWit$domainCheck]
prove;
■

theorem BClosedWitFun
$BClosedWit \in \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{P}(\mathbb{N} \to \mathbb{N}))$

proof
invoke BClosedWit;
apply FunIntro to predicate $(\lambda\ B : \mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \{\ X : \mathbb{P}\ (\mathbb{N} \to \mathbb{N}) \mid \forall\ b : B;\ F, G : X \bullet b(F, G) \in X\ \}) \in \mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}\ (\mathbb{P}\ (\mathbb{N} \to \mathbb{N}))$;
with enabled (*RelPow*, *inPower*) *prove by reduce*;
■

theorem disabled rule BClosedWitVal
$\forall\, B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet BClosedWit\ B = \{X : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \mid \forall\, b : B;\ F, G : X \bullet b(F, G) \in X\}$

proof
use BClosedWitFun;
use pairInFunction$[\mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})), \mathbb{P}\ (\mathbb{P}\ (\mathbb{N} \to \mathbb{N}))][f := BClosedWit, x := B, y := \{\ X : \mathbb{P}\ (\mathbb{N} \to \mathbb{N}) \mid \forall\ b : B;\ F, G : X \bullet b(F, G) \in X\ \}]$;
invoke BClosedWit;
prove by reduce;
■

theorem BClosedEx
$\exists\, BClosed : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{P}(\mathbb{N} \to \mathbb{N})) \bullet \forall\, B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet BClosed\ B = \{X : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \mid \forall\, b : B;\ F, G : X \bullet b(F, G) \in X\}$

proof
instantiate BClosed $==$ *BClosedWit*;
use BClosedWitFun;
with enabled (*BClosedWitVal*) *prove*;
■

Domains and applications of the curried closure function.

theorem rule ClosApp1
$\forall\, c : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \to (\mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to (\mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{N} \to \mathbb{N})));\ I : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \bullet c\ I \in \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to (\mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{N} \to \mathbb{N}))$

proof
prove;
■

theorem rule ClosApp2
$\forall\, g : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to (\mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{N} \to \mathbb{N}));\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet g\ U \in \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{N} \to \mathbb{N})$

proof
prove;
■

theorem rule ClosDom2
$\forall\, g : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to (\mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{N} \to \mathbb{N})) \bullet \operatorname{dom} g = \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}))$

proof
prove;
■

theorem rule ClosApplies3
$\forall\, h : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{N} \to \mathbb{N});\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet$ *happlies*$*toB*

proof
prove;
■

The closure, as a curried function of I, U and B.

$clos : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \to (\mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to (\mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{N} \to \mathbb{N})))$

$\langle\!\langle$ ClosDef $\rangle\!\rangle$
$\forall\, I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet clos\ I\ U\ B = \bigcap\{X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\}$

proof [clos$domainCheck]
prove;
■

Consistency: the intersection is a set of one-argument functions, because the set of all one-argument functions belongs to the family.

theorem rule ClosFamUF
$\forall\, I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \mathbb{N} \to \mathbb{N} \in \{X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\}$

proof
with enabled (*UClosedDef*, *BClosedDef*) *prove by reduce*;
■

theorem rule ClosInPower
$\forall\, I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \bigcap\{X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\} \in \mathbb{P}(\mathbb{N} \to \mathbb{N})$

proof
prove;
with enabled (*inPower*, *inBigcap*) *prove*;
■

theorem rule ClosL3
$\forall\, I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet (\lambda\, B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \bigcap\{X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\}) \in \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{N} \to \mathbb{N})$

proof
apply FunIntro to predicate $(\lambda\ B : \mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \bigcap \{\ X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\ \}) \in \mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}\ (\mathbb{N} \to \mathbb{N})$;
with enabled $(RelPow,\ inPower)$ *prove*;
■

theorem rule ClosL2
$\forall I : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \bullet (\lambda U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet (\lambda B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \bigcap\{X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\})) \in \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to (\mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{N} \to \mathbb{N}))$

proof
apply FunIntro to predicate $(\lambda\ U : \mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet (\lambda\ B : \mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \bigcap \{\ X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\ \})) \in \mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to (\mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}\ (\mathbb{N} \to \mathbb{N}))$;
with enabled $(RelPow,\ inPower)$ *prove*;
■

$ClosWit == (\lambda I : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \bullet (\lambda U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet (\lambda B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \bigcap\{X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\})))$

proof [ClosWit$domainCheck]
prove;
■

theorem ClosWitFun
$ClosWit \in \mathbb{P}(\mathbb{N} \to \mathbb{N}) \to (\mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to (\mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{N} \to \mathbb{N})))$

proof
invoke ClosWit;
apply FunIntro to predicate $(\lambda\ I : \mathbb{P}\ (\mathbb{N} \to \mathbb{N}) \bullet (\lambda\ U : \mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet (\lambda\ B : \mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \bigcap \{\ X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\ \})))\ \in \mathbb{P}\ (\mathbb{N} \to \mathbb{N}) \to (\mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to (\mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}\ (\mathbb{N} \to \mathbb{N})))$;
with enabled $(RelPow,\ inPower)$ *prove*;
■

theorem ClosWitApp1
$\forall I : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \bullet ClosWit\ I = (\lambda U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet (\lambda B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \bigcap\{X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\}))$

proof
use ClosWitFun;
use pairInFunction$[\mathbb{P}\ (\mathbb{N} \to \mathbb{N}),\ \mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to (\mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}\ (\mathbb{N} \to \mathbb{N}))][f := ClosWit,\ x := I,\ y := (\lambda\ U : \mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet (\lambda\ B : \mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \bigcap \{\ X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\ \}))]$;
invoke ClosWit;
prove;
■

theorem ClosWitApp2
$\forall I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet ClosWit\ I\ U = (\lambda B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \bigcap\{X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\})$

proof
use *ClosWitApp*1[$I := I$];
use *pairInFunction*[$\mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})), \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{N} \to \mathbb{N})$][$f := (\lambda\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet (\lambda\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \bigcap \{\ X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\ \}), x := U, y := (\lambda\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \bigcap \{\ X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\ \})$];
prove;
■

theorem rule ClosWitVal
$\forall I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet$
$ClosWit\ I\ U\ B = \bigcap\{X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\}$

proof
use *ClosWitApp*2[$I := I, U := U$];
use *pairInFunction*[$\mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})), \mathbb{P}(\mathbb{N} \to \mathbb{N})$][$f := (\lambda\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \bigcap \{\ X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\ \}), x := B, y := \bigcap \{\ X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\ \}$];
prove;
■

theorem ClosEx
$\exists\ clos : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \to (\mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to (\mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{N} \to \mathbb{N}))) \bullet \forall I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet$
$clos\ I\ U\ B = \bigcap\{X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\}$

proof
instantiate *clos* == *ClosWit*;
use *ClosWitFun*;
prove;
■

2.6 The closure rules

theorem rule ClosInit
$\forall I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ F : \mathbb{N} \to \mathbb{N} \mid F \in I \bullet F \in clos\ I\ U\ B$

proof
prove;
use *ClosDef*[$I := I, U := U, B := B$];
prove;
with enabled (*inBigcap*, *subsetDef*, *inPower*) *prove*;
instantiate *e__2* == *F*;
prove;
■

theorem disabled rule ClosMem
$\forall I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ F : \mathbb{P}(\mathbb{Z} \times \mathbb{Z}) \bullet F \in clos\ I\ U\ B \Leftrightarrow (\forall\ Y : \{X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\} \bullet F \in Y)$

proof
use *ClosDef*[$I := I, U := U, B := B$];
prove;
■

theorem rule ClosUnop
$\forall I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ u : (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N});\ F : \mathbb{N} \to \mathbb{N} \mid u \in U \wedge F \in clos\ I\ U\ B \bullet u\ F \in clos\ I\ U\ B$

proof
with enabled (*ClosMem*) *prove*;
instantiate $Y__0 == Y$;
with enabled (*UClosedDef*) *prove*;
instantiate $u__0 == u$, $F__0 == F$;
prove;
■

theorem rule ClosBinop
$\forall I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ b : (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N});\ F, G : \mathbb{N} \to \mathbb{N} \mid b \in B \land F \in clos\ I\ U\ B \land G \in clos\ I\ U\ B \bullet b(F, G) \in clos\ I\ U\ B$

proof
with enabled (*ClosMem*) *prove*;
instantiate $Y__0 == Y$, $Y__1 == Y$;
with enabled (*BClosedDef*) *prove*;
instantiate $b__0 == b$, $F__0 == F$, $G__0 == G$;
prove;
■

The closure is contained in every closed set containing I.

theorem ClosInPow
$\forall I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet clos\ I\ U\ B \in \mathbb{P}(\mathbb{N} \to \mathbb{N})$

proof
use *applyInRanFun*$[\mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})),\ \mathbb{P}\ (\mathbb{N} \to \mathbb{N})][f := clos\ I\ U,\ a := B]$;
prove;
■

theorem ClosElemUF
$\forall I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ F : \mathbb{P}(\mathbb{Z} \times \mathbb{Z}) \mid F \in clos\ I\ U\ B \bullet F \in \mathbb{N} \to \mathbb{N}$

proof
use *applyInRanFun*$[\mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})),\ \mathbb{P}\ (\mathbb{N} \to \mathbb{N})][f := clos\ I\ U,\ a := B]$;
prove;
with enabled (*inPower*) *prove*;
■

theorem ClosLeast
$\forall I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ Z : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \mid Z \in UClosed\ U \land Z \in BClosed\ B \land I \subseteq Z \bullet clos\ I\ U\ B \subseteq Z$

proof
use *ClosDef*$[I := I,\ U := U,\ B := B]$;
use *BigcapSub*$[\mathbb{N} \to \mathbb{N}][T := \{\ X :\ UClosed\ U \cap BClosed\ B \mid I \subseteq X\ \},\ Y := Z]$;
prove;
■

2.7 Arithmetic facts about Rt and E used silently in the paper

theorem RtSqAdd
$\forall\, m, k : \mathbb{N} \mid k \leq 2 * m \bullet \mathit{funRt}\,(m * m + k) = m$

proof
prove;
use $\mathit{RtI}[r := m,\ x := m * m + k]$;
prove;
■

theorem ESqAdd
$\forall\, m, k : \mathbb{N} \mid k \leq 2 * m \bullet \mathit{funE}\,(m * m + k) = k$

proof
prove;
use $\mathit{RtSqAdd}[m := m,\ k := k]$;
with enabled (EVal) *prove*;
■

theorem rule RtSq
$\forall\, m : \mathbb{N} \bullet \mathit{funRt}\,(m * m) = m$

proof
prove;
use $\mathit{RtSqAdd}[m := m,\ k := 0]$;
prove;
■

theorem ELe
$\forall\, x : \mathbb{N} \bullet \mathit{funE}\, x \leq x$

proof
prove;
use $\mathit{RtSpec}[x := x]$;
with enabled (EVal) *prove*;
■

theorem rule E0
$\mathit{funE}\, 0 = 0$

proof
use $\mathit{ESqAdd}[m := 0,\ k := 0]$;
prove;
■

theorem rule E1
$\mathit{funE}\, 1 = 0$

proof
use $\mathit{ESqAdd}[m := 1,\ k := 0]$;
prove;
■

theorem rule E2
$\mathit{funE}\, 2 = 1$

proof
use $ESqAdd[m := 1,\ k := 1]$;
prove;
■

theorem rule E3
$funE\,3 = 2$

proof
use $ESqAdd[m := 1,\ k := 2]$;
prove;
■

theorem rule E4
$funE\,4 = 0$

proof
use $ESqAdd[m := 2,\ k := 0]$;
prove;
■

3 Section 6: iteration and the unary operator F^+

The closure of the basis $\mathcal{F} = \langle \bar{1}, E, F^+, FG, F^\square \rangle$.

$C6 == clos\,\{cst\,1, funE\}\,\{opPlusI, opBox\}\,\{comp\}$

proof [C6$domainCheck]
prove;
■

theorem disabled rule C6Def
$C6 = clos\,\{cst\,1, funE\}\,\{opPlusI, opBox\}\,\{comp\}$

proof
invoke $C6$;
prove;
■

theorem disabled rule C6UF
$\forall F : \mathbb{P}(\mathbb{Z} \times \mathbb{Z}) \mid F \in C6 \bullet F \in \mathbb{N} \to \mathbb{N}$

proof
prove;
use $ClosElemUF[I := \{\ cst\,1,\ funE\ \},\ U := \{\ opPlusI,\ opBox\ \},\ B := \{\ comp\ \},\ F := F]$;
with enabled $(C6Def)$ *prove*;
■

theorem rule C6One
$cst\,1 \in C6$

proof
with enabled $(C6Def)$ *prove*;
■

theorem rule C6E
$funE \in C6$

proof
with enabled $(C6Def)$ *prove*;
■

theorem rule C6PlusI
$\forall F : \mathbb{N} \to \mathbb{N} \mid F \in C6 \bullet opPlusI\ F \in C6$

proof
with enabled $(C6Def)$ *prove*;
■

theorem rule C6Box
$\forall F : \mathbb{N} \to \mathbb{N} \mid F \in C6 \bullet opBox\ F \in C6$

proof
with enabled $(C6Def)$ *prove*;
■

theorem rule C6Comp
$\forall F, G : \mathbb{N} \to \mathbb{N} \mid F \in C6 \land G \in C6 \bullet comp(F, G) \in C6$

proof
with enabled $(C6Def)$ *prove*;
■

3.1 Auxiliary functions of Section 6

$funM3 : \mathbb{N} \to \mathbb{N}$

$\langle\!\langle$ rule M3Val $\rangle\!\rangle$
$\forall x : \mathbb{N} \bullet funM3\ x = 3 * x$

proof [funM3$domainCheck]
prove;
■

$M3Wit == (\lambda x : \mathbb{N} \bullet 3 * x)$

theorem M3WitFun
$M3Wit \in \mathbb{N} \to \mathbb{N}$

proof
invoke $M3Wit$;
apply FunIntro to predicate $(\lambda\ x : \mathbb{N} \bullet 3 * x) \in \mathbb{N} \to \mathbb{N}$;
prove by reduce;
with enabled $(RelPow)$ *prove by reduce*;
■

theorem rule M3WitVal
$\forall x : \mathbb{N} \bullet M3Wit\ x = 3 * x$

proof
prove;
use $M3\,WitFun$;
use $pairInFunction[\mathbb{N},\ \mathbb{N}][f\ :=\ M3\,Wit,\ x\ :=\ x,\ y\ :=\ 3\ *\ x]$;
invoke $M3\,Wit$;
prove by reduce;
■

theorem M3Ex
$\exists\, funM3 : \mathbb{N} \to \mathbb{N} \bullet \forall\, x : \mathbb{N} \bullet funM3\, x = 3 * x$

proof
instantiate $funM3\ ==\ M3\,Wit$;
use $M3\,WitFun$;
prove;
■

$R(x) = x + 2\,\mathrm{Rt}(x)$, as in Lemma 5.7.

$funR : \mathbb{N} \to \mathbb{N}$

$\langle\!\langle$ rule RVal $\rangle\!\rangle$
$\forall\, x : \mathbb{N} \bullet funR\, x = x + 2 * funRt\, x$

proof [funR$domainCheck]
prove;
■

$RWit == (\lambda\, x : \mathbb{N} \bullet x + 2 * funRt\, x)$

proof [RWit$domainCheck]
prove;
■

theorem RWitFun
$RWit \in \mathbb{N} \to \mathbb{N}$

proof
invoke $RWit$;
apply FunIntro to predicate $(\lambda\ x :\ \mathbb{N}\ \bullet\ x\ +\ 2\ *\ funRt\, x)\ \in\ \mathbb{N}\ \to\ \mathbb{N}$;
prove by reduce;
with enabled $(RelPow)$ *prove by reduce*;
■

theorem rule RWitVal
$\forall\, x : \mathbb{N} \bullet RWit\, x = x + 2 * funRt\, x$

proof
prove;
use $RWitFun$;
use $pairInFunction[\mathbb{N},\ \mathbb{N}][f\ :=\ RWit,\ x\ :=\ x,\ y\ :=\ x\ +\ 2\ *\ funRt\, x]$;
invoke $RWit$;
prove by reduce;
■

theorem REx
$\exists\, funR : \mathbb{N} \to \mathbb{N} \bullet \forall\, x : \mathbb{N} \bullet funR\, x = x + 2 * funRt\, x$

proof
instantiate funR == RWit;
use RWitFun;
prove;
■

The excess $E(x)$ vanishes exactly on the squares.

theorem ESquareIf
$\forall\, k : \mathbb{N} \bullet \mathit{funE}\,(k * k) = 0$

proof
prove;
use ESqAdd[$m := k,\ k := 0$];
prove;
■

theorem ESquareOnly
$\forall\, x : \mathbb{N} \mid \mathit{funE}\, x = 0 \bullet x = \mathit{funRt}\, x * \mathit{funRt}\, x$

proof
with enabled (*EVal*) *prove*;
■

theorem ESquare
$\forall\, x : \mathbb{N} \bullet \mathit{funE}\, x = 0 \Leftrightarrow (\exists\, k : \mathbb{N} \bullet x = k * k)$

proof
prove;
split funE x = 0;
cases;
use ESquareOnly[$x := x$];
prove;
instantiate k == funRt x;
prove;
next;
prove;
use ESquareIf[$k := k$];
prove;
next;
■

theorem rule OEVal
$\forall\, x : \mathbb{N} \bullet \mathit{funO}\,(\mathit{funE}\, x) = \textbf{if } (\exists\, k : \mathbb{N} \bullet x = k * k) \textbf{ then } 1 \textbf{ else } 0$

proof
prove;
use ESquare[$x := x$];
prove;
instantiate k__1 == k;
prove;
■

Q is the characteristic function of the squares.

$\mathit{funQ} : \mathbb{N} \to \mathbb{N}$

$\langle\!\langle$ rule QVal $\rangle\!\rangle$
$\forall\, x : \mathbb{N} \bullet \mathit{funQ}\, x = \textbf{if } (\exists\, k : \mathbb{N} \bullet x = k * k) \textbf{ then } 1 \textbf{ else } 0$

proof [funQ$domainCheck]
prove;
■

theorem QEx
$\exists\, funQ : \mathbb{N} \to \mathbb{N} \bullet \forall\, x : \mathbb{N} \bullet funQ\, x = \textbf{if } (\exists\, k : \mathbb{N} \bullet x = k * k) \textbf{ then } 1 \textbf{ else } 0$

proof
instantiate $funQ == comp(funO,\ funE)$;
prove;
■

3.2 First functions in $\mathrm{clos}\,\mathcal{F}$

theorem SExpr
$opPlusI\ (cst\ 1) = funS$

proof
apply UFExt to predicate $opPlusI\ (cst\ 1) = funS$;
prove;
■

theorem ZeroExpr
$comp(funE, cst\ 1) = cst\ 0$

proof
apply UFExt to predicate $comp(funE,\ cst\ 1) = cst\ 0$;
prove;
■

theorem DExpr
$opPlusI\ (opPlusI\ (cst\ 0)) = funD$

proof
apply UFExt to predicate $opPlusI\ (opPlusI\ (cst\ 0)) = funD$;
prove;
■

theorem M3Expr
$opPlusI\ funD = funM3$

proof
apply UFExt to predicate $opPlusI\ funD = funM3$;
prove;
■

theorem OExpr
$comp(comp(comp(funE, funD), funS), funSgn) = funO$

proof
apply UFExt to predicate $comp(comp(comp(funE,\ funD),\ funS),\ funSgn) = funO$;
prove;
■

theorem QExpr
$comp(funO, funE) = funQ$

proof
apply UFExt to predicate $comp(funO,\ funE) = funQ$;
prove;
■

Robinson's difference formula: if $b \leq a$ then $a - b = E((a+b)^2 + 3a + b + 1)$.

theorem RobDiffE
$\forall a, b : \mathbb{N} \mid b \leq a \bullet funE\,((a+b) * (a+b) + 3 * a + b + 1) = a - b$

proof
prove;
use ESqAdd$[m := a + b + 1,\ k := a - b]$;
prove;
■

3.3 Division with remainder

theorem DivLe
$\forall x, d, q : \mathbb{N} \mid 0 < d \land x < d * q + d \bullet x \operatorname{\mathsf{div}} d \leq q$

proof
prove;
■

theorem DivGe
$\forall x, d, q : \mathbb{N} \mid 0 < d \land d * q \leq x \bullet q \leq x \operatorname{\mathsf{div}} d$

proof
prove;
■

theorem DivMod
$\forall x, d, q, r : \mathbb{N} \mid 0 < d \land r < d \land x = d * q + r \bullet x \operatorname{\mathsf{div}} d = q \land x \operatorname{\mathsf{mod}} d = r$

proof
prove;
use DivLe$[x := x,\ d := d,\ q := q]$;
use DivGe$[x := x,\ d := d,\ q := q]$;
use divModRelation$[x := x,\ y := d]$;
with disabled (*divLowerBound*, *divUpperBound*) *prove*;
■

x mod 3.

$funMod3 : \mathbb{N} \to \mathbb{N}$

$\langle\!\langle$ rule Mod3Val $\rangle\!\rangle$
$\forall x : \mathbb{N} \bullet funMod3\, x = x \operatorname{\mathsf{mod}} 3$

proof [funMod3$domainCheck]
prove;
■

$Mod3Wit == (\lambda x : \mathbb{N} \bullet x \operatorname{\mathsf{mod}} 3)$

proof [Mod3Wit$domainCheck]
prove;
■

theorem Mod3WitFun
$Mod3\,Wit \in \mathbb{N} \to \mathbb{N}$

proof
invoke $Mod3\,Wit$;
apply FunIntro to predicate $(\lambda\ x : \mathbb{N} \bullet x \bmod 3) \in \mathbb{N} \to \mathbb{N}$;
prove by reduce;
with enabled (*RelPow*) *prove by reduce*;
■

theorem rule Mod3WitVal
$\forall x : \mathbb{N} \bullet Mod3\,Wit\ x = x \bmod 3$

proof
prove;
use $Mod3\,WitFun$;
use pairInFunction$[\mathbb{N}, \mathbb{N}][f := Mod3\,Wit, x := x, y := x \bmod 3]$;
invoke $Mod3\,Wit$;
prove by reduce;
■

theorem Mod3Ex
$\exists\, funMod3 : \mathbb{N} \to \mathbb{N} \bullet \forall x : \mathbb{N} \bullet funMod3\ x = x \bmod 3$

proof
instantiate $funMod3 == Mod3\,Wit$;
use $Mod3\,WitFun$;
prove;
■

theorem DivModOf
$\forall x, d : \mathbb{N} \mid 0 < d \bullet x = d * (x \operatorname{div} d) + x \bmod d \land 0 \leq x \bmod d \land x \bmod d < d$

proof
prove;
■

theorem Mod3Cases
$\forall n : \mathbb{N} \bullet (n \bmod 3 = 0 \land (1 + n) \bmod 3 = 1) \lor (n \bmod 3 = 1 \land (1 + n) \bmod 3 = 2) \lor (n \bmod 3 = 2 \land (1 + n) \bmod 3 = 0)$

proof
prove;
use $DivModOf[x := n, d := 3]$;
split $n \bmod 3 = 2$;
cases;
use $DivMod[x := 1 + n, d := 3, q := n \operatorname{div} 3 + 1, r := 0]$;
prove;
next;
use $DivMod[x := 1 + n, d := 3, q := n \operatorname{div} 3, r := n \bmod 3 + 1]$;
prove;
next;
■

3.4 First functions in $\mathrm{clos}\,\mathcal{F}$, continued

theorem SgnExpr
$opBox\,(cst\,1) = funSgn$

proof
apply UFExt to predicate $opBox\,(cst\,1) = funSgn$;
prove;
split $x = 0$;
cases;
prove;
next;
use $BoxSuc[F__0 := cst\,1,\ x := x - 1]$;
prove;
next;
■

$Mod3Set == \{n : \mathbb{N} \mid opBox\,(comp(comp(funE, funS), funS))\,n = n \bmod 3\}$

proof [Mod3Set$domainCheck]
prove;
■

theorem rule Mod3Base
$0 \in Mod3Set$

proof
invoke $Mod3Set$;
prove;
■

theorem rule Mod3Step
$\forall\, n : Mod3Set \bullet 1 + n \in Mod3Set$

proof
invoke $Mod3Set$;
prove;
use $Mod3Cases[n := n]$;
split $n \bmod 3 = 0$;
cases;
prove;
next;
split $n \bmod 3 = 1$;
cases;
prove;
next;
prove;
next;
■

theorem Mod3Sub
$\mathbb{N} \subseteq Mod3Set$

proof
apply natInduction to predicate $\mathbb{N} \subseteq Mod3Set$;
prove;
invoke $Mod3Set$;
prove;
■

theorem Mod3All
$\forall\, n : \mathbb{N} \bullet opBox\,(comp(comp(funE, funS), funS))\, n = n \bmod 3$

proof
prove;
use Mod3Sub;
invoke Mod3Set;
with enabled (*inPower*) *prove*;
instantiate e == n;
prove;
■

theorem Mod3Expr
$opBox\,(comp(comp(funE, funS), funS)) = funMod3$

proof
apply UFExt to predicate $opBox\,(comp(comp(funE,\ funS),\ funS)) = funMod3$;
prove;
use $Mod3All[n := x]$;
prove;
■

3.5 Membership in $\mathrm{clos}\,\mathcal{F}$

theorem SInExpr
$opPlusI\,(cst\,1) \in C6$

proof
prove;
■

theorem rule SIn
$funS \in C6$

proof
use SInExpr;
use SExpr;
prove;
■

theorem SgnInExpr
$opBox\,(cst\,1) \in C6$

proof
prove;
■

theorem rule SgnIn
$funSgn \in C6$

proof
use SgnInExpr;
use SgnExpr;
prove;
■

theorem ZeroInExpr
$comp(funE, cst\,1) \in C6$

proof
prove;
■

theorem rule ZeroIn
$cst\,0 \in C6$

proof
use ZeroInExpr;
use ZeroExpr;
prove;
■

theorem DInExpr
$opPlusI\,(opPlusI\,(cst\,0)) \in C6$

proof
prove;
■

theorem rule DIn
$funD \in C6$

proof
use DInExpr;
use DExpr;
prove;
■

theorem M3InExpr
$opPlusI\,funD \in C6$

proof
prove;
■

theorem rule M3In
$funM3 \in C6$

proof
use M3InExpr;
use M3Expr;
prove;
■

theorem OInExpr
$comp(comp(comp(funE, funD), funS), funSgn) \in C6$

proof
prove;
■

theorem rule OIn
$funO \in C6$

proof
use *OInExpr*;
use *OExpr*;
prove;
■

theorem QInExpr
$comp(funO, funE) \in C6$

proof
prove;
■

theorem rule QIn
$funQ \in C6$

proof
use *QInExpr*;
use *QExpr*;
prove;
■

theorem Mod3InExpr
$opBox\,(comp(comp(funE, funS), funS)) \in C6$

proof
prove;
■

theorem rule Mod3In
$funMod3 \in C6$

proof
use *Mod3InExpr*;
use *Mod3Expr*;
prove;
■

3.6 $R(x) = x + 2\,\mathrm{Rt}(x)$ and Sq (Lemma 5.7 technique)

theorem QSq
$\forall\, x, k : \mathbb{N} \mid x = k * k \bullet funQ\, x = 1$

proof
prove;
instantiate $k__0 == k$;
prove;
■

theorem NotSquare
$\forall\, m, x : \mathbb{N} \mid m * m < x \wedge x < (m+1) * (m+1) \bullet \neg\, (\exists\, k : \mathbb{N} \bullet x = k * k)$

proof
prove;
split $k \leq m$;
cases;
use *timesMonotonic2*$[a := k,\ b := k,\ c := m,\ d := m]$;
prove;
next;
use *timesMonotonic2*$[a := m + 1,\ b := m + 1,\ c := k,\ d := k]$;
prove;
next;
■

theorem QNonSq
$\forall\, m, x : \mathbb{N} \mid m * m < x \wedge x < (m+1) * (m+1) \bullet \mathit{funQ}\ x = 0$

proof
prove;
use *NotSquare*$[m := m,\ x := x]$;
prove;
instantiate $k__0 == k$;
prove;
■

theorem RStep
$\forall\, n : \mathbb{N} \bullet \mathit{opPlusI}\ (\mathit{comp}(\mathit{comp}(\mathit{comp}(\mathit{comp}(\mathit{funD}, \mathit{funQ}), \mathit{funS}), \mathit{funS}),$
$\mathit{funS}))\ (\mathit{funS}\ (\mathit{funR}\ n)) = \mathit{funR}\ (1 + n)$

proof
prove;
use *RtSpec*$[x := n]$;
split $(\mathit{funRt}\ n + 1) * (\mathit{funRt}\ n + 1) = n + 1$;
cases;
use *RtI*$[r := \mathit{funRt}\ n + 1,\ x := n + 1]$;
prove;
instantiate $k == 2 + \mathit{funRt}\ n$;
prove;
next;
use *RtI*$[r := \mathit{funRt}\ n,\ x := n + 1]$;
use *QNonSq*$[m := \mathit{funRt}\ n + 1,\ x := 4 + n + 2 * \mathit{funRt}\ n]$;
prove;
next;
■

$\mathit{RBoxSet} == \{n : \mathbb{N} \mid \mathit{opBox}\ (\mathit{comp}(\mathit{opPlusI}\ (\mathit{comp}(\mathit{comp}(\mathit{comp}(\mathit{comp}(\mathit{funD}, \mathit{funQ}), \mathit{funS}), \mathit{funS}), \mathit{funS})),$
$\mathit{funS}))\ n = \mathit{funR}\ n\}$

proof [RBoxSet$domainCheck]
prove;
■

theorem rule RBoxBase
$0 \in \mathit{RBoxSet}$

proof
invoke *RBoxSet*;
prove;
use *RtSq*$[m := 0]$;
prove;
■

theorem rule RBoxStep
$\forall\, n : RBoxSet \bullet 1 + n \in RBoxSet$

proof
invoke RBoxSet;
prove;
use RStep[$n := n$];
prove;
■

theorem RBoxSub
$\mathbb{N} \subseteq RBoxSet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq RBoxSet$;
prove;
invoke RBoxSet;
prove;
■

theorem RBoxAll
$\forall\, n : \mathbb{N} \bullet opBox\ (comp(opPlusI\ (comp(comp(comp(comp(funD, funQ), funS), funS), funS)), funS))\ n = funR\ n$

proof
prove;
use RBoxSub;
invoke RBoxSet;
with enabled (*inPower*) *prove*;
instantiate e $==$ n;
prove;
■

theorem RExpr
$opBox\ (comp(opPlusI\ (comp(comp(comp(comp(funD, funQ), funS), funS), funS)), funS)) = funR$

proof
apply UFExt to predicate $opBox\ (comp(opPlusI\ (comp(comp(comp(comp(funD,\ funQ),\ funS),\ funS),\ funS)),\ funS)) = funR$;
prove;
use RBoxAll[$n := x$];
prove;
■

$SqBoxSet == \{n : \mathbb{N} \mid opBox\ (comp(funS, funR))\ n = n * n\}$

proof [SqBoxSet$domainCheck]
prove;
■

theorem rule SqBoxBase
$0 \in SqBoxSet$

proof
invoke SqBoxSet;
prove;
■

theorem rule SqBoxStep
$\forall\, n : SqBoxSet \bullet 1 + n \in SqBoxSet$

proof
invoke SqBoxSet;
prove;
equality substitute opBox $(comp(funS,\ funR))\ n$;
rewrite;
■

theorem SqBoxSub
$\mathbb{N} \subseteq SqBoxSet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq SqBoxSet$;
prove;
invoke SqBoxSet;
prove;
■

theorem SqBoxAll
$\forall\, n : \mathbb{N} \bullet opBox\ (comp(funS, funR))\ n = n * n$

proof
prove;
use SqBoxSub;
invoke SqBoxSet;
with enabled (*inPower*) *prove*;
instantiate e $==$ n;
prove;
■

theorem SqExpr
$opBox\ (comp(funS, funR)) = funSq$

proof
apply UFExt to predicate opBox $(comp(funS,\ funR)) = funSq$;
prove;
use SqBoxAll$[n := x]$;
prove;
■

theorem RInExpr
$opBox\ (comp(opPlusI\ (comp(comp(comp(comp(funD, funQ), funS), funS), funS)), funS)) \in C6$

proof
prove;
■

theorem rule RIn
$funR \in C6$

proof
use RInExpr;
use RExpr;
prove;
■

theorem SqInExpr
$opBox\,(comp(funS, funR)) \in C6$

proof
prove;
■

theorem rule SqIn
$funSq \in C6$

proof
use SqInExpr;
use SqExpr;
prove;
■

Inflationary functions ($F(x) \geq x$ for all x), and pairs with $G(x)^2 \leq F(x)$.

$Infl == \{F : \mathbb{N} \to \mathbb{N} \mid \forall\, x : \mathbb{N} \bullet x \leq F\, x\}$

proof [Infl$domainCheck]
prove;
■

theorem rule InflLe
$\forall\, F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \mid F \in Infl \bullet x \leq F\, x$

proof
prove;
invoke Infl;
prove;
instantiate $x__0 == x$;
prove;
■

$SqBelow == \{F, G : \mathbb{N} \to \mathbb{N} \mid \forall\, x : \mathbb{N} \bullet G\, x * G\, x \leq F\, x\}$

proof [SqBelow$domainCheck]
prove;
■

theorem rule SqBelowLe
$\forall\, F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \mid (F, G) \in SqBelow \bullet G\, x * G\, x \leq F\, x$

proof
prove;
invoke SqBelow;
prove;
instantiate $x__0 == x$;
prove;
■

3.7 Lemma 6.2: $F^- = E\,S\,(M_3FE)^+(\mathrm{Sq}\,F^+)^+$

Definition 6.1 defines $F^-(x) = F(x) - x$, meaningful when $F(x) \geq x$; Lemmas 6.2 and 6.4 are stated accordingly: the closure contains a function with those values.

exprMinusI == $(\lambda\, F : \mathbb{N} \to \mathbb{N} \bullet$ *comp*(*comp*(*comp*(*funE*, *funS*), *opPlusI* (*comp*(*comp*(*funM*3, *F*), *funE*))), *opPlusI* (*comp*(*funSq*, *opPlusI F*))))

proof [exprMinusI$domainCheck]
prove;
■

theorem rule exprMinusIFun
$\mathit{exprMinusI} \in (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
invoke exprMinusI;
apply FunIntro to predicate (λ *F* : $\mathbb{N} \to \mathbb{N} \bullet$ *comp*(*comp*(*comp*(*funE*, *funS*), *opPlusI* (*comp*(*comp*(*funM*3, *F*), *funE*))), *opPlusI* (*comp*(*funSq*, *opPlusI F*)))) $\in (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$;
with enabled (*RelPow*, *inPower*) *prove by reduce*;
■

theorem disabled rule exprMinusIApp
$\forall\, F : \mathbb{N} \to \mathbb{N} \bullet \mathit{exprMinusI}\ F = \mathit{comp}(\mathit{comp}(\mathit{comp}(\mathit{funE}, \mathit{funS}), \mathit{opPlusI}\ (\mathit{comp}(\mathit{comp}(\mathit{funM}3, F), \mathit{funE}))), \mathit{opPlusI}\ (\mathit{comp}(\mathit{funSq}, \mathit{opPlusI}\ F)))$

proof
use exprMinusIFun;
use pairInFunction[$\mathbb{N} \to \mathbb{N}$, $\mathbb{N} \to \mathbb{N}$][*f* := *exprMinusI*, *x* := *F*, *y* := *comp*(*comp*(*comp*(*funE*, *funS*), *opPlusI* (*comp*(*comp*(*funM*3, *F*), *funE*))), *opPlusI* (*comp*(*funSq*, *opPlusI F*)))];
invoke exprMinusI;
prove by reduce;
■

theorem rule MinusIIn
$\forall\, F : \mathbb{N} \to \mathbb{N} \mid F \in C6 \bullet \mathit{exprMinusI}\ F \in C6$

proof
with enabled (*exprMinusIApp*) *prove*;
■

theorem rule MinusIVal
$\forall\, F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \mid x \leq F\,x \bullet \mathit{exprMinusI}\ F\ x = F\,x - x$

proof
with enabled (*exprMinusIApp*) *prove*;
use ESqAdd[*m* := *F x* + *x*, *k* := *x*];
use RobDiffE[*a* := *F x*, *b* := *x*];
prove;
■

theorem Lemma62
$\forall\, F : \mathbb{N} \to \mathbb{N} \mid F \in C6 \wedge F \in \mathit{Infl} \bullet \exists\, G : \mathbb{N} \to \mathbb{N} \mid G \in C6 \bullet \forall\, x : \mathbb{N} \bullet G\,x = F\,x - x$

proof
prove;
instantiate G == *exprMinusI F*;
prove;
■

3.8 Lemma 6.3: $\mathrm{Hf} = ((S\,\mathrm{Mod}_3^+)^\Box)^-$ and $\mathrm{Rt} = \mathrm{Hf}\,R^-$

HfAuxSet == $\{q : \mathbb{N} \mid opBox\,(comp(funS, opPlusI\,funMod3))\,(2 * q) = 3 * q \wedge opBox\,(comp(funS, opPlusI\,funMod3))\,(1 + 2 * q) = 1 + 3 * q\}$

proof [HfAuxSet$domainCheck]
prove;
■

theorem rule HfAuxBase
$0 \in$ *HfAuxSet*

proof
invoke HfAuxSet;
prove;
use DivMod[$x := 0,\ d := 3,\ q := 0,\ r := 0$];
prove;
■

theorem rule HfAuxStep
$\forall\, q : HfAuxSet \bullet 1 + q \in HfAuxSet$

proof
invoke HfAuxSet;
prove;
use BoxSuc[$F__0 := comp(funS,\ opPlusI\,funMod3),\ x := 1 + 2 * q$];
use BoxSuc[$F__0 := comp(funS,\ opPlusI\,funMod3),\ x := 2 + 2 * q$];
use DivMod[$x := 1 + 3 * q,\ d := 3,\ q := q,\ r := 1$];
use DivMod[$x := 3 + 3 * q,\ d := 3,\ q := q + 1,\ r := 0$];
prove;
■

theorem HfAuxSub
$\mathbb{N} \subseteq$ *HfAuxSet*

proof
apply natInduction to predicate $\mathbb{N} \subseteq HfAuxSet$;
prove;
invoke HfAuxSet;
prove;
■

theorem HfAuxAll
$\forall\, q : \mathbb{N} \bullet opBox\,(comp(funS, opPlusI\,funMod3))\,(2 * q) = 3 * q \wedge opBox\,(comp(funS, opPlusI\,funMod3))\,(1 + 2 * q) = 1 + 3 * q$

proof
prove;
use HfAuxSub;
invoke HfAuxSet;
with enabled (*inPower*) *prove*;
instantiate $e == q$;
prove;
■

theorem HfBoxVal
$\forall\, x : \mathbb{N} \bullet opBox\,(comp(funS, opPlusI\,funMod3))\,x = x + x \operatorname{div} 2$

proof
prove;
use DivModOf[$x := x,\ d := 2$];
use HfAuxAll[$q := x$ div 2];
split x mod $2 = 0$;
cases;
prove;
next;
prove;
next;
■

theorem HfExpr
$exprMinusI\ (opBox\ (comp(funS, opPlusI\ funMod3))) = funHf$

proof
apply UFExt to predicate $exprMinusI\ (opBox\ (comp(funS,\ opPlusI\ funMod3))) = funHf$;
prove;
use HfBoxVal[$x := x$];
use MinusIVal[$F := opBox\ (comp(funS,\ opPlusI\ funMod3)),\ x := x$];
prove;
■

theorem RtExpr
$comp(funHf, exprMinusI\ funR) = funRt$

proof
apply UFExt to predicate $comp(funHf,\ exprMinusI\ funR) = funRt$;
prove;
use MinusIVal[$F := funR,\ x := x$];
use DivMod[$x := 2 * funRt\ x,\ d := 2,\ q := funRt\ x,\ r := 0$];
use UFNonneg[$f := funRt,\ x := x$];
prove;
■

theorem HfInExpr
$exprMinusI\ (opBox\ (comp(funS, opPlusI\ funMod3))) \in C6$

proof
prove;
■

theorem rule HfIn
$funHf \in C6$

proof
use HfInExpr;
use HfExpr;
prove;
■

theorem RtInExpr
$comp(funHf, exprMinusI\ funR) \in C6$

proof
prove;
■

theorem rule RtIn
funRt $\in C6$

proof
use RtInExpr;
use RtExpr;
prove;
■

3.9 Lemma 6.4: $F \oplus G = (F^- E)^+ (\mathrm{Sq}\ G)^+$ and $F \ominus G = (F^+ E)^- (\mathrm{Sq}\ G)^+$

$(F \oplus G)(x) = F(x) + G(x)^2$ (Definition 6.1); $F \ominus G$, meaningful when $F(x) \geq G(x)^2$, is treated like F^-.

$opOplus : (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

$\langle\!\langle$ rule OplusVal $\rangle\!\rangle$
$\forall F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet opOplus(F, G)\, x = F\, x + G\, x * G\, x$

proof [opOplus$domainCheck]
prove;
■

theorem rule OplusInner
$\forall F, G : \mathbb{N} \to \mathbb{N} \bullet (\lambda x : \mathbb{N} \bullet F\, x + G\, x * G\, x) \in \mathbb{N} \to \mathbb{N}$

proof
apply FunIntro to predicate $(\lambda\ x : \mathbb{N} \bullet F\, x + G\, x * G\, x) \in \mathbb{N} \to \mathbb{N}$;
with enabled (*RelPow*, *inPower*) *prove by reduce*;
■

$OplusWit == (\lambda F, G : \mathbb{N} \to \mathbb{N} \bullet (\lambda x : \mathbb{N} \bullet F\, x + G\, x * G\, x))$

proof [OplusWit$domainCheck]
prove;
■

theorem OplusWitFun
$OplusWit \in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
invoke OplusWit;
apply FunIntro to predicate $(\lambda\ F,\ G : \mathbb{N} \to \mathbb{N} \bullet (\lambda\ x : \mathbb{N} \bullet F\, x + G\, x * G\, x)) \in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$;
with enabled (*RelPow*, *inPower*) *prove by reduce*;
instantiate $F == x.1,\ G == x.2$;
prove;
■

theorem OplusWitApp
$\forall F, G : \mathbb{N} \to \mathbb{N} \bullet OplusWit(F, G) = (\lambda x : \mathbb{N} \bullet F\, x + G\, x * G\, x)$

proof
use OplusWitFun;
use pairInFunction$[(\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}), \mathbb{N} \to \mathbb{N}][f := OplusWit,\ x := (F,\ G),\ y := (\lambda\ x : \mathbb{N} \bullet F\, x + G\, x * G\, x)]$;
invoke OplusWit;
prove by reduce;
■

theorem rule OplusWitVal
$\forall F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet \mathit{OplusWit}(F, G)\, x = F\, x + G\, x * G\, x$

proof
prove;
use $\mathit{OplusWitApp}[F := F,\ G := G]$;
use $\mathit{pairInFunction}[\mathbb{N},\ \mathbb{N}][f := (\lambda\ x : \mathbb{N} \bullet F\, x + G\, x * G\, x),\ x := x,\ y := F\, x + G\, x * G\, x]$;
prove;
■

theorem OplusEx
$\exists\, \mathit{opOplus} : (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}) \bullet \forall F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet \mathit{opOplus}(F, G)\, x = F\, x + G\, x * G\, x$

proof
instantiate $\mathit{opOplus} == \mathit{OplusWit}$;
use *OplusWitFun*;
prove;
■

$\mathit{exprOplus} == (\lambda F, G : \mathbb{N} \to \mathbb{N} \bullet \mathit{comp}(\mathit{opPlusI}\,(\mathit{comp}(\mathit{exprMinusI}\, F, \mathit{funE})), \mathit{opPlusI}\,(\mathit{comp}(\mathit{funSq}, G))))$

proof [exprOplus$domainCheck]
prove;
■

theorem rule exprOplusFun
$\mathit{exprOplus} \in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
invoke *exprOplus*;
apply *FunIntro to predicate* $(\lambda\ F,\ G : \mathbb{N} \to \mathbb{N} \bullet \mathit{comp}(\mathit{opPlusI}\,(\mathit{comp}(\mathit{exprMinusI}\, F,\ \mathit{funE})),\ \mathit{opPlusI}\,(\mathit{comp}(\mathit{funSq},\ G)))) \in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$;
with enabled $(\mathit{RelPow},\ \mathit{inPower})$ *prove*;
instantiate $F == x.1,\ G == x.2$;
prove;
■

theorem disabled rule exprOplusApp
$\forall F, G : \mathbb{N} \to \mathbb{N} \bullet \mathit{exprOplus}(F, G) = \mathit{comp}(\mathit{opPlusI}\,(\mathit{comp}(\mathit{exprMinusI}\, F, \mathit{funE})), \mathit{opPlusI}\,(\mathit{comp}(\mathit{funSq}, G)))$

proof
use *exprOplusFun*;
use $\mathit{pairInFunction}[(\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}),\ \mathbb{N} \to \mathbb{N}][f := \mathit{exprOplus},\ x := (F,\ G),\ y := \mathit{comp}(\mathit{opPlusI}\,(\mathit{comp}(\mathit{exprMinusI}\, F,\ \mathit{funE})),\ \mathit{opPlusI}\,(\mathit{comp}(\mathit{funSq},\ G)))]$;
invoke *exprOplus*;
prove by reduce;
■

$\mathit{exprOminus} == (\lambda F, G : \mathbb{N} \to \mathbb{N} \bullet \mathit{comp}(\mathit{exprMinusI}\,(\mathit{comp}(\mathit{opPlusI}\, F, \mathit{funE})), \mathit{opPlusI}\,(\mathit{comp}(\mathit{funSq}, G))))$

proof [exprOminus$domainCheck]
prove;
■

theorem rule exprOminusFun
$exprOminus \in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
invoke exprOminus;
apply FunIntro to predicate $(\lambda\ F,\ G : \mathbb{N} \to \mathbb{N} \bullet\ comp(exprMinusI\,(comp(opPlusI\,F,\ funE)),$
$opPlusI\,(comp(funSq,\ G)))) \in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$;
with enabled $(RelPow,\ inPower)$ *prove*;
instantiate $F == x.1,\ G == x.2$;
prove;
■

theorem disabled rule exprOminusApp
$\forall\, F, G : \mathbb{N} \to \mathbb{N} \bullet exprOminus(F, G) = comp(exprMinusI\,(comp(opPlusI\,F, funE)),$
$opPlusI\,(comp(funSq, G)))$

proof
use exprOminusFun;
use pairInFunction$[(\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}),\ \mathbb{N} \to \mathbb{N}][f := exprOminus,\ x := (F,\ G),\ y :=$
$comp(exprMinusI\,(comp(opPlusI\,F,\ funE)),\ opPlusI\,(comp(funSq,\ G)))]$;
invoke exprOminus;
prove by reduce;
■

theorem rule OplusIn
$\forall\, F, G : \mathbb{N} \to \mathbb{N} \mid F \in C6 \wedge G \in C6 \bullet exprOplus(F, G) \in C6$

proof
with enabled $(exprOplusApp)$ *prove*;
■

theorem rule OminusIn
$\forall\, F, G : \mathbb{N} \to \mathbb{N} \mid F \in C6 \wedge G \in C6 \bullet exprOminus(F, G) \in C6$

proof
with enabled $(exprOminusApp)$ *prove*;
■

theorem rule OplusExprVal
$\forall\, F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \mid x \leq F\,x \wedge x \leq G\,x \bullet exprOplus(F, G)\,x = F\,x + G\,x * G\,x$

proof
with enabled $(exprOplusApp)$ *prove*;
use ESqAdd$[m := G\,x,\ k := x]$;
use MinusIVal$[F := F,\ x := x]$;
prove;
■

theorem rule OminusExprVal
$\forall\, F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \mid x \leq G\,x \wedge G\,x * G\,x \leq F\,x \bullet exprOminus(F, G)\,x = F\,x - G\,x * G\,x$

proof
with enabled $(exprOminusApp)$ *prove*;
use ESqAdd$[m := G\,x,\ k := x]$;
use MinusIVal$[F := comp(opPlusI\,F,\ funE),\ x := G\,x * G\,x + x]$;
prove;
■

theorem OplusEq
$\forall F, G : \mathbb{N} \to \mathbb{N} \mid F \in \mathit{Infl} \land G \in \mathit{Infl} \bullet \mathit{exprOplus}(F, G) = \mathit{opOplus}(F, G)$

proof
apply UFExt to predicate $\mathit{exprOplus}(F,\ G) = \mathit{opOplus}(F,\ G)$;
prove;
■

theorem Lemma64Plus
$\forall F, G : \mathbb{N} \to \mathbb{N} \mid F \in C6 \land G \in C6 \land F \in \mathit{Infl} \land G \in \mathit{Infl} \bullet \mathit{opOplus}(F, G) \in C6$

proof
use $\mathit{OplusEq}[F := F,\ G := G]$;
prove;
equality substitute $\mathit{opOplus}(F,\ G)$;
rewrite;
■

theorem Lemma64Minus
$\forall F, G : \mathbb{N} \to \mathbb{N} \mid F \in C6 \land G \in C6 \land G \in \mathit{Infl} \land (F,$
$G) \in \mathit{SqBelow} \bullet \exists H : \mathbb{N} \to \mathbb{N} \mid H \in C6 \bullet \forall x : \mathbb{N} \bullet H\, x = F\, x - G\, x * G\, x$

proof
prove;
instantiate $H == \mathit{exprOminus}(F,\ G)$;
prove;
■

theorem Lemma64
$\forall F, G : \mathbb{N} \to \mathbb{N} \mid F \in C6 \land G \in C6 \land F \in \mathit{Infl} \land G \in \mathit{Infl} \bullet \mathit{opOplus}(F, G) \in C6 \land ((F,$
$G) \in \mathit{SqBelow} \Rightarrow (\exists H : \mathbb{N} \to \mathbb{N} \mid H \in C6 \bullet \forall x : \mathbb{N} \bullet H\, x = F\, x - G\, x * G\, x))$

proof
use $\mathit{Lemma64Plus}[F := F,\ G := G]$;
use $\mathit{Lemma64Minus}[F := F,\ G := G]$;
prove;
instantiate $H__0 == H$;
prove;
instantiate $x__0 == x$;
prove;
■

3.10 Lemma 6.5: the predecessor $\hat{P}$ and the product $F \otimes G$

theorem MulNonneg
$\forall a, b : \mathbb{N} \bullet a * b \geq 0$

proof
prove;
■

theorem SqProd
$\forall a, b : \mathbb{Z} \bullet (a * b) * (a * b) = a * (a * (b * b))$

proof
prove;
■

theorem RtSqProd
$\forall\, a, b : \mathbb{N} \bullet \mathit{funRt}\,(a * (a * (b * b))) = a * b$

proof
prove;
use $\mathit{RtI}[r := a * b,\ x := a * (a * (b * b))]$;
use $\mathit{MulNonneg}[a := a,\ b := b]$;
prove;
■

theorem LeSq
$\forall\, n : \mathbb{N} \bullet n \leq n * n$

proof
prove;
split $n = 0$;
cases;
prove;
next;
use $\mathit{timesMonotonic}1[a := n,\ b := 1,\ c := n]$;
prove;
next;
■

$\hat{P} = ((\mathrm{Sq}^+)^+)^+ \ominus S$ is a predecessor: $\hat{P}(x+1) = x$.

$\mathit{exprPhat} == \mathit{exprOminus}(\mathit{opPlusI}\,(\mathit{opPlusI}\,(\mathit{opPlusI}\,\mathit{funSq})), \mathit{funS})$

proof [exprPhat$domainCheck]
prove;
■

theorem rule PhatFun
$\mathit{exprPhat} \in \mathbb{N} \to \mathbb{N}$

proof
invoke *exprPhat*;
prove;
■

theorem rule PhatIn
$\mathit{exprPhat} \in C6$

proof
invoke *exprPhat*;
prove;
■

theorem rule PhatVal
$\forall\, x : \mathbb{N} \bullet \mathit{exprPhat}\,(1 + x) = x$

proof
prove;
invoke *exprPhat*;
use $\mathit{OminusExprVal}[F := \mathit{opPlusI}\,(\mathit{opPlusI}\,(\mathit{opPlusI}\,\mathit{funSq})),\ G := \mathit{funS},\ x := 1 + x]$;
prove;
■

$(F \otimes G)(x) = F(x)\, G(x)$ (Definition 6.1), and its expression in Lemma 6.5.

$$opOtimes : (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$$

$\langle\!\langle$ rule OtimesVal $\rangle\!\rangle$
$\forall F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet opOtimes(F, G)\, x = F\, x * G\, x$

proof [opOtimes$domainCheck]
prove;
■

theorem rule OtimesInner
$\forall F, G : \mathbb{N} \to \mathbb{N} \bullet (\lambda\, x : \mathbb{N} \bullet F\, x * G\, x) \in \mathbb{N} \to \mathbb{N}$

proof
apply FunIntro to predicate $(\lambda\ x : \mathbb{N} \bullet F\, x * G\, x) \in \mathbb{N} \to \mathbb{N}$;
with enabled $(RelPow,\ inPower)$ *prove by reduce*;
■

$OtimesWit == (\lambda\, F, G : \mathbb{N} \to \mathbb{N} \bullet (\lambda\, x : \mathbb{N} \bullet F\, x * G\, x))$

proof [OtimesWit$domainCheck]
prove;
■

theorem OtimesWitFun
$OtimesWit \in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
invoke OtimesWit;
apply FunIntro to predicate $(\lambda\ F,\ G : \mathbb{N} \to \mathbb{N} \bullet (\lambda\ x : \mathbb{N} \bullet F\, x * G\, x)) \in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$;
with enabled $(RelPow,\ inPower)$ *prove by reduce*;
instantiate $F == x.1,\ G == x.2$;
prove;
■

theorem OtimesWitApp
$\forall F, G : \mathbb{N} \to \mathbb{N} \bullet OtimesWit(F, G) = (\lambda\, x : \mathbb{N} \bullet F\, x * G\, x)$

proof
use OtimesWitFun;
use pairInFunction$[(\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}), \mathbb{N} \to \mathbb{N}][f := OtimesWit,\ x := (F,\ G),\ y := (\lambda\ x : \mathbb{N} \bullet F\, x * G\, x)]$;
invoke OtimesWit;
prove by reduce;
■

theorem rule OtimesWitVal
$\forall F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet OtimesWit(F, G)\, x = F\, x * G\, x$

proof
prove;
use OtimesWitApp$[F := F,\ G := G]$;
use pairInFunction$[\mathbb{N},\ \mathbb{N}][f := (\lambda\ x : \mathbb{N} \bullet F\, x * G\, x),\ x := x,\ y := F\, x * G\, x]$;
prove;
■

theorem OtimesEx
$\exists\, opOtimes : (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}) \bullet \forall\, F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet opOtimes(F, G)\, x = F\, x * G\, x$

proof
instantiate opOtimes == OtimesWit;
use OtimesWitFun;
prove;
■

$exprOtimes == (\lambda\, F, G : \mathbb{N} \to \mathbb{N} \bullet comp(funRt, exprOminus(comp(comp(funHf, funHf), exprOminus(comp(comp(funSq, exprPhat), exprOminus(exprOplus(comp(comp(comp(funSq, funS), funSq), F), G), comp(funSq, F))), comp(funSq, G))), comp(funSq, F))))$

proof [exprOtimes$domainCheck]
prove;
■

theorem rule exprOtimesFun
$exprOtimes \in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
invoke exprOtimes;
apply FunIntro to predicate $(\lambda\ F,\ G : \mathbb{N} \to \mathbb{N} \bullet comp(funRt,\ exprOminus(comp(comp(funHf,\ funHf),\ exprOminus(comp(comp(funSq,\ exprPhat),\ exprOminus(exprOplus(comp(comp(comp(funSq,\ funS),\ funSq),\ F),\ G),\ comp(funSq,\ G))),\ comp(funSq,\ F)))) \in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$;
with enabled (*RelPow*, *inPower*) *prove*;
instantiate $F == x.1,\ G == x.2$;
prove;
■

theorem disabled rule exprOtimesApp
$\forall\, F, G : \mathbb{N} \to \mathbb{N} \bullet exprOtimes(F, G) = comp(funRt, exprOminus(comp(comp(funHf, funHf), exprOminus(comp(comp(funSq, exprPhat), exprOminus(exprOplus(comp(comp(comp(funSq, funS), funSq), F), G), comp(funSq, F))), comp(funSq, G))), comp(funSq, F)))$

proof
use exprOtimesFun;
use pairInFunction$[(\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}), \mathbb{N} \to \mathbb{N}][f := exprOtimes,\ x := (F,\ G),\ y := comp(funRt,\ exprOminus(comp(comp(funHf,\ funHf),\ exprOminus(comp(comp(funSq,\ exprPhat),\ exprOminus(exprOplus(comp(comp(comp(funSq,\ funS),\ funSq),\ F),\ G),\ comp(funSq,\ F))),\ comp(funSq,\ G))),\ comp(funSq,\ F)))]$;
invoke exprOtimes;
prove by reduce;
■

theorem rule OtimesIn
$\forall\, F, G : \mathbb{N} \to \mathbb{N} \mid F \in C6 \wedge G \in C6 \bullet exprOtimes(F, G) \in C6$

proof
with enabled (*exprOtimesApp*) *prove*;
■

theorem rule OtimesExprVal
$\forall\, F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \mid x \leq F\, x \wedge x \leq G\, x \bullet exprOtimes(F, G)\, x = F\, x * G\, x$

proof
use $LeSq[n := F\,x]$;
use $LeSq[n := G\,x]$;
use $LeSq[n := 1 + F\,x * F\,x]$;
use $MulNonneg[a := F\,x,\ b := F\,x]$;
use $MulNonneg[a := G\,x,\ b := G\,x]$;
use $MulNonneg[a := F\,x * F\,x,\ b := G\,x * G\,x]$;
use $MulNonneg[a := F\,x * F\,x,\ b := F\,x * F\,x]$;
with enabled (*exprOtimesApp*) *prove*;
use $DivMod[x := 4 * (F\,x * F\,x * F\,x * F\,x + F\,x * F\,x * G\,x * G\,x),\ d := 2,$
$q := 2 * (F\,x * F\,x * F\,x * F\,x + F\,x * F\,x * G\,x * G\,x),\ r := 0]$;
prove;
use $DivMod[x := 2 * (F\,x * F\,x * F\,x * F\,x + F\,x * F\,x * G\,x * G\,x),\ d := 2,$
$q := F\,x * F\,x * F\,x * F\,x + F\,x * F\,x * G\,x * G\,x,\ r := 0]$;
prove;
use $RtSqProd[a := F\,x,\ b := G\,x]$;
prove;
■

theorem OtimesEq
$\forall\, F, G : \mathbb{N} \to \mathbb{N} \mid F \in \mathit{Infl} \wedge G \in \mathit{Infl} \bullet \mathit{exprOtimes}(F, G) = \mathit{opOtimes}(F, G)$

proof
apply UFExt to predicate $\mathit{exprOtimes}(F,\ G) = \mathit{opOtimes}(F,\ G)$;
prove;
■

theorem Lemma65
$\forall\, F, G : \mathbb{N} \to \mathbb{N} \mid F \in C6 \wedge G \in C6 \wedge F \in \mathit{Infl} \wedge G \in \mathit{Infl} \bullet \mathit{opOtimes}(F, G) \in C6$

proof
use $OtimesEq[F := F,\ G := G]$;
prove;
equality substitute $\mathit{opOtimes}(F,\ G)$;
rewrite;
■

3.11 Lemma 6.6: $F + G$ and the difference $F - G$

theorem LeProd
$\forall\, x, a, b : \mathbb{N} \mid x \le a \wedge x \le b \bullet x \le a * b$

proof
prove;
split $x = 0$;
cases;
use $MulNonneg[a := a,\ b := b]$;
prove;
next;
use $timesMonotonic1[a := a,\ b := 1,\ c := b]$;
prove;
next;
■

theorem RtSqSum
$\forall\, a, b : \mathbb{N} \bullet \mathit{funRt}\,(2 * (a * b) + a * a + b * b) = a + b$

proof
prove;
use RtI[$r := a + b,\ x := 2 * (a * b) + a * a + b * b$];
prove;
■

$F + G = ((\mathrm{Rt}((D(F^+ \otimes G^+) \oplus F^+) \oplus G^+))^-)^-.$

exprPlus == $(\lambda\, F, G : \mathbb{N} \to \mathbb{N} \bullet$ *exprMinusI* (*exprMinusI* (*comp*(*funRt*,
exprOplus(*exprOplus*(*comp*(*funD*, *exprOtimes*(*opPlusI F*, *opPlusI G*)), *opPlusI F*), *opPlusI G*)))))

proof [exprPlus$domainCheck]
prove;
■

theorem rule exprPlusFun
exprPlus $\in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
invoke exprPlus;
apply FunIntro to predicate ($\lambda\ F,\ G : \mathbb{N} \to \mathbb{N} \bullet$ *exprMinusI* (*exprMinusI* (*comp*(*funRt*,
exprOplus(*exprOplus*(*comp*(*funD*, *exprOtimes*(*opPlusI F*, *opPlusI G*)), *opPlusI F*),
opPlusI G))))) $\in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$;
with enabled (*RelPow*, *inPower*) *prove*;
instantiate F == *x*.1, *G* == *x*.2;
prove;
■

theorem disabled rule exprPlusApp
$\forall\, F, G : \mathbb{N} \to \mathbb{N} \bullet$ *exprPlus*(F, G) = *exprMinusI* (*exprMinusI* (*comp*(*funRt*,
exprOplus(*exprOplus*(*comp*(*funD*, *exprOtimes*(*opPlusI F*, *opPlusI G*)), *opPlusI F*), *opPlusI G*))))

proof
use exprPlusFun;
use pairInFunction[$(\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}), \mathbb{N} \to \mathbb{N}$][$f :=$ *exprPlus*, $x := (F,\ G),\ y :=$
exprMinusI (*exprMinusI* (*comp*(*funRt*, *exprOplus*(*exprOplus*(*comp*(*funD*,
exprOtimes(*opPlusI F*, *opPlusI G*)), *opPlusI F*), *opPlusI G*))))];
invoke exprPlus;
prove by reduce;
■

theorem rule PlusExprIn
$\forall\, F, G : \mathbb{N} \to \mathbb{N} \mid F \in C6 \wedge G \in C6 \bullet$ *exprPlus*$(F, G) \in C6$

proof
with enabled (*exprPlusApp*) *prove*;
■

theorem rule PlusExprVal
$\forall\, F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet$ *exprPlus*$(F, G)\, x = F\, x + G\, x$

proof
use UFNonneg[$f := F,\ x := x$];
use UFNonneg[$f := G,\ x := x$];
use LeProd[$x := x,\ a := F\, x + x,\ b := G\, x + x$];
use MulNonneg[$a := F\, x + x,\ b := G\, x + x$];
use MulNonneg[$a := F\, x + x,\ b := F\, x + x$];
use RtSqSum[$a := F\, x + x,\ b := G\, x + x$];
with enabled (*exprPlusApp*) *prove*;
■

theorem PlusExprEq
$\forall F, G : \mathbb{N} \to \mathbb{N} \bullet \mathit{exprPlus}(F, G) = \mathit{opPlus}(F, G)$

proof
apply UFExt to predicate $\mathit{exprPlus}(F,\ G) = \mathit{opPlus}(F,\ G)$;
prove;
■

The difference function of footnote 8: it equals $F - G$ where $F \geq G$.

$\mathit{opDiffE} : (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

$\langle\!\langle$ rule DiffEVal $\rangle\!\rangle$
$\forall F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet \mathit{opDiffE}(F, G)\, x = \mathit{funE}\,((F\, x + G\, x) * (F\, x + G\, x) + 3 * F\, x + G\, x + 1)$

proof [opDiffE$domainCheck]
prove;
■

theorem rule DiffEInner
$\forall F, G : \mathbb{N} \to \mathbb{N} \bullet (\lambda\, x : \mathbb{N} \bullet \mathit{funE}\,((F\, x + G\, x) * (F\, x + G\, x) + 3 * F\, x + G\, x + 1)) \in \mathbb{N} \to \mathbb{N}$

proof
apply FunIntro to predicate $(\lambda\ x : \mathbb{N} \bullet \mathit{funE}\,((F\, x + G\, x) * (F\, x + G\, x) + 3 * F\, x + G\, x + 1)) \in \mathbb{N} \to \mathbb{N}$;
with enabled $(\mathit{RelPow},\ \mathit{inPower})$ *prove by reduce*;
■

$\mathit{DiffEWit} == (\lambda\, F, G : \mathbb{N} \to \mathbb{N} \bullet (\lambda\, x : \mathbb{N} \bullet \mathit{funE}\,((F\, x + G\, x) * (F\, x + G\, x) + 3 * F\, x + G\, x + 1)))$

proof [DiffEWit$domainCheck]
prove;
■

theorem DiffEWitFun
$\mathit{DiffEWit} \in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
invoke DiffEWit;
apply FunIntro to predicate $(\lambda\ F,\ G : \mathbb{N} \to \mathbb{N} \bullet (\lambda\ x : \mathbb{N} \bullet \mathit{funE}\,((F\, x + G\, x) * (F\, x + G\, x) + 3 * F\, x + G\, x + 1))) \in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$;
with enabled $(\mathit{RelPow},\ \mathit{inPower})$ *prove by reduce*;
instantiate $F == x.1,\ G == x.2$;
prove;
■

theorem DiffEWitApp
$\forall F, G : \mathbb{N} \to \mathbb{N} \bullet \mathit{DiffEWit}(F, G) = (\lambda\, x : \mathbb{N} \bullet \mathit{funE}\,((F\, x + G\, x) * (F\, x + G\, x) + 3 * F\, x + G\, x + 1))$

proof
use DiffEWitFun;
use pairInFunction$[(\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}), \mathbb{N} \to \mathbb{N}][f := \mathit{DiffEWit},\ x := (F,\ G),\ y := (\lambda\ x : \mathbb{N} \bullet \mathit{funE}\,((F\, x + G\, x) * (F\, x + G\, x) + 3 * F\, x + G\, x + 1))]$;
invoke DiffEWit;
prove by reduce;
■

theorem rule DiffEWitVal
$\forall F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet \mathit{DiffEWit}(F, G)\, x = \mathit{funE}\,((F\, x + G\, x) * (F\, x + G\, x) + 3 * F\, x + G\, x + 1)$

proof
prove;
use DiffEWitApp[$F := F,\ G := G$];
use pairInFunction[$\mathbb{N},\ \mathbb{N}$][$f := (\lambda\ x : \mathbb{N} \bullet \mathit{funE}\,((F\, x + G\, x) * (F\, x + G\, x) + 3 * F\, x + G\, x + 1)),\ x := x,\ y := \mathit{funE}\,((F\, x + G\, x) * (F\, x + G\, x) + 3 * F\, x + G\, x + 1)$];
prove;
■

theorem DiffEEx
$\exists\, \mathit{opDiffE} : (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}) \bullet \forall F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet \mathit{opDiffE}(F, G)\, x = \mathit{funE}\,((F\, x + G\, x) * (F\, x + G\, x) + 3 * F\, x + G\, x + 1)$

proof
instantiate opDiffE $==$ *DiffEWit*;
use DiffEWitFun;
prove;
■

$F - G = E\, S\, (\mathrm{Sq}(F + G) + M_3 F + G).$

$\mathit{exprMinus} == (\lambda\, F, G : \mathbb{N} \to \mathbb{N} \bullet \mathit{comp}(\mathit{comp}(\mathit{funE}, \mathit{funS}), \mathit{exprPlus}(\mathit{exprPlus}(\mathit{comp}(\mathit{funSq}, \mathit{exprPlus}(F, G)), \mathit{comp}(\mathit{funM3}, F)), G)))$

proof [exprMinus$domainCheck]
prove;
■

theorem rule exprMinusFun
$\mathit{exprMinus} \in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
invoke exprMinus;
apply FunIntro to predicate $(\lambda\ F,\ G : \mathbb{N} \to \mathbb{N} \bullet \mathit{comp}(\mathit{comp}(\mathit{funE},\ \mathit{funS}),\ \mathit{exprPlus}(\mathit{exprPlus}(\mathit{comp}(\mathit{funSq},\ \mathit{exprPlus}(F,\ G)),\ \mathit{comp}(\mathit{funM3},\ F)),\ G))) \in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$;
with enabled (*RelPow*, *inPower*) *prove*;
instantiate $F == x.1,\ G == x.2$;
prove;
■

theorem disabled rule exprMinusApp
$\forall F, G : \mathbb{N} \to \mathbb{N} \bullet \mathit{exprMinus}(F, G) = \mathit{comp}(\mathit{comp}(\mathit{funE}, \mathit{funS}), \mathit{exprPlus}(\mathit{exprPlus}(\mathit{comp}(\mathit{funSq}, \mathit{exprPlus}(F, G)), \mathit{comp}(\mathit{funM3}, F)), G))$

proof
use exprMinusFun;
use pairInFunction[$(\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}),\ \mathbb{N} \to \mathbb{N}$][$f := \mathit{exprMinus},\ x := (F,\ G),\ y := \mathit{comp}(\mathit{comp}(\mathit{funE},\ \mathit{funS}),\ \mathit{exprPlus}(\mathit{exprPlus}(\mathit{comp}(\mathit{funSq},\ \mathit{exprPlus}(F,\ G)),\ \mathit{comp}(\mathit{funM3},\ F)),\ G))$];
invoke exprMinus;
prove by reduce;
■

theorem rule MinusExprIn
$\forall F, G : \mathbb{N} \to \mathbb{N} \mid F \in C6 \wedge G \in C6 \bullet \mathit{exprMinus}(F, G) \in C6$

proof
with enabled (*exprMinusApp*) *prove*;
■

theorem MinusExprEq
$\forall F, G : \mathbb{N} \to \mathbb{N} \bullet \mathit{exprMinus}(F, G) = \mathit{opDiffE}(F, G)$

proof
apply UFExt to predicate $\mathit{exprMinus}(F,\ G) = \mathit{opDiffE}(F,\ G)$;
with enabled (*exprMinusApp*) *prove*;
■

theorem DiffEIsDifference
$\forall F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \mid G\, x \leq F\, x \bullet \mathit{opDiffE}(F, G)\, x = F\, x - G\, x$

proof
prove;
use RobDiffE$[a\ :=\ F\, x,\ b\ :=\ G\, x]$;
prove;
■

theorem Lemma66
$\forall F, G : \mathbb{N} \to \mathbb{N} \mid F \in C6 \land G \in C6 \bullet \mathit{opPlus}(F, G) \in C6 \land \mathit{opDiffE}(F, G) \in C6$

proof
use PlusExprEq$[F\ :=\ F,\ G\ :=\ G]$;
use MinusExprEq$[F\ :=\ F,\ G\ :=\ G]$;
prove;
equality substitute $\mathit{opPlus}(F,\ G)$;
equality substitute $\mathit{opDiffE}(F,\ G)$;
rewrite;
■

theorem rule C6Plus
$\forall F, G : \mathbb{N} \to \mathbb{N} \mid F \in C6 \land G \in C6 \bullet \mathit{opPlus}(F, G) \in C6$

proof
*use Lemma*66$[F\ :=\ F,\ G\ :=\ G]$;
prove;
■

theorem rule C6DiffE
$\forall F, G : \mathbb{N} \to \mathbb{N} \mid F \in C6 \land G \in C6 \bullet \mathit{opDiffE}(F, G) \in C6$

proof
*use Lemma*66$[F\ :=\ F,\ G\ :=\ G]$;
prove;
■

4 Theorem 6.7 (closure part)

Robinson [15, Sec. 7, Thm. 3] proved that $\langle S, E, F + G, FG, F^{\Box} \rangle$ is suitable; here we show that it generates exactly $\mathrm{clos}\,\mathcal{F}$, so $\mathrm{clos}\,\mathcal{F} = \mathrm{Prim}(\mathbb{N}, \mathbb{N})$ follows from Robinson's theorem.

$\mathit{RobinsonClos} == \mathit{clos}\,\{\mathit{funS}, \mathit{funE}\}\,\{\mathit{opBox}\}\,\{\mathit{opPlus}, \mathit{comp}\}$

proof [RobinsonClos$domainCheck]
prove;
■

theorem disabled rule RobDef
$RobinsonClos = clos\,\{funS, funE\}\,\{opBox\}\,\{opPlus, comp\}$

proof
invoke RobinsonClos;
prove;
■

theorem disabled rule RobUF
$\forall F : \mathbb{P}(\mathbb{Z} \times \mathbb{Z}) \mid F \in RobinsonClos \bullet F \in \mathbb{N} \to \mathbb{N}$

proof
prove;
use ClosElemUF[$I := \{ funS, funE \}$, $U := \{ opBox \}$, $B := \{ opPlus, comp \}$, $F := F$];
with enabled (*RobDef*) *prove*;
■

theorem rule RobS
$funS \in RobinsonClos$

proof
with enabled (*RobDef*) *prove*;
■

theorem rule RobE
$funE \in RobinsonClos$

proof
with enabled (*RobDef*) *prove*;
■

theorem rule RobBox
$\forall F : \mathbb{N} \to \mathbb{N} \mid F \in RobinsonClos \bullet opBox\, F \in RobinsonClos$

proof
with enabled (*RobDef*) *prove*;
■

theorem rule RobPlus
$\forall F, G : \mathbb{N} \to \mathbb{N} \mid F \in RobinsonClos \land G \in RobinsonClos \bullet opPlus(F, G) \in RobinsonClos$

proof
with enabled (*RobDef*) *prove*;
■

theorem rule RobComp
$\forall F, G : \mathbb{N} \to \mathbb{N} \mid F \in RobinsonClos \land G \in RobinsonClos \bullet comp(F, G) \in RobinsonClos$

proof
with enabled (*RobDef*) *prove*;
■

$BoxESet == \{n : \mathbb{N} \mid opBox\ funE\ n = 0\}$

proof [BoxESet$domainCheck]
prove;
■

theorem rule BoxEBase
$0 \in BoxESet$

proof
invoke BoxESet;
prove;
■

theorem rule BoxEStep
$\forall n : BoxESet \bullet 1 + n \in BoxESet$

proof
invoke BoxESet;
prove;
■

theorem BoxESub
$\mathbb{N} \subseteq BoxESet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq BoxESet$;
prove;
invoke BoxESet;
prove;
■

theorem BoxEAll
$\forall n : \mathbb{N} \bullet opBox\ funE\ n = 0$

proof
prove;
use BoxESub;
invoke BoxESet;
with enabled (*inPower*) *prove*;
instantiate $e == n$;
prove;
■

theorem BoxEExpr
$opBox\ funE = cst\ 0$

proof
apply UFExt to predicate $opBox\ funE = cst\ 0$;
prove;
use $BoxEAll[n := x]$;
prove;
■

$BoxSSet == \{n : \mathbb{N} \mid opBox\ funS\ n = n\}$

proof [BoxSSet$domainCheck]
prove;
■

theorem rule BoxSBase
$0 \in BoxSSet$

proof
invoke BoxSSet;
prove;
■

theorem rule BoxSStep
$\forall\, n : BoxSSet \bullet 1 + n \in BoxSSet$

proof
invoke BoxSSet;
prove;
■

theorem BoxSSub
$\mathbb{N} \subseteq BoxSSet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq BoxSSet$;
prove;
invoke BoxSSet;
prove;
■

theorem BoxSAll
$\forall\, n : \mathbb{N} \bullet opBox\, funS\, n = n$

proof
prove;
use BoxSSub;
invoke BoxSSet;
with enabled (*inPower*) *prove*;
instantiate e == *n*;
prove;
■

theorem BoxSExpr
$opBox\, funS = funI$

proof
apply UFExt to predicate $opBox\, funS = funI$;
prove;
use BoxSAll[$n := x$];
prove;
■

theorem OneExpr
$comp(funS, cst\, 0) = cst\, 1$

proof
apply UFExt to predicate $comp(funS,\ cst\,0) = cst\,1$;
prove;
■

theorem PlusIEq
$\forall F : \mathbb{N} \to \mathbb{N} \bullet opPlusI\ F = opPlus(F, funI)$

proof
apply UFExt to predicate $opPlusI\ F = opPlus(F,\ funI)$;
prove;
■

theorem RobIExpr
$opBox\ funS \in RobinsonClos$

proof
prove;
■

theorem rule RobI
$funI \in RobinsonClos$

proof
use RobIExpr;
use BoxSExpr;
prove;
■

theorem RobZeroExpr
$opBox\ funE \in RobinsonClos$

proof
prove;
■

theorem rule RobZero
$cst\,0 \in RobinsonClos$

proof
use BoxEExpr;
equality substitute $cst\,0$;
rewrite;
■

theorem RobOneExpr
$comp(funS, cst\,0) \in RobinsonClos$

proof
prove;
■

theorem rule RobOne
$cst\,1 \in RobinsonClos$

proof
use *RobOneExpr*;
use *OneExpr*;
prove;
■

theorem rule RobPlusI
$\forall F : \mathbb{N} \to \mathbb{N} \mid F \in RobinsonClos \bullet opPlusI\ F \in RobinsonClos$

proof
use *PlusIEq*[$F := F$];
prove;
■

theorem C6InPower
$C6 \in \mathbb{P}(\mathbb{N} \to \mathbb{N})$

proof
use *ClosInPow*[$I := \{ cst\,1,\ funE \},\ U := \{ opPlusI,\ opBox \},\ B := \{ comp \}$];
with enabled ($C6Def$) *prove*;
■

theorem RobInPower
$RobinsonClos \in \mathbb{P}(\mathbb{N} \to \mathbb{N})$

proof
use *ClosInPow*[$I := \{ funS,\ funE \},\ U := \{ opBox \},\ B := \{ opPlus,\ comp \}$];
with enabled ($RobDef$) *prove*;
■

theorem RobUClosed
$RobinsonClos \in UClosed\,\{opPlusI, opBox\}$

proof
use *RobInPower*;
with enabled ($UClosedDef,\ C6UF,\ RobUF$) *prove*;
split $u = opPlusI$;
cases;
prove;
next;
prove;
next;
■

theorem RobBClosed
$RobinsonClos \in BClosed\,\{comp\}$

proof
use *RobInPower*;
with enabled ($BClosedDef,\ C6UF,\ RobUF$) *prove*;
■

theorem C6UClosed
$C6 \in UClosed\,\{opBox\}$

proof
use $C6InPower$;
with enabled $(UClosedDef,\ C6UF,\ RobUF)$ *prove*;
■

theorem C6BClosed
$C6 \in BClosed\,\{opPlus, comp\}$

proof
use $C6InPower$;
with enabled $(BClosedDef,\ C6UF,\ RobUF)$ *prove*;
split $b = opPlus$;
cases;
prove;
next;
prove;
next;
■

theorem C6SubRob
$C6 \subseteq RobinsonClos$

proof
use $ClosLeast[I := \{\ cst\,1,\ funE\ \},\ U := \{\ opPlusI,\ opBox\ \},\ B := \{\ comp\ \},\ Z := RobinsonClos]$;
use $RobInPower$;
use $RobUClosed$;
use $RobBClosed$;
with enabled $(C6Def)$ *prove*;
■

theorem RobSubC6
$RobinsonClos \subseteq C6$

proof
use $ClosLeast[I := \{\ funS,\ funE\ \},\ U := \{\ opBox\ \},\ B := \{\ opPlus,\ comp\ \},\ Z := C6]$;
use $C6InPower$;
use $C6UClosed$;
use $C6BClosed$;
with enabled $(RobDef)$ *prove*;
■

theorem Theorem67
$C6 = RobinsonClos$

proof
use $C6SubRob$;
use $RobSubC6$;
use $C6InPower$;
use $RobInPower$;
use $SubAntisym[\mathbb{N} \to \mathbb{N}][A := C6,\ B := RobinsonClos]$;
prove;
■

5 Cantor pairing functions

$A(x) = x(x+1)/2$, V its inverse ($A(V(x)) \leq x < A(V(x)+1)$), $K(x) = x - A(V(x))$ and $L(x) = A(V(x)+1) - x - 1$.

$$
\begin{array}{|l}
funA : \mathbb{N} \to \mathbb{N} \\
\hline
\langle\!\langle \text{ disabled rule AVal } \rangle\!\rangle \\
\forall\, x : \mathbb{N} \bullet funA\, x = (x * x + x) \operatorname{div} 2
\end{array}
$$

proof [funA$domainCheck]
prove;
■

$AWit == (\lambda\, x : \mathbb{N} \bullet (x * x + x) \operatorname{div} 2)$

theorem AWitFun
$AWit \in \mathbb{N} \to \mathbb{N}$

proof
invoke AWit;
apply FunIntro to predicate $(\lambda\ x : \mathbb{N} \bullet (x * x + x) \operatorname{div} 2) \in \mathbb{N} \to \mathbb{N}$;
prove by reduce;
with enabled (*RelPow*) *prove by reduce*;
■

theorem rule AWitVal
$\forall\, x : \mathbb{N} \bullet AWit\, x = (x * x + x) \operatorname{div} 2$

proof
prove;
use AWitFun;
use pairInFunction$[\mathbb{N}, \mathbb{N}][f := AWit, x := x, y := (x * x + x) \operatorname{div} 2]$;
invoke AWit;
prove by reduce;
■

theorem AEx
$\exists\, funA : \mathbb{N} \to \mathbb{N} \bullet \forall\, x : \mathbb{N} \bullet funA\, x = (x * x + x) \operatorname{div} 2$

proof
instantiate funA $==$ *AWit*;
use AWitFun;
prove;
■

proof [AWit$domainCheck]
prove;
■

$TwoASet == \{n : \mathbb{N} \mid 2 * funA\, n = n * n + n\}$

proof [TwoASet$domainCheck]
prove;
■

theorem rule TwoABase
$0 \in TwoASet$

proof
invoke $TwoASet$;
prove;
use $DivMod[x := 0,\ d := 2,\ q := 0,\ r := 0]$;
with enabled $(AVal)$ *prove*;
■

theorem rule TwoAStep
$\forall n : TwoASet \bullet 1 + n \in TwoASet$

proof
invoke $TwoASet$;
prove;
use $DivMod[x := (1 + n) * (1 + n) + (1 + n),\ d := 2,\ q := funA\, n + n + 1,\ r := 0]$;
with enabled $(AVal)$ *prove*;
■

theorem TwoASub
$\mathbb{N} \subseteq TwoASet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq TwoASet$;
prove;
invoke $TwoASet$;
prove;
■

theorem TwoAAll
$\forall n : \mathbb{N} \bullet 2 * funA\, n = n * n + n$

proof
prove;
use $TwoASub$;
invoke $TwoASet$;
with enabled $(inPower)$ *prove*;
instantiate $e == n$;
prove;
■

theorem TwoA
$\forall x : \mathbb{N} \bullet 2 * funA\, x = x * x + x$

proof
prove;
use $TwoAAll[n := x]$;
prove;
■

theorem ASuc
$\forall x : \mathbb{N} \bullet funA\, (1 + x) = funA\, x + x + 1$

proof
prove;
use *TwoA*[$x := x$];
use *TwoA*[$x := 1 + x$];
use *UFNonneg*[$f := funA,\ x := x$];
use *UFNonneg*[$f := funA,\ x := 1 + x$];
prove;
■

theorem AZero
$funA\,0 = 0 \land funA\,1 = 1$

proof
use *TwoA*[$x := 0$];
use *TwoA*[$x := 1$];
use *UFNonneg*[$f := funA,\ x := 0$];
use *UFNonneg*[$f := funA,\ x := 1$];
prove;
■

theorem AMono
$\forall\, m, n : \mathbb{N} \mid m \leq n \bullet funA\,m \leq funA\,n$

proof
prove;
use *TwoA*[$x := m$];
use *TwoA*[$x := n$];
use *timesMonotonic2*[$a := m,\ b := m,\ c := n,\ d := n$];
use *UFNonneg*[$f := funA,\ x := m$];
use *UFNonneg*[$f := funA,\ x := n$];
prove;
■

theorem AIndexUq
$\forall\, r, s, x : \mathbb{N} \mid funA\,r \leq x \land x < funA\,(1 + r) \land funA\,s \leq x \land x < funA\,(1 + s) \bullet r = s$

proof
prove;
split $r < s$;
cases;
use *AMono*[$m := 1 + r,\ n := s$];
prove;
next;
split $s < r$;
cases;
use *AMono*[$m := 1 + s,\ n := r$];
prove;
next;
prove;
next;
■

$AIdxSet == \{x : \mathbb{N} \mid \exists\, t : \mathbb{N} \bullet funA\,t \leq x \land x < funA\,(1 + t)\}$

proof [AIdxSet$domainCheck]
prove;
■

theorem rule AIdxBase
$0 \in AIdxSet$

proof
invoke AIdxSet;
prove;
use AZero;
instantiate $t == 0$;
prove;
■

theorem rule AIdxStep
$\forall x : AIdxSet \bullet 1 + x \in AIdxSet$

proof
invoke AIdxSet;
prove;
use ASuc$[x := 1 + t]$;
use UFNonneg$[f := funA,\ x := t]$;
use UFNonneg$[f := funA,\ x := 1 + t]$;
use UFNonneg$[f := funA,\ x := 2 + t]$;
use LeOrGt$[a := funA\,(1 + t),\ b := 1 + x]$;
split $funA\,(1 + t) \leq 1 + x$;
cases;
instantiate $t_0 == 1 + t$;
prove;
next;
instantiate $t_1 == t$;
prove;
next;
■

theorem AIdxSub
$\mathbb{N} \subseteq AIdxSet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq AIdxSet$;
prove;
invoke AIdxSet;
prove;
■

theorem AIndexEx
$\forall x : \mathbb{N} \bullet \exists t : \mathbb{N} \bullet funA\ t \leq x \land x < funA\,(1 + t)$

proof
prove;
use AIdxSub;
with enabled (*inPower*) *prove*;
instantiate $e == x$;
invoke AIdxSet;
prove;
■

$funV : \mathbb{N} \to \mathbb{N}$

$\langle\!\langle$ VSpec $\rangle\!\rangle$
$\forall x : \mathbb{N} \bullet funA\,(funV\ x) \leq x \land x < funA\,(1 + funV\ x)$

proof [funV$domainCheck]
prove;
■

$VWit == \{x, t : \mathbb{N} \mid funA\, t \leq x \wedge x < funA\,(1 + t) \bullet (x, t)\}$

proof [VWit$domainCheck]
prove;
■

theorem VWitFun
$VWit \in \mathbb{N} \to \mathbb{N}$

proof
invoke VWit;
apply FunIntro to predicate $\{\ x,\ t : \mathbb{N} \mid funA\, t \leq x \wedge x < funA\,(1 + t) \bullet (x,\ t)\ \}$
$\in \mathbb{N} \to \mathbb{N}$;
with enabled (*RelPow*) *prove by reduce*;
use AIndexEx[$x := x$];
use AIndexUq[$r := y,\ s := z,\ x := x__0$];
prove;
instantiate $y__0 == t$;
prove;
■

theorem rule VWitLe
$\forall x : \mathbb{N} \bullet funA\,(VWit\, x) \leq x$

proof
prove;
use AIndexEx[$x := x$];
use VWitFun;
prove;
use pairInFunction[$\mathbb{N},\ \mathbb{N}$][$f := VWit,\ x := x,\ y := t$];
invoke VWit;
prove by reduce;
■

theorem rule VWitLt
$\forall x : \mathbb{N} \bullet x < funA\,(1 + VWit\, x)$

proof
prove;
use AIndexEx[$x := x$];
use VWitFun;
prove;
use pairInFunction[$\mathbb{N},\ \mathbb{N}$][$f := VWit,\ x := x,\ y := t$];
invoke VWit;
prove by reduce;
■

theorem VEx
$\exists funV : \mathbb{N} \to \mathbb{N} \bullet \forall x : \mathbb{N} \bullet funA\,(funV\, x) \leq x \wedge x < funA\,(1 + funV\, x)$

proof
instantiate $funV == VWit$;
use VWitFun;
prove;
■

theorem VChar
$\forall\, t, x : \mathbb{N} \mid funA\, t \leq x \wedge x < funA\,(1 + t) \bullet funV\, x = t$

proof
prove;
use $VSpec[x := x]$;
use $AIndexUq[r := funV\, x,\ s := t,\ x := x]$;
use $UFNonneg[f := funV,\ x := x]$;
prove;
■

$funK : \mathbb{N} \to \mathbb{N}$

⟪ disabled rule KVal ⟫
$\forall\, x : \mathbb{N} \bullet funK\, x = x - funA\,(funV\, x)$

proof [funK$domainCheck]
prove;
■

$KWit == (\lambda\, x : \mathbb{N} \bullet x - funA\,(funV\, x))$

proof [KWit$domainCheck]
prove;
■

theorem KWitRel
$(\lambda\, x : \mathbb{N} \bullet x - funA\,(funV\, x)) \in \mathbb{P}(\mathbb{N} \times \mathbb{N})$

proof
with enabled (*inPower*) *prove by reduce*;
use $VSpec[x := x]$;
prove;
■

theorem KWitFun
$KWit \in \mathbb{N} \to \mathbb{N}$

proof
invoke KWit;
use KWitRel;
apply FunIntro to predicate $(\lambda\, x : \mathbb{N} \bullet x - funA\,(funV\, x)) \in \mathbb{N} \to \mathbb{N}$;
with enabled (*RelPow*) *prove by reduce*;
use $VSpec[x := x]$;
prove;
■

theorem rule KWitVal
$\forall\, x : \mathbb{N} \bullet KWit\, x = x - funA\,(funV\, x)$

proof
prove;
use KWitFun;
use $pairInFunction[\mathbb{N},\ \mathbb{N}][f := KWit,\ x := x,\ y := x - funA\,(funV\, x)]$;
invoke KWit;
prove by reduce;
■

theorem KEx
$\exists\, funK : \mathbb{N} \to \mathbb{N} \bullet \forall\, x : \mathbb{N} \bullet funK\, x = x - funA\,(funV\, x)$

proof
instantiate $funK == KWit$;
use *KWitFun*;
prove;
■

$funL : \mathbb{N} \to \mathbb{N}$

$\langle\!\langle$ disabled rule LVal $\rangle\!\rangle$
$\forall\, x : \mathbb{N} \bullet funL\, x = funA\,(1 + funV\, x) - x - 1$

proof [funL$domainCheck]
prove;
■

$LWit == (\lambda\, x : \mathbb{N} \bullet funA\,(1 + funV\, x) - x - 1)$

proof [LWit$domainCheck]
prove;
■

theorem LWitRel
$(\lambda\, x : \mathbb{N} \bullet funA\,(1 + funV\, x) - x - 1) \in \mathbb{P}(\mathbb{N} \times \mathbb{N})$

proof
with enabled (*inPower*) *prove by reduce*;
use $VSpec[x := x]$;
prove;
■

theorem LWitFun
$LWit \in \mathbb{N} \to \mathbb{N}$

proof
invoke *LWit*;
use *LWitRel*;
apply FunIntro to predicate $(\lambda\, x : \mathbb{N} \bullet funA\,(1 + funV\, x) - x - 1) \in \mathbb{N} \to \mathbb{N}$;
with enabled (*RelPow*) *prove by reduce*;
use $VSpec[x := x]$;
prove;
■

theorem rule LWitVal
$\forall\, x : \mathbb{N} \bullet LWit\, x = funA\,(1 + funV\, x) - x - 1$

proof
prove;
use *LWitFun*;
use *pairInFunction*$[\mathbb{N}, \mathbb{N}][f := LWit,\ x := x,\ y := funA\,(1 + funV\, x) - x - 1]$;
invoke *LWit*;
prove by reduce;
■

theorem LEx
$\exists\, funL : \mathbb{N} \to \mathbb{N} \bullet \forall\, x : \mathbb{N} \bullet funL\, x = funA\,(1 + funV\, x) - x - 1$

proof
instantiate funL == LWit;
use LWitFun;
prove;
■

Values on a diagonal of the Cantor table: for $k \leq t$, $V(A(t)+k) = t$, $K(A(t)+k) = k$ and $L(A(t)+k) = t-k$.

theorem Diag
$\forall\, k, t : \mathbb{N} \mid k \leq t \bullet funV\,(funA\, t + k) = t \wedge funK\,(funA\, t + k) = k \wedge funL\,(funA\, t + k) = t - k$

proof
prove;
use ASuc[$x := t$];
use UFNonneg[$f := funA,\ x := t$];
use VChar[$t := t,\ x := funA\, t + k$];
with enabled (*KVal*, *LVal*) *prove*;
■

theorem Decomp
$\forall\, x : \mathbb{N} \bullet x = funA\,(funV\, x) + funK\, x \wedge funK\, x \leq funV\, x$

proof
prove;
use VSpec[$x := x$];
use ASuc[$x := funV\, x$];
use UFNonneg[$f := funA,\ x := funV\, x$];
use UFNonneg[$f := funA,\ x := 1 + funV\, x$];
use UFNonneg[$f := funV,\ x := x$];
with enabled (*KVal*) *prove*;
■

5.1 A, V, K and L are in $\operatorname{clos}\mathcal{F}$

theorem AExpr
$comp(funHf, opPlusI\, funSq) = funA$

proof
apply UFExt to predicate $comp(funHf,\ opPlusI\, funSq) = funA$;
prove;
use TwoA[$x := x$];
use UFNonneg[$f := funA,\ x := x$];
use DivMod[$x := x * x + x,\ d := 2,\ q := funA\, x,\ r := 0$];
prove;
■

theorem RtPos
$\forall\, y : \mathbb{N} \mid 1 \leq y \bullet 1 \leq funRt\, y$

proof
prove;
use RtSpec[$x := y$];
use UFNonneg[$f := funRt,\ x := y$];
split $funRt\, y = 0$;
cases;
prove;
next;
prove;
next;
■

theorem PhatPos
$\forall\, y : \mathbb{N} \mid 1 \leq y \bullet \mathit{exprPhat}\ y = y - 1$

proof
prove;
use $\mathit{PhatVal}[x := y - 1]$;
prove;
■

theorem VFormula
$\forall\, x : \mathbb{N} \bullet \mathit{funV}\ x = (\mathit{funRt}\ (1 + 8 * x) - 1)\ \mathsf{div}\ 2$

proof
prove;
use $\mathit{VSpec}[x := x]$;
use $\mathit{TwoA}[x := \mathit{funV}\ x]$;
use $\mathit{TwoA}[x := 1 + \mathit{funV}\ x]$;
use $\mathit{UFNonneg}[f := \mathit{funA},\ x := \mathit{funV}\ x]$;
use $\mathit{UFNonneg}[f := \mathit{funA},\ x := 1 + \mathit{funV}\ x]$;
use $\mathit{UFNonneg}[f := \mathit{funV},\ x := x]$;
use $\mathit{LeOrGt}[a := (2 * \mathit{funV}\ x + 2) * (2 * \mathit{funV}\ x + 2),\ b := 1 + 8 * x]$;
split $(2 * \mathit{funV}\ x + 2) * (2 * \mathit{funV}\ x + 2) \leq 1 + 8 * x$;
cases;
use $\mathit{RtI}[r := 2 * \mathit{funV}\ x + 2,\ x := 1 + 8 * x]$;
prove;
use $\mathit{DivMod}[x := 2 * \mathit{funV}\ x + 1,\ d := 2,\ q := \mathit{funV}\ x,\ r := 1]$;
prove;
next;
use $\mathit{RtI}[r := 2 * \mathit{funV}\ x + 1,\ x := 1 + 8 * x]$;
prove;
use $\mathit{DivMod}[x := 2 * \mathit{funV}\ x,\ d := 2,\ q := \mathit{funV}\ x,\ r := 0]$;
prove;
next;
■

theorem VExpr
$\mathit{comp}(\mathit{comp}(\mathit{comp}(\mathit{comp}(\mathit{comp}(\mathit{comp}(\mathit{funHf}, \mathit{exprPhat}), \mathit{funRt}), \mathit{funS}), \mathit{funD}), \mathit{funD}), \mathit{funD}) = \mathit{funV}$

proof
apply UFExt to predicate $\mathit{comp}(\mathit{comp}(\mathit{comp}(\mathit{comp}(\mathit{comp}(\mathit{comp}(\mathit{funHf},\ \mathit{exprPhat}),\ \mathit{funRt}),\ \mathit{funS}),\ \mathit{funD}),\ \mathit{funD}),\ \mathit{funD}) = \mathit{funV}$;
prove;
use $\mathit{VFormula}[x := x]$;
use $\mathit{RtPos}[y := 1 + 8 * x]$;
use $\mathit{PhatPos}[y := \mathit{funRt}\ (1 + 8 * x)]$;
prove;
■

theorem IInExpr
$\mathit{opBox}\ \mathit{funS} \in C6$

proof
prove;
■

theorem rule IIn
$\mathit{funI} \in C6$

proof
use IInExpr;
use BoxSExpr;
prove;
■

theorem AInExpr
$comp(funHf, opPlusI\ funSq) \in C6$

proof
prove;
■

theorem rule AIn
$funA \in C6$

proof
use AInExpr;
use AExpr;
prove;
■

theorem VInExpr
$comp(comp(comp(comp(comp(comp(funHf, exprPhat), funRt), funS), funD), funD), funD) \in C6$

proof
prove;
■

theorem rule VIn
$funV \in C6$

proof
use VInExpr;
use VExpr;
prove;
■

theorem KExpr
$opDiffE(funI, comp(funA, funV)) = funK$

proof
apply UFExt to predicate $opDiffE(funI,\ comp(funA,\ funV)) = funK$;
prove;
use VSpec$[x := x]$;
use DiffEIsDifference$[F := funI,\ G := comp(funA,\ funV),\ x := x]$;
with enabled $(KVal)$ *prove*;
■

theorem LExpr
$opDiffE(comp(funA, comp(funS, funV)), funS) = funL$

proof
apply UFExt to predicate $opDiffE(comp(funA,\ comp(funS,\ funV)),\ funS) = funL$;
prove;
use VSpec$[x := x]$;
use DiffEIsDifference$[F := comp(funA,\ comp(funS,\ funV)),\ G := funS,\ x := x]$;
with enabled $(LVal)$ *prove*;
■

theorem KInExpr
$opDiffE(funI, comp(funA, funV)) \in C6$

proof
prove;
■

theorem rule KIn
$funK \in C6$

proof
use KInExpr;
use KExpr;
prove;
■

theorem LInExpr
$opDiffE(comp(funA, comp(funS, funV)), funS) \in C6$

proof
prove;
■

theorem rule LIn
$funL \in C6$

proof
use LInExpr;
use LExpr;
prove;
■

6 Theorem 6.8: the bases with K and L

6.1 Auxiliary functions (Robinson [16, p. 664])

theorem KAt
$\forall m, t, k : \mathbb{N} \mid m = funA\, t + k \wedge k \leq t \bullet funK\, m = k$

proof
prove;
use Diag$[k := k,\ t := t]$;
prove;
■

theorem RtLe
$\forall x : \mathbb{N} \bullet funRt\, x \leq x$

proof
prove;
use RtSpec$[x := x]$;
use LeSq$[n := funRt\, x]$;
prove;
■

$Z(x) = x(x+3)/2 = A(x) + x$ and $Y(x) = 2x - \mathrm{Rt}(x)$.

$$
\begin{array}{|l}
funZ : \mathbb{N} \to \mathbb{N} \\
\hline
\langle\!\langle \text{rule ZVal} \rangle\!\rangle \\
\forall\, x : \mathbb{N} \bullet funZ\, x = funA\, x + x
\end{array}
$$

proof [funZ$domainCheck]
prove;
■

$ZWit == (\lambda\, x : \mathbb{N} \bullet funA\, x + x)$

proof [ZWit$domainCheck]
prove;
■

theorem ZWitFun
$ZWit \in \mathbb{N} \to \mathbb{N}$

proof
invoke ZWit;
apply FunIntro to predicate $(\lambda\, x : \mathbb{N} \bullet funA\, x + x) \in \mathbb{N} \to \mathbb{N}$;
prove by reduce;
with enabled (*RelPow*) *prove by reduce*;
■

theorem rule ZWitVal
$\forall\, x : \mathbb{N} \bullet ZWit\, x = funA\, x + x$

proof
prove;
use ZWitFun;
use pairInFunction$[\mathbb{N}, \mathbb{N}][f := ZWit, x := x, y := funA\, x + x]$;
invoke ZWit;
prove by reduce;
■

theorem ZEx
$\exists\, funZ : \mathbb{N} \to \mathbb{N} \bullet \forall\, x : \mathbb{N} \bullet funZ\, x = funA\, x + x$

proof
instantiate funZ $==$ *ZWit*;
use ZWitFun;
prove;
■

$$
\begin{array}{|l}
funY : \mathbb{N} \to \mathbb{N} \\
\hline
\langle\!\langle \text{disabled rule YVal} \rangle\!\rangle \\
\forall\, x : \mathbb{N} \bullet funY\, x = 2 * x - funRt\, x
\end{array}
$$

proof [funY$domainCheck]
prove;
■

$YWit == (\lambda\, x : \mathbb{N} \bullet 2 * x - funRt\, x)$

proof [YWit$domainCheck]
prove;
■

theorem YWitRel
$(\lambda\, x : \mathbb{N} \bullet 2 * x - funRt\, x) \in \mathbb{P}(\mathbb{N} \times \mathbb{N})$

proof
with enabled (*inPower*) *prove by reduce*;
use $RtLe[x := x]$;
prove;
■

theorem YWitFun
$YWit \in \mathbb{N} \to \mathbb{N}$

proof
invoke YWit;
use YWitRel;
apply FunIntro to predicate $(\lambda\, x : \mathbb{N} \bullet 2 * x - funRt\, x) \in \mathbb{N} \to \mathbb{N}$;
with enabled (*RelPow*) *prove by reduce*;
use $RtLe[x := x]$;
prove;
■

theorem rule YWitVal
$\forall\, x : \mathbb{N} \bullet YWit\, x = 2 * x - funRt\, x$

proof
prove;
use YWitFun;
use pairInFunction $[\mathbb{N}, \mathbb{N}][f := YWit,\ x := x,\ y := 2 * x - funRt\, x]$;
invoke YWit;
prove by reduce;
■

theorem YEx
$\exists\, funY : \mathbb{N} \to \mathbb{N} \bullet \forall\, x : \mathbb{N} \bullet funY\, x = 2 * x - funRt\, x$

proof
instantiate $funY == YWit$;
use YWitFun;
prove;
■

$ZBoxSet == \{n : \mathbb{N} \mid opBox\, (comp(comp(funS, funS), opPlusI\, funK))\, n = funZ\, n\}$

proof [ZBoxSet$domainCheck]
prove;
■

theorem rule ZBoxBase
$0 \in ZBoxSet$

proof
invoke ZBoxSet;
prove;
use AZero;
prove;
■

theorem rule ZBoxStep
$\forall\, n : ZBoxSet \bullet 1 + n \in ZBoxSet$

proof
invoke ZBoxSet;
prove;
use Diag$[k := n,\ t := n]$;
use ASuc$[x := n]$;
prove;
■

theorem ZBoxSub
$\mathbb{N} \subseteq ZBoxSet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq ZBoxSet$;
prove;
invoke ZBoxSet;
prove;
■

theorem ZBoxAll
$\forall\, n : \mathbb{N} \bullet opBox\,(comp(comp(funS, funS), opPlusI\, funK))\, n = funZ\, n$

proof
prove;
use ZBoxSub;
invoke ZBoxSet;
with enabled (*inPower*) *prove*;
instantiate e $==$ n;
prove;
■

theorem ZExpr
$opBox\,(comp(comp(funS, funS), opPlusI\, funK)) = funZ$

proof
apply UFExt to predicate $opBox\,(comp(comp(funS,\ funS),\ opPlusI\, funK)) = funZ$;
prove;
use ZBoxAll$[n := x]$;
prove;
■

Robinson's formula: $K(Z(a + b) + 2a + 3) = a - b$ if $a \geq b$.

theorem RobDiffK
$\forall\, a, b : \mathbb{N} \mid b \leq a \bullet funK\,(funZ\,(a + b) + 2 * a + 3) = a - b$

proof
prove;
use $ASuc[x := a + b]$;
use $ASuc[x := 1 + a + b]$;
use $KAt[m := funA\,(a + b) + a + b + 2 * a + 3,\ t := a + b + 2,\ k := a - b]$;
prove;
■

$I - F = K\ S\ S\ S\ ((ZF^+)^+)^+$ where $F(x) \leq x$, and $D - F = K\ S\ S\ S\ ((((Z(F^+)^+)^+)^+)^+)^+$ where $F(x) \leq 2x$.

$exprIminusF == (\lambda\, F : \mathbb{N} \to \mathbb{N} \bullet comp(comp(comp(comp(funK, funS), funS), funS), funS),$
$opPlusI\ (opPlusI\ (comp(funZ, opPlusI\ F)))))$

proof [exprIminusF$domainCheck]
prove;
■

theorem rule exprIminusFFun
$exprIminusF \in (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
invoke *exprIminusF*;
apply *FunIntro* *to* *predicate* $(\lambda\ F : \mathbb{N} \to \mathbb{N} \bullet comp(comp(comp(comp(funK, funS), funS), funS), funS),$
$opPlusI\ (opPlusI\ (comp(funZ, opPlusI\ F)))))\ \in\ (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$;
with *enabled* $(RelPow, inPower)$ *prove* *by* *reduce*;
■

theorem disabled rule exprIminusFApp
$\forall\, F : \mathbb{N} \to \mathbb{N} \bullet exprIminusF\ F = comp(comp(comp(comp(funK, funS), funS), funS), funS),$
$opPlusI\ (opPlusI\ (comp(funZ, opPlusI\ F))))$

proof
use *exprIminusFFun*;
use *pairInFunction*$[\mathbb{N} \to \mathbb{N}, \mathbb{N} \to \mathbb{N}][f := exprIminusF,\ x := F,\ y :=$
$comp(comp(comp(comp(funK, funS), funS), funS), opPlusI\ (opPlusI\ (comp(funZ,$
$opPlusI\ F))))]$;
invoke *exprIminusF*;
prove *by* *reduce*;
■

$exprDminusF == (\lambda\, F : \mathbb{N} \to \mathbb{N} \bullet comp(comp(comp(comp(funK, funS), funS), funS), funS),$
$opPlusI\ (opPlusI\ (opPlusI\ (opPlusI\ (comp(funZ, opPlusI\ (opPlusI\ F)))))))$

proof [exprDminusF$domainCheck]
prove;
■

theorem rule exprDminusFFun
$exprDminusF \in (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
invoke *exprDminusF*;
apply *FunIntro* *to* *predicate* $(\lambda\ F : \mathbb{N} \to \mathbb{N} \bullet comp(comp(comp(comp(funK, funS), funS), funS), funS),$
$opPlusI\ (opPlusI\ (opPlusI\ (opPlusI\ (comp(funZ, opPlusI\ (opPlusI\ F)))))))\ \in\ (\mathbb{N} \to \mathbb{N}) \to$
$(\mathbb{N} \to \mathbb{N})$;
with *enabled* $(RelPow, inPower)$ *prove* *by* *reduce*;
■

theorem disabled rule exprDminusFApp
$\forall\, F : \mathbb{N} \to \mathbb{N} \bullet \mathit{exprDminusF}\ F = \mathit{comp}(\mathit{comp}(\mathit{comp}(\mathit{comp}(\mathit{funK}, \mathit{funS}), \mathit{funS}), \mathit{funS}),$
$\mathit{opPlusI}\ (\mathit{opPlusI}\ (\mathit{opPlusI}\ (\mathit{opPlusI}\ (\mathit{comp}(\mathit{funZ}, \mathit{opPlusI}\ (\mathit{opPlusI}\ F)))))))$

proof
use exprDminusFFun;
use pairInFunction$[\mathbb{N} \to \mathbb{N},\ \mathbb{N} \to \mathbb{N}][f := \mathit{exprDminusF},\ x := F,\ y :=$
$\mathit{comp}(\mathit{comp}(\mathit{comp}(\mathit{comp}(\mathit{funK},\ \mathit{funS}),\ \mathit{funS}),\ \mathit{funS}),$
$\mathit{opPlusI}\ (\mathit{opPlusI}\ (\mathit{opPlusI}\ (\mathit{opPlusI}\ (\mathit{comp}(\mathit{funZ},\ \mathit{opPlusI}\ (\mathit{opPlusI}\ F)))))))]$;
invoke exprDminusF;
prove by reduce;
■

theorem IminusFVal
$\forall\, F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \mid F\ x \leq x \bullet \mathit{exprIminusF}\ F\ x = x - F\ x$

proof
with enabled (*exprIminusFApp*) *prove*;
use RobDiffK$[a := x,\ b := F\ x]$;
prove;
■

theorem DminusFVal
$\forall\, F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \mid F\ x \leq 2 * x \bullet \mathit{exprDminusF}\ F\ x = 2 * x - F\ x$

proof
with enabled (*exprDminusFApp*) *prove*;
use RobDiffK$[a := 2 * x,\ b := F\ x]$;
prove;
■

$Y = ((\mathrm{Sgn}\, K)^{+} S)^{\square}$: $Y(n+1) = Y(n) + 1 + \mathrm{Sgn}\, K(Y(n)+1)$, and $K(Y(n)+1) = 0$ exactly when $n+1$ is a square ($Y(n)+1 = 2n - r + 1$ with $r = \mathrm{Rt}(n)$ lies on the diagonal of $A(2r+1)$, $A(2r)$ or $A(2r-1)$).

theorem YStep
$\forall\, n : \mathbb{N} \bullet \mathit{opPlusI}\ (\mathit{comp}(\mathit{funSgn}, \mathit{funK}))\ (\mathit{funS}\ (\mathit{funY}\ n)) = \mathit{funY}\ (1 + n)$

proof
with enabled (*YVal*) *prove*;
use *RtSpec*[$x := n$];
use *RtLe*[$x := n$];
use *UFNonneg*[$f := funRt,\ x := n$];
use *LeOrGt*[$a := (1 + funRt\ n) * (1 + funRt\ n),\ b := 1 + n$];
split $(1 + funRt\ n) * (1 + funRt\ n) \leq 1 + n$;
cases;
use *RtI*[$r := 1 + funRt\ n,\ x := 1 + n$];
use *TwoA*[$x := 1 + 2 * funRt\ n$];
use *UFNonneg*[$f := funA,\ x := 1 + 2 * funRt\ n$];
use *KAt*[$m := 1 + 2 * n - funRt\ n,\ t := 1 + 2 * funRt\ n,\ k := 0$];
prove;
next;
use *RtI*[$r := funRt\ n,\ x := 1 + n$];
use *LeOrGt*[$a := funRt\ n * funRt\ n + funRt\ n,\ b := n$];
split $funRt\ n * funRt\ n + funRt\ n \leq n$;
cases;
use *TwoA*[$x := 2 * funRt\ n$];
use *UFNonneg*[$f := funA,\ x := 2 * funRt\ n$];
use *KAt*[$m := 1 + 2 * n - funRt\ n,\ t := 2 * funRt\ n,\ k := 1 + 2 * n - 2 * funRt\ n * funRt\ n - 2 * funRt\ n$];
prove;
next;
use *TwoA*[$x := 2 * funRt\ n - 1$];
use *UFNonneg*[$f := funA,\ x := 2 * funRt\ n - 1$];
use *KAt*[$m := 1 + 2 * n - funRt\ n,\ t := 2 * funRt\ n - 1,\ k := 1 + 2 * n - 2 * funRt\ n * funRt\ n$];
prove;
next;
■

$YBoxSet == \{n : \mathbb{N} \mid opBox\ (comp(opPlusI\ (comp(funSgn, funK)), funS))\ n = funY\ n\}$

proof [YBoxSet$domainCheck]
prove;
■

theorem rule YBoxBase
$0 \in YBoxSet$

proof
invoke *YBoxSet*;
prove;
use *RtSq*[$m := 0$];
with enabled (*YVal*) *prove*;
■

theorem rule YBoxStep
$\forall\ n : YBoxSet \bullet 1 + n \in YBoxSet$

proof
invoke *YBoxSet*;
prove;
use *YStep*[$n := n$];
prove;
■

theorem YBoxSub
$\mathbb{N} \subseteq YBoxSet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq YBoxSet$;
prove;
invoke YBoxSet;
prove;
■

theorem YBoxAll
$\forall n : \mathbb{N} \bullet opBox\,(comp(opPlusI\,(comp(funSgn, funK)), funS))\,n = funY\,n$

proof
prove;
use YBoxSub;
invoke YBoxSet;
with enabled (*inPower*) *prove*;
instantiate $e == n$;
prove;
■

theorem YExpr
$opBox\,(comp(opPlusI\,(comp(funSgn, funK)), funS)) = funY$

proof
apply UFExt to predicate $opBox\,(comp(opPlusI\,(comp(funSgn,\ funK)),\ funS)) = funY$;
prove;
use $YBoxAll[n := x]$;
prove;
■

$\mathrm{Rt} = D - Y$, $\mathrm{Sq} = (((\bar{1}^{+})^{+}\mathrm{Rt})^{+})^{\square}$ and $E = I - \mathrm{Sq}\,\mathrm{Rt}$.

theorem RtExprK
$exprDminusF\,funY = funRt$

proof
apply UFExt to predicate $exprDminusF\,funY = funRt$;
prove;
use $RtLe[x := x]$;
use $UFNonneg[f := funRt,\ x := x]$;
use $DminusFVal[F := funY,\ x := x]$;
with enabled (*YVal*) *prove*;
■

$SqKSet == \{n : \mathbb{N} \mid opBox\,(opPlusI\,(comp(opPlusI\,(opPlusI\,(cst\,1)), funRt)))\,n = n * n\}$

proof [SqKSet$domainCheck]
prove;
■

theorem rule SqKBase
$0 \in SqKSet$

proof
invoke SqKSet;
prove;
■

theorem rule SqKStep
$\forall\, n : SqKSet \bullet 1 + n \in SqKSet$

proof
invoke SqKSet;
prove;
equality substitute opBox (*opPlusI* (*comp*(*opPlusI* (*opPlusI* (*cst* 1)), *funRt*))) *n*;
rewrite;
■

theorem SqKSub
$\mathbb{N} \subseteq SqKSet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq SqKSet$;
prove;
invoke SqKSet;
prove;
■

theorem SqKAll
$\forall\, n : \mathbb{N} \bullet opBox\,(opPlusI\,(comp(opPlusI\,(opPlusI\,(cst\,1)), funRt)))\,n = n * n$

proof
prove;
use SqKSub;
invoke SqKSet;
with enabled (*inPower*) *prove*;
instantiate e == *n*;
prove;
■

theorem SqExprK
$opBox\,(opPlusI\,(comp(opPlusI\,(opPlusI\,(cst\,1)), funRt))) = funSq$

proof
apply UFExt to predicate opBox (*opPlusI* (*comp*(*opPlusI* (*opPlusI* (*cst* 1)), *funRt*))) = *funSq*;
prove;
use SqKAll[*n* := *x*];
prove;
■

theorem EExprK
$exprIminusF\,(comp(funSq, funRt)) = funE$

proof
apply UFExt to predicate exprIminusF (*comp*(*funSq*, *funRt*)) = *funE*;
prove;
use RtSpec[*x* := *x*];
use IminusFVal[*F* := *comp*(*funSq*, *funRt*), *x* := *x*];
with enabled (*EVal*) *prove*;
■

6.2 The basis $\mathcal{F}' = \langle \bar{1}, K, F^+, FG, F^\square \rangle$

$C6K == clos\,\{cst\,1, funK\}\,\{opPlusI, opBox\}\,\{comp\}$

proof [C6K$domainCheck]
prove;
■

theorem disabled rule C6KDef
$C6K = clos\,\{cst\,1, funK\}\,\{opPlusI, opBox\}\,\{comp\}$

proof
invoke $C6K$;
prove;
■

theorem C6KInPower
$C6K \in \mathbb{P}(\mathbb{N} \to \mathbb{N})$

proof
use $ClosInPow[I := \{ cst\,1,\ funK \},\ U := \{ opPlusI,\ opBox \},\ B := \{ comp \}]$;
with enabled $(C6KDef)$ *prove*;
■

theorem disabled rule C6KUF
$\forall F : \mathbb{P}(\mathbb{Z} \times \mathbb{Z}) \mid F \in C6K \bullet F \in \mathbb{N} \to \mathbb{N}$

proof
prove;
use $ClosElemUF[I := \{ cst\,1,\ funK \},\ U := \{ opPlusI,\ opBox \},\ B := \{ comp \},\ F := F]$;
with enabled $(C6KDef)$ *prove*;
■

theorem rule C6KInit0
$cst\,1 \in C6K$

proof
with enabled $(C6KDef)$ *prove*;
■

theorem rule C6KInit1
$funK \in C6K$

proof
with enabled $(C6KDef)$ *prove*;
■

theorem rule C6KUnop0
$\forall F : \mathbb{N} \to \mathbb{N} \mid F \in C6K \bullet opPlusI\ F \in C6K$

proof
with enabled $(C6KDef)$ *prove*;
■

theorem rule C6KUnop1
$\forall F : \mathbb{N} \to \mathbb{N} \mid F \in C6K \bullet opBox\ F \in C6K$

proof
with enabled $(C6KDef)$ *prove*;
■

theorem rule C6KBinop0
$\forall F, G : \mathbb{N} \to \mathbb{N} \mid F \in C6K \land G \in C6K \bullet comp(F, G) \in C6K$

proof
with enabled $(C6KDef)$ *prove*;
■

theorem SInKExpr
$opPlusI\ (cst\ 1) \in C6K$

proof
prove;
■

theorem rule SInK
$funS \in C6K$

proof
use SInKExpr;
use SExpr;
prove;
■

theorem SgnInKExpr
$opBox\ (cst\ 1) \in C6K$

proof
prove;
■

theorem rule SgnInK
$funSgn \in C6K$

proof
use SgnInKExpr;
use SgnExpr;
prove;
■

theorem ZInKExpr
$opBox\ (comp(comp(funS, funS), opPlusI\ funK)) \in C6K$

proof
prove;
■

theorem rule ZInK
funZ ∈ *C*6*K*

proof
use ZInKExpr;
use ZExpr;
prove;
■

theorem YInKExpr
opBox (*comp*(*opPlusI* (*comp*(*funSgn*, *funK*)), *funS*)) ∈ *C*6*K*

proof
prove;
■

theorem rule YInK
funY ∈ *C*6*K*

proof
use YInKExpr;
use YExpr;
prove;
■

theorem RtInKExpr
exprDminusF funY ∈ *C*6*K*

proof
with enabled (*exprDminusFApp*) *prove*;
■

theorem rule RtInK
funRt ∈ *C*6*K*

proof
use RtInKExpr;
use RtExprK;
prove;
■

theorem SqInKExpr
opBox (*opPlusI* (*comp*(*opPlusI* (*opPlusI* (*cst* 1)), *funRt*))) ∈ *C*6*K*

proof
prove;
■

theorem rule SqInK
funSq ∈ *C*6*K*

proof
use SqInKExpr;
use SqExprK;
prove;
■

theorem EInKExpr
$exprIminusF\,(comp(funSq, funRt)) \in C6K$

proof
with enabled (*exprIminusFApp*) *prove*;
■

theorem rule EInK
$funE \in C6K$

proof
use EInKExpr;
use EExprK;
prove;
■

theorem Theorem68KSub1U
$C6 \in UClosed\,\{opPlusI, opBox\}$

proof
use C6InPower;
with enabled (*UClosedDef*, *C6UF*) *prove*;
split u = opPlusI;
cases;
prove;
next;
prove;
next;
■

theorem Theorem68KSub1B
$C6 \in BClosed\,\{comp\}$

proof
use C6InPower;
with enabled (*BClosedDef*, *C6UF*) *prove*;
prove;
■

theorem Theorem68KSub1
$C6K \subseteq C6$

proof
use ClosLeast[$I := \{ cst\,1, funK \}$, $U := \{ opPlusI, opBox \}$, $B := \{ comp \}$, $Z := C6$];
use C6InPower;
use Theorem68KSub1U;
use Theorem68KSub1B;
with enabled (*C6KDef*) *prove*;
■

theorem Theorem68KSub2U
$C6K \in UClosed\,\{opPlusI, opBox\}$

proof
use $C6KInPower$;
with enabled $(UClosedDef,\ C6KUF)$ *prove*;
split $u\ =\ opPlusI$;
cases;
prove;
next;
prove;
next;
■

theorem Theorem68KSub2B
$C6K \in BClosed\,\{comp\}$

proof
use $C6KInPower$;
with enabled $(BClosedDef,\ C6KUF)$ *prove*;
prove;
■

theorem Theorem68KSub2
$C6 \subseteq C6K$

proof
use $ClosLeast[I\ :=\ \{\ cst\,1,\ funE\ \},\ U\ :=\ \{\ opPlusI,\ opBox\ \},\ B\ :=\ \{\ comp\ \},\ Z\ :=\ C6K]$;
use $C6KInPower$;
use $Theorem68KSub2U$;
use $Theorem68KSub2B$;
with enabled $(C6Def)$ *prove*;
■

theorem Theorem68K
$C6K = C6$

proof
use $Theorem68KSub1$;
use $Theorem68KSub2$;
use $C6KInPower$;
use $C6InPower$;
use $SubAntisym[\mathbb{N}\ \to\ \mathbb{N}][A\ :=\ C6K,\ B\ :=\ C6]$;
prove;
■

6.3 The basis $\mathcal{F}'' = \langle \bar{1}, L, F^+, FG, F^\Box \rangle$

theorem LAt
$\forall\, m, t, k : \mathbb{N} \mid m = funA\,t + k \wedge k \leq t \bullet funL\,m = t - k$

proof
prove;
use $Diag[k\ :=\ k,\ t\ :=\ t]$;
prove;
■

$K = L\,\bar{1}^+((\bar{1}^+)^+L)^+$, i.e. $K(x) = L(x + 2L(x) + 2)$.

theorem KExprL
$comp(comp(funL, opPlusI\,(cst\,1)), opPlusI\,(comp(opPlusI\,(opPlusI\,(cst\,1)), funL))) = funK$

proof
apply UFExt to predicate $comp(comp(funL,\ opPlusI\,(cst\,1)),$
$opPlusI\,(comp(opPlusI\,(opPlusI\,(cst\,1)),\ funL)))\ =\ funK$;
prove;
use $Decomp[x\ :=\ x]$;
use $UFNonneg[f\ :=\ funV,\ x\ :=\ x]$;
use $UFNonneg[f\ :=\ funA,\ x\ :=\ funV\,x]$;
use $ASuc[x\ :=\ funV\,x]$;
use $LAt[m\ :=\ x,\ t\ :=\ funV\,x,\ k\ :=\ funK\,x]$;
prove;
use $UFNonneg[f\ :=\ funK,\ x\ :=\ x]$;
use $LAt[m\ :=\ 2\ +\ x\ -\ 2\ *\ funK\,x\ +\ 2\ *\ funV\,x,\ t\ :=\ 1\ +\ funV\,x,\ k\ :=\ funV\,x\ -\ funK\,x$
$+\ 1]$;
prove;
■

$C6L == clos\,\{cst\,1, funL\}\,\{opPlusI, opBox\}\,\{comp\}$

proof [C6L$domainCheck]
prove;
■

theorem disabled rule C6LDef
$C6L = clos\,\{cst\,1, funL\}\,\{opPlusI, opBox\}\,\{comp\}$

proof
invoke $C6L$;
prove;
■

theorem C6LInPower
$C6L \in \mathbb{P}(\mathbb{N} \to \mathbb{N})$

proof
use $ClosInPow[I\ :=\ \{\ cst\,1,\ funL\ \},\ U\ :=\ \{\ opPlusI,\ opBox\ \},\ B\ :=\ \{\ comp\ \}]$;
with enabled $(C6LDef)$ *prove*;
■

theorem disabled rule C6LUF
$\forall\, F : \mathbb{P}(\mathbb{Z} \times \mathbb{Z}) \mid F \in C6L \bullet F \in \mathbb{N} \to \mathbb{N}$

proof
prove;
use $ClosElemUF[I\ :=\ \{\ cst\,1,\ funL\ \},\ U\ :=\ \{\ opPlusI,\ opBox\ \},\ B\ :=\ \{\ comp\ \},\ F\ :=\ F]$;
with enabled $(C6LDef)$ *prove*;
■

theorem rule C6LInit0
$cst\,1 \in C6L$

proof
with enabled $(C6LDef)$ *prove*;
■

theorem rule C6LInit1
$funL \in C6L$

proof
with enabled (*C6LDef*) *prove*;
■

theorem rule C6LUnop0
$\forall F : \mathbb{N} \to \mathbb{N} \mid F \in C6L \bullet opPlusI\ F \in C6L$

proof
with enabled (*C6LDef*) *prove*;
■

theorem rule C6LUnop1
$\forall F : \mathbb{N} \to \mathbb{N} \mid F \in C6L \bullet opBox\ F \in C6L$

proof
with enabled (*C6LDef*) *prove*;
■

theorem rule C6LBinop0
$\forall F, G : \mathbb{N} \to \mathbb{N} \mid F \in C6L \land G \in C6L \bullet comp(F, G) \in C6L$

proof
with enabled (*C6LDef*) *prove*;
■

theorem KInLExpr
$comp(comp(funL, opPlusI\ (cst\ 1)), opPlusI\ (comp(opPlusI\ (opPlusI\ (cst\ 1)), funL))) \in C6L$

proof
prove;
■

theorem rule KInL
$funK \in C6L$

proof
use KInLExpr;
use KExprL;
prove;
■

theorem Theorem68LSub1U
$C6 \in UClosed\ \{opPlusI, opBox\}$

proof
use C6InPower;
with enabled (*UClosedDef*, *C6UF*) *prove*;
split u = opPlusI;
cases;
prove;
next;
prove;
next;
■

theorem Theorem68LSub1B
$C6 \in BClosed\,\{comp\}$

proof
use $C6InPower$;
with enabled $(BClosedDef,\ C6UF)$ *prove*;
prove;
■

theorem Theorem68LSub1
$C6L \subseteq C6$

proof
use $ClosLeast[I := \{\ cst\,1,\ funL\ \},\ U := \{\ opPlusI,\ opBox\ \},\ B := \{\ comp\ \},\ Z := C6]$;
use $C6InPower$;
use $Theorem68LSub1U$;
use $Theorem68LSub1B$;
with enabled $(C6LDef)$ *prove*;
■

theorem Theorem68LSubKU
$C6L \in UClosed\,\{opPlusI, opBox\}$

proof
use $C6LInPower$;
with enabled $(UClosedDef,\ C6LUF)$ *prove*;
split $u = opPlusI$;
cases;
prove;
next;
prove;
next;
■

theorem Theorem68LSubKB
$C6L \in BClosed\,\{comp\}$

proof
use $C6LInPower$;
with enabled $(BClosedDef,\ C6LUF)$ *prove*;
prove;
■

theorem Theorem68LSubK
$C6K \subseteq C6L$

proof
use $ClosLeast[I := \{\ cst\,1,\ funK\ \},\ U := \{\ opPlusI,\ opBox\ \},\ B := \{\ comp\ \},\ Z := C6L]$;
use $C6LInPower$;
use $Theorem68LSubKU$;
use $Theorem68LSubKB$;
with enabled $(C6KDef)$ *prove*;
■

theorem Theorem68LSub2
$C6 \subseteq C6L$

proof
use $Theorem68LSubK$;
use $Theorem68K$;
prove;
■

theorem Theorem68L
$C6L = C6$

proof
use $Theorem68LSub1$;
use $Theorem68LSub2$;
use $C6LInPower$;
use $C6InPower$;
use $SubAntisym[\mathbb{N} \to \mathbb{N}][A := C6L,\ B := C6]$;
prove;
■

theorem Theorem68
$C6K = C6 \wedge C6L = C6$

proof
use $Theorem68K$;
use $Theorem68L$;
prove;
■

7 Remark 6.9 and iteration from any $a \geq 1$

7.1 The product, multiples and iteration from a in $\operatorname{clos}\mathcal{F}$

$FG = \mathrm{Hf}((F+G)^2 - (F^2+G^2))$ for arbitrary F, G.

theorem TimesEq
$\forall F, G : \mathbb{N} \to \mathbb{N} \bullet comp(funHf, opDiffE(comp(funSq, opPlus(F, G)), opPlus(comp(funSq, F), comp(funSq, G)))) = opOtimes(F, G)$

proof
apply UFExt to predicate $comp(funHf,\ opDiffE(comp(funSq,\ opPlus(F,\ G)),\ opPlus(comp(funSq,\ F),\ comp(funSq,\ G)))) = opOtimes(F,\ G)$;
prove;
use $MulNonneg[a := F\,x,\ b := G\,x]$;
use $DiffEIsDifference[F := comp(funSq,\ opPlus(F,\ G)),\ G := opPlus(comp(funSq,\ F), comp(funSq,\ G)),\ x := x]$;
prove;
use $DivMod[x := 2 * (F\,x * G\,x),\ d := 2,\ q := F\,x * G\,x,\ r := 0]$;
prove;
■

theorem rule C6Times
$\forall F, G : \mathbb{N} \to \mathbb{N} \mid F \in C6 \wedge G \in C6 \bullet opOtimes(F, G) \in C6$

proof
use $TimesEq[F := F,\ G := G]$;
prove;
■

Multiples $M_r(x) = rx$, and $M_r = (\ldots(\bar{0}^+)^+\ldots)^+$ (r times) in every closure containing $\bar{0}$ and closed under F^+.

$$
\begin{array}{|l}
funM : \mathbb{N} \to (\mathbb{N} \to \mathbb{N}) \\
\hline
\langle\!\langle \text{rule MVal} \rangle\!\rangle \\
\forall\, r, x : \mathbb{N} \bullet funM\ r\ x = r * x
\end{array}
$$

proof [funM$domainCheck]
prove;
■

theorem rule MInner
$\forall\, r : \mathbb{N} \bullet (\lambda\, x : \mathbb{N} \bullet r * x) \in \mathbb{N} \to \mathbb{N}$

proof
apply FunIntro to predicate $(\lambda\ x : \mathbb{N} \bullet r * x) \in \mathbb{N} \to \mathbb{N}$;
with enabled (*RelPow*, *inPower*) *prove by reduce*;
■

$MWit == (\lambda\, r : \mathbb{N} \bullet (\lambda\, x : \mathbb{N} \bullet r * x))$

theorem MWitFun
$MWit \in \mathbb{N} \to (\mathbb{N} \to \mathbb{N})$

proof
invoke MWit;
apply FunIntro to predicate $(\lambda\ r : \mathbb{N} \bullet (\lambda\ x : \mathbb{N} \bullet r * x)) \in \mathbb{N} \to (\mathbb{N} \to \mathbb{N})$;
with enabled (*RelPow*, *inPower*) *prove by reduce*;
■

theorem MWitApp
$\forall\, r : \mathbb{N} \bullet MWit\ r = (\lambda\, x : \mathbb{N} \bullet r * x)$

proof
use MWitFun;
use pairInFunction$[\mathbb{N},\ \mathbb{N} \to \mathbb{N}][f := MWit,\ x := r,\ y := (\lambda\ x : \mathbb{N} \bullet r * x)]$;
invoke MWit;
prove by reduce;
■

theorem rule MWitVal
$\forall\, r, x : \mathbb{N} \bullet MWit\ r\ x = r * x$

proof
prove;
use MWitApp$[r := r]$;
use pairInFunction$[\mathbb{N},\ \mathbb{N}][f := (\lambda\ x : \mathbb{N} \bullet r * x),\ x := x,\ y := r * x]$;
prove;
■

theorem MEx
$\exists\, funM : \mathbb{N} \to (\mathbb{N} \to \mathbb{N}) \bullet \forall\, r, x : \mathbb{N} \bullet funM\ r\ x = r * x$

proof
instantiate funM == MWit;
use MWitFun;
prove;
■

theorem rule NatUFApp
$\forall\, g : \mathbb{N} \to (\mathbb{N} \to \mathbb{N});\ r : \mathbb{N} \bullet g\ r \in \mathbb{N} \to \mathbb{N}$

proof
prove;
■

theorem MZero
$funM\ 0 = cst\ 0$

proof
apply UFExt to predicate funM 0 = cst 0;
prove;
■

theorem MSuc
$\forall\, r : \mathbb{N} \bullet opPlusI\ (funM\ r) = funM\ (1 + r)$

proof
apply UFExt to predicate opPlusI (funM r) = funM (1 + r);
prove;
■

$MSet == \{r : \mathbb{N} \mid \forall\, I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \mid$
$cst\ 0 \in clos\ I\ U\ B \wedge opPlusI \in U \bullet funM\ r \in clos\ I\ U\ B\}$

proof [MSet$domainCheck]
prove;
■

theorem rule MBase
$0 \in MSet$

proof
invoke MSet;
use MZero;
prove;
■

theorem rule MStep
$\forall\, r : MSet \bullet 1 + r \in MSet$

proof
invoke MSet;
prove;
instantiate I__0 == I, U__0 == U, B__0 == B;
use MSuc[$r := r$];
prove;
equality substitute funM (1 + r);
rewrite;
■

theorem MSub
$\mathbb{N} \subseteq MSet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq MSet$;
prove;
invoke MSet;
prove;
■

theorem MInClos
$\forall I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ r : \mathbb{N} \mid cst\, 0 \in clos\ I\ U\ B \wedge opPlusI \in U \bullet funM\ r \in clos\ I\ U\ B$

proof
prove;
use MSub;
apply subsetDef to predicate $\mathbb{N} \subseteq MSet$;
apply inPower to predicate $\mathbb{N} \in \mathbb{P}\ MSet$;
prove;
instantiate $e == r$;
invoke MSet;
prove;
instantiate $I__1 == I,\ U__1 == U,\ B__1 == B$;
prove;
■

theorem rule MInC6
$\forall r : \mathbb{N} \bullet funM\ r \in C6$

proof
prove;
use MInClos$[I := \{ cst\, 1,\ funE \},\ U := \{ opPlusI,\ opBox \},\ B := \{ comp \},\ r := r]$;
use ZeroIn;
with enabled $(C6Def)$ *prove*;
■

Soundness for iteration from a: $F^{\Box(a)} = \hat{P}\, T^{\Box} S$ with $T(0) = a + 1$ and $T(y+1) = F(y) + 1$, namely $T = \mathrm{Sgn} \cdot (SF\hat{P}) + M_{a+1} O$.

theorem BoxATVal
$\forall a : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N};\ y : \mathbb{N} \bullet opPlus(opOtimes(funSgn, comp(comp(funS, F), exprPhat)), comp(funM\ (1 + a), funO))\ 0 = 1 + a \wedge opPlus(opOtimes(funSgn, comp(comp(funS, F), exprPhat)), comp(funM\ (1 + a), funO))\ (1 + y) = 1 + F\ y$

proof
prove;
■

$BoxASet == \{n : \mathbb{N} \mid \forall a : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N} \bullet opBox\ (opPlus(opOtimes(funSgn, comp(comp(funS, F), exprPhat)), comp(funM\ (1 + a), funO)))\ (1 + n) = 1 + opBoxA\ a\ F\ n\}$

proof [BoxASet$domainCheck]
prove;
■

theorem rule BoxABase
$0 \in BoxASet$

proof
invoke $BoxASet$;
prove;
■

theorem rule BoxAStep
$\forall\, n : BoxASet \bullet 1 + n \in BoxASet$

proof
invoke $BoxASet$;
with disabled ($BoxSuc$) *prove*;
instantiate $a__0 == a,\ F__0 == F$;
use $BoxSuc[F__0 := opPlus(opOtimes(funSgn,\ comp(comp(funS,\ F),\ exprPhat)),\ comp(funM\ (1 + a),\ funO)),\ x := 1 + n]$;
use $BoxATVal[a := a,\ F := F,\ y := opBoxA\ a\ F\ n]$;
with disabled ($BoxSuc,\ PlusVal$) *prove*;
■

theorem BoxASub
$\mathbb{N} \subseteq BoxASet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq BoxASet$;
prove;
invoke $BoxASet$;
prove;
■

theorem BoxAExprEq
$\forall\, a : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N} \bullet comp(comp(exprPhat, opBox\ (opPlus(opOtimes(funSgn, comp(comp(funS, F), exprPhat)), comp(funM\ (1 + a), funO)))), funS) = opBoxA\ a\ F$

proof
apply UFExt to predicate $comp(comp(exprPhat,\ opBox\ (opPlus(opOtimes(funSgn,\ comp(comp(funS,\ F),\ exprPhat)),\ comp(funM\ (1 + a),\ funO)))),\ funS) = opBoxA\ a\ F$;
with disabled ($BoxSuc$) *prove*;
use $BoxASub$;
apply subsetDef to predicate $\mathbb{N} \subseteq BoxASet$;
apply inPower to predicate $\mathbb{N} \in \mathbb{P}\ BoxASet$;
with disabled ($BoxSuc$) *prove*;
instantiate $e == x$;
invoke $BoxASet$;
with disabled ($BoxSuc$) *prove*;
instantiate $a__1 == a,\ F__1 == F$;
with disabled ($BoxSuc$) *prove*;
equality substitute $opBox\ (opPlus(opOtimes(funSgn,\ comp(comp(funS,\ F),\ exprPhat)),\ comp(funM\ (1 + a),\ funO)))\ (1 + x)$;
with disabled ($BoxSuc$) *rewrite*;
■

theorem rule BoxAIn
$\forall\, a : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N} \mid F \in C6 \bullet opBoxA\ a\ F \in C6$

proof
use $BoxAExprEq[a := a,\ F := F]$;
prove;
■

7.2 Powers under composition

F^k is the Toolkit's iterate $F^k =$ `iter k F`.

$$IterSSet == \{k : \mathbb{N} \mid \forall x : \mathbb{N} \bullet iter[\mathbb{N}]\, k\, funS\, x = x + k\}$$

proof [IterSSet$domainCheck]
prove;
■

theorem rule IterSBase
$0 \in IterSSet$

proof
invoke IterSSet;
prove;
■

theorem rule IterSStep
$\forall k : IterSSet \bullet 1 + k \in IterSSet$

proof
invoke IterSSet;
prove;
instantiate $x__0 == x$;
prove;
■

theorem IterSSub
$\mathbb{N} \subseteq IterSSet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq IterSSet$;
prove;
invoke IterSSet;
prove;
■

theorem IterSAll
$\forall k : \mathbb{N} \bullet \forall x : \mathbb{N} \bullet iter[\mathbb{N}]\, k\, funS\, x = x + k$

proof
prove;
use IterSSub;
invoke IterSSet;
with enabled (*inPower*) *prove*;
instantiate $e == k$;
prove;
instantiate $x__1 == x$;
prove;
■

Predecessors ($P(x+1) = x$) undo successors: $P^k(x+k) = x$.

$$Preds == \{P : \mathbb{N} \to \mathbb{N} \mid \forall x : \mathbb{N} \bullet P\,(1 + x) = x\}$$

proof [Preds$domainCheck]
prove;
■

theorem rule PredsVal
$\forall P : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \mid P \in \mathit{Preds} \bullet P\,(1 + x) = x$

proof
prove;
invoke Preds;
prove;
instantiate $x__0 == x$;
prove;
■

$\mathit{IterPSet} == \{k : \mathbb{N} \mid \forall P : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \mid P \in \mathit{Preds} \bullet \mathit{iter}[\mathbb{N}]\,k\,P\,(x + k) = x\}$

proof [IterPSet$domainCheck]
prove;
■

theorem rule IterPBase
$0 \in \mathit{IterPSet}$

proof
invoke IterPSet;
prove;
■

theorem rule IterPStep
$\forall k : \mathit{IterPSet} \bullet 1 + k \in \mathit{IterPSet}$

proof
invoke IterPSet;
prove;
instantiate $P__0 == P,\ x__0 == 1 + x$;
use IterSuc$[F := P,\ n := k,\ a := 1 + x + k]$;
prove;
equality substitute $\mathit{iter}[\mathbb{N}]\,k\,P\,(1 + (k + x))$;
rewrite;
■

theorem IterPSub
$\mathbb{N} \subseteq \mathit{IterPSet}$

proof
apply natInduction to predicate $\mathbb{N} \subseteq \mathit{IterPSet}$;
prove;
invoke IterPSet;
prove;
■

theorem IterPAll
$\forall k : \mathbb{N} \bullet \forall P : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \mid P \in \mathit{Preds} \bullet \mathit{iter}[\mathbb{N}]\,k\,P\,(x + k) = x$

proof
prove;
use IterPSub;
invoke IterPSet;
with enabled (*inPower*) *prove*;
instantiate $e == k$;
prove;
instantiate $P__1 == P,\ x__1 == x$;
prove;
■

$F^k\, G$ stays in every closure containing composition.

$PowSet == \{k : \mathbb{N} \mid \forall\, I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ F, G : \mathbb{N} \to \mathbb{N} \mid comp \in B \wedge F \in clos\ I\ U\ B \wedge G \in clos\ I\ U\ B \bullet comp(iter[\mathbb{N}]\, k\, F, G) \in clos\ I\ U\ B\}$

proof [PowSet$domainCheck]
prove;
■

theorem IterCompZero
$\forall\, F, G : \mathbb{N} \to \mathbb{N} \bullet comp(iter[\mathbb{N}]\, 0\, F, G) = G$

proof
apply UFExt to predicate $comp(iter[\mathbb{N}]\, 0\, F,\ G) = G$;
prove;
■

theorem IterCompSuc
$\forall\, F, G : \mathbb{N} \to \mathbb{N};\ k : \mathbb{N} \bullet comp(iter[\mathbb{N}]\, (1 + k)\, F, G) = comp(F, comp(iter[\mathbb{N}]\, k\, F, G))$

proof
apply UFExt to predicate $comp(iter[\mathbb{N}]\, (1 + k)\, F,\ G) = comp(F,\ comp(iter[\mathbb{N}]\, k\, F,\ G))$;
prove;
■

theorem rule PowBase
$0 \in PowSet$

proof
invoke PowSet;
prove;
use IterCompZero$[F := F,\ G := G]$;
prove;
■

theorem rule PowStep
$\forall\, k : PowSet \bullet 1 + k \in PowSet$

proof
invoke PowSet;
prove;
instantiate $I__0 == I,\ U__0 == U,\ B__0 == B,\ F__0 == F,\ G__0 == G$;
use IterCompSuc$[F := F,\ G := G,\ k := k]$;
prove;
■

theorem PowSub
$\mathbb{N} \subseteq PowSet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq PowSet$;
prove;
invoke PowSet;
prove;
■

theorem PowIn
$\forall I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ F,$
$G : \mathbb{N} \to \mathbb{N};\ k : \mathbb{N} \mid comp \in B \wedge F \in clos\ I\ U\ B \wedge G \in clos\ I\ U\ B \bullet comp(iter[\mathbb{N}]\ k\ F,$
$G) \in clos\ I\ U\ B$

proof
prove;
use PowSub;
apply subsetDef to predicate $\mathbb{N} \subseteq PowSet$;
apply inPower to predicate $\mathbb{N} \in \mathbb{P}\ PowSet$;
prove;
instantiate $e == k$;
invoke PowSet;
prove;
instantiate $I__1 == I,\ U__1 == U,\ B__1 == B,\ F__1 == F,\ G__1 == G$;
prove;
■

Recovering iteration from 0 out of iteration from a (as in Remark 5.10 of the paper): $F^{\square} = P^a(S^a F P^a)^{\square(a)}$ for a predecessor P.

$BoxFromASet == \{n : \mathbb{N} \mid \forall a : \mathbb{N};\ F, P : \mathbb{N} \to \mathbb{N} \mid P \in Preds \bullet opBoxA\ a\ (comp(comp(iter[\mathbb{N}]\ a\ funS, F),$
$iter[\mathbb{N}]\ a\ P))\ n = opBox\ F\ n + a\}$

proof [BoxFromASet$domainCheck]
prove;
■

theorem rule BoxFromABase
$0 \in BoxFromASet$

proof
invoke BoxFromASet;
prove;
■

theorem rule BoxFromAStep
$\forall n : BoxFromASet \bullet 1 + n \in BoxFromASet$

proof
invoke BoxFromASet;
prove;
instantiate $a__0 == a,\ F__0 == F,\ P__0 == P$;
use IterPAll$[k := a,\ P := P,\ x := opBox\ F\ n]$;
use IterSAll$[k := a,\ x := F\ (opBox\ F\ n)]$;
prove;
■

theorem BoxFromASub
$\mathbb{N} \subseteq \mathit{BoxFromASet}$

proof
apply natInduction to predicate $\mathbb{N} \subseteq \mathit{BoxFromASet}$;
prove;
invoke BoxFromASet;
prove;
■

theorem BoxFromAAll
$\forall n : \mathbb{N} \bullet \forall a : \mathbb{N};\ F, P : \mathbb{N} \to \mathbb{N} \mid P \in \mathit{Preds} \bullet \mathit{opBoxA}\ a\ (\mathit{comp}(\mathit{comp}(\mathit{iter}[\mathbb{N}]\ a\ \mathit{funS}, F),$
$\mathit{iter}[\mathbb{N}]\ a\ P))\ n = \mathit{opBox}\ F\ n + a$

proof
prove;
use BoxFromASub;
invoke BoxFromASet;
with enabled (*inPower*) *prove*;
instantiate $e == n$;
prove;
instantiate $a__1 == a,\ F__1 == F,\ P__1 == P$;
prove;
■

theorem BoxFromBoxA
$\forall a : \mathbb{N};\ F, P : \mathbb{N} \to \mathbb{N} \mid P \in \mathit{Preds} \bullet \mathit{comp}(\mathit{iter}[\mathbb{N}]\ a\ P, \mathit{opBoxA}\ a\ (\mathit{comp}(\mathit{comp}(\mathit{iter}[\mathbb{N}]\ a\ \mathit{funS}, F),$
$\mathit{iter}[\mathbb{N}]\ a\ P))) = \mathit{opBox}\ F$

proof
apply UFExt to predicate $\mathit{comp}(\mathit{iter}[\mathbb{N}]\ a\ P,\ \mathit{opBoxA}\ a\ (\mathit{comp}(\mathit{comp}(\mathit{iter}[\mathbb{N}]\ a\ \mathit{funS},\ F),\ \mathit{iter}[\mathbb{N}]\ a\ P)))$
$= \mathit{opBox}\ F$;
prove;
use $\mathit{BoxFromAAll}[n := x,\ a := a,\ F := F,\ P := P]$;
use $\mathit{IterPAll}[k := a,\ P := P,\ x := \mathit{opBox}\ F\ x]$;
prove;
■

7.3 The bases $\langle X, F^+, FG, F^{\square(a)} \rangle$: the common part

X admits multipliers if $X(0) = 0$ and for every $i \geq 1$ some r has $X(r(i+1)) = i$; then $T_j = XM_{r(j)}$ maps j to $j-1$ and 0 to 0.

$\mathit{Mult} == \{X : \mathbb{N} \to \mathbb{N} \mid X\ 0 = 0 \land (\forall i : \mathbb{N} \mid 1 \leq i \bullet \exists r : \mathbb{N} \bullet X\ (r * (1+i)) = i)\}$

proof [Mult$domainCheck]
prove;
■

theorem MultZero
$\forall X : \mathbb{N} \to \mathbb{N} \mid X \in \mathit{Mult} \bullet X\ 0 = 0$

proof
invoke Mult;
prove;
■

theorem MultEx
$\forall X : \mathbb{N} \to \mathbb{N};\ i : \mathbb{N} \mid X \in \mathit{Mult} \wedge 1 \leq i \bullet \exists r : \mathbb{N} \bullet X\,(r * (1 + i)) = i$

proof
prove;
invoke Mult;
prove;
instantiate $i__0 == i$;
prove;
instantiate $r__1 == r$;
prove;
■

The chain $N = T_2 \cdots T_{k+1}$ maps $k + 1$ to 1 and 0 to 0.

$\mathit{ChainSet} == \{k : \mathbb{N} \mid \forall X : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \mid X \in \mathit{Mult} \wedge \mathit{cst}\,0 \in \mathit{clos}\,\{X\}\,\{\mathit{opPlusI},\ \mathit{opBoxA}\,a\}\,\{\mathit{comp}\} \bullet \exists W : \mathbb{N} \to \mathbb{N} \mid W \in \mathit{clos}\,\{X\}\,\{\mathit{opPlusI},\ \mathit{opBoxA}\,a\}\,\{\mathit{comp}\} \bullet W\,(1 + k) = 1 \wedge W\,0 = 0\}$

proof [ChainSet$domainCheck]
prove;
■

theorem rule ChainBase
$0 \in \mathit{ChainSet}$

proof
invoke ChainSet;
prove;
use MInClos[$I := \{ X \},\ U := \{ \mathit{opPlusI},\ \mathit{opBoxA}\,a \},\ B := \{ \mathit{comp} \},\ r := 1$];
prove;
instantiate $W == \mathit{funM}\,1$;
prove;
■

theorem rule ChainStep
$\forall k : \mathit{ChainSet} \bullet 1 + k \in \mathit{ChainSet}$

proof
invoke ChainSet;
prove;
instantiate $X__0 == X,\ a__0 == a$;
prove;
use MultEx[$X := X,\ i := 1 + k$];
use MultZero[$X := X$];
prove;
use MInClos[$I := \{ X \},\ U := \{ \mathit{opPlusI},\ \mathit{opBoxA}\,a \},\ B := \{ \mathit{comp} \},\ r := r$];
prove;
instantiate $W__1 == \mathit{comp}(W,\ \mathit{comp}(X,\ \mathit{funM}\,r))$;
prove;
■

theorem ChainSub
$\mathbb{N} \subseteq \mathit{ChainSet}$

proof
apply natInduction to predicate $\mathbb{N} \subseteq \mathit{ChainSet}$;
prove;
invoke ChainSet;
prove;
■

theorem ChainAll
$\forall X : \mathbb{N} \to \mathbb{N};\ a, k : \mathbb{N} \mid X \in Mult \wedge cst\, 0 \in clos\, \{X\}\, \{opPlusI,$
$opBoxA\, a\}\, \{comp\} \bullet \exists W : \mathbb{N} \to \mathbb{N} \mid W \in clos\, \{X\}\, \{opPlusI,$
$opBoxA\, a\}\, \{comp\} \bullet W\, (1 + k) = 1 \wedge W\, 0 = 0$

proof
prove;
use ChainSub;
apply subsetDef to predicate $\mathbb{N} \subseteq ChainSet$;
apply inPower to predicate $\mathbb{N} \in \mathbb{P}\ ChainSet$;
prove;
instantiate $e == k$;
invoke ChainSet;
prove;
instantiate $X__1 == X,\ a__1 == a$;
prove;
instantiate $W__2 == W$;
prove;
■

theorem MOne
$funM\, 1 = funI \wedge funM\, 2 = funD$

proof
use $UFExt[F := funM\, 1,\ G := funI]$;
use $UFExt[F := funM\, 2,\ G := funD]$;
prove;
■

$BoxAISet == \{n : \mathbb{N} \mid \forall a : \mathbb{N} \bullet opBoxA\, a\, funI\, n = a\}$

proof [BoxAISet$domainCheck]
prove;
■

theorem rule BoxAIBase
$0 \in BoxAISet$

proof
invoke BoxAISet;
prove;
■

theorem rule BoxAIStep
$\forall n : BoxAISet \bullet 1 + n \in BoxAISet$

proof
invoke BoxAISet;
prove;
■

theorem BoxAISub
$\mathbb{N} \subseteq BoxAISet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq BoxAISet$;
prove;
invoke BoxAISet;
prove;
■

theorem BoxAIAll
$\forall n : \mathbb{N} \bullet \forall a : \mathbb{N} \bullet opBoxA\ a\ funI\ n = a$

proof
prove;
use BoxAISub;
invoke BoxAISet;
with enabled (*inPower*) *prove*;
instantiate $e == n$;
prove;
instantiate $a__1 == a$;
prove;
■

theorem OneFromW
$\forall W : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \mid W\ a = 1 \bullet comp(W, opBoxA\ a\ funI) = cst\ 1$

proof
apply UFExt to predicate $comp(W,\ opBoxA\ a\ funI) = cst\ 1$;
prove;
use $BoxAIAll[n := x]$;
prove;
■

theorem OFromW
$\forall W : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \mid W\ a = 1 \land W\ 0 = 0 \bullet comp(W, opBoxA\ a\ (cst\ 0)) = funO$

proof
apply UFExt to predicate $comp(W,\ opBoxA\ a\ (cst\ 0)) = funO$;
prove;
split $x = 0$;
cases;
prove;
next;
use $BoxASuc[a__0 := a,\ F__0 := cst\ 0,\ x := x - 1]$;
prove;
next;
■

theorem ExtAI
$\forall X : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \mid X \in Mult \land cst\ 0 \in clos\ \{X\}\ \{opPlusI,$
$opBoxA\ a\}\ \{comp\} \bullet funI \in clos\ \{X\}\ \{opPlusI, opBoxA\ a\}\ \{comp\} \land funD \in clos\ \{X\}\ \{opPlusI,$
$opBoxA\ a\}\ \{comp\}$

proof
prove;
use $MInClos[I := \{\ X\ \},\ U := \{\ opPlusI,\ opBoxA\ a\ \},\ B := \{\ comp\ \},\ r := 1]$;
use $MInClos[I := \{\ X\ \},\ U := \{\ opPlusI,\ opBoxA\ a\ \},\ B := \{\ comp\ \},\ r := 2]$;
use MOne;
prove;
■

theorem OneInCB
$\forall X, W : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \mid W \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\} \wedge funI \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\} \wedge W\ a = 1 \bullet cst\ 1 \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\}$

proof
use $OneFromW[W := W,\ a := a]$;
prove;
■

theorem OExprInCB
$\forall X, W : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \mid W \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\} \wedge cst\ 0 \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\} \bullet comp(W, opBoxA\ a\ (cst\ 0)) \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\}$

proof
prove;
■

theorem OInCB
$\forall X, W : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \mid W \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\} \wedge cst\ 0 \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\} \wedge W\ a = 1 \wedge W\ 0 = 0 \bullet funO \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\}$

proof
use $OExprInCB[X := X,\ W := W,\ a := a]$;
use $OFromW[W := W,\ a := a]$;
prove;
■

theorem ExtAOne
$\forall X : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \mid 1 \leq a \wedge X \in Mult \wedge cst\ 0 \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\} \bullet cst\ 1 \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\} \wedge funO \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\}$

proof
prove;
use $ExtAI[X := X,\ a := a]$;
use $ChainAll[X := X,\ a := a,\ k := a - 1]$;
prove;
use $OneInCB[X := X,\ W := W,\ a := a]$;
use $OInCB[X := X,\ W := W,\ a := a]$;
prove;
■

theorem SExprInCB
$\forall X : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \mid cst\ 1 \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\} \bullet opPlusI\ (cst\ 1) \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\}$

proof
prove;
■

theorem ExtAS
$\forall X : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \mid 1 \leq a \wedge X \in Mult \wedge cst\ 0 \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\} \bullet funS \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\}$

proof
prove;
use ExtAOne[$X := X,\ a := a$];
prove;
use SExprInCB[$X := X,\ a := a$];
use SExpr;
prove;
■

theorem IterCompI
$\forall F : \mathbb{N} \to \mathbb{N};\ k : \mathbb{N} \bullet comp(iter[\mathbb{N}]\, k\, F, funI) = iter[\mathbb{N}]\, k\, F$

proof
apply UFExt to predicate $comp(iter[\mathbb{N}]\, k\, F,\ funI) = iter[\mathbb{N}]\, k\, F$;
prove;
■

theorem IterIn
$\forall I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ F : \mathbb{N} \to \mathbb{N};\ k : \mathbb{N} \mid comp \in B \wedge F \in clos\, I\ U\ B \wedge funI \in clos\, I\ U\ B \bullet iter[\mathbb{N}]\, k\, F \in clos\, I\ U\ B$

proof
prove;
use PowIn[$I := I,\ U := U,\ B := B,\ F := F,\ G := funI,\ k := k$];
use IterCompI[$F := F,\ k := k$];
prove;
■

With a predecessor in the closure, iteration from 0 is available.

theorem ExtABox
$\forall X, P, F : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \mid 1 \leq a \wedge X \in Mult \wedge cst\, 0 \in clos\, \{X\}\, \{opPlusI,$
$opBoxA\ a\}\, \{comp\} \wedge P \in clos\, \{X\}\, \{opPlusI,$
$opBoxA\ a\}\, \{comp\} \wedge P \in Preds \wedge F \in clos\, \{X\}\, \{opPlusI,$
$opBoxA\ a\}\, \{comp\} \bullet opBox\, F \in clos\, \{X\}\, \{opPlusI, opBoxA\ a\}\, \{comp\}$

proof
prove;
use ExtAI[$X := X,\ a := a$];
use ExtAS[$X := X,\ a := a$];
use IterIn[$I := \{\ X\ \},\ U := \{\ opPlusI,\ opBoxA\ a\ \},\ B := \{\ comp\ \},\ F := P,\ k := a$];
use IterIn[$I := \{\ X\ \},\ U := \{\ opPlusI,\ opBoxA\ a\ \},\ B := \{\ comp\ \},\ F := funS,\ k := a$];
use BoxFromBoxA[$a := a,\ F := F,\ P := P$];
prove;
■

7.4 Simulation

theorem CBUClosed
$\forall X, P : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \mid 1 \leq a \wedge X \in Mult \wedge cst\, 0 \in clos\, \{X\}\, \{opPlusI,$
$opBoxA\ a\}\, \{comp\} \wedge P \in clos\, \{X\}\, \{opPlusI, opBoxA\ a\}\, \{comp\} \wedge P \in Preds \bullet clos\, \{X\}\, \{opPlusI,$
$opBoxA\ a\}\, \{comp\} \in UClosed\, \{opPlusI, opBox\}$

proof
prove;
use ClosInPow[$I := \{ X \}$, $U := \{ opPlusI,\ opBoxA\ a \}$, $B := \{ comp \}$];
with enabled (*UClosedDef*) *prove*;
use ClosElemUF[$I := \{ X \}$, $U := \{ opPlusI,\ opBoxA\ a \}$, $B := \{ comp \}$, $F := F$];
split $u = opPlusI$;
cases;
prove;
next;
use ExtABox[$X := X$, $P := P$, $F := F$, $a := a$];
prove;
next;
■

theorem CBBClosed
$\forall X : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \bullet clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\} \in BClosed\,\{comp\}$

proof
prove;
use ClosInPow[$I := \{ X \}$, $U := \{ opPlusI,\ opBoxA\ a \}$, $B := \{ comp \}$];
with enabled (*BClosedDef*) *prove*;
■

theorem ExtASup
$\forall X, P : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \mid 1 \leq a \wedge X \in Mult \wedge cst\,0 \in clos\,\{X\}\,\{opPlusI,$
$opBoxA\ a\}\,\{comp\} \wedge P \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\} \wedge P \in Preds \bullet clos\,\{cst\,1,$
$X\}\,\{opPlusI, opBox\}\,\{comp\} \subseteq clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\}$

proof
prove;
use ClosLeast[$I := \{ cst\,1,\ X \}$, $U := \{ opPlusI,\ opBox \}$, $B := \{ comp \}$, $Z := clos\,\{ X \}\,\{$
$opPlusI,\ opBoxA\ a \}\,\{ comp \}$];
use ClosInPow[$I := \{ X \}$, $U := \{ opPlusI,\ opBoxA\ a \}$, $B := \{ comp \}$];
use CBUClosed[$X := X$, $P := P$, $a := a$];
use CBBClosed[$X := X$, $a := a$];
use ExtAOne[$X := X$, $a := a$];
prove;
■

theorem C6UClosedA
$\forall a : \mathbb{N} \bullet C6 \in UClosed\,\{opPlusI, opBoxA\ a\}$

proof
prove;
use C6InPower;
with enabled (*UClosedDef*, *C6UF*) *prove*;
use InPair[$\mathbb{P}\,(\mathbb{P}\,(\mathbb{Z} \times \mathbb{Z}) \times \mathbb{P}\,(\mathbb{Z} \times \mathbb{Z}))$][$u := u$, $p := opPlusI$, $q := opBoxA\ a$];
split $u = opBoxA\ a$;
cases;
prove;
next;
prove;
next;
■

theorem C6BClosedComp
$C6 \in BClosed\,\{comp\}$

proof
use $C6InPower$;
with enabled $(BClosedDef,\ C6UF)$ *prove*;
■

theorem ExtASub6
$\forall X : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \mid X \in C6 \bullet clos\,\{X\}\,\{opPlusI, opBoxA\,a\}\,\{comp\} \subseteq C6$

proof
prove;
use $ClosLeast[I := \{ X \},\ U := \{ opPlusI,\ opBoxA\,a \},\ B := \{ comp \},\ Z := C6]$;
use $C6InPower$;
use $C6UClosedA[a := a]$;
use $C6BClosedComp$;
prove;
■

7.5 Zero for a decreasing X: $\bar{0} = X^{a} X^{\square(a)}$

$Decr == \{X : \mathbb{N} \to \mathbb{N} \mid X\,0 = 0 \land (\forall y : \mathbb{N} \mid 0 < y \bullet X\,y < y)\}$

proof [Decr$domainCheck]
prove;
■

theorem DecrZero
$\forall X : \mathbb{N} \to \mathbb{N} \mid X \in Decr \bullet X\,0 = 0$

proof
invoke $Decr$;
prove;
■

theorem DecrLt
$\forall X : \mathbb{N} \to \mathbb{N};\ y : \mathbb{N} \mid X \in Decr \land 0 < y \bullet X\,y < y \land X\,0 = 0$

proof
prove;
invoke $Decr$;
prove;
instantiate $y_0 == y$;
prove;
■

$DecBoundSet == \{n : \mathbb{N} \mid \forall a : \mathbb{N};\ X : \mathbb{N} \to \mathbb{N} \mid X \in Decr \bullet opBoxA\,a\,X\,n \leq a\}$

proof [DecBoundSet$domainCheck]
prove;
■

theorem rule DecBoundBase
$0 \in DecBoundSet$

proof
invoke $DecBoundSet$;
prove;
■

theorem rule DecBoundStep
$\forall\, n : \mathit{DecBoundSet} \bullet 1 + n \in \mathit{DecBoundSet}$

proof
invoke *DecBoundSet*;
prove;
instantiate $a__0 == a,\ X__0 == X$;
use $\mathit{DecrZero}[X := X]$;
use $\mathit{DecrLt}[X := X,\ y := \mathit{opBoxA}\ a\ X\ n]$;
use $\mathit{UFNonneg}[f := X,\ x := \mathit{opBoxA}\ a\ X\ n]$;
use $\mathit{UFNonneg}[f := \mathit{opBoxA}\ a\ X,\ x := n]$;
split $\mathit{opBoxA}\ a\ X\ n = 0$;
cases;
prove;
next;
prove;
next;
■

theorem DecBoundSub
$\mathbb{N} \subseteq \mathit{DecBoundSet}$

proof
apply natInduction to predicate $\mathbb{N} \subseteq \mathit{DecBoundSet}$;
prove;
invoke *DecBoundSet*;
prove;
■

theorem DecBoundAll
$\forall\, n : \mathbb{N} \bullet \forall\, a : \mathbb{N};\ X : \mathbb{N} \to \mathbb{N} \mid X \in \mathit{Decr} \bullet \mathit{opBoxA}\ a\ X\ n \leq a$

proof
prove;
use *DecBoundSub*;
invoke *DecBoundSet*;
with enabled (*inPower*) *prove*;
instantiate $e == n$;
prove;
instantiate $a__1 == a,\ X__1 == X$;
prove;
■

$\mathit{DecIterSet} == \{k : \mathbb{N} \mid \forall\, X : \mathbb{N} \to \mathbb{N};\ y : \mathbb{N} \mid X \in \mathit{Decr} \land y \leq k \bullet \mathit{iter}[\mathbb{N}]\ k\ X\ y = 0\}$

proof [DecIterSet$domainCheck]
prove;
■

theorem rule DecIterBase
$0 \in \mathit{DecIterSet}$

proof
invoke *DecIterSet*;
prove;
■

theorem rule DecIterStep
$\forall\, k : \mathit{DecIterSet} \bullet 1 + k \in \mathit{DecIterSet}$

proof
invoke DecIterSet;
prove;
instantiate $X__0 == X,\ y__0 == X\ y$;
use IterSucR$[F := X,\ k := k,\ a := y]$;
use DecrZero$[X := X]$;
use DecrLt$[X := X,\ y := y]$;
use UFNonneg$[f := X,\ x := y]$;
split $y = 0$;
cases;
prove;
next;
prove;
next;
■

theorem DecIterSub
$\mathbb{N} \subseteq \mathit{DecIterSet}$

proof
apply natInduction to predicate $\mathbb{N} \subseteq \mathit{DecIterSet}$;
prove;
invoke DecIterSet;
prove;
■

theorem DecIterAll
$\forall\, k : \mathbb{N} \bullet \forall\, X : \mathbb{N} \to \mathbb{N};\ y : \mathbb{N} \mid X \in \mathit{Decr} \land y \leq k \bullet \mathit{iter}[\mathbb{N}]\ k\ X\ y = 0$

proof
prove;
use DecIterSub;
invoke DecIterSet;
with enabled (*inPower*) *prove*;
instantiate $e == k$;
prove;
instantiate $X__1 == X,\ y__1 == y$;
prove;
■

theorem DecZeroEq
$\forall\, X : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \mid X \in \mathit{Decr} \bullet \mathit{comp}(\mathit{iter}[\mathbb{N}]\ a\ X, \mathit{opBoxA}\ a\ X) = \mathit{cst}\ 0$

proof
apply UFExt to predicate $\mathit{comp}(\mathit{iter}[\mathbb{N}]\ a\ X,\ \mathit{opBoxA}\ a\ X) = \mathit{cst}\ 0$;
prove;
use DecBoundAll$[n := x,\ a := a,\ X := X]$;
use DecIterAll$[k := a,\ X := X,\ y := \mathit{opBoxA}\ a\ X\ x]$;
prove;
■

theorem ZeroInCB
$\forall\, X : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \mid X \in \mathit{Decr} \bullet \mathit{cst}\ 0 \in \mathit{clos}\ \{X\}\ \{\mathit{opPlusI}, \mathit{opBoxA}\ a\}\ \{\mathit{comp}\}$

proof
prove;
use PowIn[$I := \{ X \}$, $U := \{ opPlusI,\ opBoxA\ a \}$, $B := \{ comp \}$, $F := X$, $G := opBoxA\ a\ X$, $k := a$];
use DecZeroEq[$X := X$, $a := a$];
prove;
■

7.6 The case $X = E$

theorem ELt
$\forall\, y : \mathbb{N} \mid 0 < y \bullet funE\ y < y$

proof
prove;
use RtSpec[$x := y$];
use RtPos[$y := y$];
use timesMonotonic2[$a := 1$, $b := 1$, $c := funRt\ y$, $d := funRt\ y$];
with enabled (*EVal*) *prove*;
■

theorem EDecr
$funE \in Decr$

proof
invoke Decr;
prove;
use ELt[$y := y$];
prove;
■

theorem EMult
$funE \in Mult$

proof
invoke Mult;
prove;
instantiate $r == i$;
use ESqAdd[$m := i$, $k := i$];
prove;
■

theorem rule IterSVal
$\forall\, k, x : \mathbb{N} \bullet iter[\mathbb{N}]\ k\ funS\ x = x + k$

proof
prove;
use IterSAll[$k := k$, $x := x$];
prove;
■

theorem RRec
$\forall\, t : \mathbb{N} \bullet funR\ (1 + t) = 2 * funQ\ (4 + funR\ t) + funR\ t + 1$

proof
prove;
use RStep[$n := t$];
prove;
■

With $c = a^2 + a + 4$: $B = (S(DQS^c)^+)^{\Box(a)}$ satisfies $B(n) = R(n + a^2) - a^2 - a$, $G = (S^{a+1}B)^{\Box(a)}$ satisfies $G(n) = n^2 + 2an + a$, and $\hat{P} = E\, S^{a^2-a-1} G^+$ is a predecessor.

$BEValSet == \{n : \mathbb{N} \mid \forall\, a : \mathbb{N} \bullet opBoxA\ a\ (comp(funS, opPlusI\ (comp(comp(funD, funQ),$
$iter[\mathbb{N}]\ (a * a + a + 4)\ funS)))\ n = funR\ (n + a * a) - a * a - a\}$

proof [BEValSet$domainCheck]
prove;
■

theorem rule BEValBase
$0 \in BEValSet$

proof
invoke BEValSet;
prove;
■

theorem rule BEValStep
$\forall\, n : BEValSet \bullet 1 + n \in BEValSet$

proof
invoke BEValSet;
with disabled (*QVal*, *RVal*) *prove*;
instantiate a__0 == a;
use RRec[$t := n + a * a$];
with disabled (*QVal*, *RVal*) *prove*;
■

theorem BEValSub
$\mathbb{N} \subseteq BEValSet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq BEValSet$;
prove;
invoke BEValSet;
prove;
■

theorem BEValAll
$\forall\, n : \mathbb{N} \bullet \forall\, a : \mathbb{N} \bullet opBoxA\ a\ (comp(funS, opPlusI\ (comp(comp(funD, funQ),$
$iter[\mathbb{N}]\ (a * a + a + 4)\ funS)))\ n = funR\ (n + a * a) - a * a - a$

proof
prove;
use BEValSub;
invoke BEValSet;
with enabled (*inPower*) *prove*;
instantiate e == n;
prove;
instantiate a__1 == a;
prove;
■

$GEValSet == \{n : \mathbb{N} \mid \forall\, a : \mathbb{N} \bullet opBoxA\ a\ (comp(iter[\mathbb{N}]\ (1 + a)\ funS, opBoxA\ a\ (comp(funS,$
$opPlusI\ (comp(comp(funD, funQ), iter[\mathbb{N}]\ (a * a + a + 4)\ funS))))))\ n = n * n + 2 * a * n + a\}$

proof [GEValSet$domainCheck]
prove;
■

theorem rule GEValBase
$0 \in GEValSet$

proof
invoke GEValSet;
prove;
■

theorem rule GEValStep
$\forall\, n : GEValSet \bullet 1 + n \in GEValSet$

proof
invoke GEValSet;
prove;
instantiate a__0 == a;
use BEValAll[$n := n * n + 2 * a * n + a,\ a := a$];
use RtSqAdd[$m := n + a,\ k := a$];
prove;
■

theorem GEValSub
$\mathbb{N} \subseteq GEValSet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq GEValSet$;
prove;
invoke GEValSet;
prove;
■

theorem GEValAll
$\forall\, n : \mathbb{N} \bullet \forall\, a : \mathbb{N} \bullet opBoxA\ a\ (comp(iter[\mathbb{N}]\ (1 + a)\ funS, opBoxA\ a\ (comp(funS,$
$opPlusI\ (comp(comp(funD, funQ), iter[\mathbb{N}]\ (a * a + a + 4)\ funS))))))\ n = n * n + 2 * a * n + a$

proof
prove;
use GEValSub;
invoke GEValSet;
with enabled (*inPower*) *prove*;
instantiate e == n;
prove;
instantiate a__1 == a;
prove;
■

theorem PEVal
$\forall\, a, k : \mathbb{N} \mid 2 \leq a \bullet comp(comp(funE, iter[\mathbb{N}]\ (a * a - a - 1)\ funS),$
$opPlusI\ (opBoxA\ a\ (comp(iter[\mathbb{N}]\ (1 + a)\ funS, opBoxA\ a\ (comp(funS, opPlusI\ (comp(comp(funD,$
$funQ), iter[\mathbb{N}]\ (a * a + a + 4)\ funS)))))))))\ (1 + k) = k$

proof
prove;
use $timesMonotonic2[a := 2,\ b := a,\ c := a,\ d := a]$;
use $GEValAll[n := 1 + k,\ a := a]$;
use $ESqAdd[m := 1 + k + a,\ k := k]$;
prove;
■

theorem CBComp
$\forall X : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N};\ F, G : \mathbb{P}(\mathbb{Z} \times \mathbb{Z}) \mid F \in clos\,\{X\}\,\{opPlusI,$
$opBoxA\ a\}\,\{comp\} \wedge G \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\} \bullet comp(F,$
$G) \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\}$

proof
prove;
use $ClosElemUF[I := \{\ X\ \},\ U := \{\ opPlusI,\ opBoxA\ a\ \},\ B := \{\ comp\ \},\ F := F]$;
use $ClosElemUF[I := \{\ X\ \},\ U := \{\ opPlusI,\ opBoxA\ a\ \},\ B := \{\ comp\ \},\ F := G]$;
prove;
■

theorem CBPlusI
$\forall X : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N};\ F : \mathbb{P}(\mathbb{Z} \times \mathbb{Z}) \mid F \in clos\,\{X\}\,\{opPlusI,$
$opBoxA\ a\}\,\{comp\} \bullet opPlusI\ F \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\}$

proof
prove;
use $ClosElemUF[I := \{\ X\ \},\ U := \{\ opPlusI,\ opBoxA\ a\ \},\ B := \{\ comp\ \},\ F := F]$;
prove;
■

theorem CBBoxA
$\forall X : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N};\ F : \mathbb{P}(\mathbb{Z} \times \mathbb{Z}) \mid F \in clos\,\{X\}\,\{opPlusI,$
$opBoxA\ a\}\,\{comp\} \bullet opBoxA\ a\ F \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\}$

proof
prove;
use $ClosElemUF[I := \{\ X\ \},\ U := \{\ opPlusI,\ opBoxA\ a\ \},\ B := \{\ comp\ \},\ F := F]$;
prove;
■

theorem QExprInCE
$\forall a : \mathbb{N} \mid 1 \leq a \bullet comp(funO, funE) \in clos\,\{funE\}\,\{opPlusI, opBoxA\ a\}\,\{comp\}$

proof
prove;
use $ZeroInCB[X := funE,\ a := a]$;
use $EDecr$;
use $EMult$;
use $ExtAOne[X := funE,\ a := a]$;
prove;
■

theorem QInCE
$\forall a : \mathbb{N} \mid 1 \leq a \bullet funQ \in clos\,\{funE\}\,\{opPlusI, opBoxA\ a\}\,\{comp\}$

proof
prove;
use $QExprInCE[a := a]$;
use $QExpr$;
prove;
■

theorem PEInCE
$\forall\, a : \mathbb{N} \mid 2 \leq a \bullet comp(comp(funE, iter[\mathbb{N}]\,(a * a - a - 1)\,funS),$
$opPlusI\,(opBoxA\,a\,(comp(iter[\mathbb{N}]\,(1 + a)\,funS, opBoxA\,a\,(comp(funS, opPlusI\,(comp(comp(funD,$
$funQ), iter[\mathbb{N}]\,(a * a + a + 4)\,funS)))))))) \in clos\,\{funE\}\,\{opPlusI, opBoxA\,a\}\,\{comp\}$

proof
prove;
use $timesMonotonic2[a := 2,\ b := a,\ c := a,\ d := a]$;
use $ZeroInCB[X := funE,\ a := a]$;
use $EDecr$;
use $EMult$;
use $ExtAI[X := funE,\ a := a]$;
use $ExtAS[X := funE,\ a := a]$;
use $QInCE[a := a]$;
prove;
use $IterIn[I := \{\ funE\ \},\ U := \{\ opPlusI,\ opBoxA\,a\ \},\ B := \{\ comp\ \},\ F := funS,\ k := a * a + a + 4]$;
use $IterIn[I := \{\ funE\ \},\ U := \{\ opPlusI,\ opBoxA\,a\ \},\ B := \{\ comp\ \},\ F := funS,\ k := 1 + a]$;
use $IterIn[I := \{\ funE\ \},\ U := \{\ opPlusI,\ opBoxA\,a\ \},\ B := \{\ comp\ \},\ F := funS,\ k := a * a - a - 1]$;
use $CBComp[X := funE,\ a := a,\ F := funD,\ G := funQ]$;
prove;
use $CBComp[X := funE,\ a := a,\ F := comp(funD,\ funQ),\ G := iter[\mathbb{N}]\,(a * a + a + 4)\,funS]$;
prove;
use $CBPlusI[X := funE,\ a := a,\ F := comp(comp(funD,\ funQ),\ iter[\mathbb{N}]\,(a * a + a + 4)\,funS)]$;
prove;
■

theorem PEUF
$\forall\, a : \mathbb{N} \mid 2 \leq a \bullet comp(comp(funE, iter[\mathbb{N}]\,(a * a - a - 1)\,funS),$
$opPlusI\,(opBoxA\,a\,(comp(iter[\mathbb{N}]\,(1 + a)\,funS, opBoxA\,a\,(comp(funS, opPlusI\,(comp(comp(funD,$
$funQ), iter[\mathbb{N}]\,(a * a + a + 4)\,funS)))))))) \in \mathbb{N} \to \mathbb{N}$

proof
prove;
use $PEInCE[a := a]$;
use $ClosInPow[I := \{\ funE\ \},\ U := \{\ opPlusI,\ opBoxA\,a\ \},\ B := \{\ comp\ \}]$;
apply inPower to predicate $clos\,\{\ funE\ \}\,\{\ opPlusI,\ opBoxA\,a\ \}\,\{\ comp\ \} \in \mathbb{P}\,(\mathbb{N} \to \mathbb{N})$;
instantiate e $== comp(comp(funE,\ iter[\mathbb{N}]\,(a * a - a - 1)\,funS),$
$opPlusI\,(opBoxA\,a\,(comp(iter[\mathbb{N}]\,(1 + a)\,funS,\ opBoxA\,a\,(comp(funS,$
$opPlusI\,(comp(comp(funD,\ funQ),\ iter[\mathbb{N}]\,(a * a + a + 4)\,funS)))))))$;
prove;
■

theorem PEPred
$\forall\, a : \mathbb{N} \mid 2 \leq a \bullet comp(comp(funE, iter[\mathbb{N}]\,(a * a - a - 1)\,funS),$
$opPlusI\,(opBoxA\,a\,(comp(iter[\mathbb{N}]\,(1 + a)\,funS, opBoxA\,a\,(comp(funS, opPlusI\,(comp(comp(funD,$
$funQ), iter[\mathbb{N}]\,(a * a + a + 4)\,funS)))))))) \in Preds$

proof
prove;
use PEInCE[$a := a$];
use ClosInPow[$I := \{ funE \}$, $U := \{ opPlusI,\ opBoxA\ a \}$, $B := \{ comp \}$];
apply inPower to predicate clos $\{ funE \} \{ opPlusI,\ opBoxA\ a \} \{ comp \} \in \mathbb{P}(\mathbb{N} \to \mathbb{N})$;
instantiate e $==$ $comp(comp(funE,\ iter[\mathbb{N}]\,(a * a - a - 1)\,funS),$
$opPlusI\,(opBoxA\ a\,(comp(iter[\mathbb{N}]\,(1 + a)\,funS,\ opBoxA\ a\,(comp(funS,$
$opPlusI\,(comp(comp(funD,\ funQ),\ iter[\mathbb{N}]\,(a * a + a + 4)\,funS))))))))$;
prove;
invoke Preds;
prove;
use PEVal[$a := a,\ k := x$];
prove;
■

theorem ExtAE
$\forall\, a : \mathbb{N} \mid 2 \le a \bullet clos\,\{funE\}\,\{opPlusI, opBoxA\ a\}\,\{comp\} = C6$

proof
prove;
use EDecr;
use EMult;
use ZeroInCB[$X := funE,\ a := a$];
use PEInCE[$a := a$];
use PEUF[$a := a$];
use PEPred[$a := a$];
use ExtASup[$X := funE,\ P := comp(comp(funE,\ iter[\mathbb{N}]\,(a * a - a - 1)\,funS),$
$opPlusI\,(opBoxA\ a\,(comp(iter[\mathbb{N}]\,(1 + a)\,funS,\ opBoxA\ a\,(comp(funS,$
$opPlusI\,(comp(comp(funD,\ funQ),\ iter[\mathbb{N}]\,(a * a + a + 4)\,funS)))))))),\ a := a$];
*use ExtASub*6[$X := funE,\ a := a$];
use C6Def;
use ClosInPow[$I := \{ funE \},\ U := \{ opPlusI,\ opBoxA\ a \},\ B := \{ comp \}$];
use C6InPower;
use SubAntisym[$\mathbb{N} \to \mathbb{N}$][$A := clos\,\{ funE \}\,\{ opPlusI,\ opBoxA\ a \}\,\{ comp \},\ B := C6$];
prove;
■

7.7 The case $X = K$

theorem KLt
$\forall\, y : \mathbb{N} \mid 0 < y \bullet funK\ y < y$

proof
prove;
use Decomp[$x := y$];
use UFNonneg[$f := funV,\ x := y$];
use UFNonneg[$f := funK,\ x := y$];
prove;
split $funV\ y = 0$;
cases;
use AZero;
prove;
next;
use AMono[$m := 1,\ n := funV\ y$];
use AZero;
prove;
next;
■

theorem KZero
$funK\ 0 = 0$

proof
use $KAt[m := 0,\ t := 0,\ k := 0]$;
use *AZero*;
prove;
■

theorem KDecr
$funK \in Decr$

proof
invoke *Decr*;
use *KZero*;
prove;
use $KLt[y := y]$;
prove;
■

$K(2i(i+1)) = i$, since $2i(i+1) = A(2i) + i$.

theorem KMult
$funK \in Mult$

proof
invoke *Mult*;
use *KZero*;
prove;
instantiate $r == 2 * i$;
use $TwoA[x := 2 * i]$;
use $KAt[m := 2 * i * (1 + i),\ t := 2 * i,\ k := i]$;
prove;
■

With $c = A(a)$: $B = (S^2(KS^c)^+)^{\Box(a)}$ satisfies $B(n) + A(a) + 1 = A(n + a + 1)$, and $\hat{P} = KS^cB^+$ is a predecessor.

$BKValSet == \{n : \mathbb{N} \mid \forall\, a : \mathbb{N} \bullet (opBoxA\ a\ (comp(iter[\mathbb{N}]\ 2\ funS, opPlusI\ (comp(funK,$
$iter[\mathbb{N}]\ (funA\ a)\ funS))))\ n + funA\ a + 1 = funA\ (n + a + 1)\}$

proof [BKValSet$domainCheck]
prove;
■

theorem rule BKValBase
$0 \in BKValSet$

proof
invoke *BKValSet*;
prove;
use $ASuc[x := a]$;
prove;
■

theorem rule BKValStep
$\forall\, n : BKValSet \bullet 1 + n \in BKValSet$

proof
invoke BKValSet;
prove;
instantiate $a_0 == a$;
use $ASuc[x := n + a]$;
use $ASuc[x := 1 + n + a]$;
use $KAt[m := (opBoxA\, a\, (comp(iter[\mathbb{N}]\, 2\, funS,\ opPlusI\, (comp(funK,\ iter[\mathbb{N}]\, (funA\, a)\, funS))))\, n +$
$funA\, a,\ t := n + a,\ k := n + a]$;
prove;
■

theorem BKValSub
$\mathbb{N} \subseteq BKValSet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq BKValSet$;
prove;
invoke BKValSet;
prove;
■

theorem BKValAll
$\forall\, n, a : \mathbb{N} \bullet (opBoxA\, a\, (comp(iter[\mathbb{N}]\, 2\, funS, opPlusI\, (comp(funK,$
$iter[\mathbb{N}]\, (funA\, a)\, funS))))\, n + funA\, a + 1 = funA\, (n + a + 1)$

proof
prove;
use BKValSub;
invoke BKValSet;
with enabled (*inPower*) *prove*;
instantiate $e == n$;
prove;
instantiate $a_1 == a$;
prove;
■

theorem PKVal
$\forall\, a, k : \mathbb{N} \bullet comp(comp(funK, iter[\mathbb{N}]\, (funA\, a)\, funS), opPlusI\, (opBoxA\, a\, (comp(iter[\mathbb{N}]\, 2\, funS,$
$opPlusI\, (comp(funK, iter[\mathbb{N}]\, (funA\, a)\, funS))))))\, (1 + k) = k$

proof
prove;
use $BKValAll[n := 1 + k,\ a := a]$;
use $KAt[m := (opBoxA\, a\, (comp(iter[\mathbb{N}]\, 2\, funS,\ opPlusI\, (comp(funK,\ iter[\mathbb{N}]\, (funA\, a)\, funS))))\, (1$
$+ k) + (1 + k) + funA\, a,\ t := k + a + 2,\ k := k]$;
prove;
■

theorem PKInCK
$\forall\, a : \mathbb{N} \mid 1 \leq a \bullet comp(comp(funK, iter[\mathbb{N}]\, (funA\, a)\, funS), opPlusI\, (opBoxA\, a\, (comp(iter[\mathbb{N}]\, 2\, funS,$
$opPlusI\, (comp(funK, iter[\mathbb{N}]\, (funA\, a)\, funS))))) \in clos\, \{funK\}\, \{opPlusI, opBoxA\, a\}\, \{comp\}$

proof
prove;
use *ZeroInCB*[$X := funK,\ a := a$];
use *KDecr*;
use *KMult*;
use *ExtAI*[$X := funK,\ a := a$];
use *ExtAS*[$X := funK,\ a := a$];
prove;
use *IterIn*[$I := \{ funK \},\ U := \{ opPlusI,\ opBoxA\ a \},\ B := \{ comp \},\ F := funS,\ k :=$ $funA\ a$];
use *IterIn*[$I := \{ funK \},\ U := \{ opPlusI,\ opBoxA\ a \},\ B := \{ comp \},\ F := funS,\ k := 2$];
prove;
■

theorem PKUF
$\forall\, a : \mathbb{N} \mid 1 \leq a \bullet comp(comp(funK, iter[\mathbb{N}]\,(funA\ a)\,funS), opPlusI\ (opBoxA\ a\ (comp(iter[\mathbb{N}]\,2\,funS,$ $opPlusI\ (comp(funK, iter[\mathbb{N}]\,(funA\ a)\,funS))))) \in \mathbb{N} \to \mathbb{N}$

proof
prove;
■

theorem PKPred
$\forall\, a : \mathbb{N} \mid 1 \leq a \bullet comp(comp(funK, iter[\mathbb{N}]\,(funA\ a)\,funS), opPlusI\ (opBoxA\ a\ (comp(iter[\mathbb{N}]\,2\,funS,$ $opPlusI\ (comp(funK, iter[\mathbb{N}]\,(funA\ a)\,funS))))) \in Preds$

proof
prove;
use *PKUF*[$a := a$];
invoke *Preds*;
prove;
use *PKVal*[$a := a,\ k := x$];
prove;
■

theorem ExtAK
$\forall\, a : \mathbb{N} \mid 1 \leq a \bullet clos\,\{funK\}\,\{opPlusI, opBoxA\ a\}\,\{comp\} = C6$

proof
prove;
use *KDecr*;
use *KMult*;
use *ZeroInCB*[$X := funK,\ a := a$];
use *PKInCK*[$a := a$];
use *PKUF*[$a := a$];
use *PKPred*[$a := a$];
use *ExtASup*[$X := funK,\ P := comp(comp(funK,\ iter[\mathbb{N}]\,(funA\ a)\,funS),$ $opPlusI\ (opBoxA\ a\ (comp(iter[\mathbb{N}]\,2\,funS,\ opPlusI\ (comp(funK,\ iter[\mathbb{N}]\,(funA\ a)\,funS))))),\ a :=$ a];
use *ExtASub*6[$X := funK,\ a := a$];
use *Theorem*68K;
use *C*6*KDef*;
use *ClosInPow*[$I := \{ funK \},\ U := \{ opPlusI,\ opBoxA\ a \},\ B := \{ comp \}$];
use *C*6*InPower*;
use *SubAntisym*[$\mathbb{N} \to \mathbb{N}$][$A := clos\,\{ funK \}\,\{ opPlusI,\ opBoxA\ a \}\,\{ comp \},\ B := C6$];
prove;
■

7.8 The case $X = L$

$\bar{0} = LL^+$, because $L(x + L(x)) = 0$.

theorem LZeroExpr
$comp(funL, opPlusI\ funL) = cst\ 0$

proof
apply UFExt to predicate $comp(funL,\ opPlusI\ funL)\ =\ cst\ 0$;
prove;
use Decomp$[x\ :=\ x]$;
prove;
use LAt$[m\ :=\ x,\ t\ :=\ funV\ x,\ k\ :=\ funK\ x]$;
prove;
use LAt$[m\ :=\ funA\ (funV\ x)\ +\ funV\ x,\ t\ :=\ funV\ x,\ k\ :=\ funV\ x]$;
prove;
■

theorem LZeroExprInCL
$\forall\ a : \mathbb{N} \bullet comp(funL, opPlusI\ funL) \in clos\ \{funL\}\ \{opPlusI, opBoxA\ a\}\ \{comp\}$

proof
prove;
■

theorem ZeroInCL
$\forall\ a : \mathbb{N} \bullet cst\ 0 \in clos\ \{funL\}\ \{opPlusI, opBoxA\ a\}\ \{comp\}$

proof
prove;
use LZeroExprInCL$[a\ :=\ a]$;
use LZeroExpr;
prove;
■

$L(2(i + 1)(i + 1)) = i$, since $2(i + 1)(i + 1) = A(2i + 1) + i + 1$.

theorem LMult
$funL \in Mult$

proof
invoke Mult;
use LAt$[m\ :=\ 0,\ t\ :=\ 0,\ k\ :=\ 0]$;
use AZero;
prove;
instantiate $r\ ==\ 2\ +\ 2\ *\ i$;
use TwoA$[x\ :=\ 1\ +\ 2\ *\ i]$;
use LAt$[m\ :=\ (2\ +\ 2\ *\ i)\ *\ (1\ +\ i),\ t\ :=\ 1\ +\ 2\ *\ i,\ k\ :=\ 1\ +\ i]$;
prove;
■

$K = L\bar{1}^+((\bar{1}^+)^+L)^+$ (Theorem 6.8), so the case of L reduces to that of K.

theorem KExprInCL
$\forall\ a : \mathbb{N} \mid 1 \leq a \bullet comp(comp(funL, opPlusI\ (cst\ 1)), opPlusI\ (comp(opPlusI\ (opPlusI\ (cst\ 1)),$
$funL))) \in clos\ \{funL\}\ \{opPlusI, opBoxA\ a\}\ \{comp\}$

proof
prove;
use *ZeroInCL*[$a := a$];
use *LMult*;
use *ExtAOne*[$X := funL,\ a := a$];
prove;
■

theorem KInCL
$\forall\, a : \mathbb{N} \mid 1 \leq a \bullet funK \in clos\,\{funL\}\,\{opPlusI, opBoxA\ a\}\,\{comp\}$

proof
prove;
use *KExprInCL*[$a := a$];
use *KExprL*;
prove;
■

theorem CBUClosedA
$\forall\, X : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \bullet clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\} \in UClosed\,\{opPlusI, opBoxA\ a\}$

proof
prove;
use *ClosInPow*[$I := \{\ X\ \},\ U := \{\ opPlusI,\ opBoxA\ a\ \},\ B := \{\ comp\ \}$];
with enabled (*UClosedDef*) *prove*;
■

theorem CKSubCL
$\forall\, a : \mathbb{N} \mid 1 \leq a \bullet clos\,\{funK\}\,\{opPlusI, opBoxA\ a\}\,\{comp\} \subseteq clos\,\{funL\}\,\{opPlusI, opBoxA\ a\}\,\{comp\}$

proof
prove;
use *KInCL*[$a := a$];
use *ClosLeast*[$I := \{\ funK\ \},\ U := \{\ opPlusI,\ opBoxA\ a\ \},\ B := \{\ comp\ \},\ Z := clos\,\{\ funL\ \}\,\{\ opPlusI,\ opBoxA\ a\ \}\,\{\ comp\ \}$];
use *ClosInPow*[$I := \{\ funL\ \},\ U := \{\ opPlusI,\ opBoxA\ a\ \},\ B := \{\ comp\ \}$];
use *CBUClosedA*[$X := funL,\ a := a$];
use *CBBClosed*[$X := funL,\ a := a$];
prove;
■

theorem ExtAL
$\forall\, a : \mathbb{N} \mid 1 \leq a \bullet clos\,\{funL\}\,\{opPlusI, opBoxA\ a\}\,\{comp\} = C6$

proof
prove;
use *ExtAK*[$a := a$];
use *CKSubCL*[$a := a$];
use *ExtASub*6[$X := funL,\ a := a$];
use *ClosInPow*[$I := \{\ funL\ \},\ U := \{\ opPlusI,\ opBoxA\ a\ \},\ B := \{\ comp\ \}$];
use *C*6*InPower*;
use *SubAntisym*[$\mathbb{N} \to \mathbb{N}$][$A := clos\,\{\ funL\ \}\,\{\ opPlusI,\ opBoxA\ a\ \}\,\{\ comp\ \},\ B := C6$];
prove;
■

7.9 Every starting value $a \geq 1$ (Remark 6.9)

For $a \geq 1$ the constructions for K and L above already work (ExtAK, ExtAL). For E and $a = 1$: $G = (S^2B)^{\square(1)}$ (the G above with $a = 1$) satisfies $G(n) = (n+1)^2$, and $\hat{P} = ES^2((G^+)^+)^+$ is a predecessor, since $(x+2)^2 + 3(x+1) + 2 = (x+3)^2 + x$.

theorem PEOneVal
$\forall\, k : \mathbb{N} \bullet comp(comp(funE, iter[\mathbb{N}]\, 2\, funS), opPlusI\ (opPlusI\ (opPlusI\ (opBoxA\, 1\ (comp(iter[\mathbb{N}]\, 2\, funS,$
$opBoxA\, 1\ (comp(funS, opPlusI\ (comp(comp(funD, funQ), iter[\mathbb{N}]\, 6\, funS)))))))))))\ (1 + k) = k$

proof
prove;
use $GEValAll[n := 1 + k,\ a := 1]$;
use $ESqAdd[m := 3 + k,\ k := k]$;
prove;
■

theorem PEOneInCE
$comp(comp(funE, iter[\mathbb{N}]\, 2\, funS), opPlusI\ (opPlusI\ (opPlusI\ (opBoxA\, 1\ (comp(iter[\mathbb{N}]\, 2\, funS,$
$opBoxA\, 1\ (comp(funS, opPlusI\ (comp(comp(funD, funQ),$
$iter[\mathbb{N}]\, 6\, funS)))))))))) \in clos\ \{funE\}\ \{opPlusI, opBoxA\, 1\}\ \{comp\}$

proof
use $ZeroInCB[X := funE,\ a := 1]$;
use $EDecr$;
use $EMult$;
use $ExtAI[X := funE,\ a := 1]$;
use $ExtAS[X := funE,\ a := 1]$;
use $QInCE[a := 1]$;
prove;
use $IterIn[I := \{ funE \},\ U := \{ opPlusI,\ opBoxA\,1 \},\ B := \{ comp \},\ F := funS,\ k := 6]$;
use $IterIn[I := \{ funE \},\ U := \{ opPlusI,\ opBoxA\,1 \},\ B := \{ comp \},\ F := funS,\ k := 2]$;
prove;
use $CBComp[X := funE,\ a := 1,\ F := funD,\ G := funQ]$;
prove;
use $CBComp[X := funE,\ a := 1,\ F := comp(funD,\ funQ),\ G := iter[\mathbb{N}]\,6\,funS]$;
prove;
use $CBPlusI[X := funE,\ a := 1,\ F := comp(comp(funD,\ funQ),\ iter[\mathbb{N}]\,6\,funS)]$;
prove;
use $CBComp[X := funE,\ a := 1,\ F := funS,\ G := opPlusI\,(comp(comp(funD,\ funQ),\ iter[\mathbb{N}]\,6\,funS))]$;
prove;
use $CBBoxA[X := funE,\ a := 1,\ F := comp(funS,\ opPlusI\,(comp(comp(funD,\ funQ),\ iter[\mathbb{N}]\,6\,funS)))]$;
prove;
use $CBComp[X := funE,\ a := 1,\ F := iter[\mathbb{N}]\,2\,funS,\ G := opBoxA\,1\,(comp(funS,\ opPlusI\,(comp(comp(funD,\ funQ),\ iter[\mathbb{N}]\,6\,funS))))]$;
prove;
use $CBBoxA[X := funE,\ a := 1,\ F := comp(iter[\mathbb{N}]\,2\,funS,\ opBoxA\,1\,(comp(funS,\ opPlusI\,(comp(comp(funD,\ funQ),\ iter[\mathbb{N}]\,6\,funS)))))]$;
prove;
use $CBPlusI[X := funE,\ a := 1,\ F := opBoxA\,1\,(comp(iter[\mathbb{N}]\,2\,funS,\ opBoxA\,1\,(comp(funS,\ opPlusI\,(comp(comp(funD,\ funQ),\ iter[\mathbb{N}]\,6\,funS))))))]$;
prove;
use $CBPlusI[X := funE,\ a := 1,\ F := opPlusI\,(opBoxA\,1\,(comp(iter[\mathbb{N}]\,2\,funS,\ opBoxA\,1\,(comp(funS,\ opPlusI\,(comp(comp(funD,\ funQ),\ iter[\mathbb{N}]\,6\,funS))))))),]$;
prove;
use $CBPlusI[X := funE,\ a := 1,\ F := opPlusI\,(opPlusI\,(opBoxA\,1\,(comp(iter[\mathbb{N}]\,2\,funS,\ opBoxA\,1\,(comp(funS,\ opPlusI\,(comp(comp(funD,\ funQ),\ iter[\mathbb{N}]\,6\,funS))))))))]$;
prove;
use $CBComp[X := funE,\ a := 1,\ F := funE,\ G := iter[\mathbb{N}]\,2\,funS]$;
prove;
use $CBComp[X := funE,\ a := 1,\ F := comp(funE,\ iter[\mathbb{N}]\,2\,funS),\ G := opPlusI\,(opPlusI\,(opPlusI\,(opBoxA\,1\,(comp(iter[\mathbb{N}]\,2\,funS,\ opBoxA\,1\,(comp(funS,\ opPlusI\,(comp(comp(funD,\ funQ),\ iter[\mathbb{N}]\,6\,funS))))))))))]$;
prove;
■

theorem PEOneUF
$comp(comp(funE, iter[\mathbb{N}]\,2\,funS), opPlusI\,(opPlusI\,(opPlusI\,(opBoxA\,1\,(comp(iter[\mathbb{N}]\,2\,funS,\ opBoxA\,1\,(comp(funS, opPlusI\,(comp(comp(funD, funQ), iter[\mathbb{N}]\,6\,funS))))))))))) \in \mathbb{N} \to \mathbb{N}$

proof
use $PEOneInCE$;
use $ClosInPow[I := \{ funE \},\ U := \{ opPlusI,\ opBoxA\,1 \},\ B := \{ comp \}]$;
apply inPower to predicate $clos\,\{ funE \}\,\{ opPlusI,\ opBoxA\,1 \}\,\{ comp \} \in \mathbb{P}\,(\mathbb{N} \to \mathbb{N})$;
instantiate $e == comp(comp(funE,\ iter[\mathbb{N}]\,2\,funS),\ opPlusI\,(opPlusI\,(opPlusI\,(opBoxA\,1\,(comp(iter[\mathbb{N}]\,2\,funS,\ opBoxA\,1\,(comp(funS,\ opPlusI\,(comp(comp(funD,\ funQ),\ iter[\mathbb{N}]\,6\,funS))))))))))$;
prove;
■

theorem PEOnePred
$comp(comp(funE, iter[\mathbb{N}]\,2\,funS), opPlusI\,(opPlusI\,(opPlusI\,(opBoxA\,1\,(comp(iter[\mathbb{N}]\,2\,funS,$
$opBoxA\,1\,(comp(funS, opPlusI\,(comp(comp(funD, funQ), iter[\mathbb{N}]\,6\,funS))))))))))) \in Preds$

proof
use $PEOneUF$;
invoke $Preds$;
prove;
use $PEOneVal[k := x]$;
prove;
■

theorem ExtAEOne
$clos\,\{funE\}\,\{opPlusI, opBoxA\,1\}\,\{comp\} = C6$

proof
use $EDecr$;
use $EMult$;
use $ZeroInCB[X := funE,\ a := 1]$;
use $PEOneInCE$;
use $PEOneUF$;
use $PEOnePred$;
use $ExtASup[X := funE,\ P := comp(comp(funE,\ iter[\mathbb{N}]\,2\,funS),$
$opPlusI\,(opPlusI\,(opPlusI\,(opBoxA\,1\,(comp(iter[\mathbb{N}]\,2\,funS,\ opBoxA\,1\,(comp(funS,$
$opPlusI\,(comp(comp(funD,\ funQ),\ iter[\mathbb{N}]\,6\,funS))))))))),\ a := 1]$;
use $ExtASub6[X := funE,\ a := 1]$;
use $C6Def$;
use $ClosInPow[I := \{\ funE\ \},\ U := \{\ opPlusI,\ opBoxA\,1\ \},\ B := \{\ comp\ \}]$;
use $C6InPower$;
use $SubAntisym[\mathbb{N} \to \mathbb{N}][A := clos\,\{\ funE\ \}\,\{\ opPlusI,\ opBoxA\,1\ \}\,\{\ comp\ \},\ B := C6]$;
prove;
■

theorem Remark69
$clos\,\{funE\}\,\{opPlusI, opBoxA\,1\}\,\{comp\} = C6 \wedge clos\,\{funK\}\,\{opPlusI,$
$opBoxA\,1\}\,\{comp\} = C6 \wedge clos\,\{funL\}\,\{opPlusI, opBoxA\,1\}\,\{comp\} = C6$

proof
use $ExtAEOne$;
use $ExtAK[a := 1]$;
use $ExtAL[a := 1]$;
prove;
■

theorem ExtAAll
$\forall\, a : \mathbb{N};\ X : \mathbb{N} \to \mathbb{N} \mid 1 \leq a \wedge X \in \{funE, funK, funL\} \bullet clos\,\{X\}\,\{opPlusI, opBoxA\,a\}\,\{comp\} = C6$

proof
prove;
use ExtAK[$a := a$];
use ExtAL[$a := a$];
split $a = 1$;
cases;
use ExtAEOne;
prove;
split $X = funE$;
cases;
prove;
next;
split $X = funK$;
cases;
prove;
next;
prove;
next;
next;
use ExtAE[$a := a$];
prove;
split $X = funE$;
cases;
prove;
next;
split $X = funK$;
cases;
prove;
next;
prove;
next;
next;
■

theorem rule UFInPow
$\forall F : \mathbb{N} \to \mathbb{N} \bullet F \in \mathbb{P}(\mathbb{Z} \times \mathbb{Z})$

proof
prove;
■

8 The shifted operator F^{+b} (open question 2.2)

For fixed b, $F^{+b}(x) = F(x) + x + b$. The basis $\langle E, F^{+b}, FG, F^{\square} \rangle$ generates the same functions as Robinson's basis iff $b \geq 1$.

$opPlusIb : \mathbb{N} \to ((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}))$

$\langle\!\langle$ rule PlusIbVal $\rangle\!\rangle$
$\forall b : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet opPlusIb\ b\ F\ x = F\ x + x + b$

proof [opPlusIb$domainCheck]
prove;
■

theorem rule PlusIbInner
$\forall b : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N} \bullet (\lambda x : \mathbb{N} \bullet F\ x + x + b) \in \mathbb{N} \to \mathbb{N}$

proof
apply FunIntro to predicate $(\lambda\ x : \mathbb{N} \bullet F\,x + x + b) \in \mathbb{N} \to \mathbb{N}$;
with enabled (*RelPow*, *inPower*) *prove by reduce*;
■

theorem rule PlusIbMid
$\forall\, b : \mathbb{N} \bullet (\lambda\, F : \mathbb{N} \to \mathbb{N} \bullet (\lambda\, x : \mathbb{N} \bullet F\,x + x + b)) \in (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
apply FunIntro to predicate $(\lambda\ F : \mathbb{N} \to \mathbb{N} \bullet (\lambda\ x : \mathbb{N} \bullet F\,x + x + b)) \in (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$;
with enabled (*RelPow*, *inPower*) *prove by reduce*;
■

$PlusIbWit == (\lambda\, b : \mathbb{N} \bullet (\lambda\, F : \mathbb{N} \to \mathbb{N} \bullet (\lambda\, x : \mathbb{N} \bullet F\,x + x + b)))$

proof [PlusIbWit$domainCheck]
prove;
■

theorem PlusIbWitFun
$PlusIbWit \in \mathbb{N} \to ((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}))$

proof
invoke PlusIbWit;
apply FunIntro to predicate $(\lambda\ b : \mathbb{N} \bullet (\lambda\ F : \mathbb{N} \to \mathbb{N} \bullet (\lambda\ x : \mathbb{N} \bullet F\,x + x + b))) \in \mathbb{N} \to ((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}))$;
with enabled (*RelPow*, *inPower*) *prove by reduce*;
■

theorem PlusIbWitApp
$\forall\, b : \mathbb{N} \bullet PlusIbWit\ b = (\lambda\, F : \mathbb{N} \to \mathbb{N} \bullet (\lambda\, x : \mathbb{N} \bullet F\,x + x + b))$

proof
use PlusIbWitFun;
use pairInFunction$[\mathbb{N}, (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})][f := PlusIbWit, x := b, y := (\lambda\ F : \mathbb{N} \to \mathbb{N} \bullet (\lambda\ x : \mathbb{N} \bullet F\,x + x + b))]$;
invoke PlusIbWit;
prove by reduce;
■

theorem PlusIbWitApp2
$\forall\, b : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N} \bullet PlusIbWit\ b\ F = (\lambda\, x : \mathbb{N} \bullet F\,x + x + b)$

proof
prove;
use PlusIbWitApp$[b := b]$;
use pairInFunction$[\mathbb{N} \to \mathbb{N}, \mathbb{N} \to \mathbb{N}][f := (\lambda\ F : \mathbb{N} \to \mathbb{N} \bullet (\lambda\ x : \mathbb{N} \bullet F\,x + x + b)), x := F, y := (\lambda\ x : \mathbb{N} \bullet F\,x + x + b)]$;
prove;
■

theorem rule PlusIbWitVal
$\forall\, b : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet PlusIbWit\ b\ F\ x = F\,x + x + b$

proof
prove;
use *PlusIbWitApp2*[$b := b$, $F := F$];
use *pairInFunction*[$\mathbb{N}$, $\mathbb{N}$][$f := (\lambda\ x : \mathbb{N} \bullet F\ x + x + b)$, $x := x$, $y := F\ x + x + b$];
prove;
■

theorem PlusIbEx
$\exists\ opPlusIb : \mathbb{N} \to ((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \forall\ b : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet opPlusIb\ b\ F\ x = F\ x + x + b$

proof
instantiate *opPlusIb* $==$ *PlusIbWit*;
use *PlusIbWitFun*;
prove;
■

8.1 Shifts

$Shift == \{F : \mathbb{N} \to \mathbb{N};\ c : \mathbb{N} \mid \forall\ y : \mathbb{N} \bullet F\ y = y + c\}$

proof [Shift$domainCheck]
prove;
■

$ShiftPowSet == \{k : \mathbb{N} \mid \forall\ F : \mathbb{N} \to \mathbb{N};\ c, x : \mathbb{N} \mid (F, c) \in Shift \bullet iter[\mathbb{N}]\ k\ F\ x = x + k * c\}$

proof [ShiftPowSet$domainCheck]
prove;
■

theorem rule ShiftPowBase
$0 \in ShiftPowSet$

proof
invoke *ShiftPowSet*;
prove;
■

theorem rule ShiftPowStep
$\forall\ k : ShiftPowSet \bullet 1 + k \in ShiftPowSet$

proof
invoke *ShiftPowSet*;
prove;
instantiate $F__0 == F$, $c__0 == c$, $x__0 == x$;
invoke *Shift*;
prove;
instantiate $y__0 == iter[\mathbb{N}]\ k\ F\ x$;
prove;
■

theorem ShiftPowSub
$\mathbb{N} \subseteq ShiftPowSet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq \mathit{ShiftPowSet}$;
prove;
invoke ShiftPowSet;
prove;
■

theorem ShiftPow
$\forall\, k, c, x : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N} \mid (F, c) \in \mathit{Shift} \bullet \mathit{iter}[\mathbb{N}]\, k\, F\, x = x + k * c$

proof
prove;
use ShiftPowSub;
invoke ShiftPowSet;
with enabled (*inPower*) *prove*;
instantiate $e == k$;
prove;
instantiate $F__1 == F,\ c__1 == c,\ x__1 == x$;
prove;
■

8.2 The construction, with $b = c + 1$

$\bar{0} = E^{\Box}$, $A = \bar{0}^{+b}$ ($A(x) = x+b$), $C = EA^{b-1}\bar{0}$ (the constant $b-1$), $B = C^{+b}$ ($B(x) = x+2b-1$), $T = B^{2b-2}AA$ ($T(x) = x + 4b^2 - 4b + 2$), $H = A^3(ET)^{+b}$, $G = H^{\Box}$ ($G(n) = n^2 + 4bn$), $\hat{P} = EA^{4b+1}((G^{+b})^{+b})^{+b}$ (a predecessor), $\bar{1} = \hat{P}^{b-1}A\bar{0}$, and $F^{+} = \hat{P}^{b}F^{+b}$.

theorem AShift
$\forall\, c : \mathbb{N} \bullet (\mathit{opPlusIb}\,(1 + c)\,(\mathit{cst}\,0), 1 + c) \in \mathit{Shift}$

proof
invoke Shift;
prove;
■

theorem rule CbVal
$\forall\, c, x : \mathbb{N} \bullet \mathit{comp}(\mathit{funE}, \mathit{comp}(\mathit{iter}[\mathbb{N}]\, c\,(\mathit{opPlusIb}\,(1 + c)\,(\mathit{cst}\,0)), \mathit{cst}\,0))\, x = c$

proof
prove;
use AShift$[c := c]$;
use ShiftPow$[k := c,\ c := 1 + c,\ x := 0,\ F := \mathit{opPlusIb}\,(1 + c)\,(\mathit{cst}\,0)]$;
use ESqAdd$[m := c,\ k := c]$;
prove;
■

theorem BShift
$\forall\, c : \mathbb{N} \bullet (\mathit{opPlusIb}\,(1+c)\,(\mathit{comp}(\mathit{funE}, \mathit{comp}(\mathit{iter}[\mathbb{N}]\, c\,(\mathit{opPlusIb}\,(1+c)\,(\mathit{cst}\,0)), \mathit{cst}\,0))), 1+2*c) \in \mathit{Shift}$

proof
invoke Shift;
prove;
■

theorem rule TbVal
$\forall\, c, x : \mathbb{N} \bullet \mathit{comp}(\mathit{iter}[\mathbb{N}]\,(2 * c)\,(\mathit{opPlusIb}\,(1 + c)\,(\mathit{comp}(\mathit{funE}, \mathit{comp}(\mathit{iter}[\mathbb{N}]\, c\,(\mathit{opPlusIb}\,(1 + c)\,(\mathit{cst}\,0)),$
$\mathit{cst}\,0)))), \mathit{comp}(\mathit{opPlusIb}\,(1 + c)\,(\mathit{cst}\,0), \mathit{opPlusIb}\,(1 + c)\,(\mathit{cst}\,0)))\, x = x + 4 * c * c + 4 * c + 2$

proof
prove;
use BShift[$c := c$];
use ShiftPow[$k := 2 * c$, $c := 1 + 2 * c$, $x := 2 + 2 * c + x$, $F := opPlusIb\,(1 + c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1 + c)\,(cst\,0)),\ cst\,0)))$];
prove;
■

theorem rule HbVal
$\forall\, c, y : \mathbb{N} \bullet comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1 + c)\,(cst\,0)), opPlusIb\,(1 + c)\,(comp(funE,$
$comp(iter[\mathbb{N}]\,(2 * c)\,(opPlusIb\,(1 + c)\,(comp(funE, comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1 + c)\,(cst\,0)),$
$cst\,0)))), comp(opPlusIb\,(1 + c)\,(cst\,0),$
$opPlusIb\,(1 + c)\,(cst\,0))))))\ y = funE\,(y + 4 * c * c + 4 * c + 2) + y + 4 * c + 4$

proof
prove;
use AShift[$c := c$];
use ShiftPow[$k := 3$, $c := 1 + c$, $x := funE\,(y + 4 * c * c + 4 * c + 2) + y + 1 + c$, $F := opPlusIb\,(1 + c)\,(cst\,0)$];
prove;
■

$GbSet == \{n : \mathbb{N} \mid \forall\, c : \mathbb{N} \bullet opBox\,(comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1 + c)\,(cst\,0)), opPlusIb\,(1 + c)\,(comp(funE,$
$comp(iter[\mathbb{N}]\,(2 * c)\,(opPlusIb\,(1 + c)\,(comp(funE, comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1 + c)\,(cst\,0)), cst\,0)))),$
$comp(opPlusIb\,(1 + c)\,(cst\,0), opPlusIb\,(1 + c)\,(cst\,0)))))))\ n = n * n + 4 * (1 + c) * n\}$

proof [GbSet$domainCheck]
prove;
■

theorem rule GbBase
$0 \in GbSet$

proof
invoke GbSet;
prove;
■

theorem rule GbStep
$\forall\, n : GbSet \bullet 1 + n \in GbSet$

proof
invoke GbSet;
prove;
instantiate c__0 == c;
use ESqAdd[$m := n + 2 * c + 1$, $k := 2 * n + 1$];
prove;
■

theorem GbSub
$\mathbb{N} \subseteq GbSet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq GbSet$;
prove;
invoke GbSet;
prove;
■

theorem rule GbVal

$\forall\, n, c : \mathbb{N} \bullet opBox\,(comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1+c)\,(cst\,0)), opPlusIb\,(1+c)\,(comp(funE,$
$comp(iter[\mathbb{N}]\,(2*c)\,(opPlusIb\,(1+c)\,(comp(funE, comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1+c)\,(cst\,0)),$
$cst\,0)))), comp(opPlusIb\,(1+c)\,(cst\,0), opPlusIb\,(1+c)\,(cst\,0)))))))\,n = n*n+4*(1+c)*n$

proof

prove;
use GbSub;
invoke GbSet;
with enabled (*inPower*) *prove*;
instantiate $e == n$;
prove;
instantiate $c__1 == c$;
prove;
■

theorem PbVal

$\forall\, c, k : \mathbb{N} \bullet comp(funE, comp(iter[\mathbb{N}]\,(4*c+5)\,(opPlusIb\,(1+c)\,(cst\,0)),$
$opPlusIb\,(1+c)\,(opPlusIb\,(1+c)\,(opPlusIb\,(1+c)\,(opBox\,(comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1+c)\,(cst\,0)),$
$opPlusIb\,(1+c)\,(comp(funE, comp(iter[\mathbb{N}]\,(2*c)\,(opPlusIb\,(1+c)\,(comp(funE,$
$comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1+c)\,(cst\,0)), cst\,0)))), comp(opPlusIb\,(1+c)\,(cst\,0),$
$opPlusIb\,(1+c)\,(cst\,0))))))))))))\,(1+k) = k$

proof

with disabled (*BoxSuc*, *HbVal*) *prove*;
use AShift$[c := c]$;
use ShiftPow$[k := 4*c+5,\ c := 1+c,\ x := 6+3*c+3*k+(1+k)*(1+k)+$
$4*(1+c)*(1+k),\ F := opPlusIb\,(1+c)\,(cst\,0)]$;
use ESqAdd$[m := k+2*c+4,\ k := k]$;
with disabled (*BoxSuc*, *HbVal*) *prove*;
■

theorem PbPred

$\forall\, c : \mathbb{N} \bullet comp(funE, comp(iter[\mathbb{N}]\,(4*c+5)\,(opPlusIb\,(1+c)\,(cst\,0)),$
$opPlusIb\,(1+c)\,(opPlusIb\,(1+c)\,(opPlusIb\,(1+c)\,(opBox\,(comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1+c)\,(cst\,0)),$
$opPlusIb\,(1+c)\,(comp(funE, comp(iter[\mathbb{N}]\,(2*c)\,(opPlusIb\,(1+c)\,(comp(funE,$
$comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1+c)\,(cst\,0)), cst\,0)))), comp(opPlusIb\,(1+c)\,(cst\,0),$
$opPlusIb\,(1+c)\,(cst\,0))))))))))))) \in Preds$

proof

prove;
invoke Preds;
prove;
use PbVal$[c := c,\ k := x]$;
prove;
■

theorem OneEqVb

$\forall\, c : \mathbb{N} \bullet comp(iter[\mathbb{N}]\,c\,(comp(funE, comp(iter[\mathbb{N}]\,(4*c+5)\,(opPlusIb\,(1+c)\,(cst\,0)),$
$opPlusIb\,(1+c)\,(opPlusIb\,(1+c)\,(opPlusIb\,(1+c)\,(opBox\,(comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1+c)\,(cst\,0)),$
$opPlusIb\,(1+c)\,(comp(funE, comp(iter[\mathbb{N}]\,(2*c)\,(opPlusIb\,(1+c)\,(comp(funE,$
$comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1+c)\,(cst\,0)), cst\,0)))), comp(opPlusIb\,(1+c)\,(cst\,0),$
$opPlusIb\,(1+c)\,(cst\,0))))))))))))), comp(opPlusIb\,(1+c)\,(cst\,0), cst\,0)) = cst\,1$

proof

apply UFExt to predicate $comp(iter[\mathbb{N}]\,c\,(comp(funE,\ comp(iter[\mathbb{N}]\,(4\ *\ c\ +\ 5)\,(opPlusIb\,(1\ +\ c)\,(cst\,0)),\ opPlusIb\,(1\ +\ c)\,(opPlusIb\,(1\ +\ c)\,(opPlusIb\,(1\ +\ c)\,(opBox\,(comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1\ +\ c)\,(cst\,0)),\ opPlusIb\,(1\ +\ c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,(2\ *\ c)\,(opPlusIb\,(1\ +\ c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1\ +\ c)\,(cst\,0)),\ cst\,0)))),\ comp(opPlusIb\,(1\ +\ c)\,(cst\,0),\ opPlusIb\,(1\ +\ c)\,(cst\,0))))))))))))),\ comp(opPlusIb\,(1\ +\ c)\,(cst\,0),\ cst\,0))\ =\ cst\,1$;

prove;

use $PbPred[c\ :=\ c]$;

use $IterPAll[k\ :=\ c,\ P\ :=\ comp(funE,\ comp(iter[\mathbb{N}]\,(4\ *\ c\ +\ 5)\,(opPlusIb\,(1\ +\ c)\,(cst\,0)),\ opPlusIb\,(1\ +\ c)\,(opPlusIb\,(1\ +\ c)\,(opPlusIb\,(1\ +\ c)\,(opBox\,(comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1\ +\ c)\,(cst\,0)),\ opPlusIb\,(1\ +\ c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,(2\ *\ c)\,(opPlusIb\,(1\ +\ c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1\ +\ c)\,(cst\,0)),\ cst\,0)))),\ comp(opPlusIb\,(1\ +\ c)\,(cst\,0),\ opPlusIb\,(1\ +\ c)\,(cst\,0))))))))))))),\ x\ :=\ 1]$;

prove;

■

theorem PlusIEqVb

$\forall\, c : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N} \bullet comp(iter[\mathbb{N}]\,(1+c)\,(comp(funE, comp(iter[\mathbb{N}]\,(4*c+5)\,(opPlusIb\,(1+c)\,(cst\,0)), opPlusIb\,(1+c)\,(opPlusIb\,(1+c)\,(opPlusIb\,(1+c)\,(opBox\,(comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1+c)\,(cst\,0)), opPlusIb\,(1+c)\,(comp(funE, comp(iter[\mathbb{N}]\,(2*c)\,(opPlusIb\,(1+c)\,(comp(funE, comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1+c)\,(cst\,0)), cst\,0)))), comp(opPlusIb\,(1+c)\,(cst\,0), opPlusIb\,(1+c)\,(cst\,0))))))))))))), opPlusIb\,(1+c)\,F) = opPlusI\ F$

proof

apply UFExt to predicate $comp(iter[\mathbb{N}]\,(1\ +\ c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,(4\ *\ c\ +\ 5)\,(opPlusIb\,(1\ +\ c)\,(cst\,0)),\ opPlusIb\,(1\ +\ c)\,(opPlusIb\,(1\ +\ c)\,(opPlusIb\,(1\ +\ c)\,(opBox\,(comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1\ +\ c)\,(cst\,0)),\ opPlusIb\,(1\ +\ c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,(2\ *\ c)\,(opPlusIb\,(1\ +\ c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1\ +\ c)\,(cst\,0)),\ cst\,0)))),\ comp(opPlusIb\,(1\ +\ c)\,(cst\,0),\ opPlusIb\,(1\ +\ c)\,(cst\,0))))))))))))),\ opPlusIb\,(1\ +\ c)\,F)\ =\ opPlusI\ F$;

prove;

use $PbPred[c\ :=\ c]$;

use $IterPAll[k\ :=\ 1\ +\ c,\ P\ :=\ comp(funE,\ comp(iter[\mathbb{N}]\,(4\ *\ c\ +\ 5)\,(opPlusIb\,(1\ +\ c)\,(cst\,0)),\ opPlusIb\,(1\ +\ c)\,(opPlusIb\,(1\ +\ c)\,(opPlusIb\,(1\ +\ c)\,(opBox\,(comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1\ +\ c)\,(cst\,0)),\ opPlusIb\,(1\ +\ c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,(2\ *\ c)\,(opPlusIb\,(1\ +\ c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1\ +\ c)\,(cst\,0)),\ cst\,0)))),\ comp(opPlusIb\,(1\ +\ c)\,(cst\,0),\ opPlusIb\,(1\ +\ c)\,(cst\,0))))))))))))),\ x\ :=\ x\ +\ F\,x]$;

prove;

■

8.3 Closure steps in $\langle E, F^{+b}, FG, F^{\Box} \rangle$

theorem CVComp

$\forall\, b : \mathbb{N};\ F, G : \mathbb{P}(\mathbb{Z} \times \mathbb{Z}) \mid F \in clos\,\{funE\}\,\{opPlusIb\ b, opBox\}\,\{comp\} \wedge G \in clos\,\{funE\}\,\{opPlusIb\ b, opBox\}\,\{comp\} \bullet comp(F, G) \in clos\,\{funE\}\,\{opPlusIb\ b, opBox\}\,\{comp\}$

proof

prove;

use $ClosElemUF[I\ :=\ \{\ funE\ \},\ U\ :=\ \{\ opPlusIb\ b,\ opBox\ \},\ B\ :=\ \{\ comp\ \},\ F\ :=\ F]$;

use $ClosElemUF[I\ :=\ \{\ funE\ \},\ U\ :=\ \{\ opPlusIb\ b,\ opBox\ \},\ B\ :=\ \{\ comp\ \},\ F\ :=\ G]$;

prove;

■

theorem CVPlusIb

$\forall\, b : \mathbb{N};\ F : \mathbb{P}(\mathbb{Z} \times \mathbb{Z}) \mid F \in clos\,\{funE\}\,\{opPlusIb\ b, opBox\}\,\{comp\} \bullet opPlusIb\ b\ F \in clos\,\{funE\}\,\{opPlusIb\ b, opBox\}\,\{comp\}$

proof

prove;

use $ClosElemUF[I\ :=\ \{\ funE\ \},\ U\ :=\ \{\ opPlusIb\ b,\ opBox\ \},\ B\ :=\ \{\ comp\ \},\ F\ :=\ F]$;

prove;

■

theorem CVBox
$\forall\, b : \mathbb{N};\ F : \mathbb{P}(\mathbb{Z} \times \mathbb{Z}) \mid F \in clos\,\{funE\}\,\{opPlusIb\ b,$
$opBox\}\,\{comp\} \bullet opBox\ F \in clos\,\{funE\}\,\{opPlusIb\ b, opBox\}\,\{comp\}$

proof
prove;
use $ClosElemUF[I := \{\ funE\ \},\ U := \{\ opPlusIb\ b,\ opBox\ \},\ B := \{\ comp\ \},\ F := F]$;
prove;
■

theorem CVIter
$\forall\, b, k : \mathbb{N};\ F, G : \mathbb{P}(\mathbb{Z} \times \mathbb{Z}) \mid F \in clos\,\{funE\}\,\{opPlusIb\ b,$
$opBox\}\,\{comp\} \wedge G \in clos\,\{funE\}\,\{opPlusIb\ b, opBox\}\,\{comp\} \bullet comp(iter[\mathbb{N}]\ k\ F,$
$G) \in clos\,\{funE\}\,\{opPlusIb\ b, opBox\}\,\{comp\}$

proof
prove;
use $ClosElemUF[I := \{\ funE\ \},\ U := \{\ opPlusIb\ b,\ opBox\ \},\ B := \{\ comp\ \},\ F := F]$;
use $ClosElemUF[I := \{\ funE\ \},\ U := \{\ opPlusIb\ b,\ opBox\ \},\ B := \{\ comp\ \},\ F := G]$;
use $PowIn[I := \{\ funE\ \},\ U := \{\ opPlusIb\ b,\ opBox\ \},\ B := \{\ comp\ \},\ F := F,\ G := G,\ k$
$:= k]$;
prove;
■

theorem ZeroExprInCV
$\forall\, b : \mathbb{N} \bullet opBox\ funE \in clos\,\{funE\}\,\{opPlusIb\ b, opBox\}\,\{comp\}$

proof
prove;
■

theorem ZeroInCV
$\forall\, b : \mathbb{N} \bullet cst\ 0 \in clos\,\{funE\}\,\{opPlusIb\ b, opBox\}\,\{comp\}$

proof
prove;
use $ZeroExprInCV[b := b]$;
use *BoxEExpr*;
prove;
equality substitute $cst\ 0$;
prove;
■

theorem PbInCV
$\forall\, c : \mathbb{N} \bullet comp(funE, comp(iter[\mathbb{N}]\ (4 * c + 5)\ (opPlusIb\ (1 + c)\ (cst\ 0)),$
$opPlusIb\ (1 + c)\ (opPlusIb\ (1 + c)\ (opPlusIb\ (1 + c)\ (opBox\ (comp(iter[\mathbb{N}]\ 3\ (opPlusIb\ (1 + c)\ (cst\ 0)),$
$opPlusIb\ (1 + c)\ (comp(funE, comp(iter[\mathbb{N}]\ (2 * c)\ (opPlusIb\ (1 + c)\ (comp(funE,$
$comp(iter[\mathbb{N}]\ c\ (opPlusIb\ (1 + c)\ (cst\ 0)), cst\ 0)))), comp(opPlusIb\ (1 + c)\ (cst\ 0),$
$opPlusIb\ (1 + c)\ (cst\ 0)))))))))))) \in clos\,\{funE\}\,\{opPlusIb\ (1 + c), opBox\}\,\{comp\}$

proof

prove;
use *ZeroInCV*[$b := 1 + c$];
prove;
use *CVPlusIb*[$b := 1 + c,\ F := cst\,0$];
prove;
use *CVIter*[$b := 1 + c,\ k := c,\ F := opPlusIb\,(1 + c)\,(cst\,0),\ G := cst\,0$];
prove;
use *CVComp*[$b := 1 + c,\ F := funE,\ G := comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1 + c)\,(cst\,0)),\ cst\,0)$];
prove;
use *CVPlusIb*[$b := 1 + c,\ F := comp(funE,\ comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1 + c)\,(cst\,0)),\ cst\,0))$];
prove;
use *CVComp*[$b := 1 + c,\ F := opPlusIb\,(1 + c)\,(cst\,0),\ G := opPlusIb\,(1 + c)\,(cst\,0)$];
prove;
use *CVIter*[$b := 1 + c,\ k := 2 * c,\ F := opPlusIb\,(1 + c)\,(comp(funE,$
$comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1 + c)\,(cst\,0)),\ cst\,0))),\ G := comp(opPlusIb\,(1 + c)\,(cst\,0),$
$opPlusIb\,(1 + c)\,(cst\,0))$];
prove;
use *CVComp*[$b := 1 + c,\ F := funE,\ G := comp(iter[\mathbb{N}]\,(2 * c)\,(opPlusIb\,(1 + c)\,(comp(funE,$
$comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1 + c)\,(cst\,0)),\ cst\,0)))),\ comp(opPlusIb\,(1 + c)\,(cst\,0),\ opPlusIb\,(1$
$+ c)\,(cst\,0)))$];
prove;
use *CVPlusIb*[$b := 1 + c,\ F := comp(funE,\ comp(iter[\mathbb{N}]\,(2 * c)\,(opPlusIb\,(1 + c)\,(comp(funE,$
$comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1 + c)\,(cst\,0)),\ cst\,0)))),\ comp(opPlusIb\,(1 + c)\,(cst\,0),\ opPlusIb\,(1$
$+ c)\,(cst\,0))))$];
prove;
use *CVIter*[$b := 1 + c,\ k := 3,\ F := opPlusIb\,(1 + c)\,(cst\,0),\ G := opPlusIb\,(1 +$
$c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,(2 * c)\,(opPlusIb\,(1 + c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1$
$+ c)\,(cst\,0)),\ cst\,0)))),\ comp(opPlusIb\,(1 + c)\,(cst\,0),\ opPlusIb\,(1 + c)\,(cst\,0)))))$];
prove;
use *CVBox*[$b := 1 + c,\ F := comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1 + c)\,(cst\,0)),\ opPlusIb\,(1 +$
$c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,(2 * c)\,(opPlusIb\,(1 + c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1$
$+ c)\,(cst\,0)),\ cst\,0)))),\ comp(opPlusIb\,(1 + c)\,(cst\,0),\ opPlusIb\,(1 + c)\,(cst\,0))))))$];
prove;
use *CVPlusIb*[$b := 1 + c,\ F := opBox\,(comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1 + c)\,(cst\,0)),\ opPlusIb\,(1 +$
$c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,(2 * c)\,(opPlusIb\,(1 + c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1$
$+ c)\,(cst\,0)),\ cst\,0)))),\ comp(opPlusIb\,(1 + c)\,(cst\,0),\ opPlusIb\,(1 + c)\,(cst\,0)))))))$];
prove;
use *CVPlusIb*[$b := 1 + c,\ F := opPlusIb\,(1 + c)\,(opBox\,(comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1 +$
$c)\,(cst\,0)),\ opPlusIb\,(1 + c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,(2 * c)\,(opPlusIb\,(1 + c)\,(comp(funE,$
$comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1 + c)\,(cst\,0)),\ cst\,0)))),\ comp(opPlusIb\,(1 + c)\,(cst\,0),\ opPlusIb\,(1$
$+ c)\,(cst\,0)))))))))$];
prove;
use *CVPlusIb*[$b := 1 + c,\ F := opPlusIb\,(1 + c)\,(opPlusIb\,(1 +$
$c)\,(opBox\,(comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1 + c)\,(cst\,0)),\ opPlusIb\,(1 + c)\,(comp(funE,$
$comp(iter[\mathbb{N}]\,(2 * c)\,(opPlusIb\,(1 + c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1 + c)\,(cst\,0)),$
$cst\,0)))),\ comp(opPlusIb\,(1 + c)\,(cst\,0),\ opPlusIb\,(1 + c)\,(cst\,0))))))))))$];
prove;
use *CVIter*[$b := 1 + c,\ k := 4 * c + 5,\ F := opPlusIb\,(1 + c)\,(cst\,0),\ G := opPlusIb\,(1 +$
$c)\,(opPlusIb\,(1 + c)\,(opPlusIb\,(1 + c)\,(opBox\,(comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1 + c)\,(cst\,0)),$
$opPlusIb\,(1 + c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,(2 * c)\,(opPlusIb\,(1 + c)\,(comp(funE,$
$comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1 + c)\,(cst\,0)),\ cst\,0)))),\ comp(opPlusIb\,(1 + c)\,(cst\,0),\ opPlusIb\,(1$
$+ c)\,(cst\,0))))))))))$];
prove;
■

theorem OneInCV

$\forall\, c : \mathbb{N} \bullet cst\,1 \in clos\,\{funE\}\,\{opPlusIb\,(1 + c), opBox\}\,\{comp\}$

proof
prove;
use $ZeroInCV[b := 1 + c]$;
use $PbInCV[c := c]$;
prove;
use $CVPlusIb[b := 1 + c,\ F := cst\,0]$;
prove;
use $CVComp[b := 1 + c,\ F := opPlusIb\,(1 + c)\,(cst\,0),\ G := cst\,0]$;
prove;
use $CVIter[b := 1 + c,\ k := c,\ F := comp(funE,\ comp(iter[\mathbb{N}]\,(4 * c + 5)\,(opPlusIb\,(1 + c)\,(cst\,0)),\ opPlusIb\,(1 + c)\,(opPlusIb\,(1 + c)\,(opPlusIb\,(1 + c)\,(opBox\,(comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1 + c)\,(cst\,0)),\ opPlusIb\,(1 + c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,(2 * c)\,(opPlusIb\,(1 + c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1 + c)\,(cst\,0)),\ cst\,0)))),\ comp(opPlusIb\,(1 + c)\,(cst\,0),\ opPlusIb\,(1 + c)\,(cst\,0))))))))))),\ G := comp(opPlusIb\,(1 + c)\,(cst\,0),\ cst\,0)]$;
prove;
use $OneEqVb[c := c]$;
prove;
■

theorem PlusIExprInCV
$\forall\, c : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N} \mid F \in clos\,\{funE\}\,\{opPlusIb\,(1 + c),\ opBox\}\,\{comp\} \bullet comp(iter[\mathbb{N}]\,(1+c)\,(comp(funE, comp(iter[\mathbb{N}]\,(4*c+5)\,(opPlusIb\,(1+c)\,(cst\,0)),\ opPlusIb\,(1+c)\,(opPlusIb\,(1+c)\,(opPlusIb\,(1+c)\,(opBox\,(comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1+c)\,(cst\,0)),\ opPlusIb\,(1+c)\,(comp(funE, comp(iter[\mathbb{N}]\,(2*c)\,(opPlusIb\,(1+c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1+c)\,(cst\,0)),\ cst\,0)))),\ comp(opPlusIb\,(1+c)\,(cst\,0),\ opPlusIb\,(1+c)\,(cst\,0))))))))))),\ opPlusIb\,(1+c)\,F) \in clos\,\{funE\}\,\{opPlusIb\,(1+c),\ opBox\}\,\{comp\}$

proof
prove;
use $PbInCV[c := c]$;
use $CVPlusIb[b := 1 + c,\ F := F]$;
prove;
use $CVIter[b := 1 + c,\ k := 1 + c,\ F := comp(funE,\ comp(iter[\mathbb{N}]\,(4 * c + 5)\,(opPlusIb\,(1 + c)\,(cst\,0)),\ opPlusIb\,(1 + c)\,(opPlusIb\,(1 + c)\,(opPlusIb\,(1 + c)\,(opBox\,(comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1 + c)\,(cst\,0)),\ opPlusIb\,(1 + c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,(2 * c)\,(opPlusIb\,(1 + c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1 + c)\,(cst\,0)),\ cst\,0)))),\ comp(opPlusIb\,(1 + c)\,(cst\,0),\ opPlusIb\,(1 + c)\,(cst\,0))))))))))),\ G := opPlusIb\,(1 + c)\,F]$;
prove;
■

theorem PlusIInCV
$\forall\, c : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N} \mid F \in clos\,\{funE\}\,\{opPlusIb\,(1 + c),\ opBox\}\,\{comp\} \bullet opPlusI\ F \in clos\,\{funE\}\,\{opPlusIb\,(1 + c), opBox\}\,\{comp\}$

proof
prove;
use $PlusIExprInCV[c := c,\ F := F]$;
use $PlusIEqVb[c := c,\ F := F]$;
prove;
■

8.4 $\langle E, F^{+b}, FG, F^{\Box} \rangle$ for $b \geq 1$

theorem CVUClosed
$\forall\, c : \mathbb{N} \bullet clos\,\{funE\}\,\{opPlusIb\,(1 + c), opBox\}\,\{comp\} \in UClosed\,\{opPlusI, opBox\}$

proof
prove;
use ClosInPow[$I := \{ funE \}$, $U := \{ opPlusIb\,(1 + c),\ opBox \}$, $B := \{ comp \}$];
with enabled (*UClosedDef*) *prove*;
use ClosElemUF[$I := \{ funE \}$, $U := \{ opPlusIb\,(1 + c),\ opBox \}$, $B := \{ comp \}$, $F := F$];
split $u = opPlusI$;
cases;
use PlusIInCV[$c := c$, $F := F$];
prove;
next;
prove;
next;
■

theorem CVBClosed
$\forall\, b : \mathbb{N} \bullet clos\,\{funE\}\,\{opPlusIb\ b, opBox\}\,\{comp\} \in BClosed\,\{comp\}$

proof
prove;
use ClosInPow[$I := \{ funE \}$, $U := \{ opPlusIb\ b,\ opBox \}$, $B := \{ comp \}$];
with enabled (*BClosedDef*) *prove*;
■

theorem C6SubCV
$\forall\, c : \mathbb{N} \bullet C6 \subseteq clos\,\{funE\}\,\{opPlusIb\,(1 + c), opBox\}\,\{comp\}$

proof
prove;
use C6Def;
use ClosLeast[$I := \{ cst\,1,\ funE \}$, $U := \{ opPlusI,\ opBox \}$, $B := \{ comp \}$, $Z := clos\,\{$ *funE* $\}\,\{$ $opPlusIb\,(1 + c),\ opBox \}\,\{$ *comp* $\}$];
use ClosInPow[$I := \{ funE \}$, $U := \{ opPlusIb\,(1 + c),\ opBox \}$, $B := \{ comp \}$];
use CVUClosed[$c := c$];
use CVBClosed[$b := 1 + c$];
use OneInCV[$c := c$];
prove;
■

$F^{+b} = (S^b F)^+$, so C_6 is closed under F^{+b}.

theorem PlusIbExpr
$\forall\, b : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N} \bullet opPlusIb\ b\ F = opPlusI\,(comp(iter[\mathbb{N}]\ b\ funS, F))$

proof
prove;
apply UFExt to predicate $opPlusIb\ b\ F = opPlusI\,(comp(iter[\mathbb{N}]\ b\ funS,\ F))$;
prove;
■

theorem C6PlusIbExpr
$\forall\, b : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N} \mid F \in C6 \bullet opPlusI\,(comp(iter[\mathbb{N}]\ b\ funS, F)) \in C6$

proof
prove;
use C6Def;
use PowIn[$I := \{ cst\,1,\ funE \}$, $U := \{ opPlusI,\ opBox \}$, $B := \{ comp \}$, $F := funS$, $G := F$, $k := b$];
prove;
■

theorem C6PlusIb
$\forall\, b : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N} \mid F \in C6 \bullet opPlusIb\ b\ F \in C6$

proof
prove;
use $C6PlusIbExpr[b := b,\ F := F]$;
use $PlusIbExpr[b := b,\ F := F]$;
prove;
■

theorem C6UClosedB
$\forall\, b : \mathbb{N} \bullet C6 \in UClosed\ \{opPlusIb\ b, opBox\}$

proof
prove;
use $C6InPower$;
with enabled $(UClosedDef,\ C6\,UF)$ *prove*;
use $InPair[\mathbb{P}\ (\mathbb{P}\ (\mathbb{Z} \times \mathbb{Z}) \times \mathbb{P}\ (\mathbb{Z} \times \mathbb{Z}))][u := u,\ p := opPlusIb\ b,\ q := opBox]$;
split $u = opBox$;
cases;
prove;
next;
use $C6PlusIb[b := b,\ F := F]$;
prove;
next;
■

theorem CVSubC6
$\forall\, b : \mathbb{N} \bullet clos\ \{funE\}\ \{opPlusIb\ b, opBox\}\ \{comp\} \subseteq C6$

proof
prove;
use $ClosLeast[I := \{\ funE\ \},\ U := \{\ opPlusIb\ b,\ opBox\ \},\ B := \{\ comp\ \},\ Z := C6]$;
use $C6InPower$;
use $C6\,UClosedB[b := b]$;
use $C6BClosedComp$;
prove;
■

theorem ExtVbSucc
$\forall\, c : \mathbb{N} \bullet clos\ \{funE\}\ \{opPlusIb\ (1 + c), opBox\}\ \{comp\} = C6$

proof
prove;
use $C6SubCV[c := c]$;
use $CVSubC6[b := 1 + c]$;
use $ClosInPow[I := \{\ funE\ \},\ U := \{\ opPlusIb\ (1 + c),\ opBox\ \},\ B := \{\ comp\ \}]$;
use $C6InPower$;
use $SubAntisym[\mathbb{N} \to \mathbb{N}][A := clos\ \{\ funE\ \}\ \{\ opPlusIb\ (1 + c),\ opBox\ \}\ \{\ comp\ \},\ B := C6]$;
prove;
■

theorem ExtVb
$\forall\, b : \mathbb{N} \mid 1 \leq b \bullet clos\ \{funE\}\ \{opPlusIb\ b, opBox\}\ \{comp\} = C6$

proof
prove;
use $ExtVbSucc[c := b - 1]$;
prove;
■

8.5 $b = 0$: every generated function fixes 0, so $\bar{1}$ is not generated

$FixZero == \{F : \mathbb{N} \to \mathbb{N} \mid F\,0 = 0\}$

proof [FixZero$domainCheck]
prove;
■

theorem FixZeroInPower
$FixZero \in \mathbb{P}(\mathbb{N} \to \mathbb{N})$

proof
invoke FixZero;
with enabled (*inPower*) *prove*;
■

theorem FixZeroUClosed
$FixZero \in UClosed\,\{opPlusIb\,0, opBox\}$

proof
use FixZeroInPower;
with enabled (*UClosedDef*) *prove*;
invoke FixZero;
prove;
use InPair$[\mathbb{P}\,(\mathbb{P}\,(\mathbb{Z} \times \mathbb{Z}) \times \mathbb{P}\,(\mathbb{Z} \times \mathbb{Z}))][u := u,\ p := opPlusIb\,0,\ q := opBox]$;
split $u = opBox$;
cases;
prove;
next;
prove;
next;
■

theorem FixZeroBClosed
$FixZero \in BClosed\,\{comp\}$

proof
use FixZeroInPower;
with enabled (*BClosedDef*) *prove*;
invoke FixZero;
prove;
■

theorem CVZeroFix
$clos\,\{funE\}\,\{opPlusIb\,0, opBox\}\,\{comp\} \subseteq FixZero$

proof
use ClosLeast$[I := \{\ funE\ \},\ U := \{\ opPlusIb\,0,\ opBox\ \},\ B := \{\ comp\ \},\ Z := FixZero]$;
use FixZeroInPower;
use FixZeroUClosed;
use FixZeroBClosed;
invoke FixZero;
prove;
■

theorem OneNotCVZero
$\neg\ cst\,1 \in clos\,\{funE\}\,\{opPlusIb\,0, opBox\}\,\{comp\}$

proof
use CVZeroFix;
apply subsetDef to predicate clos { *funE* } { *opPlusIb* 0, *opBox* } { *comp* } ⊆ *FixZero*;
apply inPower to predicate clos { *funE* } { *opPlusIb* 0, *opBox* } { *comp* } ∈ ℙ *FixZero*;
instantiate e == *cst* 1;
invoke FixZero;
prove;
■

theorem OneInEqC6
$\forall\, Y : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \mid Y = C6 \bullet \mathit{cst}\, 1 \in Y$

proof
prove;
■

theorem ExtVbZero
$\neg\ \mathit{clos}\,\{\mathit{funE}\}\,\{\mathit{opPlusIb}\,0, \mathit{opBox}\}\,\{\mathit{comp}\} = C6$

proof
use OneNotCVZero;
use ClosInPow[*I* := { *funE* }, *U* := { *opPlusIb* 0, *opBox* }, *B* := { *comp* }];
use OneInEqC6[*Y* := *clos* { *funE* } { *opPlusIb* 0, *opBox* } { *comp* }];
prove;
■

theorem ExtVbIff
$\forall\, b : \mathbb{N} \bullet \mathit{clos}\,\{\mathit{funE}\}\,\{\mathit{opPlusIb}\, b, \mathit{opBox}\}\,\{\mathit{comp}\} = C6 \Leftrightarrow 1 \leq b$

proof
prove;
split b = 0;
cases;
use ExtVbZero;
prove;
next;
use ExtVb[*b* := *b*];
prove;
next;
■

9 General recursive functions (README §2.3)

J. Robinson (Proc. AMS 1 (1950), Sec. 4, Thm. 3) proved that the unary general recursive functions are the closure of $\langle S, E, F + G, FG, F^{-1}\rangle$, where $F^{-1}(x) = \mu\, y\, [F(y) = x]$ for F onto. Here $F + G$ is replaced by F^{+}.

9.1 Least preimages and inversion

Every value that is taken is taken at a least argument.

$\mathit{LeastPreSet} == \{k : \mathbb{N} \mid \forall\, F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet (\forall\, j : \mathbb{N} \mid j \leq k \bullet \neg\ F\, j = x) \lor (\exists\, m : \mathbb{N} \bullet F\, m = x \land (\forall\, n : \mathbb{N} \mid F\, n = x \bullet m \leq n))\}$

proof [LeastPreSet$domainCheck]
prove;
■

theorem rule LeastPreBase
$0 \in \mathit{LeastPreSet}$

proof
invoke *LeastPreSet*;
prove;
instantiate $m == 0$;
prove;
■

theorem rule LeastPreStep
$\forall\, k : \mathit{LeastPreSet} \bullet 1 + k \in \mathit{LeastPreSet}$

proof
invoke *LeastPreSet*;
prove;
split $j \leq k$;
cases;
instantiate $F__0 == F$, $j__0 == j$;
prove;
instantiate $m__1 == m$;
prove;
instantiate $n__1 == n$;
prove;
next;
split $\exists\, n : \mathbb{N} \mid n \leq k \bullet F\, n = F\, j$;
cases;
prove;
instantiate $F__0 == F$, $j__0 == n$;
prove;
instantiate $m__1 == m$;
prove;
instantiate $n__2 == n__0$;
prove;
next;
prove;
instantiate $m__0 == j$;
prove;
instantiate $n__0 == n$;
prove;
next;
■

theorem LeastPreSub
$\mathbb{N} \subseteq \mathit{LeastPreSet}$

proof
apply *natInduction* *to predicate* $\mathbb{N} \subseteq \mathit{LeastPreSet}$;
prove;
invoke *LeastPreSet*;
prove;
■

theorem LeastPre
$\forall\, F : \mathbb{N} \to \mathbb{N};\ y : \mathbb{N} \bullet \exists\, m : \mathbb{N} \bullet F\, m = F\, y \wedge (\forall\, n : \mathbb{N} \mid F\, n = F\, y \bullet m \leq n)$

proof
prove;
use LeastPreSub;
invoke LeastPreSet;
with enabled (*inPower*) *prove*;
instantiate $e == y$;
prove;
instantiate $F__1 == F$, $j__0 == y$;
prove;
instantiate $m__2 == m$;
prove;
instantiate $n__2 == n$;
prove;
■

y is the least preimage of x under F (or 0 if x is not a value).

$$IsLeast == \{F : \mathbb{N} \to \mathbb{N};\ x,\ y : \mathbb{N} \mid (\forall v : \mathbb{N} \mid F\ v = x \bullet y \leq v) \land (F\ y = x \lor ((\forall w : \mathbb{N} \bullet \neg\ F\ w = x) \land y = 0))\}$$

proof [IsLeast$domainCheck]
prove;
■

theorem LinvUnique
$\forall F : \mathbb{N} \to \mathbb{N};\ x, y, z : \mathbb{N} \mid (F, x, y) \in IsLeast \land (F, x, z) \in IsLeast \bullet y = z$

proof
invoke IsLeast;
prove;
instantiate $v == z$, $v__0 == y$, $w == z$, $w__0 == y$;
prove;
split $F\ y = x$;
cases;
with normalization prove;
next;
prove;
next;
■

theorem LinvExists
$\forall F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet \exists y : \mathbb{N} \bullet (F, x, y) \in IsLeast$

proof
invoke IsLeast;
prove;
split $\exists\ w : \mathbb{N} \bullet F\ w = x$;
cases;
prove;
use LeastPre$[F := F,\ y := w]$;
prove;
instantiate $y == m$;
prove;
instantiate $n == v$;
prove;
next;
prove;
next;
■

theorem LinvInner
$\forall F : \mathbb{N} \to \mathbb{N} \bullet \{x, y : \mathbb{N} \mid (F, x, y) \in \mathit{IsLeast} \bullet (x, y)\} \in \mathbb{N} \to \mathbb{N}$

proof
apply FunIntro to predicate $\{ x, y : \mathbb{N} \mid (F, x, y) \in \mathit{IsLeast} \bullet (x, y) \} \in \mathbb{N} \to \mathbb{N}$;
with enabled (*RelPow*) *prove*;
use LinvExists$[F := F, x := x]$;
use LinvUnique$[F := F, x := x__0, y := y, z := z]$;
prove;
instantiate $y__1 == y__0$;
prove;
■

$\mathit{LinvWit} == (\lambda F : \mathbb{N} \to \mathbb{N} \bullet \{x, y : \mathbb{N} \mid (F, x, y) \in \mathit{IsLeast} \bullet (x, y)\})$

theorem LinvWitFun
$\mathit{LinvWit} \in (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
invoke LinvWit;
apply FunIntro to predicate $(\lambda F : \mathbb{N} \to \mathbb{N} \bullet \{ x, y : \mathbb{N} \mid (F, x, y) \in \mathit{IsLeast} \bullet (x, y) \}) \in (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$;
with enabled (*RelPow*, *inPower*) *prove*;
use LinvInner$[F := F]$;
use LinvInner$[F := x]$;
prove;
■

theorem LinvWitApp
$\forall F : \mathbb{N} \to \mathbb{N} \bullet \mathit{LinvWit}\ F = \{x, y : \mathbb{N} \mid (F, x, y) \in \mathit{IsLeast} \bullet (x, y)\}$

proof
use LinvWitFun;
use pairInFunction$[\mathbb{N} \to \mathbb{N}, \mathbb{N} \to \mathbb{N}][f := \mathit{LinvWit}, x := F, y := \{ x, y : \mathbb{N} \mid (F, x, y) \in \mathit{IsLeast} \bullet (x, y) \}]$;
invoke LinvWit;
prove;
■

theorem LinvWitVal
$\forall F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet (F, x, \mathit{LinvWit}\ F\ x) \in \mathit{IsLeast}$

proof
prove;
use LinvWitApp$[F := F]$;
use LinvInner$[F := F]$;
use LinvExists$[F := F, x := x]$;
prove;
use pairInFunction$[\mathbb{N}, \mathbb{N}][f := \{ x, y : \mathbb{N} \mid (F, x, y) \in \mathit{IsLeast} \bullet (x, y) \}, x := x, y := y]$;
prove;
■

The least preimage of x (an arbitrary value if x is not a value of F).

$\mathit{linv} : (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

$\langle\!\langle$ LinvSpec $\rangle\!\rangle$
$\forall F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet (F, x, \mathit{linv}\ F\ x) \in \mathit{IsLeast}$

proof [linv$domainCheck]
prove;
■

theorem LinvEx
$\exists\, linv : (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}) \bullet \forall\, F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet (F, x, linv\ F\ x) \in IsLeast$

proof
instantiate linv == LinvWit;
use LinvWitFun;
prove;
use LinvWitVal[$F := F,\ x := x$];
prove;
■

theorem LinvProp
$\forall\, F : \mathbb{N} \to \mathbb{N};\ x, y : \mathbb{N} \mid F\ y = x \bullet F\ (linv\ F\ x) = x \land linv\ F\ x \leq y$

proof
prove;
use LinvSpec[$F := F,\ x := x$];
invoke IsLeast;
prove;
instantiate v == y, w == y;
prove;
■

F is onto. Inversion: $F^{-1}(x) = \mu\, y\, [F(y) = x]$ if F is onto, and $F^{-1} = F$ otherwise, so that a closure under *opInv* is a closure under Robinson's inversion.

$Surj == \{F : \mathbb{N} \to \mathbb{N} \mid \forall\, x : \mathbb{N} \bullet \exists\, y : \mathbb{N} \bullet F\ y = x\}$

proof [Surj$domainCheck]
prove;
■

$opInv : (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

⟨⟨ disabled rule InvVal ⟩⟩
$\forall\, F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet opInv\ F\ x = \textbf{if}\ F \in Surj\ \textbf{then}\ linv\ F\ x\ \textbf{else}\ F\ x$

proof [opInv$domainCheck]
prove;
■

theorem rule InvInner
$\forall\, F : \mathbb{N} \to \mathbb{N} \bullet (\lambda\, x : \mathbb{N} \bullet \textbf{if}\ F \in Surj\ \textbf{then}\ linv\ F\ x\ \textbf{else}\ F\ x) \in \mathbb{N} \to \mathbb{N}$

proof
apply FunIntro to predicate ($\lambda\ x : \mathbb{N} \bullet$ **if** $F \in Surj$ **then** $linv\ F\ x$ **else** $F\ x$) $\in \mathbb{N} \to \mathbb{N}$;
with enabled (*RelPow*, *inPower*) *prove*;
split $F \in Surj$;
cases;
prove;
next;
prove;
next;
■

$InvWit == (\lambda\, F : \mathbb{N} \to \mathbb{N} \bullet (\lambda\, x : \mathbb{N} \bullet \textbf{if } F \in Surj \textbf{ then } linv\, F\, x \textbf{ else } F\, x))$

proof [InvWit$domainCheck]
prove;
■

theorem InvWitFun
$InvWit \in (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
invoke InvWit;
apply FunIntro to predicate $(\lambda\, F : \mathbb{N} \to \mathbb{N} \bullet (\lambda\, x : \mathbb{N} \bullet \textbf{if } F \in Surj \textbf{ then } linv\, F\, x \textbf{ else } F\, x)) \in (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$;
with enabled (*RelPow*, *inPower*) *prove*;
■

theorem InvWitApp
$\forall\, F : \mathbb{N} \to \mathbb{N} \bullet InvWit\, F = (\lambda\, x : \mathbb{N} \bullet \textbf{if } F \in Surj \textbf{ then } linv\, F\, x \textbf{ else } F\, x)$

proof
use InvWitFun;
use pairInFunction$[\mathbb{N} \to \mathbb{N}, \mathbb{N} \to \mathbb{N}][f := InvWit, x := F, y := (\lambda\, x : \mathbb{N} \bullet \textbf{if } F \in Surj \textbf{ then } linv\, F\, x \textbf{ else } F\, x)]$;
invoke InvWit;
prove by reduce;
■

theorem rule InvWitVal
$\forall\, F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet InvWit\, F\, x = \textbf{if } F \in Surj \textbf{ then } linv\, F\, x \textbf{ else } F\, x$

proof
prove;
use InvWitApp$[F := F]$;
use pairInFunction$[\mathbb{N}, \mathbb{N}][f := (\lambda\, x : \mathbb{N} \bullet \textbf{if } F \in Surj \textbf{ then } linv\, F\, x \textbf{ else } F\, x), x := x, y := \textbf{if } F \in Surj \textbf{ then } linv\, F\, x \textbf{ else } F\, x]$;
prove;
■

theorem InvEx
$\exists\, opInv : (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}) \bullet \forall\, F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet opInv\, F\, x = \textbf{if } F \in Surj \textbf{ then } linv\, F\, x \textbf{ else } F\, x$

proof
instantiate opInv == *InvWit*;
use InvWitFun;
prove;
■

theorem InvNot
$\forall\, F : \mathbb{N} \to \mathbb{N} \mid \neg\, F \in Surj \bullet opInv\, F = F$

proof
apply UFExt to predicate opInv $F = F$;
prove;
use InvVal$[F := F, x := x]$;
prove;
■

theorem rule InvRight
$\forall F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \mid F \in \mathit{Surj} \bullet F\,(\mathit{opInv}\,F\,x) = x$

proof
use $\mathit{InvVal}[F := F,\ x := x]$;
invoke *Surj*;
prove;
instantiate $x__0 == x$;
prove;
use $\mathit{LinvProp}[F := F,\ x := x,\ y := y]$;
prove;
■

theorem InvLeast
$\forall F : \mathbb{N} \to \mathbb{N};\ x, y : \mathbb{N} \mid F \in \mathit{Surj} \land F\,y = x \bullet \mathit{opInv}\,F\,x \leq y$

proof
use $\mathit{InvVal}[F := F,\ x := x]$;
use $\mathit{LinvProp}[F := F,\ x := x,\ y := y]$;
prove;
■

theorem InvFix0
$\forall F : \mathbb{N} \to \mathbb{N} \mid F\,0 = 0 \bullet \mathit{opInv}\,F\,0 = 0$

proof
use $\mathit{InvVal}[F := F,\ x := 0]$;
use $\mathit{LinvProp}[F := F,\ x := 0,\ y := 0]$;
use $\mathit{UFNonneg}[f := \mathit{linv}\,F,\ x := 0]$;
prove;
■

9.2 Lemmas 6.2–6.6 in a rich set

The expressions of Lemmas 6.2–6.6 only use $I, E, S, D, M_3, \mathrm{Sq}, \mathrm{Hf}, \mathrm{Rt},\ F^+$ and composition, so they work in every set of functions that contains these and is closed under F^+ and composition.

$\mathit{Rich} == \{\, Y : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \mid \mathit{funI} \in Y \land \mathit{funE} \in Y \land \mathit{funS} \in Y \land \mathit{funD} \in Y \land \mathit{funM3} \in Y \land \mathit{funSq} \in Y \land \mathit{funHf} \in Y \land \mathit{funRt} \in Y \land (\forall F : Y \bullet \mathit{opPlusI}\,F \in Y) \land (\forall F, G : Y \bullet \mathit{comp}(F, G) \in Y)\}$

proof [Rich$domainCheck]
prove;
■

theorem rule RichI
$\forall Y : \mathit{Rich} \bullet \mathit{funI} \in Y$

proof
invoke *Rich*;
prove;
■

theorem rule RichE
$\forall Y : \mathit{Rich} \bullet \mathit{funE} \in Y$

proof
invoke Rich;
prove;
■

theorem rule RichS
$\forall\ Y : Rich \bullet funS \in Y$

proof
invoke Rich;
prove;
■

theorem rule RichD
$\forall\ Y : Rich \bullet funD \in Y$

proof
invoke Rich;
prove;
■

theorem rule RichM3
$\forall\ Y : Rich \bullet funM3 \in Y$

proof
invoke Rich;
prove;
■

theorem rule RichSq
$\forall\ Y : Rich \bullet funSq \in Y$

proof
invoke Rich;
prove;
■

theorem rule RichHf
$\forall\ Y : Rich \bullet funHf \in Y$

proof
invoke Rich;
prove;
■

theorem rule RichRt
$\forall\ Y : Rich \bullet funRt \in Y$

proof
invoke Rich;
prove;
■

theorem rule RichPlusI
$\forall\ Y : Rich;\ F : \mathbb{N} \to \mathbb{N} \mid F \in Y \bullet opPlusI\ F \in Y$

proof
invoke Rich;
prove;
instantiate $F__0 == F$;
prove;
■

theorem rule RichComp
$\forall Y : Rich;\ F, G : \mathbb{N} \to \mathbb{N} \mid F \in Y \land G \in Y \bullet comp(F, G) \in Y$

proof
invoke Rich;
prove;
instantiate $F__1 == F,\ G__0 == G$;
prove;
■

theorem rule RMinusIIn
$\forall Y : Rich;\ F : \mathbb{N} \to \mathbb{N} \mid F \in Y \bullet exprMinusI\ F \in Y$

proof
with enabled (*exprMinusIApp*) *prove*;
■

theorem rule ROplusIn
$\forall Y : Rich;\ F, G : \mathbb{N} \to \mathbb{N} \mid F \in Y \land G \in Y \bullet exprOplus(F, G) \in Y$

proof
with enabled (*exprOplusApp*) *prove*;
■

theorem rule ROminusIn
$\forall Y : Rich;\ F, G : \mathbb{N} \to \mathbb{N} \mid F \in Y \land G \in Y \bullet exprOminus(F, G) \in Y$

proof
with enabled (*exprOminusApp*) *prove*;
■

theorem rule RPhatIn
$\forall Y : Rich \bullet exprPhat \in Y$

proof
invoke exprPhat;
prove;
■

theorem rule ROtimesIn
$\forall Y : Rich;\ F, G : \mathbb{N} \to \mathbb{N} \mid F \in Y \land G \in Y \bullet exprOtimes(F, G) \in Y$

proof
with enabled (*exprOtimesApp*) *prove*;
■

theorem rule RPlusExprIn
$\forall Y : Rich;\ F, G : \mathbb{N} \to \mathbb{N} \mid F \in Y \land G \in Y \bullet exprPlus(F, G) \in Y$

proof
with enabled (*exprPlusApp*) *prove*;
■

theorem rule RMinusExprIn
$\forall\, Y : Rich;\ F, G : \mathbb{N} \to \mathbb{N} \mid F \in Y \wedge G \in Y \bullet exprMinus(F, G) \in Y$

proof
with enabled (*exprMinusApp*) *prove*;
■

Lemma 6.6 in a rich set.

theorem rule RichPlus
$\forall\, Y : Rich;\ F, G : \mathbb{N} \to \mathbb{N} \mid F \in Y \wedge G \in Y \bullet opPlus(F, G) \in Y$

proof
use PlusExprEq[$F := F,\ G := G$];
prove;
equality substitute opPlus($F,\ G$);
prove;
■

theorem rule RichDiffE
$\forall\, Y : Rich;\ F, G : \mathbb{N} \to \mathbb{N} \mid F \in Y \wedge G \in Y \bullet opDiffE(F, G) \in Y$

proof
use MinusExprEq[$F := F,\ G := G$];
prove;
equality substitute opDiffE($F,\ G$);
prove;
■

$A,\ V,\ K$ and L are in every rich set.

theorem RichAExpr
$\forall\, Y : Rich \bullet comp(funHf, opPlusI\ funSq) \in Y$

proof
prove;
■

theorem rule RichA
$\forall\, Y : Rich \bullet funA \in Y$

proof
prove;
use RichAExpr[$Y := Y$];
use AExpr;
prove;
■

theorem RichVExpr
$\forall\, Y : Rich \bullet comp(comp(comp(comp(comp(comp(funHf, exprPhat), funRt), funS), funD), funD), funD) \in Y$

proof
prove;
■

theorem rule RichV
$\forall\, Y : Rich \bullet funV \in Y$

proof
prove;
use $RichVExpr[Y := Y]$;
use *VExpr*;
prove;
■

theorem RichKExpr
$\forall\, Y : Rich \bullet opDiffE(funI, comp(funA, funV)) \in Y$

proof
prove;
■

theorem rule RichK
$\forall\, Y : Rich \bullet funK \in Y$

proof
prove;
use $RichKExpr[Y := Y]$;
use *KExpr*;
prove;
■

theorem RichLExpr
$\forall\, Y : Rich \bullet opDiffE(comp(funA, comp(funS, funV)), funS) \in Y$

proof
prove;
■

theorem rule RichL
$\forall\, Y : Rich \bullet funL \in Y$

proof
prove;
use $RichLExpr[Y := Y]$;
use *LExpr*;
prove;
■

9.3 The bases with inversion

J. Robinson's basis $\langle S, E, F + G, FG, F^{-1}\rangle$ and $\langle \bar{1}, X, F^{+}, FG, F^{-1}\rangle$ for $X \in \{E, K, L\}$.

$ClosJR == clos\,\{funS, funE\}\,\{opInv\}\,\{opPlus, comp\}$

proof [ClosJR$domainCheck]
prove;
■

theorem disabled rule ClosJRDef
$ClosJR = clos\,\{funS, funE\}\,\{opInv\}\,\{opPlus, comp\}$

proof
invoke $ClosJR$;
prove;
■

theorem ClosJRInPower
$ClosJR \in \mathbb{P}(\mathbb{N} \to \mathbb{N})$

proof
use $ClosInPow[I := \{ funS, funE \}, U := \{ opInv \}, B := \{ opPlus, comp \}]$;
with enabled $(ClosJRDef)$ *prove*;
■

theorem disabled rule ClosJRUF
$\forall F : \mathbb{P}(\mathbb{Z} \times \mathbb{Z}) \mid F \in ClosJR \bullet F \in \mathbb{N} \to \mathbb{N}$

proof
prove;
use $ClosElemUF[I := \{ funS, funE \}, U := \{ opInv \}, B := \{ opPlus, comp \}, F := F]$;
with enabled $(ClosJRDef)$ *prove*;
■

theorem rule ClosJRInit0
$funS \in ClosJR$

proof
with enabled $(ClosJRDef)$ *prove*;
■

theorem rule ClosJRInit1
$funE \in ClosJR$

proof
with enabled $(ClosJRDef)$ *prove*;
■

theorem rule ClosJRUnop0
$\forall F : \mathbb{N} \to \mathbb{N} \mid F \in ClosJR \bullet opInv\ F \in ClosJR$

proof
with enabled $(ClosJRDef)$ *prove*;
■

theorem rule ClosJRBinop0
$\forall F, G : \mathbb{N} \to \mathbb{N} \mid F \in ClosJR \land G \in ClosJR \bullet opPlus(F, G) \in ClosJR$

proof
with enabled $(ClosJRDef)$ *prove*;
■

theorem rule ClosJRBinop1
$\forall F, G : \mathbb{N} \to \mathbb{N} \mid F \in ClosJR \land G \in ClosJR \bullet comp(F, G) \in ClosJR$

proof
with enabled (*ClosJRDef*) *prove*;
■

$CGE == clos\,\{cst\,1, funE\}\,\{opPlusI, opInv\}\,\{comp\}$

proof [CGE$domainCheck]
prove;
■

theorem disabled rule CGEDef
$CGE = clos\,\{cst\,1, funE\}\,\{opPlusI, opInv\}\,\{comp\}$

proof
invoke CGE;
prove;
■

theorem CGEInPower
$CGE \in \mathbb{P}(\mathbb{N} \to \mathbb{N})$

proof
use ClosInPow[$I := \{ cst\,1, funE \}, U := \{ opPlusI, opInv \}, B := \{ comp \}$];
with enabled (*CGEDef*) *prove*;
■

theorem disabled rule CGEUF
$\forall F : \mathbb{P}(\mathbb{Z} \times \mathbb{Z}) \mid F \in CGE \bullet F \in \mathbb{N} \to \mathbb{N}$

proof
prove;
use ClosElemUF[$I := \{ cst\,1, funE \}, U := \{ opPlusI, opInv \}, B := \{ comp \}, F := F$];
with enabled (*CGEDef*) *prove*;
■

theorem rule CGEInit0
$cst\,1 \in CGE$

proof
with enabled (*CGEDef*) *prove*;
■

theorem rule CGEInit1
$funE \in CGE$

proof
with enabled (*CGEDef*) *prove*;
■

theorem rule CGEUnop0
$\forall F : \mathbb{N} \to \mathbb{N} \mid F \in CGE \bullet opPlusI\ F \in CGE$

proof
with enabled (*CGEDef*) *prove*;
■

theorem rule CGEUnop1
$\forall F : \mathbb{N} \to \mathbb{N} \mid F \in CGE \bullet opInv\ F \in CGE$

proof
with enabled (*CGEDef*) *prove*;
■

theorem rule CGEBinop0
$\forall F, G : \mathbb{N} \to \mathbb{N} \mid F \in CGE \wedge G \in CGE \bullet comp(F, G) \in CGE$

proof
with enabled (*CGEDef*) *prove*;
■

9.4 The case $X = E$: arithmetic

theorem EBound
$\forall y : \mathbb{N} \bullet funE\ y \leq 2 * funRt\ y$

proof
prove;
use RtSpec[$x := y$];
with enabled (*EVal*) *prove*;
■

theorem EDecomp
$\forall y : \mathbb{N} \bullet funE\ y + funRt\ y * funRt\ y = y$

proof
prove;
use RtSpec[$x := y$];
with enabled (*EVal*) *prove*;
■

theorem TwoLeSq
$\forall a : \mathbb{N} \bullet 2 * a \leq a * a + 1$

proof
prove;
split $a = 0$;
cases;
prove;
next;
use MulNonneg[$a := a - 1,\ b := a - 1$];
prove;
next;
■

theorem TwoELe
$\forall y : \mathbb{N} \bullet 2 * funE\ y \leq y + 1$

proof
prove;
use EBound[$y := y$];
use TwoLeSq[$a := funRt\ y$];
use RtSpec[$x := y$];
use UFNonneg[$f := funRt,\ x := y$];
with enabled (*EVal*) *prove*;
■

theorem ESurj
$funE \in Surj$

proof
invoke Surj;
prove;
instantiate $y == x * x + x$;
use ESqAdd[$m := x,\ k := x$];
prove;
■

$E^{-1}(x) = c^2 + x$ where $c = \lceil x/2 \rceil$.

theorem EinvVal
$\forall\, x, c : \mathbb{N} \mid x \leq 2 * c \wedge 2 * c \leq x + 1 \bullet opInv\ funE\ x = c * c + x$

proof
prove;
use ESurj;
use ESqAdd[$m := c,\ k := x$];
use InvLeast[$F := funE,\ x := x,\ y := c * c + x$];
use InvRight[$F := funE,\ x := x$];
use EBound[$y := opInv\ funE\ x$];
use EDecomp[$y := opInv\ funE\ x$];
use UFNonneg[$f := funRt,\ x := opInv\ funE\ x$];
*use timesMonotonic*2[$a := c,\ b := c,\ c := funRt\ (opInv\ funE\ x),\ d := funRt\ (opInv\ funE\ x)$];
prove;
■

theorem rule EinvD
$\forall\, z : \mathbb{N} \bullet opInv\ funE\ (2 * z) = z * z + 2 * z$

proof
prove;
use EinvVal[$x := 2 * z,\ c := z$];
prove;
■

theorem rule EinvOdd
$\forall\, m : \mathbb{N} \bullet opInv\ funE\ (1 + 2 * m) = (m + 1) * (m + 1) + 2 * m + 1$

proof
prove;
use EinvVal[$x := 1 + 2 * m,\ c := m + 1$];
prove;
■

9.5 The case $X = E$: S, $\bar{0}$, I, D, M_3, Sq

theorem IExprZero
$opPlusI\,(cst\,0) = funI$

proof
apply UFExt to predicate $opPlusI\,(cst\,0) = funI$;
prove;
■

theorem GESExpr
$opPlusI\,(cst\,1) \in CGE$

proof
prove;
■

theorem rule GES
$funS \in CGE$

proof
use GESExpr;
use SExpr;
prove;
■

theorem GEZeroExpr
$comp(funE, cst\,1) \in CGE$

proof
prove;
■

theorem rule GEZero
$cst\,0 \in CGE$

proof
use GEZeroExpr;
use ZeroExpr;
prove;
■

theorem GEIExpr
$opPlusI\,(cst\,0) \in CGE$

proof
prove;
■

theorem rule GEI
$funI \in CGE$

proof
use GEIExpr;
use IExprZero;
prove;
■

theorem GEDExpr
$opPlusI\ (opPlusI\ (cst\ 0)) \in CGE$

proof
prove;
■

theorem rule GED
$funD \in CGE$

proof
use GEDExpr;
use DExpr;
prove;
■

theorem GEM3Expr
$opPlusI\ funD \in CGE$

proof
prove;
■

theorem rule GEM3
$funM3 \in CGE$

proof
use GEM3Expr;
use M3Expr;
prove;
■

$F^- = E\,(SSSM_3FE)^+S\,(E^{-1}DF^+)^+$, without Sq.

$exprMinusIE == (\lambda\, F : \mathbb{N} \to \mathbb{N} \bullet comp(comp(comp(funE, opPlusI\ (comp(comp(comp(comp(comp(funS, funS), funS), funS), funM3), F), funE))), funS), opPlusI\ (comp(comp(opInv\ funE, funD), opPlusI\ F))))$

proof [exprMinusIE$domainCheck]
prove;
■

theorem rule exprMinusIEFun
$exprMinusIE \in (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
invoke exprMinusIE;
apply FunIntro to predicate $(\lambda\ F : \mathbb{N} \to \mathbb{N} \bullet comp(comp(comp(funE, opPlusI\ (comp(comp(comp(comp(comp(funS, funS), funS), funM3), F), funE))), funS), opPlusI\ (comp(comp(opInv\ funE, funD), opPlusI\ F)))) \in (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$;
with enabled $(RelPow, inPower)$ *prove*;
■

theorem disabled rule exprMinusIEApp
$\forall\, F : \mathbb{N} \to \mathbb{N} \bullet exprMinusIE\ F = comp(comp(comp(funE, opPlusI\ (comp(comp(comp(comp(comp(funS, funS), funS), funM3), F), funE))), funS), opPlusI\ (comp(comp(opInv\ funE, funD), opPlusI\ F)))$

proof
use exprMinusIEFun;
use pairInFunction[$\mathbb{N} \to \mathbb{N}$, $\mathbb{N} \to \mathbb{N}$][$f := exprMinusIE$, $x := F$, $y := comp(comp(comp(funE,$ $opPlusI\ (comp(comp(comp(comp(comp(funS,\ funS),\ funS),\ funM3),\ F),\ funE))),\ funS),$ $opPlusI\ (comp(comp(opInv\ funE,\ funD),\ opPlusI\ F)))$];
invoke exprMinusIE;
prove;
■

theorem rule MinusIEIn
$\forall F : \mathbb{N} \to \mathbb{N} \mid F \in CGE \bullet exprMinusIE\ F \in CGE$

proof
with enabled (*exprMinusIEApp*) *prove*;
■

theorem rule MinusIEVal
$\forall F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \mid x \leq F\ x \bullet exprMinusIE\ F\ x = F\ x - x$

proof
with enabled (*exprMinusIEApp*) *prove*;
use EinvD[$z := F\ x\ +\ x$];
use ESqAdd[$m := F\ x\ +\ x\ +\ 1,\ k := x$];
use ESqAdd[$m := F\ x\ +\ x\ +\ 2,\ k := F\ x\ -\ x$];
prove;
■

theorem SqExprGE
$exprMinusIE\ (exprMinusIE\ (comp(opInv\ funE, funD))) = funSq$

proof
apply UFExt to predicate $exprMinusIE\ (exprMinusIE\ (comp(opInv\ funE,\ funD)))\ =\ funSq$;
prove;
■

theorem GESqExpr
$exprMinusIE\ (exprMinusIE\ (comp(opInv\ funE, funD))) \in CGE$

proof
prove;
■

theorem rule GESq
$funSq \in CGE$

proof
use GESqExpr;
use SqExprGE;
prove;
■

$I - F = E\ (S\ (F^+)^+ E)^+ (\mathrm{Sq}\ F^+)^+$ where $F(x) \leq x$.

$exprIminusE == (\lambda F : \mathbb{N} \to \mathbb{N} \bullet comp(comp(funE, opPlusI\ (comp(comp(funS, opPlusI\ (opPlusI\ F)),$ $funE))), opPlusI\ (comp(funSq, opPlusI\ F))))$

proof [exprIminusE$domainCheck]
prove;
■

theorem rule exprIminusEFun
$exprIminusE \in (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
invoke exprIminusE;
apply FunIntro to predicate $(\lambda\ F : \mathbb{N} \to \mathbb{N} \bullet comp(comp(funE,\ opPlusI\ (comp(comp(funS,\ opPlusI\ (opPlusI\ F)),\ funE))),\ opPlusI\ (comp(funSq,\ opPlusI\ F)))) \in (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$;
with enabled $(RelPow,\ inPower)$ *prove*;
■

theorem disabled rule exprIminusEApp
$\forall F : \mathbb{N} \to \mathbb{N} \bullet exprIminusE\ F = comp(comp(funE, opPlusI\ (comp(comp(funS, opPlusI\ (opPlusI\ F)), funE))), opPlusI\ (comp(funSq, opPlusI\ F)))$

proof
use exprIminusEFun;
use pairInFunction$[\mathbb{N} \to \mathbb{N},\ \mathbb{N} \to \mathbb{N}][f := exprIminusE,\ x := F,\ y := comp(comp(funE,\ opPlusI\ (comp(comp(funS,\ opPlusI\ (opPlusI\ F)),\ funE))),\ opPlusI\ (comp(funSq,\ opPlusI\ F)))]$;
invoke exprIminusE;
prove;
■

theorem rule IminusEIn
$\forall F : \mathbb{N} \to \mathbb{N} \mid F \in CGE \bullet exprIminusE\ F \in CGE$

proof
with enabled $(exprIminusEApp)$ *prove*;
■

theorem rule IminusEVal
$\forall F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \mid F\ x \leq x \bullet exprIminusE\ F\ x = x - F\ x$

proof
with enabled $(exprIminusEApp)$ *prove*;
use UFNonneg$[f := F,\ x := x]$;
use ESqAdd$[m := F\ x + x,\ k := x]$;
use RobDiffE$[a := x,\ b := F\ x]$;
prove;
■

$W = E\,(SSDEE)^{+}E^{-1}$: $W(2q) = 2E(2q) + 1$ and $W(2q+1) = 2E(2q+1)$.

$exprW == comp(comp(funE, opPlusI\ (comp(comp(comp(comp(funS, funS), funD), funE), funE))), opInv\ funE)$

proof [exprW$domainCheck]
prove;
■

theorem WEven
$\forall q : \mathbb{N} \bullet exprW\ (2 * q) = 2 * funE\ (2 * q) + 1$

proof
invoke exprW;
prove;
use ESqAdd[$m := q,\ k := 2 * q$];
use TwoELe[$y := 2 * q$];
use ESqAdd[$m := q + 1,\ k := 2 * funE\,(2 * q) + 1$];
prove;
■

theorem WOdd
$\forall\, q : \mathbb{N} \bullet exprW\,(1 + 2 * q) = 2 * funE\,(1 + 2 * q)$

proof
invoke exprW;
prove;
use ESqAdd[$m := q + 1,\ k := 1 + 2 * q$];
use TwoELe[$y := 1 + 2 * q$];
use ESqAdd[$m := q + 2,\ k := 2 * funE\,(1 + 2 * q)$];
prove;
■

theorem EAt
$\forall\, y, m, k : \mathbb{N} \mid y = m * m + k \wedge k \leq 2 * m \bullet funE\ y = k$

proof
prove;
use ESqAdd[$m := m,\ k := k$];
prove;
■

theorem WFun
$exprW \in \mathbb{N} \to \mathbb{N}$

proof
invoke exprW;
prove;
■

theorem WSurj
$exprW \in Surj$

proof
invoke Surj;
use WFun;
prove;
use DivModOf[$x := x,\ d := 2$];
use TwoA[$x := x \ \mathsf{div}\ 2$];
use TwoA[$x := 1 + x \ \mathsf{div}\ 2$];
prove;
split $x \ \mathsf{mod}\ 2 = 0$;
cases;
instantiate $y == 1 + 2 * (funA\,(x \ \mathsf{div}\ 2) + x \ \mathsf{div}\ 2)$;
use WOdd[$q := funA\,(x \ \mathsf{div}\ 2) + x \ \mathsf{div}\ 2$];
use EAt[$y := 1 + 2 * (funA\,(x \ \mathsf{div}\ 2) + x \ \mathsf{div}\ 2),\ m := 1 + x \ \mathsf{div}\ 2,\ k := x \ \mathsf{div}\ 2$];
prove;
next;
instantiate $y__0 == 2 * (funA\,(1 + x \ \mathsf{div}\ 2) + x \ \mathsf{div}\ 2 + 1)$;
use WEven[$q := funA\,(1 + x \ \mathsf{div}\ 2) + x \ \mathsf{div}\ 2 + 1$];
use EAt[$y := 2 * (funA\,(1 + x \ \mathsf{div}\ 2) + x \ \mathsf{div}\ 2 + 1),\ m := 2 + x \ \mathsf{div}\ 2,\ k := x \ \mathsf{div}\ 2$];
prove;
next;
■

9.6 The case $X = E$: $\mathrm{Hf} = E\ W^{-1}$ and Rt

theorem WPar
$\forall\, y : \mathbb{N} \bullet (y \bmod 2 = 0 \Rightarrow exprW\ y = 2 * funE\ y + 1) \wedge (y \bmod 2 = 1 \Rightarrow exprW\ y = 2 * funE\ y)$

proof
prove;
use DivModOf[$x := y,\ d := 2$];
use WEven[$q := y \operatorname{div} 2$];
use WOdd[$q := y \operatorname{div} 2$];
prove;
■

theorem HfExprGE
$comp(funE, opInv\ exprW) = funHf$

proof
apply UFExt to predicate $comp(funE,\ opInv\ exprW) = funHf$;
prove;
use WSurj;
use WFun;
use InvRight[$F := exprW,\ x := x$];
use UFNonneg[$f := opInv\ exprW,\ x := x$];
use WPar[$y := opInv\ exprW\ x$];
use DivModOf[$x := opInv\ exprW\ x,\ d := 2$];
use UFNonneg[$f := funE,\ x := opInv\ exprW\ x$];
prove;
split $opInv\ exprW\ x \bmod 2 = 0$;
cases;
use DivMod[$x := x,\ d := 2,\ q := funE\,(opInv\ exprW\ x),\ r := 1$];
prove;
next;
use DivMod[$x := x,\ d := 2,\ q := funE\,(opInv\ exprW\ x),\ r := 0$];
prove;
next;
■

theorem GEHfExpr
$comp(funE, opInv\ exprW) \in CGE$

proof
use WFun;
invoke exprW;
prove;
■

theorem rule GEHf
$funHf \in CGE$

proof
use GEHfExpr;
use HfExprGE;
prove;
■

$\mathrm{Rt}(4x+1) \in \{2\,\mathrm{Rt}(x), 2\,\mathrm{Rt}(x)+1\}$, so $\mathrm{Rt} = \mathrm{Hf}\,\mathrm{Hf}\,S\,S\,E\,(I - \bar{1})(I - E)\,S\,M_3^+$.

theorem HfRt4
$\forall\, x : \mathbb{N} \bullet funHf\ (funRt\ (1 + 4 * x)) = funRt\ x$

proof
prove;
use $RtSpec[x := x]$;
use $UFNonneg[f := funRt,\ x := x]$;
split $1 + 4 * x < (2 * funRt\, x + 1) * (2 * funRt\, x + 1)$;
cases;
use $RtI[r := 2 * funRt\, x,\ x := 1 + 4 * x]$;
use $DivMod[x := 2 * funRt\, x,\ d := 2,\ q := funRt\, x,\ r := 0]$;
prove;
next;
use $RtI[r := 2 * funRt\, x + 1,\ x := 1 + 4 * x]$;
use $DivMod[x := 2 * funRt\, x + 1,\ d := 2,\ q := funRt\, x,\ r := 1]$;
prove;
next;
■

theorem rule IminusEE
$\forall z : \mathbb{N} \bullet exprIminusE\, funE\, z = funRt\, z * funRt\, z$

proof
prove;
use $ELe[x := z]$;
use $IminusEVal[F := funE,\ x := z]$;
use $EDecomp[y := z]$;
prove;
■

theorem IminusEOne
$\forall n : \mathbb{N} \mid 1 \leq n \bullet exprIminusE\, (cst\, 1)\, (n * n) = n * n - 1$

proof
prove;
■

theorem ESqPred
$\forall n : \mathbb{N} \mid 1 \leq n \bullet funE\, (n * n - 1) = 2 * n - 2$

proof
prove;
use $EAt[y := n * n - 1,\ m := n - 1,\ k := 2 * n - 2]$;
prove;
■

theorem HfTwo
$\forall n : \mathbb{N} \bullet funHf\, (2 * n) = n$

proof
prove;
■

$exprRtGE == comp(comp(comp(comp(comp(comp(comp(funHf, funHf), funS), funS), funE),$
$exprIminusE\, (cst\, 1)), exprIminusE\, funE), funS), opPlusI\, funM3)$

proof [exprRtGE$domainCheck]
prove;
■

theorem RtExprGE
exprRtGE = *funRt*

proof
invoke exprRtGE;
apply UFExt to predicate comp(*comp*(*comp*(*comp*(*comp*(*comp*(*comp*(*funHf*, *funHf*), *funS*), *funS*), *funE*), *exprIminusE* (*cst* 1)), *exprIminusE funE*), *funS*), *opPlusI funM*3) = *funRt*;
prove;
use RtPos[*y* := 1 + 4 ∗ *x*];
use IminusEOne[*n* := *funRt* (1 + 4 ∗ *x*)];
use ESqPred[*n* := *funRt* (1 + 4 ∗ *x*)];
use HfTwo[*n* := *funRt* (1 + 4 ∗ *x*)];
*use HfRt*4[*x* := *x*];
prove;
■

theorem GERtExpr
exprRtGE ∈ *CGE*

proof
invoke exprRtGE;
prove;
■

theorem rule GERt
funRt ∈ *CGE*

proof
use GERtExpr;
use RtExprGE;
prove;
■

theorem GERich
CGE ∈ *Rich*

proof
use CGEInPower;
invoke Rich;
with enabled (*CGEUF*) *prove*;
■

9.7 The case $X = E$: the closure is Robinson's

theorem JRUClosedGE
CGE ∈ *UClosed* {*opInv*}

proof
use CGEInPower;
with enabled (*UClosedDef*, *CGEUF*) *prove*;
■

theorem JRBClosedGE
CGE ∈ *BClosed* {*opPlus*, *comp*}

proof
use CGEInPower;
use GERich;
with enabled (*BClosedDef*, *CGEUF*) *prove*;
split b = *opPlus*;
cases;
prove;
next;
prove;
next;
■

theorem JRSubGE
$ClosJR \subseteq CGE$

proof
use ClosLeast[I := { *funS*, *funE* }, U := { *opInv* }, B := { *opPlus*, *comp* }, Z := *CGE*];
use CGEInPower;
use JRUClosedGE;
use JRBClosedGE;
with enabled (*ClosJRDef*) *prove*;
■

In Robinson's closure: $I = E\,E^{-1}$, $\bar{0} = E\,S\,E^{-1}(I + I)$, $\bar{1} = S\,\bar{0}$ and $F^{+} = F + I$.

theorem EEinv
$comp(funE, opInv\,funE) = funI$

proof
apply UFExt to predicate $comp(funE,\ opInv\,funE) = funI$;
prove;
use ESurj;
use InvRight[F := *funE*, x := x];
prove;
■

theorem JRIExpr
$comp(funE, opInv\,funE) \in ClosJR$

proof
prove;
■

theorem rule JRI
$funI \in ClosJR$

proof
use JRIExpr;
use EEinv;
prove;
■

theorem ZeroExprJR
$comp(comp(comp(funE, funS), opInv\,funE), opPlus(funI, funI)) = cst\,0$

proof
apply UFExt to predicate $comp(comp(comp(funE,\ funS),\ opInv\, funE),\ opPlus(funI,\ funI)) = cst\, 0$;
prove;
use $EinvD[z := x]$;
use $EAt[y := 1 + x * x + 2 * x,\ m := x + 1,\ k := 0]$;
prove;
■

theorem JRZeroExpr
$comp(comp(comp(funE, funS), opInv\, funE), opPlus(funI, funI)) \in ClosJR$

proof
prove;
■

theorem rule JRZero
$cst\, 0 \in ClosJR$

proof
use *JRZeroExpr*;
use *ZeroExprJR*;
prove;
■

theorem JROneExpr
$comp(funS, cst\, 0) \in ClosJR$

proof
prove;
■

theorem rule JROne
$cst\, 1 \in ClosJR$

proof
use *JROneExpr*;
use *OneExpr*;
prove;
■

theorem rule JRPlusI
$\forall F : \mathbb{N} \to \mathbb{N} \mid F \in ClosJR \bullet opPlusI\ F \in ClosJR$

proof
use $PlusIEq[F := F]$;
prove;
■

theorem GEUClosedJR
$ClosJR \in UClosed\, \{opPlusI, opInv\}$

proof
use ClosJRInPower;
with enabled (*UClosedDef*, *ClosJRUF*) *prove*;
split $u = opPlusI$;
cases;
prove;
next;
prove;
next;
■

theorem GEBClosedJR
$ClosJR \in BClosed\,\{comp\}$

proof
use ClosJRInPower;
with enabled (*BClosedDef*, *ClosJRUF*) *prove*;
■

theorem GESubJR
$CGE \subseteq ClosJR$

proof
use ClosLeast$[I := \{\ cst\,1,\ funE\ \},\ U := \{\ opPlusI,\ opInv\ \},\ B := \{\ comp\ \},\ Z := ClosJR]$;
use ClosJRInPower;
use GEUClosedJR;
use GEBClosedJR;
with enabled (*CGEDef*) *prove*;
■

theorem GREqJRE
$CGE = ClosJR$

proof
use GESubJR;
use JRSubGE;
use CGEInPower;
use ClosJRInPower;
use SubAntisym$[\mathbb{N} \to \mathbb{N}][A := CGE,\ B := ClosJR]$;
prove;
■

9.8 The case $X = K$

theorem Halve
$\forall\, a, b : \mathbb{Z} \mid 2 * a = 2 * b \bullet a = b$

proof
prove;
■

$Z = K^{-1}$, $A = KSSS((ZZ^{+}K)^{+}ZK)^{+}Z$, $F^{-} = K(SSSM_3FK)^{+}(AF^{+})^{+}$, Sq $= (((DZ)^{-})^{-})^{-}$, $P = K(KS)^{-1}$, $U = I - AKP(I - K)$ $(U = V + K)$,

theorem Div2Bound
$\forall\, x : \mathbb{N} \bullet 2 * (x \operatorname{div} 2) \leq x \wedge x \leq 2 * (x \operatorname{div} 2) + 1$

proof
use DivModOf[$x := x,\ d := 2$];
prove;
■

Hf $= KU^{-1}$, $B = I - (I - \mathrm{Sq}(I - \mathrm{Hf})\,U)(I - K)$ ($B = K + \lceil V/2 \rceil^2$) and $E = KB^{-1}$.

$CGK == clos\,\{cst\,1, funK\}\,\{opPlusI, opInv\}\,\{comp\}$

proof [CGK$domainCheck]
prove;
■

theorem disabled rule CGKDef
$CGK = clos\,\{cst\,1, funK\}\,\{opPlusI, opInv\}\,\{comp\}$

proof
invoke CGK;
prove;
■

theorem CGKInPower
$CGK \in \mathbb{P}(\mathbb{N} \to \mathbb{N})$

proof
use ClosInPow[$I := \{\ cst\,1,\ funK\ \},\ U := \{\ opPlusI,\ opInv\ \},\ B := \{\ comp\ \}$];
with enabled (*CGKDef*) *prove*;
■

theorem disabled rule CGKUF
$\forall\, F : \mathbb{P}(\mathbb{Z} \times \mathbb{Z}) \mid F \in CGK \bullet F \in \mathbb{N} \to \mathbb{N}$

proof
prove;
use ClosElemUF[$I := \{\ cst\,1,\ funK\ \},\ U := \{\ opPlusI,\ opInv\ \},\ B := \{\ comp\ \},\ F := F$];
with enabled (*CGKDef*) *prove*;
■

theorem rule CGKInit0
$cst\,1 \in CGK$

proof
with enabled (*CGKDef*) *prove*;
■

theorem rule CGKInit1
$funK \in CGK$

proof
with enabled (*CGKDef*) *prove*;
■

theorem rule CGKUnop0
$\forall\, F : \mathbb{N} \to \mathbb{N} \mid F \in CGK \bullet opPlusI\ F \in CGK$

proof
with enabled (*CGKDef*) *prove*;
■

theorem rule CGKUnop1
$\forall F : \mathbb{N} \to \mathbb{N} \mid F \in CGK \bullet opInv\, F \in CGK$

proof
with enabled (*CGKDef*) *prove*;
■

theorem rule CGKBinop0
$\forall F, G : \mathbb{N} \to \mathbb{N} \mid F \in CGK \wedge G \in CGK \bullet comp(F, G) \in CGK$

proof
with enabled (*CGKDef*) *prove*;
■

theorem ZeroExprK
$comp(funK, cst\, 1) = cst\, 0$

proof
apply UFExt to predicate $comp(funK,\ cst\, 1) = cst\, 0$;
prove;
use $KAt[m := 1,\ t := 1,\ k := 0]$;
use AZero;
prove;
■

theorem GKSExpr
$opPlusI\ (cst\, 1) \in CGK$

proof
prove;
■

theorem rule GKS
$funS \in CGK$

proof
use GKSExpr;
use SExpr;
prove;
■

theorem GKZeroExpr
$comp(funK, cst\, 1) \in CGK$

proof
prove;
■

theorem rule GKZero
$cst\, 0 \in CGK$

proof
use GKZeroExpr;
use ZeroExprK;
prove;
■

theorem GKIExpr
$opPlusI\,(cst\,0) \in CGK$

proof
prove;
■

theorem rule GKI
$funI \in CGK$

proof
use GKIExpr;
use IExprZero;
prove;
■

theorem GKDExpr
$opPlusI\,(opPlusI\,(cst\,0)) \in CGK$

proof
prove;
■

theorem rule GKD
$funD \in CGK$

proof
use GKDExpr;
use DExpr;
prove;
■

theorem GKM3Expr
$opPlusI\,funD \in CGK$

proof
prove;
■

theorem rule GKM3
$funM3 \in CGK$

proof
use GKM3Expr;
use M3Expr;
prove;
■

theorem KSurj
$funK \in Surj$

proof
invoke $Surj$;
prove;
instantiate $y == funA\, x + x$;
use $KAt[m := funA\, x + x,\ t := x,\ k := x]$;
use $UFNonneg[f := funA,\ x := x]$;
prove;
■

theorem KinvZ
$opInv\, funK = funZ$

proof
apply $UFExt$ *to predicate* $opInv\, funK = funZ$;
prove;
use $KSurj$;
use $KAt[m := funA\, x + x,\ t := x,\ k := x]$;
use $UFNonneg[f := funA,\ x := x]$;
use $InvLeast[F := funK,\ x := x,\ y := funA\, x + x]$;
use $InvRight[F := funK,\ x := x]$;
use $UFNonneg[f := opInv\, funK,\ x := x]$;
use $Decomp[x := opInv\, funK\, x]$;
use $UFNonneg[f := funV,\ x := opInv\, funK\, x]$;
use $AMono[m := x,\ n := funV\, (opInv\, funK\, x)]$;
prove;
■

theorem GKZExpr
$opInv\, funK \in CGK$

proof
prove;
■

theorem rule GKZ
$funZ \in CGK$

proof
use $GKZExpr$;
use $KinvZ$;
prove;
■

theorem KZ
$\forall\, x : \mathbb{N} \bullet funK\, (funA\, x + x) = x$

proof
prove;
use $KAt[m := funA\, x + x,\ t := x,\ k := x]$;
use $UFNonneg[f := funA,\ x := x]$;
prove;
■

$exprAK == comp(comp(comp(comp(comp(funK, funS), funS), funS),$
$\quad opPlusI\ (comp(comp(opPlusI\ (comp(comp(funZ, opPlusI\ funZ), funK)), funZ), funK))), funZ)$

proof [exprAK$domainCheck]
prove;
■

theorem AExprK
$exprAK = funA$

proof
invoke exprAK;
apply UFExt to predicate $comp(comp(comp(comp(comp(funK,\ funS),\ funS),\ funS),$
$\quad opPlusI\ (comp(comp(opPlusI\ (comp(comp(funZ,\ opPlusI\ funZ),\ funK)),\ funZ),\ funK))),$
$\quad funZ)\ =\ funA$;
prove;
use $KZ[x\ :=\ x]$;
use $UFNonneg[f\ :=\ funA,\ x\ :=\ x]$;
use $RobDiffK[a\ :=\ funA\,x\ +\ x,\ b\ :=\ x]$;
prove;
■

theorem GKAExpr
$exprAK \in CGK$

proof
invoke exprAK;
prove;
■

theorem rule GKA
$funA \in CGK$

proof
use GKAExpr;
use AExprK;
prove;
■

$exprMinusIK == (\lambda\, F : \mathbb{N} \to \mathbb{N} \bullet comp(comp(funK, opPlusI\ (comp(comp(comp(comp(comp(funS, funS),$
$\quad funS), funM3), F), funK))), opPlusI\ (comp(funA, opPlusI\ F))))$

proof [exprMinusIK$domainCheck]
prove;
■

theorem rule exprMinusIKFun
$exprMinusIK \in (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
invoke exprMinusIK;
apply FunIntro to predicate $(\lambda\ F :\ \mathbb{N}\ \to\ \mathbb{N}\ \bullet\ comp(comp(funK,$
$\quad opPlusI\ (comp(comp(comp(comp(comp(funS,\ funS),\ funS),\ funM3),\ F),\ funK))),$
$\quad opPlusI\ (comp(funA,\ opPlusI\ F)))) \in (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$;
with enabled $(RelPow,\ inPower)$ *prove*;
■

theorem disabled rule exprMinusIKApp
$\forall F : \mathbb{N} \to \mathbb{N} \bullet exprMinusIK\ F = comp(comp(funK, opPlusI\ (comp(comp(comp(comp(comp(funS, funS), funS), funS), funM3), F), funK))), opPlusI\ (comp(funA, opPlusI\ F)))$

proof
use exprMinusIKFun;
use pairInFunction[$\mathbb{N} \to \mathbb{N}$, $\mathbb{N} \to \mathbb{N}$][$f := exprMinusIK$, $x := F$, $y := comp(comp(funK, opPlusI\ (comp(comp(comp(comp(comp(funS, funS), funS), funM3), F), funK))), opPlusI\ (comp(funA, opPlusI\ F)))$];
invoke exprMinusIK;
prove;
■

theorem rule MinusIKIn
$\forall F : \mathbb{N} \to \mathbb{N} \mid F \in CGK \bullet exprMinusIK\ F \in CGK$

proof
with enabled (*exprMinusIKApp*) *prove*;
■

theorem rule MinusIKVal
$\forall F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \mid x \leq F\ x \bullet exprMinusIK\ F\ x = F\ x - x$

proof
with enabled (*exprMinusIKApp*) *prove*;
use UFNonneg[$f := F$, $x := x$];
use UFNonneg[$f := funA$, $x := F\ x + x$];
use KAt[$m := funA\ (F\ x + x) + x$, $t := F\ x + x$, $k := x$];
use ASuc[$x := F\ x + x$];
use ASuc[$x := 1 + F\ x + x$];
use KAt[$m := 3 + (funA\ (F\ x + x) + x + 3 * F\ x)$, $t := 2 + F\ x + x$, $k := F\ x - x$];
prove;
■

theorem SqExprGK
$exprMinusIK\ (exprMinusIK\ (exprMinusIK\ (comp(funD, funZ)))) = funSq$

proof
apply UFExt to predicate $exprMinusIK\ (exprMinusIK\ (exprMinusIK\ (comp(funD, funZ)))) = funSq$;
prove;
use TwoA[$x := x$];
prove;
■

theorem GKSqExpr
$exprMinusIK\ (exprMinusIK\ (exprMinusIK\ (comp(funD, funZ)))) \in CGK$

proof
prove;
■

theorem rule GKSq
$funSq \in CGK$

proof
use GKSqExpr;
use SqExprGK;
prove;
■

The predecessor $P = K(KS)^{-1}$.

theorem KSSurj
$comp(funK, funS) \in Surj$

proof
invoke Surj;
prove;
instantiate $y == funA\, x + 2 * x$;
use $ASuc[x := x]$;
use $UFNonneg[f := funA,\ x := x]$;
use $KAt[m := 1 + funA\, x + 2 * x,\ t := 1 + x,\ k := x]$;
prove;
■

$exprPK == comp(funK, opInv\,(comp(funK, funS)))$

proof [exprPK$domainCheck]
prove;
■

theorem rule PKFun
$exprPK \in \mathbb{N} \to \mathbb{N}$

proof
invoke exprPK;
prove;
■

theorem KSInvRight
$\forall\, x : \mathbb{N} \bullet funK\,(1 + opInv\,(comp(funK, funS))\,x) = x$

proof
prove;
use KSSurj;
use $InvRight[F := comp(funK,\ funS),\ x := x]$;
with disabled (*InvRight*) *prove*;
■

theorem PKZero
$exprPK\, 0 = 0$

proof
invoke exprPK;
prove;
use KSSurj;
use $KAt[m := 1,\ t := 1,\ k := 0]$;
use AZero;
use $InvLeast[F := comp(funK,\ funS),\ x := 0,\ y := 0]$;
use $UFNonneg[f := opInv\,(comp(funK,\ funS)),\ x := 0]$;
use KZero;
prove;
■

theorem PKSuc
$\forall x : \mathbb{N} \bullet exprPK\,(1 + x) = x$

proof
invoke exprPK;
prove;
use KSInvRight[$x := 1 + x$];
use UFNonneg[$f := opInv\,(comp(funK,\ funS)),\ x := 1 + x$];
use Decomp[$x := 1 + opInv\,(comp(funK,\ funS))\,(1 + x)$];
use UFNonneg[$f := funA,\ x := funV\,(1 + opInv\,(comp(funK,\ funS))\,(1 + x))$];
use KAt[$m := opInv\,(comp(funK,\ funS))\,(1 + x),\ t := funV\,(1 + opInv\,(comp(funK,\ funS))\,(1 + x)),\ k := x$];
prove;
■

theorem GKPExpr
$comp(funK, opInv\,(comp(funK, funS))) \in CGK$

proof
prove;
■

theorem rule GKP
$exprPK \in CGK$

proof
use GKPExpr;
invoke exprPK;
prove;
■

theorem rule GKIminusF
$\forall F : \mathbb{N} \to \mathbb{N} \mid F \in CGK \bullet exprIminusF\ F \in CGK$

proof
with enabled (*exprIminusFApp*) *prove*;
■

theorem rule IminusKVal
$\forall y : \mathbb{N} \bullet exprIminusF\ funK\ y = funA\,(funV\ y)$

proof
prove;
use Decomp[$x := y$];
use UFNonneg[$f := funA,\ x := funV\ y$];
use IminusFVal[$F := funK,\ x := y$];
prove;
■

$exprU == exprIminusF\,(comp(comp(comp(funA, funK), exprPK), exprIminusF\ funK))$

proof [exprU$domainCheck]
use PKFun;
prove;
■

theorem UIn
$exprU \in CGK$

proof
invoke $exprU$;
use GKP;
prove;
■

theorem UPairPos
$\forall\, t, k : \mathbb{N} \mid k \leq 1 + t \bullet exprU\,(funA\,(1 + t) + k) = 1 + t + k$

proof
invoke $exprU$;
prove;
use $Diag[k := k,\ t := 1 + t]$;
use $ASuc[x := t]$;
use $UFNonneg[f := funA,\ x := t]$;
use $PKSuc[x := funA\,t + t]$;
use $KAt[m := funA\,t + t,\ t := t,\ k := t]$;
use $IminusFVal[F := comp(comp(comp(funA,\ funK),\ exprPK),\ exprIminusF\,funK),\ x :=$
$funA\,(1 + t) + k]$;
prove;
■

theorem UZero
$exprU\,0 = 0$

proof
use $IminusFVal[F := comp(comp(comp(funA,\ funK),\ exprPK),\ exprIminusF\,funK),\ x := 0]$;
use $IminusKVal[y := 0]$;
use $Diag[k := 0,\ t := 0]$;
use $AZero$;
use $PKZero$;
use $KZero$;
invoke $exprU$;
prove;
■

theorem UPair
$\forall\, k, t : \mathbb{N} \mid k \leq t \bullet exprU\,(funA\,t + k) = t + k$

proof
prove;
split $t = 0$;
cases;
use $UZero$;
use $AZero$;
prove;
next;
use $UPairPos[t := t - 1,\ k := k]$;
prove;
next;
■

theorem UVal
$\forall\, y : \mathbb{N} \bullet exprU\ y = funV\ y + funK\ y$

proof
prove;
use Decomp[$x := y$];
use UPair[$k := \mathit{funK}\ y,\ t := \mathit{funV}\ y$];
prove;
■

theorem rule UFun
$\mathit{exprU} \in \mathbb{N} \to \mathbb{N}$

proof
invoke exprU;
prove;
■

theorem USurj
$\mathit{exprU} \in \mathit{Surj}$

proof
invoke Surj;
use UFun;
prove;
use DivModOf[$x := x,\ d := 2$];
split $x \bmod 2 = 0$;
cases;
instantiate $y == \mathit{funA}\,(x \operatorname{div} 2) + x \operatorname{div} 2$;
use UPair[$k := x \operatorname{div} 2,\ t := x \operatorname{div} 2$];
prove;
next;
instantiate $y_0 == \mathit{funA}\,(1 + x \operatorname{div} 2) + x \operatorname{div} 2$;
use UPair[$k := x \operatorname{div} 2,\ t := 1 + x \operatorname{div} 2$];
prove;
next;
■

Hf $= KU^{-1}$: the least $y = A(t) + k$ with $t + k = x$ has $k = \lfloor x/2 \rfloor$.

theorem HfMin
$\forall\, x, y, t, k : \mathbb{N} \mid y = \mathit{funA}\, t + k \wedge k \leq t \wedge t + k = x \wedge y \leq \mathit{funA}\,(x - x \operatorname{div} 2) + x \operatorname{div} 2 \bullet k = x \operatorname{div} 2$

proof
prove;
use Div2Bound[$x := x$];
use ASuc[$x := x - x \operatorname{div} 2$];
use UFNonneg[$f := \mathit{funA},\ x := x - x \operatorname{div} 2$];
use AMono[$m := 1 + (x - x \operatorname{div} 2),\ n := t$];
prove;
■

theorem HfExprK
$\mathit{comp}(\mathit{funK}, \mathit{opInv}\ \mathit{exprU}) = \mathit{funHf}$

proof
apply UFExt to predicate $comp(funK,\ opInv\ exprU) = funHf$;
prove;
use USurj;
use InvRight$[F := exprU,\ x := x]$;
use UFNonneg$[f := opInv\ exprU,\ x := x]$;
use UVal$[y := opInv\ exprU\ x]$;
use Decomp$[x := opInv\ exprU\ x]$;
use Div2Bound$[x := x]$;
use UPair$[k := x \mathsf{\ div\ } 2,\ t := x - x \mathsf{\ div\ } 2]$;
use InvLeast$[F := exprU,\ x := x,\ y := funA\,(x - x \mathsf{\ div\ } 2) + x \mathsf{\ div\ } 2]$;
use HfMin$[x := x,\ y := opInv\ exprU\ x,\ t := funV\,(opInv\ exprU\ x),\ k :=$
$funK\,(opInv\ exprU\ x)]$;
prove;
■

theorem GKHfExpr
$comp(funK, opInv\ exprU) \in CGK$

proof
use UIn;
use UFun;
prove;
■

theorem rule GKHf
$funHf \in CGK$

proof
use GKHfExpr;
use HfExprK;
prove;
■

$B = K + \lceil V/2 \rceil^2$ and $E = KB^{-1}$.

theorem AHalf
$\forall\, h, r : \mathbb{N} \mid r = 0 \vee r = 1 \bullet funA\,(2 * h + r) = (h + r) * (h + r) + h * h + h$

proof
prove;
use TwoA$[x := 2 * h + r]$;
use Halve$[a := funA\,(2 * h + r),\ b := (h + r) * (h + r) + h * h + h]$;
split $r = 0$;
cases;
prove;
next;
prove;
next;
■

theorem rule IminusHfEven
$\forall\, h : \mathbb{N} \bullet exprIminusF\ funHf\ (2 * h) = h$

proof
prove;
use DivMod$[x := 2 * h,\ d := 2,\ q := h,\ r := 0]$;
use IminusFVal$[F := funHf,\ x := 2 * h]$;
prove;
■

theorem rule IminusHfOdd
$\forall\, h : \mathbb{N} \bullet \mathit{exprIminusF}\ \mathit{funHf}\ (1 + 2 * h) = 1 + h$

proof
prove;
use $\mathit{DivMod}[x := 1 + 2 * h,\ d := 2,\ q := h,\ r := 1]$;
use $\mathit{IminusFVal}[F := \mathit{funHf},\ x := 1 + 2 * h]$;
prove;
■

$\mathit{exprB} == \mathit{exprIminusF}\ (\mathit{comp}(\mathit{exprIminusF}\ (\mathit{comp}(\mathit{comp}(\mathit{funSq}, \mathit{exprIminusF}\ \mathit{funHf}), \mathit{exprU})),$
$\mathit{exprIminusF}\ \mathit{funK}))$

proof [exprB$domainCheck]
prove;
■

theorem BIn
$\mathit{exprB} \in \mathit{CGK}$

proof
invoke *exprB*;
use *UIn*;
use *UFun*;
prove;
■

theorem rule BFun
$\mathit{exprB} \in \mathbb{N} \to \mathbb{N}$

proof
invoke *exprB*;
use *UFun*;
prove;
■

theorem BVal
$\forall\, h, r, k : \mathbb{N} \mid (r = 0 \lor r = 1) \land k \leq 2 * h + r \bullet \mathit{exprB}\ (\mathit{funA}\ (2 * h + r) + k) = k + (h + r) * (h + r)$

proof
invoke *exprB*;
prove;
use $\mathit{Diag}[k := k,\ t := 2 * h + r]$;
use $\mathit{AHalf}[h := h,\ r := r]$;
use $\mathit{MulNonneg}[a := h,\ b := h]$;
use $\mathit{UPair}[k := 0,\ t := 2 * h + r]$;
use $\mathit{IminusFVal}[F := \mathit{comp}(\mathit{comp}(\mathit{funSq},\ \mathit{exprIminusF}\ \mathit{funHf}),\ \mathit{exprU}),\ x := \mathit{funA}\ (2 * h + r)]$;
use $\mathit{IminusFVal}[F := \mathit{comp}(\mathit{exprIminusF}\ (\mathit{comp}(\mathit{comp}(\mathit{funSq},\ \mathit{exprIminusF}\ \mathit{funHf}),\ \mathit{exprU})),$
$\mathit{exprIminusF}\ \mathit{funK}),\ x := \mathit{funA}\ (2 * h + r) + k]$;
prove;
split $r = 0$;
cases;
prove;
next;
prove;
next;
■

theorem BSurj
$exprB \in Surj$

proof
invoke $Surj$;
use $BFun$;
prove;
instantiate $y == funA\,(2 * funRt\,x) + funE\,x$;
use $EBound[y := x]$;
use $UFNonneg[f := funA,\ x := 2 * funRt\,x]$;
use $Diag[k := funE\,x,\ t := 2 * funRt\,x]$;
use $BVal[h := funRt\,x,\ r := 0,\ k := funE\,x]$;
use $EDecomp[y := x]$;
prove;
■

theorem Mod2Cases
$\forall\, x : \mathbb{N} \bullet x = 2 * (x \operatorname{div} 2) + x \bmod 2 \land (x \bmod 2 = 0 \lor x \bmod 2 = 1)$

proof
prove;
■

theorem Div2Nonneg
$\forall\, x : \mathbb{N} \bullet 0 \leq x \operatorname{div} 2$

proof
prove;
■

theorem EKey
$\forall\, x, y, t, k, q, r : \mathbb{N} \mid y = funA\,t + k \land k \leq t \land t = 2 * q + r \land (r = 0 \lor r = 1) \land exprB\,y = x \bullet funE\,x = k$

proof
prove;
use $BVal[h := q,\ r := r,\ k := k]$;
use $EAt[y := x,\ m := q + r,\ k := k]$;
prove;
■

theorem EExprKB
$comp(funK, opInv\ exprB) = funE$

proof
apply UFExt to predicate $comp(funK,\ opInv\ exprB) = funE$;
prove;
use $BSurj$;
use $InvRight[F := exprB,\ x := x]$;
use $UFNonneg[f := opInv\ exprB,\ x := x]$;
use $Decomp[x := opInv\ exprB\,x]$;
use $Mod2Cases[x := funV\,(opInv\ exprB\,x)]$;
use $Div2Nonneg[x := funV\,(opInv\ exprB\,x)]$;
use $EKey[x := x,\ y := opInv\ exprB\,x,\ t := funV\,(opInv\ exprB\,x),\ k := funK\,(opInv\ exprB\,x),$
$q := funV\,(opInv\ exprB\,x) \operatorname{div} 2,\ r := funV\,(opInv\ exprB\,x) \bmod 2]$;
prove;
■

theorem GKEExpr
$comp(funK, opInv\ exprB) \in CGK$

proof
use BIn;
use BFun;
prove;
■

theorem rule GKE
$funE \in CGK$

proof
use GKEExpr;
use EExprKB;
prove;
■

9.9 The cases $X = K$ and $X = L$: the closure is Robinson's

theorem CGEUClosedG
$CGE \in UClosed\,\{opPlusI, opInv\}$

proof
use CGEInPower;
with enabled (*UClosedDef*, *CGEUF*) *prove*;
split u = opPlusI;
cases;
prove;
next;
prove;
next;
■

theorem CGEBClosedG
$CGE \in BClosed\,\{comp\}$

proof
use CGEInPower;
with enabled (*BClosedDef*, *CGEUF*) *prove*;
■

theorem CGKUClosedG
$CGK \in UClosed\,\{opPlusI, opInv\}$

proof
use CGKInPower;
with enabled (*UClosedDef*, *CGKUF*) *prove*;
split u = opPlusI;
cases;
prove;
next;
prove;
next;
■

theorem CGKBClosedG
$CGK \in BClosed\,\{comp\}$

proof
use CGKInPower;
with enabled (*BClosedDef*, *CGKUF*) *prove*;
■

theorem rule GEK
$funK \in CGE$

proof
use GERich;
use RichK[$Y := CGE$];
prove;
■

theorem rule GEL
$funL \in CGE$

proof
use GERich;
use RichL[$Y := CGE$];
prove;
■

theorem GKSubGE
$CGK \subseteq CGE$

proof
use ClosLeast[$I := \{ cst\,1, funK \}, U := \{ opPlusI, opInv \}, B := \{ comp \}, Z := CGE$];
use CGEInPower;
use CGEUClosedG;
use CGEBClosedG;
with enabled (*CGKDef*) *prove*;
■

theorem GESubGK
$CGE \subseteq CGK$

proof
use ClosLeast[$I := \{ cst\,1, funE \}, U := \{ opPlusI, opInv \}, B := \{ comp \}, Z := CGK$];
use CGKInPower;
use CGKUClosedG;
use CGKBClosedG;
with enabled (*CGEDef*) *prove*;
■

$CGL == clos\,\{cst\,1, funL\}\,\{opPlusI, opInv\}\,\{comp\}$

proof [CGL$domainCheck]
prove;
■

theorem disabled rule CGLDef
$CGL = clos\,\{cst\,1, funL\}\,\{opPlusI, opInv\}\,\{comp\}$

proof
invoke CGL;
prove;
■

theorem CGLInPower
$CGL \in \mathbb{P}(\mathbb{N} \to \mathbb{N})$

proof
use $ClosInPow[I := \{ cst\,1,\ funL \},\ U := \{ opPlusI,\ opInv \},\ B := \{ comp \}]$;
with enabled $(CGLDef)$ *prove*;
■

theorem disabled rule CGLUF
$\forall F : \mathbb{P}(\mathbb{Z} \times \mathbb{Z}) \mid F \in CGL \bullet F \in \mathbb{N} \to \mathbb{N}$

proof
prove;
use $ClosElemUF[I := \{ cst\,1,\ funL \},\ U := \{ opPlusI,\ opInv \},\ B := \{ comp \},\ F := F]$;
with enabled $(CGLDef)$ *prove*;
■

theorem rule CGLInit0
$cst\,1 \in CGL$

proof
with enabled $(CGLDef)$ *prove*;
■

theorem rule CGLInit1
$funL \in CGL$

proof
with enabled $(CGLDef)$ *prove*;
■

theorem rule CGLUnop0
$\forall F : \mathbb{N} \to \mathbb{N} \mid F \in CGL \bullet opPlusI\ F \in CGL$

proof
with enabled $(CGLDef)$ *prove*;
■

theorem rule CGLUnop1
$\forall F : \mathbb{N} \to \mathbb{N} \mid F \in CGL \bullet opInv\ F \in CGL$

proof
with enabled $(CGLDef)$ *prove*;
■

theorem rule CGLBinop0
$\forall F, G : \mathbb{N} \to \mathbb{N} \mid F \in CGL \wedge G \in CGL \bullet comp(F, G) \in CGL$

proof
with enabled (*CGLDef*) *prove*;
■

theorem CGLUClosedG
$CGL \in UClosed\,\{opPlusI, opInv\}$

proof
use CGLInPower;
with enabled (*UClosedDef*, *CGLUF*) *prove*;
split u = opPlusI;
cases;
prove;
next;
prove;
next;
■

theorem CGLBClosedG
$CGL \in BClosed\,\{comp\}$

proof
use CGLInPower;
with enabled (*BClosedDef*, *CGLUF*) *prove*;
■

theorem GLKExpr
$comp(comp(funL, opPlusI\,(cst\,1)), opPlusI\,(comp(opPlusI\,(opPlusI\,(cst\,1)), funL))) \in CGL$

proof
prove;
■

theorem rule GLK
$funK \in CGL$

proof
use GLKExpr;
use KExprL;
prove;
■

theorem GKSubGL
$CGK \subseteq CGL$

proof
use ClosLeast[$I := \{ cst\,1, funK \}, U := \{ opPlusI, opInv \}, B := \{ comp \}, Z := CGL$];
use CGLInPower;
use CGLUClosedG;
use CGLBClosedG;
with enabled (*CGKDef*) *prove*;
■

theorem GLSubGE
$CGL \subseteq CGE$

proof
use $ClosLeast[I := \{ cst\,1,\ funL \},\ U := \{ opPlusI,\ opInv \},\ B := \{ comp \},\ Z := CGE]$;
use $CGEInPower$;
use $CGEUClosedG$;
use $CGEBClosedG$;
with enabled $(CGLDef)$ *prove*;
■

theorem rule GLE
$funE \in CGL$

proof
use $GKSubGL$;
use GKE;
apply $subsetDef$ *to predicate* $CGK \subseteq CGL$;
apply $inPower$ *to predicate* $CGK \in \mathbb{P}\ CGL$;
instantiate $e == funE$;
prove;
■

theorem GESubGL
$CGE \subseteq CGL$

proof
use $ClosLeast[I := \{ cst\,1,\ funE \},\ U := \{ opPlusI,\ opInv \},\ B := \{ comp \},\ Z := CGL]$;
use $CGLInPower$;
use $CGLUClosedG$;
use $CGLBClosedG$;
with enabled $(CGEDef)$ *prove*;
■

theorem GREqJR
$CGE = ClosJR \wedge CGK = ClosJR \wedge CGL = ClosJR$

proof
use $GREqJRE$;
use $GKSubGE$;
use $GESubGK$;
use $GKSubGL$;
use $GLSubGE$;
use $CGEInPower$;
use $CGKInPower$;
use $CGLInPower$;
use $SubAntisym[\mathbb{N} \to \mathbb{N}][A := CGK,\ B := CGE]$;
use $SubAntisym[\mathbb{N} \to \mathbb{N}][A := CGL,\ B := CGE]$;
use $GESubGL$;
prove;
■

9.10 The constant cannot be dropped

Without $\bar{1}$, every generated function fixes 0: $\langle X, F^+, FG, F^{-1} \rangle$ does not generate Robinson's closure for $X \in \{E, K, L\}$.

theorem rule L0
$funL\,0 = 0$

proof
use $LAt[m := 0,\ t := 0,\ k := 0]$;
use *AZero*;
prove;
■

theorem FixZeroUClosedInv
$FixZero \in UClosed\,\{opPlusI, opInv\}$

proof
use *FixZeroInPower*;
with enabled (*UClosedDef*) *prove*;
invoke *FixZero*;
prove;
split $u = opPlusI$;
cases;
prove;
next;
use $InvFix0[F := F]$;
prove;
next;
■

theorem GR0Fix
$\forall X : \mathbb{N} \to \mathbb{N} \mid X\,0 = 0 \bullet clos\,\{X\}\,\{opPlusI, opInv\}\,\{comp\} \subseteq FixZero$

proof
prove;
use $ClosLeast[I := \{ X \},\ U := \{ opPlusI,\ opInv \},\ B := \{ comp \},\ Z := FixZero]$;
use *FixZeroInPower*;
use *FixZeroUClosedInv*;
use *FixZeroBClosed*;
invoke *FixZero*;
prove;
■

theorem GR0NotOne
$\forall X : \mathbb{N} \to \mathbb{N} \mid X\,0 = 0 \bullet \neg\ cst\,1 \in clos\,\{X\}\,\{opPlusI, opInv\}\,\{comp\}$

proof
use $GR0Fix[X := X]$;
apply subsetDef to predicate $clos\,\{ X \}\,\{ opPlusI,\ opInv \}\,\{ comp \} \subseteq FixZero$;
apply inPower to predicate $clos\,\{ X \}\,\{ opPlusI,\ opInv \}\,\{ comp \} \in \mathbb{P}\ FixZero$;
instantiate $e == cst\,1$;
invoke *FixZero*;
prove;
■

theorem OneInJREq
$\forall Y : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \mid Y = ClosJR \bullet cst\,1 \in Y$

proof
prove;
■

theorem GR0Neq
$\forall X : \mathbb{N} \to \mathbb{N} \mid X\,0 = 0 \bullet \neg\ clos\,\{X\}\,\{opPlusI, opInv\}\,\{comp\} = ClosJR$

proof
use $GR0NotOne[X := X]$;
use $JROne$;
prove;
split $clos\,\{\ X\ \}\,\{\ opPlusI,\ opInv\ \}\,\{\ comp\ \} = ClosJR$;
cases;
equality substitute $clos\,\{\ X\ \}\,(\{\ opPlusI\ \} \cup \{\ opInv\ \})\,\{\ comp\ \}$;
prove;
next;
prove;
next;
■

theorem ExtGR0
$\neg\ clos\,\{funE\}\,\{opPlusI, opInv\}\,\{comp\} = ClosJR \wedge \neg\ clos\,\{funK\}\,\{opPlusI, opInv\}\,\{comp\} = ClosJR \wedge \neg\ clos\,\{funL\}\,\{opPlusI, opInv\}\,\{comp\} = ClosJR$

proof
use $GR0Neq[X := funE]$;
use $GR0Neq[X := funK]$;
use $GR0Neq[X := funL]$;
use $KZero$;
prove;
■

9.11 Remark: $\bar{1}$ and S are interchangeable

With S in place of $\bar{1}$: $\bar{0} = (ESE^{-1})(ESE^{-1})$, $\bar{0} = KSK^{-1}$ and $\bar{0} = L(LS)^{-1}S$, and $\bar{1} = S\bar{0}$.

theorem ESEEven
$\forall x, q : \mathbb{N} \mid x = 2 * q \bullet comp(comp(funE, funS), opInv\,funE)\,x = 0$

proof
prove;
use $EAt[y := 1 + q * q + 2 * q,\ m := q + 1,\ k := 0]$;
prove;
■

theorem ESEOdd
$\forall x, q : \mathbb{N} \mid x = 1 + 2 * q \bullet comp(comp(funE, funS), opInv\,funE)\,x = 2 + 2 * q$

proof
prove;
use $EAt[y := 2 + (q + 1) * (q + 1) + 2 * q,\ m := q + 1,\ k := 2 + 2 * q]$;
prove;
■

theorem ZeroExprSE
$comp(comp(comp(funE, funS), opInv\,funE), comp(comp(funE, funS), opInv\,funE)) = cst\,0$

proof
apply UFExt to predicate $comp(comp(comp(funE,\ funS),\ opInv\,funE),\ comp(comp(funE,\ funS),\ opInv\,funE)) = cst\,0$;
prove;
use $Mod2Cases[x := x]$;
use $Div2Nonneg[x := x]$;
use $ESEEven[x := 0,\ q := 0]$;
split $x \bmod 2 = 0$;
cases;
use $ESEEven[x := x,\ q := x \operatorname{div} 2]$;
prove;
next;
use $ESEOdd[x := x,\ q := x \operatorname{div} 2]$;
use $ESEEven[x := 2 + 2 * (x \operatorname{div} 2),\ q := 1 + x \operatorname{div} 2]$;
prove;
next;
■

theorem ZeroExprSK
$comp(comp(funK, funS), opInv\,funK) = cst\,0$

proof
apply UFExt to predicate $comp(comp(funK,\ funS),\ opInv\,funK) = cst\,0$;
prove;
use $KinvZ$;
use $ASuc[x := x]$;
use $UFNonneg[f := funA,\ x := x]$;
use $KAt[m := 1 + funA\,x + x,\ t := 1 + x,\ k := 0]$;
prove;
■

theorem LSSurj
$comp(funL, funS) \in Surj$

proof
invoke $Surj$;
prove;
instantiate $y == funA\,(1 + x)$;
use $UFNonneg[f := funA,\ x := 1 + x]$;
use $LAt[m := 1 + funA\,(1 + x),\ t := 1 + x,\ k := 1]$;
prove;
■

theorem LSLow
$\forall\, x, y, t, k : \mathbb{N} \mid 1 + y = funA\,t + k \wedge k \leq t \wedge t - k = 1 + x \bullet funA\,x + x \leq y$

proof
prove;
use $AMono[m := 1 + x,\ n := t]$;
use $ASuc[x := x]$;
prove;
■

theorem LSInvSuc
$\forall\, x : \mathbb{N} \bullet opInv\,(comp(funL, funS))\,(1 + x) = funA\,x + x$

proof
prove;
use LSSurj;
use InvRight[$F := comp(funL,\ funS),\ x := 1 + x$];
use UFNonneg[$f := opInv\,(comp(funL,\ funS)),\ x := 1 + x$];
use Decomp[$x := 1 + opInv\,(comp(funL,\ funS))\,(1 + x)$];
use LAt[$m := 1 + opInv\,(comp(funL,\ funS))\,(1 + x),\ t := funV\,(1 + opInv\,(comp(funL,$ $funS))\,(1 + x)),\ k := funK\,(1 + opInv\,(comp(funL,\ funS))\,(1 + x))$];
use LSLow[$x := x,\ y := opInv\,(comp(funL,\ funS))\,(1 + x),\ t := funV\,(1 + opInv\,(comp(funL,$ $funS))\,(1 + x)),\ k := funK\,(1 + opInv\,(comp(funL,\ funS))\,(1 + x))$];
use ASuc[$x := x$];
use UFNonneg[$f := funA,\ x := x$];
use LAt[$m := 1 + funA\,x + x,\ t := 1 + x,\ k := 0$];
use InvLeast[$F := comp(funL,\ funS),\ x := 1 + x,\ y := funA\,x + x$];
with disabled (*InvRight*) *prove*;
■

theorem ZeroExprSL
$comp(comp(funL, opInv\,(comp(funL, funS))), funS) = cst\,0$

proof
apply UFExt to predicate $comp(comp(funL,\ opInv\,(comp(funL,\ funS))),\ funS) = cst\,0$;
prove;
use LSInvSuc[$x := x$];
use UFNonneg[$f := funA,\ x := x$];
use LAt[$m := funA\,x + x,\ t := x,\ k := x$];
prove;
■

$CGSE == clos\,\{funS, funE\}\,\{opPlusI, opInv\}\,\{comp\}$

proof [CGSE$domainCheck]
prove;
■

theorem disabled rule CGSEDef
$CGSE = clos\,\{funS, funE\}\,\{opPlusI, opInv\}\,\{comp\}$

proof
invoke CGSE;
prove;
■

theorem CGSEInPower
$CGSE \in \mathbb{P}(\mathbb{N} \to \mathbb{N})$

proof
use ClosInPow[$I := \{\ funS,\ funE\ \},\ U := \{\ opPlusI,\ opInv\ \},\ B := \{\ comp\ \}$];
with enabled (*CGSEDef*) *prove*;
■

theorem disabled rule CGSEUF
$\forall\, F : \mathbb{P}(\mathbb{Z} \times \mathbb{Z}) \mid F \in CGSE \bullet F \in \mathbb{N} \to \mathbb{N}$

proof
prove;
use ClosElemUF[$I := \{\ funS,\ funE\ \},\ U := \{\ opPlusI,\ opInv\ \},\ B := \{\ comp\ \},\ F := F$];
with enabled (*CGSEDef*) *prove*;
■

theorem rule CGSEInit0
$funS \in CGSE$

proof
with enabled (*CGSEDef*) *prove*;
■

theorem rule CGSEInit1
$funE \in CGSE$

proof
with enabled (*CGSEDef*) *prove*;
■

theorem rule CGSEUnop0
$\forall F : \mathbb{N} \to \mathbb{N} \mid F \in CGSE \bullet opPlusI\ F \in CGSE$

proof
with enabled (*CGSEDef*) *prove*;
■

theorem rule CGSEUnop1
$\forall F : \mathbb{N} \to \mathbb{N} \mid F \in CGSE \bullet opInv\ F \in CGSE$

proof
with enabled (*CGSEDef*) *prove*;
■

theorem rule CGSEBinop0
$\forall F, G : \mathbb{N} \to \mathbb{N} \mid F \in CGSE \land G \in CGSE \bullet comp(F, G) \in CGSE$

proof
with enabled (*CGSEDef*) *prove*;
■

theorem CGSEUClosedG
$CGSE \in UClosed\,\{opPlusI, opInv\}$

proof
use CGSEInPower;
with enabled (*UClosedDef*, *CGSEUF*) *prove*;
split $u = opPlusI$;
cases;
prove;
next;
prove;
next;
■

theorem CGSEBClosedG
$CGSE \in BClosed\,\{comp\}$

proof
use CGSEInPower;
with enabled (*BClosedDef*, *CGSEUF*) *prove*;
■

theorem GSEZeroExpr
$comp(comp(comp(funE, funS), opInv\ funE), comp(comp(funE, funS), opInv\ funE)) \in CGSE$

proof
prove;
■

theorem rule GSEZero
$cst\ 0 \in CGSE$

proof
use GSEZeroExpr;
use ZeroExprSE;
prove;
■

theorem GSEOneExpr
$comp(funS, cst\ 0) \in CGSE$

proof
prove;
■

theorem rule GSEOne
$cst\ 1 \in CGSE$

proof
use GSEOneExpr;
use OneExpr;
prove;
■

theorem GRSESub1
$CGSE \subseteq CGE$

proof
use ClosLeast[$I := \{ funS, funE \}, U := \{ opPlusI, opInv \}, B := \{ comp \}, Z := CGE$];
use CGEInPower;
use CGEUClosedG;
use CGEBClosedG;
with enabled (*CGSEDef*) *prove*;
■

theorem GRSESub2
$CGE \subseteq CGSE$

proof
use ClosLeast[$I := \{ cst\ 1, funE \}, U := \{ opPlusI, opInv \}, B := \{ comp \}, Z := CGSE$];
use CGSEInPower;
use CGSEUClosedG;
use CGSEBClosedG;
with enabled (*CGEDef*) *prove*;
■

theorem GRSE
$CGSE = CGE$

proof
use $GRSESub1$;
use $GRSESub2$;
use $CGSEInPower$;
use $CGEInPower$;
use $SubAntisym[\mathbb{N} \to \mathbb{N}][A := CGSE,\ B := CGE]$;
prove;
■

$CGSK == clos\,\{funS, funK\}\,\{opPlusI, opInv\}\,\{comp\}$

proof [CGSK$domainCheck]
prove;
■

theorem disabled rule CGSKDef
$CGSK = clos\,\{funS, funK\}\,\{opPlusI, opInv\}\,\{comp\}$

proof
invoke $CGSK$;
prove;
■

theorem CGSKInPower
$CGSK \in \mathbb{P}(\mathbb{N} \to \mathbb{N})$

proof
use $ClosInPow[I := \{ funS,\ funK \},\ U := \{ opPlusI,\ opInv \},\ B := \{ comp \}]$;
with enabled $(CGSKDef)$ *prove*;
■

theorem disabled rule CGSKUF
$\forall F : \mathbb{P}(\mathbb{Z} \times \mathbb{Z}) \mid F \in CGSK \bullet F \in \mathbb{N} \to \mathbb{N}$

proof
prove;
use $ClosElemUF[I := \{ funS,\ funK \},\ U := \{ opPlusI,\ opInv \},\ B := \{ comp \},\ F := F]$;
with enabled $(CGSKDef)$ *prove*;
■

theorem rule CGSKInit0
$funS \in CGSK$

proof
with enabled $(CGSKDef)$ *prove*;
■

theorem rule CGSKInit1
$funK \in CGSK$

proof
with enabled (*CGSKDef*) *prove*;
■

theorem rule CGSKUnop0
$\forall F : \mathbb{N} \to \mathbb{N} \mid F \in CGSK \bullet opPlusI\ F \in CGSK$

proof
with enabled (*CGSKDef*) *prove*;
■

theorem rule CGSKUnop1
$\forall F : \mathbb{N} \to \mathbb{N} \mid F \in CGSK \bullet opInv\ F \in CGSK$

proof
with enabled (*CGSKDef*) *prove*;
■

theorem rule CGSKBinop0
$\forall F, G : \mathbb{N} \to \mathbb{N} \mid F \in CGSK \wedge G \in CGSK \bullet comp(F, G) \in CGSK$

proof
with enabled (*CGSKDef*) *prove*;
■

theorem CGSKUClosedG
$CGSK \in UClosed\,\{opPlusI, opInv\}$

proof
use CGSKInPower;
with enabled (*UClosedDef*, *CGSKUF*) *prove*;
split u = opPlusI;
cases;
prove;
next;
prove;
next;
■

theorem CGSKBClosedG
$CGSK \in BClosed\,\{comp\}$

proof
use CGSKInPower;
with enabled (*BClosedDef*, *CGSKUF*) *prove*;
■

theorem GSKZeroExpr
$comp(comp(funK, funS), opInv\ funK) \in CGSK$

proof
prove;
■

theorem rule GSKZero
$cst\,0 \in CGSK$

proof
use $GSKZeroExpr$;
use $ZeroExprSK$;
prove;
■

theorem GSKOneExpr
$comp(funS, cst\,0) \in CGSK$

proof
prove;
■

theorem rule GSKOne
$cst\,1 \in CGSK$

proof
use $GSKOneExpr$;
use $OneExpr$;
prove;
■

theorem GRSKSub1
$CGSK \subseteq CGK$

proof
use $ClosLeast[I := \{ funS, funK \}, U := \{ opPlusI, opInv \}, B := \{ comp \}, Z := CGK]$;
use $CGKInPower$;
use $CGKUClosedG$;
use $CGKBClosedG$;
with enabled $(CGSKDef)$ *prove*;
■

theorem GRSKSub2
$CGK \subseteq CGSK$

proof
use $ClosLeast[I := \{ cst\,1, funK \}, U := \{ opPlusI, opInv \}, B := \{ comp \}, Z := CGSK]$;
use $CGSKInPower$;
use $CGSKUClosedG$;
use $CGSKBClosedG$;
with enabled $(CGKDef)$ *prove*;
■

theorem GRSK
$CGSK = CGK$

proof
use $GRSKSub1$;
use $GRSKSub2$;
use $CGSKInPower$;
use $CGKInPower$;
use $SubAntisym[\mathbb{N} \to \mathbb{N}][A := CGSK, B := CGK]$;
prove;
■

$CGSL == clos\,\{funS, funL\}\,\{opPlusI, opInv\}\,\{comp\}$

proof [CGSL$domainCheck]
prove;
■

theorem disabled rule CGSLDef
$CGSL = clos\,\{funS, funL\}\,\{opPlusI, opInv\}\,\{comp\}$

proof
invoke CGSL;
prove;
■

theorem CGSLInPower
$CGSL \in \mathbb{P}(\mathbb{N} \to \mathbb{N})$

proof
use ClosInPow[$I := \{ funS, funL \}$, $U := \{ opPlusI, opInv \}$, $B := \{ comp \}$];
with enabled (*CGSLDef*) *prove*;
■

theorem disabled rule CGSLUF
$\forall F : \mathbb{P}(\mathbb{Z} \times \mathbb{Z}) \mid F \in CGSL \bullet F \in \mathbb{N} \to \mathbb{N}$

proof
prove;
use ClosElemUF[$I := \{ funS, funL \}$, $U := \{ opPlusI, opInv \}$, $B := \{ comp \}$, $F := F$];
with enabled (*CGSLDef*) *prove*;
■

theorem rule CGSLInit0
$funS \in CGSL$

proof
with enabled (*CGSLDef*) *prove*;
■

theorem rule CGSLInit1
$funL \in CGSL$

proof
with enabled (*CGSLDef*) *prove*;
■

theorem rule CGSLUnop0
$\forall F : \mathbb{N} \to \mathbb{N} \mid F \in CGSL \bullet opPlusI\ F \in CGSL$

proof
with enabled (*CGSLDef*) *prove*;
■

theorem rule CGSLUnop1
$\forall F : \mathbb{N} \to \mathbb{N} \mid F \in CGSL \bullet opInv\ F \in CGSL$

proof
with enabled (*CGSLDef*) *prove*;
■

theorem rule CGSLBinop0
$\forall F, G : \mathbb{N} \to \mathbb{N} \mid F \in CGSL \wedge G \in CGSL \bullet comp(F, G) \in CGSL$

proof
with enabled (*CGSLDef*) *prove*;
■

theorem CGSLUClosedG
$CGSL \in UClosed\,\{opPlusI, opInv\}$

proof
use CGSLInPower;
with enabled (*UClosedDef*, *CGSLUF*) *prove*;
split u = opPlusI;
cases;
prove;
next;
prove;
next;
■

theorem CGSLBClosedG
$CGSL \in BClosed\,\{comp\}$

proof
use CGSLInPower;
with enabled (*BClosedDef*, *CGSLUF*) *prove*;
■

theorem GSLZeroExpr
$comp(comp(funL, opInv\,(comp(funL, funS))), funS) \in CGSL$

proof
prove;
■

theorem rule GSLZero
$cst\,0 \in CGSL$

proof
use GSLZeroExpr;
use ZeroExprSL;
prove;
■

theorem GSLOneExpr
$comp(funS, cst\,0) \in CGSL$

proof
prove;
■

theorem rule GSLOne
$cst\,1 \in CGSL$

proof
use $GSLOneExpr$;
use $OneExpr$;
prove;
■

theorem GLSExpr
$opPlusI\,(cst\,1) \in CGL$

proof
prove;
■

theorem rule GLS
$funS \in CGL$

proof
use $GLSExpr$;
use $SExpr$;
prove;
■

theorem GRSLSub1
$CGSL \subseteq CGL$

proof
use $ClosLeast[I := \{\ funS,\ funL\ \},\ U := \{\ opPlusI,\ opInv\ \},\ B := \{\ comp\ \},\ Z := CGL]$;
use $CGLInPower$;
use $CGLUClosedG$;
use $CGLBClosedG$;
with enabled $(CGSLDef)$ *prove*;
■

theorem GRSLSub2
$CGL \subseteq CGSL$

proof
use $ClosLeast[I := \{\ cst\,1,\ funL\ \},\ U := \{\ opPlusI,\ opInv\ \},\ B := \{\ comp\ \},\ Z := CGSL]$;
use $CGSLInPower$;
use $CGSLUClosedG$;
use $CGSLBClosedG$;
with enabled $(CGLDef)$ *prove*;
■

theorem GRSL
$CGSL = CGL$

proof
use $GRSLSub1$;
use $GRSLSub2$;
use $CGSLInPower$;
use $CGLInPower$;
use $SubAntisym[\mathbb{N} \to \mathbb{N}][A := CGSL,\ B := CGL]$;
prove;
■

theorem GRSEq
$CGSE = ClosJR \wedge CGSK = ClosJR \wedge CGSL = ClosJR$

proof
use GRSE;
use GRSK;
use GRSL;
use GREqJR;
prove;
■

9.12 The shifted operator with inversion

$\langle X, F^{+b}, FG, F^{-1}\rangle$ generates Robinson's closure iff $b \geq 1$ ($X \in \{E, K, L\}$). A successor S is generated first; a predecessor P then gives $F^+ = P^b F^{+b}$ and $\overline{1}$.

theorem rule CbX
$\forall X : \mathbb{N} \to \mathbb{N};\ b : \mathbb{N} \bullet X \in clos\,\{X\}\,\{opPlusIb\ b, opInv\}\,\{comp\}$

proof
prove;
■

theorem CbComp
$\forall X : \mathbb{N} \to \mathbb{N};\ b : \mathbb{N};\ F, G : \mathbb{P}(\mathbb{Z} \times \mathbb{Z}) \mid F \in clos\,\{X\}\,\{opPlusIb\ b, opInv\}\,\{comp\} \wedge G \in clos\,\{X\}\,\{opPlusIb\ b, opInv\}\,\{comp\} \bullet comp(F, G) \in clos\,\{X\}\,\{opPlusIb\ b, opInv\}\,\{comp\}$

proof
prove;
use ClosElemUF[$I := \{\ X\ \},\ U := \{\ opPlusIb\ b,\ opInv\ \},\ B := \{\ comp\ \},\ F := F$];
use ClosElemUF[$I := \{\ X\ \},\ U := \{\ opPlusIb\ b,\ opInv\ \},\ B := \{\ comp\ \},\ F := G$];
prove;
■

theorem CbUn
$\forall X : \mathbb{N} \to \mathbb{N};\ b : \mathbb{N};\ F : \mathbb{P}(\mathbb{Z} \times \mathbb{Z}) \mid F \in clos\,\{X\}\,\{opPlusIb\ b, opInv\}\,\{comp\} \bullet opPlusIb\ b\ F \in clos\,\{X\}\,\{opPlusIb\ b, opInv\}\,\{comp\} \wedge opInv\ F \in clos\,\{X\}\,\{opPlusIb\ b, opInv\}\,\{comp\}$

proof
prove;
use ClosElemUF[$I := \{\ X\ \},\ U := \{\ opPlusIb\ b,\ opInv\ \},\ B := \{\ comp\ \},\ F := F$];
prove;
■

theorem CbIter
$\forall X : \mathbb{N} \to \mathbb{N};\ b, k : \mathbb{N};\ F, G : \mathbb{N} \to \mathbb{N} \mid F \in clos\,\{X\}\,\{opPlusIb\ b, opInv\}\,\{comp\} \wedge G \in clos\,\{X\}\,\{opPlusIb\ b, opInv\}\,\{comp\} \bullet comp(iter[\mathbb{N}]\ k\ F, G) \in clos\,\{X\}\,\{opPlusIb\ b, opInv\}\,\{comp\}$

proof
prove;
use PowIn[$I := \{\ X\ \},\ U := \{\ opPlusIb\ b,\ opInv\ \},\ B := \{\ comp\ \},\ F := F,\ G := G,\ k := k$];
prove;
■

$F^+ = P^b F^{+b}$ for a predecessor P.

theorem PlusIFromPred
$\forall\, b : \mathbb{N};\ P, F : \mathbb{N} \to \mathbb{N} \mid P \in Preds \bullet comp(iter[\mathbb{N}]\ b\ P, opPlusIb\ b\ F) = opPlusI\ F$

proof
apply UFExt to predicate $comp(iter[\mathbb{N}]\ b\ P,\ opPlusIb\ b\ F)\ =\ opPlusI\ F$;
prove;
use IterPAll$[k\ :=\ b,\ P\ :=\ P,\ x\ :=\ F\ x\ +\ x]$;
prove;
■

theorem CbPlusI
$\forall\, X : \mathbb{N} \to \mathbb{N};\ b : \mathbb{N};\ P, F : \mathbb{N} \to \mathbb{N} \mid P \in Preds \wedge P \in clos\,\{X\}\,\{opPlusIb\ b,$
$opInv\}\,\{comp\} \wedge F \in clos\,\{X\}\,\{opPlusIb\ b, opInv\}\,\{comp\} \bullet opPlusI\ F \in clos\,\{X\}\,\{opPlusIb\ b,$
$opInv\}\,\{comp\}$

proof
prove;
use CbUn$[X\ :=\ X,\ b\ :=\ b,\ F\ :=\ F]$;
use CbIter$[X\ :=\ X,\ b\ :=\ b,\ k\ :=\ b,\ F\ :=\ P,\ G\ :=\ opPlusIb\ b\ F]$;
use PlusIFromPred$[b\ :=\ b,\ P\ :=\ P,\ F\ :=\ F]$;
prove;
■

With $\bar{1}$ and a predecessor, the closure contains $\langle \bar{1}, X, F^+, FG, F^{-1} \rangle$.

theorem CbUClosed
$\forall\, X : \mathbb{N} \to \mathbb{N};\ b : \mathbb{N};\ P : \mathbb{N} \to \mathbb{N} \mid P \in Preds \wedge P \in clos\,\{X\}\,\{opPlusIb\ b,$
$opInv\}\,\{comp\} \bullet clos\,\{X\}\,\{opPlusIb\ b, opInv\}\,\{comp\} \in UClosed\,\{opPlusI, opInv\}$

proof
prove;
use ClosInPow$[I\ :=\ \{\ X\ \},\ U\ :=\ \{\ opPlusIb\ b,\ opInv\ \},\ B\ :=\ \{\ comp\ \}]$;
with enabled $(UClosedDef)$ *prove*;
use ClosElemUF$[I\ :=\ \{\ X\ \},\ U\ :=\ \{\ opPlusIb\ b,\ opInv\ \},\ B\ :=\ \{\ comp\ \},\ F\ :=\ F]$;
prove;
split $u\ =\ opPlusI$;
cases;
use CbPlusI$[X\ :=\ X,\ b\ :=\ b,\ P\ :=\ P,\ F\ :=\ F]$;
prove;
next;
use CbUn$[X\ :=\ X,\ b\ :=\ b,\ F\ :=\ F]$;
prove;
next;
■

theorem CbBClosed
$\forall\, X : \mathbb{N} \to \mathbb{N};\ b : \mathbb{N} \bullet clos\,\{X\}\,\{opPlusIb\ b, opInv\}\,\{comp\} \in BClosed\,\{comp\}$

proof
prove;
use ClosInPow$[I\ :=\ \{\ X\ \},\ U\ :=\ \{\ opPlusIb\ b,\ opInv\ \},\ B\ :=\ \{\ comp\ \}]$;
with enabled $(BClosedDef)$ *prove*;
■

theorem CbSub
$\forall\, X : \mathbb{N} \to \mathbb{N};\ b : \mathbb{N};\ P : \mathbb{N} \to \mathbb{N} \mid cst\ 1 \in clos\,\{X\}\,\{opPlusIb\ b,$
$opInv\}\,\{comp\} \wedge P \in Preds \wedge P \in clos\,\{X\}\,\{opPlusIb\ b, opInv\}\,\{comp\} \bullet clos\,\{cst\ 1,$
$X\}\,\{opPlusI, opInv\}\,\{comp\} \subseteq clos\,\{X\}\,\{opPlusIb\ b, opInv\}\,\{comp\}$

proof
prove;
use *ClosLeast*[$I := \{ cst\,1,\ X \},\ U := \{ opPlusI,\ opInv \},\ B := \{ comp \},\ Z := clos\{ X \}\{ opPlusIb\,b,\ opInv \}\{ comp \}$];
use *ClosInPow*[$I := \{ X \},\ U := \{ opPlusIb\,b,\ opInv \},\ B := \{ comp \}$];
use *CbUClosed*[$X := X,\ b := b,\ P := P$];
use *CbBClosed*[$X := X,\ b := b$];
prove;
■

Soundness: $F^{+b} = (S^b F)^+$.

theorem GEPlusIb
$\forall\, b : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N} \mid F \in CGE \bullet opPlusIb\,b\,F \in CGE$

proof
prove;
use *PlusIbExpr*[$b := b,\ F := F$];
use *GES*;
use *PowIn*[$I := \{ cst\,1,\ funE \},\ U := \{ opPlusI,\ opInv \},\ B := \{ comp \},\ F := funS,\ G := F,\ k := b$];
with enabled (*CGEDef*) *prove*;
■

theorem GEUClosedb
$\forall\, b : \mathbb{N} \bullet CGE \in UClosed\,\{opPlusIb\,b, opInv\}$

proof
prove;
use *CGEInPower*;
with enabled (*UClosedDef*, *CGEUF*) *prove*;
use *InPair*[$\mathbb{P}\,(\mathbb{P}\,(\mathbb{Z} \times \mathbb{Z}) \times \mathbb{P}\,(\mathbb{Z} \times \mathbb{Z}))$][$u := u,\ p := opPlusIb\,b,\ q := opInv$];
split $u = opInv$;
cases;
prove;
next;
use *GEPlusIb*[$b := b,\ F := F$];
prove;
next;
■

theorem GRbSubGE
$\forall\, X : \mathbb{N} \to \mathbb{N};\ b : \mathbb{N} \mid X \in CGE \bullet clos\,\{X\}\,\{opPlusIb\,b, opInv\}\,\{comp\} \subseteq CGE$

proof
prove;
use *ClosLeast*[$I := \{ X \},\ U := \{ opPlusIb\,b,\ opInv \},\ B := \{ comp \},\ Z := CGE$];
use *CGEInPower*;
use *CGEBClosedG*;
use *GEUClosedb*[$b := b$];
prove;
■

theorem ShiftVal
$\forall\, W : \mathbb{N} \to \mathbb{N};\ c, y : \mathbb{N} \mid (W, c) \in Shift \bullet W\ y = y + c$

proof
invoke *Shift*;
prove;
instantiate $y__0 == y$;
prove;
■

Subtraction by inversion with E: if $V(y) = y + c$ then $E\,(EV)^{-1}$ subtracts c.

theorem ESubSurj
$\forall\, W : \mathbb{N} \to \mathbb{N};\ c : \mathbb{N} \mid (W, c) \in \mathit{Shift} \bullet \mathit{comp}(\mathit{funE}, W) \in \mathit{Surj}$

proof
invoke Surj;
prove;
instantiate $y == (x + c) * (x + c) + x - c$;
use $\mathit{LeSq}[n := x + c]$;
use $\mathit{ShiftVal}[W := W,\ c := c,\ y := (x + c) * (x + c) + x - c]$;
use $\mathit{EAt}[y := (x + c) * (x + c) + x,\ m := x + c,\ k := x]$;
prove;
■

theorem ESubCore
$\forall\, y, c, x : \mathbb{N} \mid \mathit{funE}\,(y + c) = x \wedge c \leq x \bullet \mathit{funE}\, y = x - c$

proof
prove;
use $\mathit{EDecomp}[y := y + c]$;
use $\mathit{EBound}[y := y + c]$;
use $\mathit{EAt}[y := y,\ m := \mathit{funRt}\,(y + c),\ k := x - c]$;
prove;
■

theorem ESubVal
$\forall\, W : \mathbb{N} \to \mathbb{N};\ c, x : \mathbb{N} \mid (W, c) \in \mathit{Shift} \wedge c \leq x \bullet \mathit{funE}\,(\mathit{opInv}\,(\mathit{comp}(\mathit{funE}, W))\,x) = x - c$

proof
prove;
use $\mathit{ESubSurj}[W := W,\ c := c]$;
use $\mathit{InvRight}[F := \mathit{comp}(\mathit{funE},\ W),\ x := x]$;
use $\mathit{UFNonneg}[f := \mathit{opInv}\,(\mathit{comp}(\mathit{funE},\ W)),\ x := x]$;
use $\mathit{ShiftVal}[W := W,\ c := c,\ y := \mathit{opInv}\,(\mathit{comp}(\mathit{funE},\ W))\,x]$;
use $\mathit{ESubCore}[y := \mathit{opInv}\,(\mathit{comp}(\mathit{funE},\ W))\,x,\ c := c,\ x := x]$;
with disabled (*InvRight*) *prove*;
■

9.13 $\langle E, F^{+b}, FG, F^{-1}\rangle$

$E(E^{-1}(2m) + x + c + 1) = x + c$ and $E(E^{-1}(2m + 1) + x + c + 2) = x + c$.

theorem EShiftE
$\forall\, m, x, c : \mathbb{N} \mid x + c \leq 2 * m + 2 \bullet \mathit{funE}\,(\mathit{opInv}\,\mathit{funE}\,(2 * m) + x + c + 1) = x + c$

proof
prove;
use $\mathit{EAt}[y := m * m + 2 * m + x + c + 1,\ m := m + 1,\ k := x + c]$;
prove;
■

theorem EShiftO
$\forall\, m, x, c : \mathbb{N} \mid x + c \leq 2 * m + 4 \bullet \mathit{funE}\,(\mathit{opInv}\,\mathit{funE}\,(1 + 2 * m) + x + c + 2) = x + c$

proof
prove;
use $\mathit{EAt}[y := (m + 1) * (m + 1) + 2 * m + 1 + x + c + 2,\ m := m + 2,\ k := x + c]$;
prove;
■

For $b \geq 2$, $W_1 = E\,(E^{-1}\,T)^{+b}$ and $W_2 = E\,(E^{-1}\,W_1\,T)^{+b}$ $(T = I^{+b})$ are the shifts $x + b - 1$ and $x + b - 2$, in an order given by the parity of b.

theorem W1EvenVal
$\forall\, p, x : \mathbb{N} \bullet comp(funE, opPlusIb\,(2{+}2{*}p)\,(comp(opInv\,funE, opPlusIb\,(2{+}2{*}p)\,funI)))\,x = x{+}2{*}p{+}1$

proof
prove;
use $EShiftE[m := x + p + 1,\ x := x,\ c := 2 * p + 1]$;
prove;
■

theorem W2EvenVal
$\forall\, p, x : \mathbb{N} \bullet comp(funE, opPlusIb\,(2 + 2 * p)\,(comp(comp(opInv\,funE, comp(funE,$
$opPlusIb\,(2 + 2 * p)\,(comp(opInv\,funE, opPlusIb\,(2 + 2 * p)\,funI)))),$
$opPlusIb\,(2 + 2 * p)\,funI)))\,x = x + 2 * p$

proof
prove;
use $W1EvenVal[p := p,\ x := 2 + 2 * p + 2 * x]$;
use $EShiftO[m := x + 2 * p + 1,\ x := x,\ c := 2 * p]$;
prove;
■

theorem W1OddVal
$\forall\, p, x : \mathbb{N} \bullet comp(funE, opPlusIb\,(3{+}2{*}p)\,(comp(opInv\,funE, opPlusIb\,(3{+}2{*}p)\,funI)))\,x = x{+}2{*}p{+}1$

proof
prove;
use $EShiftO[m := x + p + 1,\ x := x,\ c := 2 * p + 1]$;
prove;
■

theorem W2OddVal
$\forall\, p, x : \mathbb{N} \bullet comp(funE, opPlusIb\,(3 + 2 * p)\,(comp(comp(opInv\,funE, comp(funE,$
$opPlusIb\,(3 + 2 * p)\,(comp(opInv\,funE, opPlusIb\,(3 + 2 * p)\,funI)))),$
$opPlusIb\,(3 + 2 * p)\,funI)))\,x = x + 2 * p + 2$

proof
prove;
use $W1OddVal[p := p,\ x := 3 + 2 * p + 2 * x]$;
use $EShiftE[m := x + 2 * p + 2,\ x := x,\ c := 2 * p + 2]$;
prove;
■

theorem CbEIn
$\forall\, b : \mathbb{N} \bullet funI \in clos\,\{funE\}\,\{opPlusIb\ b, opInv\}\,\{comp\} \wedge comp(funE, opPlusIb\ b\,(comp(opInv\,funE,$
$opPlusIb\ b\,funI))) \in clos\,\{funE\}\,\{opPlusIb\ b, opInv\}\,\{comp\} \wedge comp(funE,$
$opPlusIb\ b\,(comp(comp(opInv\,funE, comp(funE, opPlusIb\ b\,(comp(opInv\,funE, opPlusIb\ b\,funI))))),$
$opPlusIb\ b\,funI))) \in clos\,\{funE\}\,\{opPlusIb\ b, opInv\}\,\{comp\}$

proof
prove;
use $CbUn[X := funE,\ b := b,\ F := funE]$;
use $CbComp[X := funE,\ b := b,\ F := funE,\ G := opInv\,funE]$;
use $EEinv$;
prove;
■

$S = E\,(E\ W_2)^{-1}\,W_1$ (or with W_1 and W_2 exchanged).

theorem SFromShifts
$\forall\, G, H : \mathbb{N} \to \mathbb{N};\ d : \mathbb{N} \mid (G, d+1) \in \mathit{Shift} \wedge (H, d) \in \mathit{Shift} \bullet \mathit{comp}(\mathit{comp}(\mathit{funE}, \mathit{opInv}\,(\mathit{comp}(\mathit{funE}, H))), G) = \mathit{funS}$

proof
apply UFExt to predicate $\mathit{comp}(\mathit{comp}(\mathit{funE},\ \mathit{opInv}\,(\mathit{comp}(\mathit{funE},\ H))),\ G)\ =\ \mathit{funS}$;
prove;
use $\mathit{ShiftVal}[W\ :=\ G,\ c\ :=\ d\ +\ 1,\ y\ :=\ x]$;
use $\mathit{ESubVal}[W\ :=\ H,\ c\ :=\ d,\ x\ :=\ 1\ +\ x\ +\ d]$;
prove;
■

theorem SInCbFromShifts
$\forall\, X : \mathbb{N} \to \mathbb{N};\ b, d : \mathbb{N};\ G, H : \mathbb{N} \to \mathbb{N} \mid \mathit{funE} \in \mathit{clos}\,\{X\}\,\{\mathit{opPlusIb}\ b, \mathit{opInv}\}\,\{\mathit{comp}\} \wedge G \in \mathit{clos}\,\{X\}\,\{\mathit{opPlusIb}\ b, \mathit{opInv}\}\,\{\mathit{comp}\} \wedge H \in \mathit{clos}\,\{X\}\,\{\mathit{opPlusIb}\ b, \mathit{opInv}\}\,\{\mathit{comp}\} \wedge (G, d+1) \in \mathit{Shift} \wedge (H, d) \in \mathit{Shift} \bullet \mathit{funS} \in \mathit{clos}\,\{X\}\,\{\mathit{opPlusIb}\ b, \mathit{opInv}\}\,\{\mathit{comp}\}$

proof
use $\mathit{SFromShifts}[G\ :=\ G,\ H\ :=\ H,\ d\ :=\ d]$;
prove;
equality substitute funS;
use $\mathit{CbComp}[X\ :=\ X,\ b\ :=\ b,\ F\ :=\ \mathit{funE},\ G\ :=\ H]$;
use $\mathit{CbUn}[X\ :=\ X,\ b\ :=\ b,\ F\ :=\ \mathit{comp}(\mathit{funE},\ H)]$;
use $\mathit{CbComp}[X\ :=\ X,\ b\ :=\ b,\ F\ :=\ \mathit{funE},\ G\ :=\ \mathit{opInv}\,(\mathit{comp}(\mathit{funE},\ H))]$;
use $\mathit{CbComp}[X\ :=\ X,\ b\ :=\ b,\ F\ :=\ \mathit{comp}(\mathit{funE},\ \mathit{opInv}\,(\mathit{comp}(\mathit{funE},\ H))),\ G\ :=\ G]$;
prove;
■

theorem W12Shifts
$\forall\, p : \mathbb{N} \bullet (\mathit{comp}(\mathit{funE}, \mathit{opPlusIb}\,(2 + 2 * p)\,(\mathit{comp}(\mathit{opInv}\,\mathit{funE}, \mathit{opPlusIb}\,(2 + 2 * p)\,\mathit{funI}))),$
$2 * p + 1) \in \mathit{Shift} \wedge (\mathit{comp}(\mathit{funE}, \mathit{opPlusIb}\,(2 + 2 * p)\,(\mathit{comp}(\mathit{comp}(\mathit{opInv}\,\mathit{funE}, \mathit{comp}(\mathit{funE},$
$\mathit{opPlusIb}\,(2 + 2 * p)\,(\mathit{comp}(\mathit{opInv}\,\mathit{funE}, \mathit{opPlusIb}\,(2 + 2 * p)\,\mathit{funI})))), \mathit{opPlusIb}\,(2 + 2 * p)\,\mathit{funI}))),$
$2 * p) \in \mathit{Shift} \wedge (\mathit{comp}(\mathit{funE}, \mathit{opPlusIb}\,(3 + 2 * p)\,(\mathit{comp}(\mathit{comp}(\mathit{opInv}\,\mathit{funE}, \mathit{comp}(\mathit{funE},$
$\mathit{opPlusIb}\,(3 + 2 * p)\,(\mathit{comp}(\mathit{opInv}\,\mathit{funE}, \mathit{opPlusIb}\,(3 + 2 * p)\,\mathit{funI})))), \mathit{opPlusIb}\,(3 + 2 * p)\,\mathit{funI}))),$
$2 * p + 2) \in \mathit{Shift} \wedge (\mathit{comp}(\mathit{funE}, \mathit{opPlusIb}\,(3 + 2 * p)\,(\mathit{comp}(\mathit{opInv}\,\mathit{funE},$
$\mathit{opPlusIb}\,(3 + 2 * p)\,\mathit{funI}))), 2 * p + 1) \in \mathit{Shift}$

proof
invoke Shift;
prove;
use $W1\mathit{EvenVal}[p\ :=\ p,\ x\ :=\ y]$;
use $W2\mathit{EvenVal}[p\ :=\ p,\ x\ :=\ y__0]$;
use $W2\mathit{OddVal}[p\ :=\ p,\ x\ :=\ y__1]$;
use $W1\mathit{OddVal}[p\ :=\ p,\ x\ :=\ y__2]$;
prove;
■

theorem CbESEven
$\forall\, p : \mathbb{N} \bullet \mathit{funS} \in \mathit{clos}\,\{\mathit{funE}\}\,\{\mathit{opPlusIb}\,(2 + 2 * p), \mathit{opInv}\}\,\{\mathit{comp}\}$

proof
prove;
use $\mathit{CbEIn}[b\ :=\ 2\ +\ 2\ *\ p]$;
use $W12\mathit{Shifts}[p\ :=\ p]$;
use $\mathit{SInCbFromShifts}[X\ :=\ \mathit{funE},\ b\ :=\ 2\ +\ 2\ *\ p,\ d\ :=\ 2\ *\ p,\ G\ :=\ \mathit{comp}(\mathit{funE},\ \mathit{opPlusIb}\,(2\ +\ 2\ *\ p)\,(\mathit{comp}(\mathit{opInv}\,\mathit{funE},\ \mathit{opPlusIb}\,(2\ +\ 2\ *\ p)\,\mathit{funI}))),\ H\ :=\ \mathit{comp}(\mathit{funE},\ \mathit{opPlusIb}\,(2\ +\ 2\ *\ p)\,(\mathit{comp}(\mathit{comp}(\mathit{opInv}\,\mathit{funE},\ \mathit{comp}(\mathit{funE},\ \mathit{opPlusIb}\,(2\ +\ 2\ *\ p)\,(\mathit{comp}(\mathit{opInv}\,\mathit{funE},$
$\mathit{opPlusIb}\,(2\ +\ 2\ *\ p)\,\mathit{funI})))),\ \mathit{opPlusIb}\,(2\ +\ 2\ *\ p)\,\mathit{funI})))]$;
prove;
■

theorem CbESOdd
$\forall\, p : \mathbb{N} \bullet \mathit{funS} \in \mathit{clos}\,\{\mathit{funE}\}\,\{\mathit{opPlusIb}\,(3 + 2 * p), \mathit{opInv}\}\,\{\mathit{comp}\}$

proof
prove;
use $W12\mathit{Shifts}[p := p]$;
use $\mathit{SInCbFromShifts}[X := \mathit{funE},\ b := 3 + 2 * p,\ d := 2 * p + 1,\ G := \mathit{comp}(\mathit{funE},\ \mathit{opPlusIb}\,(3 + 2 * p)\,(\mathit{comp}(\mathit{comp}(\mathit{opInv}\,\mathit{funE},\ \mathit{comp}(\mathit{funE},\ \mathit{opPlusIb}\,(3 + 2 * p)\,(\mathit{comp}(\mathit{opInv}\,\mathit{funE},\ \mathit{opPlusIb}\,(3 + 2 * p)\,\mathit{funI})))),\ \mathit{opPlusIb}\,(3 + 2 * p)\,\mathit{funI}))),\ H := \mathit{comp}(\mathit{funE},\ \mathit{opPlusIb}\,(3 + 2 * p)\,(\mathit{comp}(\mathit{opInv}\,\mathit{funE},\ \mathit{opPlusIb}\,(3 + 2 * p)\,\mathit{funI})))]$;
prove;
use $\mathit{CbEIn}[b := 3 + 2 * p]$;
with disabled (*ClosBinop*, *ClosUnop*, *ClosInit*) *prove*;
■

For $b = 1$: $\hat{P} = E\,(E^{-1}\,T)^{+1}$ ($\hat{P}(y+1) = y$, $\hat{P}(0) = 2$) and $S = E\,\hat{P}^{-1}(E^{-1}\,\hat{P}\,T)^{+1}$.

theorem PE1Suc
$\forall\, y : \mathbb{N} \bullet \mathit{comp}(\mathit{funE}, \mathit{opPlusIb}\,1\,(\mathit{comp}(\mathit{opInv}\,\mathit{funE}, \mathit{opPlusIb}\,1\,\mathit{funI})))\,(1 + y) = y$

proof
prove;
use $\mathit{EShiftO}[m := 1 + y,\ x := y,\ c := 0]$;
prove;
■

theorem PE1Zero
$\mathit{comp}(\mathit{funE}, \mathit{opPlusIb}\,1\,(\mathit{comp}(\mathit{opInv}\,\mathit{funE}, \mathit{opPlusIb}\,1\,\mathit{funI})))\,0 = 2$

proof
prove;
use $\mathit{EinvOdd}[m := 0]$;
use $\mathit{EAt}[y := 3,\ m := 1,\ k := 2]$;
prove;
■

theorem PE1Core
$\forall\, v, y : \mathbb{N} \mid \mathit{comp}(\mathit{funE}, \mathit{opPlusIb}\,1\,(\mathit{comp}(\mathit{opInv}\,\mathit{funE}, \mathit{opPlusIb}\,1\,\mathit{funI})))\,y = v \wedge \neg\, v = 2 \bullet y = 1 + v$

proof
prove;
use $\mathit{PE1Zero}$;
use $\mathit{PE1Suc}[y := y - 1]$;
split $y = 0$;
cases;
prove;
next;
prove;
next;
■

theorem PE1Surj
$\mathit{comp}(\mathit{funE}, \mathit{opPlusIb}\,1\,(\mathit{comp}(\mathit{opInv}\,\mathit{funE}, \mathit{opPlusIb}\,1\,\mathit{funI}))) \in \mathit{Surj}$

proof
invoke *Surj*;
with disabled (*EinvOdd*, *EinvD*) *prove*;
use $\mathit{PE1Suc}[y := x]$;
instantiate $y == 1 + x$;
with disabled (*EinvOdd*, *EinvD*) *prove*;
■

theorem PE1Inv
$\forall\, v : \mathbb{N} \mid \neg\, v = 2 \bullet \mathit{opInv}\,(\mathit{comp}(\mathit{funE}, \mathit{opPlusIb}\,1\,(\mathit{comp}(\mathit{opInv}\,\mathit{funE}, \mathit{opPlusIb}\,1\,\mathit{funI})))\,v = 1 + v$

proof
prove;
use $\mathit{PE1Surj}$;
use $\mathit{InvRight}[F := \mathit{comp}(\mathit{funE},\ \mathit{opPlusIb}\,1\,(\mathit{comp}(\mathit{opInv}\,\mathit{funE},\ \mathit{opPlusIb}\,1\,\mathit{funI}))),\ x := v]$;
use $\mathit{UFNonneg}[f := \mathit{opInv}\,(\mathit{comp}(\mathit{funE},\ \mathit{opPlusIb}\,1\,(\mathit{comp}(\mathit{opInv}\,\mathit{funE},\ \mathit{opPlusIb}\,1\,\mathit{funI})))),\ x := v]$;
use $\mathit{PE1Core}[v := v,\ y := \mathit{opInv}\,(\mathit{comp}(\mathit{funE},\ \mathit{opPlusIb}\,1\,(\mathit{comp}(\mathit{opInv}\,\mathit{funE},$
$\mathit{opPlusIb}\,1\,\mathit{funI})))\,v]$;
with disabled (*InvRight*) *prove*;
■

theorem SE1Val
$\mathit{comp}(\mathit{comp}(\mathit{funE}, \mathit{opInv}\,(\mathit{comp}(\mathit{funE}, \mathit{opPlusIb}\,1\,(\mathit{comp}(\mathit{opInv}\,\mathit{funE}, \mathit{opPlusIb}\,1\,\mathit{funI}))))),$
$\mathit{opPlusIb}\,1\,(\mathit{comp}(\mathit{comp}(\mathit{opInv}\,\mathit{funE}, \mathit{comp}(\mathit{funE}, \mathit{opPlusIb}\,1\,(\mathit{comp}(\mathit{opInv}\,\mathit{funE}, \mathit{opPlusIb}\,1\,\mathit{funI})))),$
$\mathit{opPlusIb}\,1\,\mathit{funI}))) = \mathit{funS}$

proof
apply UFExt to predicate $\mathit{comp}(\mathit{comp}(\mathit{funE},\ \mathit{opInv}\,(\mathit{comp}(\mathit{funE},\ \mathit{opPlusIb}\,1\,(\mathit{comp}(\mathit{opInv}\,\mathit{funE},$
$\mathit{opPlusIb}\,1\,\mathit{funI}))))),\ \mathit{opPlusIb}\,1\,(\mathit{comp}(\mathit{comp}(\mathit{opInv}\,\mathit{funE},\ \mathit{comp}(\mathit{funE},$
$\mathit{opPlusIb}\,1\,(\mathit{comp}(\mathit{opInv}\,\mathit{funE},\ \mathit{opPlusIb}\,1\,\mathit{funI})))),\ \mathit{opPlusIb}\,1\,\mathit{funI})))\ =\ \mathit{funS}$;
prove;
use $\mathit{PE1Suc}[y := 2 * x]$;
use $\mathit{MulNonneg}[a := x,\ b := x]$;
use $\mathit{PE1Inv}[v := 1 + x * x + 3 * x]$;
use $\mathit{EAt}[y := 2 + x * x + 3 * x,\ m := x + 1,\ k := x + 1]$;
prove;
split $x = 0$;
cases;
prove;
next;
prove;
next;
■

theorem SE1In
$\mathit{comp}(\mathit{comp}(\mathit{funE}, \mathit{opInv}\,(\mathit{comp}(\mathit{funE}, \mathit{opPlusIb}\,1\,(\mathit{comp}(\mathit{opInv}\,\mathit{funE}, \mathit{opPlusIb}\,1\,\mathit{funI}))))),$
$\mathit{opPlusIb}\,1\,(\mathit{comp}(\mathit{comp}(\mathit{opInv}\,\mathit{funE}, \mathit{comp}(\mathit{funE}, \mathit{opPlusIb}\,1\,(\mathit{comp}(\mathit{opInv}\,\mathit{funE}, \mathit{opPlusIb}\,1\,\mathit{funI})))),$
$\mathit{opPlusIb}\,1\,\mathit{funI}))) \in \mathit{clos}\,\{\mathit{funE}\}\,\{\mathit{opPlusIb}\,1, \mathit{opInv}\}\,\{\mathit{comp}\}$

proof
use $\mathit{CbEIn}[b := 1]$;
prove;
■

theorem CbESOne
$\mathit{funS} \in \mathit{clos}\,\{\mathit{funE}\}\,\{\mathit{opPlusIb}\,1, \mathit{opInv}\}\,\{\mathit{comp}\}$

proof
use $\mathit{SE1In}$;
use $\mathit{SE1Val}$;
prove;
■

From S: the predecessor $E\,(ES)^{-1}$, then F^+, $\bar{0} = ESE^{-1}I^+$ and $\bar{1} = S\bar{0}$.

theorem SShift
$(\mathit{funS}, 1) \in \mathit{Shift}$

proof
invoke Shift;
prove;
■

theorem PEPreds
$comp(funE, opInv\,(comp(funE, funS))) \in Preds$

proof
invoke Preds;
prove;
use SShift;
use ESubVal[$W := funS,\ c := 1,\ x := 1 + x$];
prove;
■

theorem ZeroExprE2
$comp(comp(comp(funE, funS), opInv\,funE), opPlusI\,funI) = cst\,0$

proof
apply UFExt to predicate $comp(comp(comp(funE,\ funS),\ opInv\,funE),\ opPlusI\,funI) = cst\,0$;
prove;
use EAt[$y := 1 + x * x + 2 * x,\ m := x + 1,\ k := 0$];
prove;
■

theorem CbEPI
$\forall\, b : \mathbb{N} \mid funS \in clos\,\{funE\}\,\{opPlusIb\ b, opInv\}\,\{comp\} \bullet comp(funE, opInv\,(comp(funE,$
$funS))) \in clos\,\{funE\}\,\{opPlusIb\ b, opInv\}\,\{comp\} \land opPlusI\,funI \in clos\,\{funE\}\,\{opPlusIb\ b,$
$opInv\}\,\{comp\}$

proof
prove;
use CbEIn[$b := b$];
use PEPreds;
use CbPlusI[$X := funE,\ b := b,\ P := comp(funE,\ opInv\,(comp(funE,\ funS))),\ F := funI$];
prove;
■

theorem CbEZeroIn
$\forall\, b : \mathbb{N} \mid funS \in clos\,\{funE\}\,\{opPlusIb\ b, opInv\}\,\{comp\} \bullet comp(comp(comp(funE, funS), opInv\,funE),$
$opPlusI\,funI) \in clos\,\{funE\}\,\{opPlusIb\ b, opInv\}\,\{comp\}$

proof
prove;
use CbEPI[$b := b$];
prove;
■

theorem CbEOne
$\forall\, b : \mathbb{N} \mid funS \in clos\,\{funE\}\,\{opPlusIb\ b, opInv\}\,\{comp\} \bullet cst\,0 \in clos\,\{funE\}\,\{opPlusIb\ b,$
$opInv\}\,\{comp\} \land cst\,1 \in clos\,\{funE\}\,\{opPlusIb\ b, opInv\}\,\{comp\}$

proof
prove;
use CbEZeroIn[$b := b$];
use ZeroExprE2;
prove;
use OneExpr;
prove;
■

theorem CbEFromS
$\forall\, b : \mathbb{N} \mid funS \in clos\,\{funE\}\,\{opPlusIb\ b, opInv\}\,\{comp\} \bullet clos\,\{funE\}\,\{opPlusIb\ b,$
$opInv\}\,\{comp\} = ClosJR$

proof
prove;
use $CbEOne[b := b]$;
use $CbEPI[b := b]$;
use *PEPreds*;
use $CbSub[X := funE,\ b := b,\ P := comp(funE,\ opInv\,(comp(funE,\ funS)))]$;
use $GRbSubGE[X := funE,\ b := b]$;
use *GREqJRE*;
use $ClosInPow[I := \{\ funE\ \},\ U := \{\ opPlusIb\ b,\ opInv\ \},\ B := \{\ comp\ \}]$;
use *CGEInPower*;
use $SubAntisym[\mathbb{N} \to \mathbb{N}][A := clos\,\{\ funE\ \}\,\{\ opPlusIb\ b,\ opInv\ \}\,\{\ comp\ \},\ B := CGE]$;
with enabled $(CGEDef)$ *prove*;
■

9.14 $\langle K, F^{+b}, FG, F^{-1} \rangle$

theorem rule CbKZ
$\forall\, b : \mathbb{N} \bullet funZ \in clos\,\{funK\}\,\{opPlusIb\ b, opInv\}\,\{comp\}$

proof
prove;
use $CbUn[X := funK,\ b := b,\ F := funK]$;
use *KinvZ*;
prove;
■

theorem KZI
$comp(funK, funZ) = funI$

proof
apply UFExt to predicate $comp(funK,\ funZ) = funI$;
prove;
use $KZ[x := x]$;
prove;
■

theorem CbKIExpr
$\forall\, b : \mathbb{N} \bullet comp(funK, funZ) \in clos\,\{funK\}\,\{opPlusIb\ b, opInv\}\,\{comp\}$

proof
prove;
■

theorem rule CbKI
$\forall\, b : \mathbb{N} \bullet funI \in clos\,\{funK\}\,\{opPlusIb\ b, opInv\}\,\{comp\}$

proof
prove;
use $CbKIExpr[b := b]$;
use *KZI*;
prove;
■

theorem ALeCore
$\forall\, a, n, q : \mathbb{Z} \mid n \leq q \wedge 2 * a = n + q \bullet n \leq a$

proof
prove;
■

theorem ALe
$\forall\, n : \mathbb{N} \bullet n \leq \mathit{funA}\, n$

proof
prove;
use $\mathit{TwoA}[x := n]$;
use $\mathit{LeSq}[n := n]$;
use $\mathit{ALeCore}[a := \mathit{funA}\, n,\ n := n,\ q := n * n]$;
prove;
■

theorem ShiftComp
$\forall\, F, G : \mathbb{N} \to \mathbb{N};\ a, c : \mathbb{N} \mid (F, a) \in \mathit{Shift} \wedge (G, c) \in \mathit{Shift} \bullet (\mathit{comp}(F, G), a + c) \in \mathit{Shift}$

proof
prove;
invoke *Shift*;
prove;
instantiate $y_0 == G\, y,\ y_1 == y$;
prove;
■

Subtraction by inversion with K.

theorem KSubSurj
$\forall\, W : \mathbb{N} \to \mathbb{N};\ c : \mathbb{N} \mid (W, c) \in \mathit{Shift} \bullet \mathit{comp}(\mathit{funK}, W) \in \mathit{Surj}$

proof
invoke *Surj*;
prove;
instantiate $y == \mathit{funA}\,(x + c) + x - c$;
use $\mathit{ALe}[n := x + c]$;
use $\mathit{ShiftVal}[W := W,\ c := c,\ y := \mathit{funA}\,(x + c) + x - c]$;
use $\mathit{KAt}[m := \mathit{funA}\,(x + c) + x,\ t := x + c,\ k := x]$;
prove;
■

theorem KSubCore
$\forall\, y, c, x : \mathbb{N} \mid \mathit{funK}\,(y + c) = x \wedge c \leq x \bullet \mathit{funK}\, y = x - c$

proof
prove;
use $\mathit{Decomp}[x := y + c]$;
use $\mathit{KAt}[m := y,\ t := \mathit{funV}\,(y + c),\ k := x - c]$;
prove;
■

theorem KSubVal
$\forall\, W : \mathbb{N} \to \mathbb{N};\ c, x : \mathbb{N} \mid (W, c) \in \mathit{Shift} \wedge c \leq x \bullet \mathit{funK}\,(\mathit{opInv}\,(\mathit{comp}(\mathit{funK}, W))\, x) = x - c$

proof
prove;
use *KSubSurj*[$W := W,\ c := c$];
use *InvRight*[$F := comp(funK,\ W),\ x := x$];
use *UFNonneg*[$f := opInv\,(comp(funK,\ W)),\ x := x$];
use *ShiftVal*[$W := W,\ c := c,\ y := opInv\,(comp(funK,\ W))\,x$];
use *KSubCore*[$y := opInv\,(comp(funK,\ W))\,x,\ c := c,\ x := x$];
with disabled (*InvRight*) *prove*;
■

$U = K\,(ZT)^{+b}$ is the shift $b - 1$.

theorem UKShift
$\forall\, c : \mathbb{N} \bullet (comp(funK, opPlusIb\,(1 + c)\,(comp(funZ, opPlusIb\,(1 + c)\,funI))), c) \in Shift$

proof
invoke *Shift*;
prove;
use *ASuc*[$x := 1 + c + 2 * y$];
use *UFNonneg*[$f := funA,\ x := 1 + c + 2 * y$];
use *KAt*[$m := funA\,(2 + c + 2 * y) + y + c,\ t := 2 + c + 2 * y,\ k := y + c$];
prove;
■

theorem rule CbKUIn
$\forall\, b : \mathbb{N} \bullet comp(funK, opPlusIb\ b\,(comp(funZ, opPlusIb\ b\ funI))) \in clos\,\{funK\}\,\{opPlusIb\ b, opInv\}\,\{comp\}$

proof
prove;
■

For $b = d + 2$: the constant $C = K\,(ZP_1)^{+b}\,U^3 = 2d$ with $P_1 = K\,(KU)^{-1}$, and $S = K\,(K\ C^{+b})^{-1}\,U^3$.

theorem CKVal
$\forall\, d, x : \mathbb{N} \bullet comp(comp(funK, opPlusIb\,(2 + d)\,(comp(funZ, comp(funK, opInv\,(comp(funK, comp(funK, opPlusIb\,(2 + d)\,(comp(funZ, opPlusIb\,(2 + d)\,funI)))))))), comp(comp(comp(funK, opPlusIb\,(2 + d)\,(comp(funZ, opPlusIb\,(2 + d)\,funI))), comp(funK, opPlusIb\,(2 + d)\,(comp(funZ, opPlusIb\,(2 + d)\,funI)))), comp(funK, opPlusIb\,(2 + d)\,(comp(funZ, opPlusIb\,(2 + d)\,funI))))) x = 2 * d$

proof
prove;
use *UKShift*[$c := 1 + d$];
use *ShiftVal*[$W := comp(funK,\ opPlusIb\,(2 + d)\,(comp(funZ,\ opPlusIb\,(2 + d)\,funI))),\ c := 1 + d,\ y := x$];
use *ShiftVal*[$W := comp(funK,\ opPlusIb\,(2 + d)\,(comp(funZ,\ opPlusIb\,(2 + d)\,funI))),\ c := 1 + d,\ y := 1 + d + x$];
use *ShiftVal*[$W := comp(funK,\ opPlusIb\,(2 + d)\,(comp(funZ,\ opPlusIb\,(2 + d)\,funI))),\ c := 1 + d,\ y := 2 + 2 * d + x$];
use *KSubVal*[$W := comp(funK,\ opPlusIb\,(2 + d)\,(comp(funZ,\ opPlusIb\,(2 + d)\,funI))),\ c := 1 + d,\ x := 3 + 3 * d + x$];
use *ASuc*[$x := 2 + 2 * d + x$];
use *ASuc*[$x := 3 + 2 * d + x$];
use *UFNonneg*[$f := funA,\ x := 2 + 2 * d + x$];
use *KAt*[$m := funA\,(4 + 2 * d + x) + 2 * d,\ t := 4 + 2 * d + x,\ k := 2 * d$];
prove;
■

theorem KShifts

$\forall\, d : \mathbb{N} \bullet (comp(comp(comp(funK, opPlusIb\,(2+d)\,(comp(funZ, opPlusIb\,(2+d)\,funI))), comp(funK, opPlusIb\,(2+d)\,(comp(funZ, opPlusIb\,(2+d)\,funI)))), comp(funK, opPlusIb\,(2+d)\,(comp(funZ, opPlusIb\,(2+d)\,funI)))), 3*d+3) \in Shift \wedge (opPlusIb\,(2+d)\,(comp(comp(funK, opPlusIb\,(2+d)\,(comp(funZ, comp(funK, opInv\,(comp(funK, comp(funK, opPlusIb\,(2+d)\,(comp(funZ, opPlusIb\,(2+d)\,funI)))))))), comp(comp(comp(funK, opPlusIb\,(2+d)\,(comp(funZ, opPlusIb\,(2+d)\,funI))), comp(funK, opPlusIb\,(2+d)\,(comp(funZ, opPlusIb\,(2+d)\,funI)))), comp(funK, opPlusIb\,(2+d)\,(comp(funZ, opPlusIb\,(2+d)\,funI))))), 3*d+2) \in Shift$

proof

prove;
use $UKShift[c := 1 + d]$;
use $ShiftComp[F := comp(funK, opPlusIb\,(2 + d)\,(comp(funZ, opPlusIb\,(2 + d)\,funI))), G := comp(funK, opPlusIb\,(2 + d)\,(comp(funZ, opPlusIb\,(2 + d)\,funI))), a := 1 + d, c := 1 + d]$;
use $ShiftComp[F := comp(comp(funK, opPlusIb\,(2 + d)\,(comp(funZ, opPlusIb\,(2 + d)\,funI))), comp(funK, opPlusIb\,(2 + d)\,(comp(funZ, opPlusIb\,(2 + d)\,funI)))), G := comp(funK, opPlusIb\,(2 + d)\,(comp(funZ, opPlusIb\,(2 + d)\,funI))), a := 2 + 2 * d, c := 1 + d]$;
prove;
invoke Shift;
prove;
use $CKVal[d := d, x := y]$;
prove;
■

theorem SFromShiftsK

$\forall\, G, H : \mathbb{N} \to \mathbb{N};\ d : \mathbb{N} \mid (G, d+1) \in Shift \wedge (H, d) \in Shift \bullet comp(comp(funK, opInv\,(comp(funK, H))), G) = funS$

proof

apply UFExt to predicate $comp(comp(funK, opInv\,(comp(funK, H))), G) = funS$;
prove;
use $ShiftVal[W := G, c := d + 1, y := x]$;
use $KSubVal[W := H, c := d, x := 1 + x + d]$;
prove;
■

theorem rule CbKU3In

$\forall\, b : \mathbb{N} \bullet comp(comp(comp(funK, opPlusIb\ b\,(comp(funZ, opPlusIb\ b\,funI))), comp(funK, opPlusIb\ b\,(comp(funZ, opPlusIb\ b\,funI)))), comp(funK, opPlusIb\ b\,(comp(funZ, opPlusIb\ b\,funI)))) \in clos\,\{funK\}\,\{opPlusIb\ b, opInv\}\,\{comp\}$

proof

prove;
■

theorem rule CbKP1In

$\forall\, b : \mathbb{N} \bullet comp(funK, opInv\,(comp(funK, comp(funK, opPlusIb\ b\,(comp(funZ, opPlusIb\ b\,funI)))))) \in clos\,\{funK\}\,\{opPlusIb\ b, opInv\}\,\{comp\}$

proof

prove;
■

theorem rule CbKCIn

$\forall\, b : \mathbb{N} \bullet comp(comp(funK, opPlusIb\ b\,(comp(funZ, comp(funK, opInv\,(comp(funK, comp(funK, opPlusIb\ b\,(comp(funZ, opPlusIb\ b\,funI)))))))), comp(comp(comp(funK, opPlusIb\ b\,(comp(funZ, opPlusIb\ b\,funI))), comp(funK, opPlusIb\ b\,(comp(funZ, opPlusIb\ b\,funI)))), comp(funK, opPlusIb\ b\,(comp(funZ, opPlusIb\ b\,funI))))) \in clos\,\{funK\}\,\{opPlusIb\ b, opInv\}\,\{comp\}$

proof
prove;
■

theorem CbKSTwoIn
$\forall\, d : \mathbb{N} \bullet comp(comp(funK, opInv\,(comp(funK, opPlusIb\,(2+d)\,(comp(comp(funK, opPlusIb\,(2+d)\,(comp(funZ, comp(funK, opInv\,(comp(funK, comp(funK, opPlusIb\,(2+d)\,(comp(funZ, opPlusIb\,(2+d)\,funI)))))))))), comp(comp(comp(funK, opPlusIb\,(2+d)\,(comp(funZ, opPlusIb\,(2+d)\,funI)))), comp(funK, opPlusIb\,(2+d)\,(comp(funZ, opPlusIb\,(2+d)\,funI))))))))), comp(comp(comp(funK, opPlusIb\,(2+d)\,(comp(funZ, opPlusIb\,(2+d)\,funI))), comp(funK, opPlusIb\,(2+d)\,(comp(funZ, opPlusIb\,(2+d)\,funI))))), comp(funK, opPlusIb\,(2+d)\,(comp(funZ, opPlusIb\,(2+d)\,funI))))) \in clos\,\{funK\}\,\{opPlusIb\,(2+d), opInv\}\,\{comp\}$

proof
prove;
■

theorem CbKSTwo
$\forall\, d : \mathbb{N} \bullet funS \in clos\,\{funK\}\,\{opPlusIb\,(2+d), opInv\}\,\{comp\}$

proof
prove;
use *CbKSTwoIn*[$d := d$];
use *KShifts*[$d := d$];
use *SFromShiftsK*[$G := comp(comp(comp(funK, opPlusIb\,(2+d)\,(comp(funZ, opPlusIb\,(2+d)\,funI))), comp(funK, opPlusIb\,(2+d)\,(comp(funZ, opPlusIb\,(2+d)\,funI)))), comp(funK, opPlusIb\,(2+d)\,(comp(funZ, opPlusIb\,(2+d)\,funI)))), H := opPlusIb\,(2+d)\,(comp(comp(funK, opPlusIb\,(2+d)\,(comp(funZ, comp(funK, opInv\,(comp(funK, comp(funK, opPlusIb\,(2+d)\,(comp(funZ, opPlusIb\,(2+d)\,funI)))))))))), comp(comp(comp(funK, opPlusIb\,(2+d)\,(comp(funZ, opPlusIb\,(2+d)\,funI))), comp(funK, opPlusIb\,(2+d)\,(comp(funZ, opPlusIb\,(2+d)\,funI))))), comp(funK, opPlusIb\,(2+d)\,(comp(funZ, opPlusIb\,(2+d)\,funI)))))), d := 3*d+2$];
prove;
■

For $b = 1$: $\hat{P} = K\,(Z^{+1})^{+1}$ ($\hat{P}(y+1) = y$, $\hat{P}(0) = 1$) and $S = K\,\hat{P}^{-1}(ZT)^{+1}$.

theorem PK1Suc
$\forall\, y : \mathbb{N} \bullet comp(funK, opPlusIb\,1\,(opPlusIb\,1\,funZ))\,(1+y) = y$

proof
prove;
use *ASuc*[$x := 1 + y$];
use *ASuc*[$x := 2 + y$];
use *UFNonneg*[$f := funA,\ x := 1 + y$];
use *KAt*[$m := funA\,(3 + y) + y,\ t := 3 + y,\ k := y$];
prove;
■

theorem PK1Zero
$comp(funK, opPlusIb\,1\,(opPlusIb\,1\,funZ))\,0 = 1$

proof
prove;
use *AZero*;
use *KAt*[$m := 2,\ t := 1,\ k := 1$];
prove;
■

theorem PK1Core
$\forall\, v, y : \mathbb{N} \mid comp(funK, opPlusIb\,1\,(opPlusIb\,1\,funZ))\,y = v \wedge \neg\, v = 1 \bullet y = 1 + v$

proof
prove;
use $PK1Zero$;
use $PK1Suc[y := y - 1]$;
split $y = 0$;
cases;
prove;
next;
prove;
next;
■

theorem PK1Surj
$comp(funK, opPlusIb\,1\,(opPlusIb\,1\,funZ)) \in Surj$

proof
invoke $Surj$;
prove;
use $PK1Suc[y := x]$;
instantiate $y == 1 + x$;
prove;
■

theorem PK1Inv
$\forall\, v : \mathbb{N} \mid \neg\, v = 1 \bullet opInv\,(comp(funK, opPlusIb\,1\,(opPlusIb\,1\,funZ)))\,v = 1 + v$

proof
prove;
use $PK1Surj$;
use $InvRight[F := comp(funK,\ opPlusIb\,1\,(opPlusIb\,1\,funZ)),\ x := v]$;
use $UFNonneg[f := opInv\,(comp(funK,\ opPlusIb\,1\,(opPlusIb\,1\,funZ))),\ x := v]$;
use $PK1Core[v := v,\ y := opInv\,(comp(funK,\ opPlusIb\,1\,(opPlusIb\,1\,funZ)))\,v]$;
with disabled $(InvRight)$ *prove*;
■

theorem SK1Val
$comp(comp(funK, opInv\,(comp(funK, opPlusIb\,1\,(opPlusIb\,1\,funZ)))), opPlusIb\,1\,(comp(funZ,$
$\quad opPlusIb\,1\,funI))) = funS$

proof
apply $UFExt$ *to predicate* $comp(comp(funK,\ opInv\,(comp(funK,\ opPlusIb\,1\,(opPlusIb\,1\,funZ)))),$
$\quad opPlusIb\,1\,(comp(funZ,\ opPlusIb\,1\,funI)))\ =\ funS$;
prove;
use $ASuc[x := 1 + 2 * x]$;
use $UFNonneg[f := funA,\ x := 1 + 2 * x]$;
use $ALe[n := 2 + 2 * x]$;
use $PK1Inv[v := funA\,(2 + 2 * x) + x]$;
use $KAt[m := 1 + funA\,(2 + 2 * x) + x,\ t := 2 + 2 * x,\ k := 1 + x]$;
prove;
■

theorem SK1In
$comp(comp(funK, opInv\,(comp(funK, opPlusIb\,1\,(opPlusIb\,1\,funZ)))), opPlusIb\,1\,(comp(funZ,$
$\quad opPlusIb\,1\,funI))) \in clos\,\{funK\}\,\{opPlusIb\,1, opInv\}\,\{comp\}$

proof
prove;
use $CbUn[X := funK,\ b := 1,\ F := funZ]$;
use $CbUn[X := funK,\ b := 1,\ F := opPlusIb\,1\,funZ]$;
use $CbComp[X := funK,\ b := 1,\ F := funK,\ G := opPlusIb\,1\,(opPlusIb\,1\,funZ)]$;
prove;
■

theorem CbKSOne
$funS \in clos\,\{funK\}\,\{opPlusIb\,1, opInv\}\,\{comp\}$

proof
use $SK1In$;
use $SK1Val$;
prove;
■

From S: $P = K(KS)^{-1}$, $F^+ = P^b F^{+b}$, A, $\bar{0} = KA$ and $\bar{1} = S\bar{0}$.

theorem PKPreds
$exprPK \in Preds$

proof
invoke $Preds$;
prove;
use $PKSuc[x := x]$;
prove;
■

theorem rule CbKPlusI
$\forall\, b : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N} \mid funS \in clos\,\{funK\}\,\{opPlusIb\,b, opInv\}\,\{comp\} \wedge F \in clos\,\{funK\}\,\{opPlusIb\,b, opInv\}\,\{comp\} \bullet opPlusI\ F \in clos\,\{funK\}\,\{opPlusIb\,b, opInv\}\,\{comp\}$

proof
prove;
use $PKPreds$;
use $CbComp[X := funK,\ b := b,\ F := funK,\ G := funS]$;
use $CbUn[X := funK,\ b := b,\ F := comp(funK,\ funS)]$;
use $CbComp[X := funK,\ b := b,\ F := funK,\ G := opInv\,(comp(funK,\ funS))]$;
invoke $exprPK$;
use $CbPlusI[X := funK,\ b := b,\ P := comp(funK,\ opInv\,(comp(funK,\ funS))),\ F := F]$;
prove;
■

theorem ZeroExprKA
$comp(funK, funA) = cst\,0$

proof
apply UFExt to predicate $comp(funK,\ funA) = cst\,0$;
prove;
use $UFNonneg[f := funA,\ x := x]$;
use $KAt[m := funA\,x,\ t := x,\ k := 0]$;
prove;
■

theorem CbKAIn
$\forall\, b : \mathbb{N} \mid funS \in clos\,\{funK\}\,\{opPlusIb\,b, opInv\}\,\{comp\} \bullet exprAK \in clos\,\{funK\}\,\{opPlusIb\,b, opInv\}\,\{comp\}$

proof
invoke exprAK;
prove;
■

theorem CbKOne
$\forall\, b : \mathbb{N} \mid funS \in clos\,\{funK\}\,\{opPlusIb\ b, opInv\}\,\{comp\} \bullet cst\,0 \in clos\,\{funK\}\,\{opPlusIb\ b, opInv\}\,\{comp\} \land cst\,1 \in clos\,\{funK\}\,\{opPlusIb\ b, opInv\}\,\{comp\}$

proof
prove;
use CbKAIn[$b := b$];
use AExprK;
prove;
use ZeroExprKA;
prove;
use OneExpr;
prove;
■

theorem CbKPIn
$\forall\, b : \mathbb{N} \mid funS \in clos\,\{funK\}\,\{opPlusIb\ b, opInv\}\,\{comp\} \bullet exprPK \in clos\,\{funK\}\,\{opPlusIb\ b, opInv\}\,\{comp\}$

proof
invoke exprPK;
prove;
■

theorem CbKFromS
$\forall\, b : \mathbb{N} \mid funS \in clos\,\{funK\}\,\{opPlusIb\ b, opInv\}\,\{comp\} \bullet clos\,\{funK\}\,\{opPlusIb\ b, opInv\}\,\{comp\} = ClosJR$

proof
prove;
use CbKOne[$b := b$];
use CbKPIn[$b := b$];
use PKPreds;
use CbSub[$X := funK,\ b := b,\ P := exprPK$];
use GRbSubGE[$X := funK,\ b := b$];
use GEK;
use GREqJR;
use GKSubGE;
use GESubGK;
use ClosInPow[$I := \{\ funK\ \},\ U := \{\ opPlusIb\ b,\ opInv\ \},\ B := \{\ comp\ \}$];
use CGEInPower;
use CGKInPower;
use SubAntisym[$\mathbb{N} \to \mathbb{N}$][$A := clos\,\{\ funK\ \}\,\{\ opPlusIb\ b,\ opInv\ \}\,\{\ comp\ \},\ B := CGE$];
with enabled (*CGKDef*) *prove*;
■

9.15 $\langle L, F^{+b}, FG, F^{-1}\rangle$

$L^{-1} = A$; $U = L\,(\ldots(A^{+b})\ldots)^{+b}$ ($2b-2$ times) is the shift $b-1$, LA^{+b} subtracts $b-2$, and $S = LA^{+b}\,U$.

theorem LSurj
$funL \in Surj$

proof
invoke Surj;
prove;
instantiate $y == funA\,x$;
use $UFNonneg[f := funA,\ x := x]$;
use $LAt[m := funA\,x,\ t := x,\ k := 0]$;
prove;
■

theorem LLowCore
$\forall\, x, y, t, k : \mathbb{N} \mid y = funA\,t + k \wedge k \leq t \wedge t - k = x \bullet funA\,x \leq y$

proof
prove;
use $AMono[m := x,\ n := t]$;
prove;
■

theorem LinvA
$opInv\,funL = funA$

proof
apply UFExt to predicate $opInv\,funL = funA$;
prove;
use LSurj;
use $UFNonneg[f := funA,\ x := x]$;
use $LAt[m := funA\,x,\ t := x,\ k := 0]$;
use $InvLeast[F := funL,\ x := x,\ y := funA\,x]$;
use $InvRight[F := funL,\ x := x]$;
use $UFNonneg[f := opInv\,funL,\ x := x]$;
use $Decomp[x := opInv\,funL\,x]$;
use $LAt[m := opInv\,funL\,x,\ t := funV\,(opInv\,funL\,x),\ k := funK\,(opInv\,funL\,x)]$;
use $LLowCore[x := x,\ y := opInv\,funL\,x,\ t := funV\,(opInv\,funL\,x),\ k := funK\,(opInv\,funL\,x)]$;
with disabled $(InvRight)$ *prove*;
split $funA\,x \leq opInv\,funL\,x$;
cases;
prove;
next;
with normalization prove;
next;
■

theorem rule CbLA
$\forall\, b : \mathbb{N} \bullet funA \in clos\,\{funL\}\,\{opPlusIb\,b, opInv\}\,\{comp\}$

proof
prove;
use $CbUn[X := funL,\ b := b,\ F := funL]$;
use LinvA;
prove;
■

$A + n(I + \bar{b})$, i.e. A^{+b} iterated n times, is in the closure.

$IterLSet == \{n : \mathbb{N} \mid \forall\, b : \mathbb{N} \bullet opPlus(funA, comp(funM\ n, opPlus(funI,$
$cst\ b))) \in clos\,\{funL\}\,\{opPlusIb\,b, opInv\}\,\{comp\}\}$

proof [IterLSet$domainCheck]
prove;
■

theorem IterLEqZero
$\forall\, b : \mathbb{N} \bullet opPlus(funA, comp(funM\ 0, opPlus(funI, cst\ b))) = funA$

proof
prove;
apply UFExt to predicate $opPlus(funA,\ comp(funM\ 0,\ opPlus(funI,\ cst\ b)))\ =\ funA$;
prove;
■

theorem IterLEqSuc
$\forall\, n, b : \mathbb{N} \bullet opPlus(funA, comp(funM\ (1 + n), opPlus(funI, cst\ b))) = opPlusIb\ b\ (opPlus(funA,$
$comp(funM\ n, opPlus(funI, cst\ b))))$

proof
prove;
apply UFExt to predicate $opPlus(funA,\ comp(funM\ (1\ +\ n),\ opPlus(funI,\ cst\ b)))\ =$
$opPlusIb\ b\ (opPlus(funA,\ comp(funM\ n,\ opPlus(funI,\ cst\ b))))$;
prove;
■

theorem rule IterLBase
$0 \in IterLSet$

proof
invoke IterLSet;
prove;
use $IterLEqZero[b\ :=\ b]$;
prove;
■

theorem rule IterLStep
$\forall\, n : IterLSet \bullet 1 + n \in IterLSet$

proof
invoke IterLSet;
prove;
instantiate $b__0\ ==\ b$;
use $IterLEqSuc[n\ :=\ n,\ b\ :=\ b]$;
use $CbUn[X\ :=\ funL,\ b\ :=\ b,\ F\ :=\ opPlus(funA,\ comp(funM\ n,\ opPlus(funI,\ cst\ b)))]$;
prove;
■

theorem IterLSub
$\mathbb{N} \subseteq IterLSet$

proof
apply natInduction to predicate $\mathbb{N}\ \subseteq\ IterLSet$;
prove;
invoke IterLSet;
prove;
■

theorem rule IterLAll
$\forall\, n, b : \mathbb{N} \bullet opPlus(funA, comp(funM\ n, opPlus(funI, cst\ b))) \in clos\ \{funL\}\ \{opPlusIb\ b, opInv\}\ \{comp\}$

proof
prove;
use IterLSub;
invoke IterLSet;
with enabled (*inPower*) *prove*;
instantiate $e == n$;
prove;
instantiate $b__1 == b$;
prove;
■

theorem AAddTwo
$\forall\, x, c : \mathbb{N} \bullet \mathit{funA}\,(x + 2 * c) = \mathit{funA}\,x + 2 * c * x + 2 * c * c + c$

proof
prove;
use $\mathit{TwoA}[x := x + 2 * c]$;
use $\mathit{TwoA}[x := x]$;
use $\mathit{Halve}[a := \mathit{funA}\,(x + 2 * c),\ b := \mathit{funA}\,x + 2 * c * x + 2 * c * c + c]$;
prove;
■

theorem ULShift
$\forall\, c : \mathbb{N} \bullet (\mathit{comp}(\mathit{funL}, \mathit{opPlus}(\mathit{funA}, \mathit{comp}(\mathit{funM}\,(2 * c), \mathit{opPlus}(\mathit{funI}, \mathit{cst}\,(1 + c))))), c) \in \mathit{Shift}$

proof
invoke Shift;
prove;
use $\mathit{AAddTwo}[x := y,\ c := c]$;
use $\mathit{UFNonneg}[f := \mathit{funA},\ x := y + 2 * c]$;
use $\mathit{LAt}[m := \mathit{funA}\,(y + 2 * c) + c,\ t := y + 2 * c,\ k := c]$;
prove;
■

theorem SigmaS
$\forall\, c : \mathbb{N};\ U : \mathbb{N} \to \mathbb{N} \mid (U, c) \in \mathit{Shift} \bullet \mathit{comp}(\mathit{comp}(\mathit{funL}, \mathit{opPlusIb}\,(1 + c)\,\mathit{funA}), U) = \mathit{funS}$

proof
apply UFExt to predicate $\mathit{comp}(\mathit{comp}(\mathit{funL},\ \mathit{opPlusIb}\,(1 + c)\,\mathit{funA}),\ U) = \mathit{funS}$;
prove;
use $\mathit{ShiftVal}[W := U,\ c := c,\ y := x]$;
use $\mathit{ASuc}[x := x + c]$;
use $\mathit{UFNonneg}[f := \mathit{funA},\ x := x + c]$;
use $\mathit{LAt}[m := \mathit{funA}\,(1 + x + c) + c,\ t := 1 + x + c,\ k := c]$;
prove;
■

theorem CbLSIn
$\forall\, c : \mathbb{N} \bullet \mathit{comp}(\mathit{comp}(\mathit{funL}, \mathit{opPlusIb}\,(1 + c)\,\mathit{funA}), \mathit{comp}(\mathit{funL}, \mathit{opPlus}(\mathit{funA}, \mathit{comp}(\mathit{funM}\,(2 * c),$
$\mathit{opPlus}(\mathit{funI}, \mathit{cst}\,(1 + c)))))) \in \mathit{clos}\,\{\mathit{funL}\}\,\{\mathit{opPlusIb}\,(1 + c), \mathit{opInv}\}\,\{\mathit{comp}\}$

proof
prove;
■

theorem CbLS
$\forall\, c : \mathbb{N} \bullet \mathit{funS} \in \mathit{clos}\,\{\mathit{funL}\}\,\{\mathit{opPlusIb}\,(1 + c), \mathit{opInv}\}\,\{\mathit{comp}\}$

proof
prove;
use CbLSIn[$c := c$];
use ULShift[$c := c$];
use SigmaS[$c := c,\ U := comp(funL,\ opPlus(funA,\ comp(funM\,(2 * c),\ opPlus(funI,\ cst\,(1 + c)))))$];
prove;
■

From S: the predecessor LSA, F^+, $\bar{0} = LA^+$ and $\bar{1} = S\bar{0}$.

theorem PLPreds
$comp(comp(funL, funS), funA) \in Preds$

proof
invoke Preds;
prove;
use UFNonneg[$f := funA,\ x := 1 + x$];
use LAt[$m := 1 + funA\,(1 + x),\ t := 1 + x,\ k := 1$];
prove;
■

theorem ZeroExprLA
$comp(funL, opPlusI\ funA) = cst\,0$

proof
apply UFExt to predicate $comp(funL,\ opPlusI\ funA) = cst\,0$;
prove;
use UFNonneg[$f := funA,\ x := x$];
use LAt[$m := funA\,x + x,\ t := x,\ k := x$];
prove;
■

theorem CbLPI
$\forall\, b : \mathbb{N} \mid funS \in clos\,\{funL\}\,\{opPlusIb\ b, opInv\}\,\{comp\} \bullet comp(comp(funL, funS),$
$funA) \in clos\,\{funL\}\,\{opPlusIb\ b, opInv\}\,\{comp\} \land opPlusI\ funA \in clos\,\{funL\}\,\{opPlusIb\ b,$
$opInv\}\,\{comp\}$

proof
prove;
use PLPreds;
use CbPlusI[$X := funL,\ b := b,\ P := comp(comp(funL,\ funS),\ funA),\ F := funA$];
prove;
■

theorem CbLOne
$\forall\, b : \mathbb{N} \mid funS \in clos\,\{funL\}\,\{opPlusIb\ b, opInv\}\,\{comp\} \bullet cst\,0 \in clos\,\{funL\}\,\{opPlusIb\ b,$
$opInv\}\,\{comp\} \land cst\,1 \in clos\,\{funL\}\,\{opPlusIb\ b, opInv\}\,\{comp\}$

proof
prove;
use CbLPI[$b := b$];
use ZeroExprLA;
prove;
use OneExpr;
prove;
■

theorem CbLFromS
$\forall\, b : \mathbb{N} \mid funS \in clos\,\{funL\}\,\{opPlusIb\ b, opInv\}\,\{comp\} \bullet clos\,\{funL\}\,\{opPlusIb\ b, opInv\}\,\{comp\} = ClosJR$

proof
prove;
use *CbLOne*[$b := b$];
use *CbLPI*[$b := b$];
use *PLPreds*;
use *CbSub*[$X := funL,\ b := b,\ P := comp(comp(funL,\ funS),\ funA)$];
use *GRbSubGE*[$X := funL,\ b := b$];
use *GEL*;
use *GREqJR*;
use *GLSubGE*;
use *ClosInPow*[$I := \{\ funL\ \},\ U := \{\ opPlusIb\ b,\ opInv\ \},\ B := \{\ comp\ \}$];
use *CGEInPower*;
use *CGLInPower*;
use *SubAntisym*[$\mathbb{N} \to \mathbb{N}$][$A := clos\,\{\ funL\ \}\,\{\ opPlusIb\ b,\ opInv\ \}\,\{\ comp\ \},\ B := CGE$];
with enabled (*CGLDef*) *prove*;
■

9.16 The shifted operator with inversion: the result

theorem CbES
$\forall\, b : \mathbb{N} \mid 1 \leq b \bullet funS \in clos\,\{funE\}\,\{opPlusIb\ b, opInv\}\,\{comp\}$

proof
prove;
use *Mod2Cases*[$x := b$];
use *Div2Nonneg*[$x := b$];
split $b = 1$;
cases;
use *CbESOne*;
prove;
next;
split $b \ \mathsf{mod}\ 2 = 0$;
cases;
use *CbESEven*[$p := b \ \mathsf{div}\ 2 - 1$];
prove;
next;
use *CbESOdd*[$p := b \ \mathsf{div}\ 2 - 1$];
prove;
next;
next;
■

theorem CbKS
$\forall\, b : \mathbb{N} \mid 1 \leq b \bullet funS \in clos\,\{funK\}\,\{opPlusIb\ b, opInv\}\,\{comp\}$

proof
prove;
split $b = 1$;
cases;
use *CbKSOne*;
prove;
next;
use *CbKSTwo*[$d := b - 2$];
prove;
next;
■

theorem CbLSAll
$\forall\, b : \mathbb{N} \mid 1 \leq b \bullet \mathit{funS} \in \mathit{clos}\,\{\mathit{funL}\}\,\{\mathit{opPlusIb}\; b, \mathit{opInv}\}\,\{\mathit{comp}\}$

proof
prove;
use $\mathit{CbLS}[c := b - 1]$;
prove;
■

theorem ExtGRb
$\forall\, b : \mathbb{N} \mid 1 \leq b \bullet \mathit{clos}\,\{\mathit{funE}\}\,\{\mathit{opPlusIb}\; b, \mathit{opInv}\}\,\{\mathit{comp}\} = \mathit{ClosJR} \wedge \mathit{clos}\,\{\mathit{funK}\}\,\{\mathit{opPlusIb}\; b,$
$\mathit{opInv}\}\,\{\mathit{comp}\} = \mathit{ClosJR} \wedge \mathit{clos}\,\{\mathit{funL}\}\,\{\mathit{opPlusIb}\; b, \mathit{opInv}\}\,\{\mathit{comp}\} = \mathit{ClosJR}$

proof
prove;
use $\mathit{CbES}[b := b]$;
use $\mathit{CbKS}[b := b]$;
use $\mathit{CbLSAll}[b := b]$;
use $\mathit{CbEFromS}[b := b]$;
use $\mathit{CbKFromS}[b := b]$;
use $\mathit{CbLFromS}[b := b]$;
prove;
■

$b = 0$: every generated function fixes 0.

theorem FixZeroUClosedIb
$\mathit{FixZero} \in \mathit{UClosed}\,\{\mathit{opPlusIb}\, 0, \mathit{opInv}\}$

proof
use *FixZeroInPower*;
with enabled (*UClosedDef*) *prove*;
invoke *FixZero*;
prove;
use $\mathit{InPair}[\mathbb{P}\,(\mathbb{P}\,(\mathbb{Z} \times \mathbb{Z}) \times \mathbb{P}\,(\mathbb{Z} \times \mathbb{Z}))][u := u,\ p := \mathit{opPlusIb}\, 0,\ q := \mathit{opInv}]$;
split $u = \mathit{opInv}$;
cases;
use $\mathit{InvFix}0[F := F]$;
prove;
next;
prove;
next;
■

theorem GRbZeroFix
$\forall\, X : \mathbb{N} \to \mathbb{N} \mid X\, 0 = 0 \bullet \mathit{clos}\,\{X\}\,\{\mathit{opPlusIb}\, 0, \mathit{opInv}\}\,\{\mathit{comp}\} \subseteq \mathit{FixZero}$

proof
prove;
use $\mathit{ClosLeast}[I := \{\ X\ \},\ U := \{\ \mathit{opPlusIb}\, 0,\ \mathit{opInv}\ \},\ B := \{\ \mathit{comp}\ \},\ Z := \mathit{FixZero}]$;
use *FixZeroInPower*;
use *FixZeroUClosedIb*;
use *FixZeroBClosed*;
invoke *FixZero*;
prove;
■

theorem GRbZeroNotOne
$\forall\, X : \mathbb{N} \to \mathbb{N} \mid X\, 0 = 0 \bullet \neg\ \mathit{cst}\, 1 \in \mathit{clos}\,\{X\}\,\{\mathit{opPlusIb}\, 0, \mathit{opInv}\}\,\{\mathit{comp}\}$

proof
use $GRbZeroFix[X := X]$;
apply subsetDef to predicate $clos\{X\}\{opPlusIb\,0,\ opInv\}\{comp\} \subseteq FixZero$;
apply inPower to predicate $clos\{X\}\{opPlusIb\,0,\ opInv\}\{comp\} \in \mathbb{P}\ FixZero$;
instantiate $e == cst\,1$;
invoke $FixZero$;
prove;
■

theorem GRbZeroNeq
$\forall X : \mathbb{N} \to \mathbb{N} \mid X\,0 = 0 \bullet \neg\ clos\,\{X\}\,\{opPlusIb\,0, opInv\}\,\{comp\} = ClosJR$

proof
use $GRbZeroNotOne[X := X]$;
use $JROne$;
prove;
split $clos\{X\}\{opPlusIb\,0,\ opInv\}\{comp\} = ClosJR$;
cases;
equality substitute $clos\{X\}(\{opPlusIb\,0\} \cup \{opInv\})\{comp\}$;
prove;
next;
prove;
next;
■

theorem ExtGRbIff
$\forall b : \mathbb{N} \bullet (clos\,\{funE\}\,\{opPlusIb\,b, opInv\}\,\{comp\} = ClosJR \Leftrightarrow 1 \leq b) \wedge (clos\,\{funK\}\,\{opPlusIb\,b,$
$opInv\}\,\{comp\} = ClosJR \Leftrightarrow 1 \leq b) \wedge (clos\,\{funL\}\,\{opPlusIb\,b, opInv\}\,\{comp\} = ClosJR \Leftrightarrow 1 \leq b)$

proof
prove;
split $b = 0$;
cases;
use $GRbZeroNeq[X := funE]$;
use $GRbZeroNeq[X := funK]$;
use $GRbZeroNeq[X := funL]$;
use $KZero$;
prove;
next;
use $ExtGRb[b := b]$;
prove;
next;
■

9.17 Remark: any constant $\bar{n}$ with $n \geq 1$

For $X \in \{E, K, L\}$ and a set U of unary operators with $F^+ \in U$, $\langle \bar{n}, X, U, FG\rangle$ and $\langle \bar{1}, X, U, FG\rangle$ have the same closure if $n \geq 1$. Since $\bar{n}^+(x) = x + n$, every constant $c + kn$ is generated from c, and X steps down from c to $c - 1$: $E(c + (nc^2 + 2c)n) = c - 1$, $K(c + (2nc^2 + 3c)n) = c - 1$, $L(c + (2nc^2 + 2c)n) = nc - c$ and $L((nc - c) + 2nc^2 n) = c - 1$.

theorem CnComp
$\forall X : \mathbb{N} \to \mathbb{N};\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ n : \mathbb{N};\ F, G : \mathbb{N} \to \mathbb{N} \mid F \in clos\,\{cst\,n,$
$X\}\,U\,\{comp\} \wedge G \in clos\,\{cst\,n, X\}\,U\,\{comp\} \bullet comp(F, G) \in clos\,\{cst\,n, X\}\,U\,\{comp\}$

proof
prove;
■

theorem CstComp
$\forall F : \mathbb{N} \to \mathbb{N};\ m : \mathbb{N} \bullet comp(F, cst\, m) = cst\,(F\, m)$

proof
prove;
apply UFExt to predicate $comp(F,\ cst\, m)\ =\ cst\,(F\, m)$;
prove;
■

theorem CstShiftEq
$\forall n, c, k : \mathbb{N} \bullet comp(iter[\mathbb{N}]\, k\,(opPlusI\,(cst\, n)), cst\, c) = cst\,(c + k * n)$

proof
prove;
apply UFExt to predicate $comp(iter[\mathbb{N}]\, k\,(opPlusI\,(cst\, n)),\ cst\, c)\ =\ cst\,(c\ +\ k\ *\ n)$;
prove;
use ShiftPow$[k\ :=\ k,\ c\ :=\ n,\ x\ :=\ c,\ F\ :=\ opPlusI\,(cst\, n)]$;
invoke Shift;
prove;
■

theorem CstShift
$\forall X : \mathbb{N} \to \mathbb{N};\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ n, c, k : \mathbb{N} \mid opPlusI \in U \wedge cst\, c \in clos\,\{cst\, n,$
$X\}\ U\,\{comp\} \bullet cst\,(c + k * n) \in clos\,\{cst\, n, X\}\ U\,\{comp\} \wedge cst\,(X\,(c + k * n)) \in clos\,\{cst\, n,$
$X\}\ U\,\{comp\}$

proof
prove;
use PowIn$[I\ :=\ \{\ cst\, n,\ X\ \},\ U\ :=\ U,\ B\ :=\ \{\ comp\ \},\ F\ :=\ opPlusI\,(cst\, n),\ G\ :=\ cst\, c,\ k$
$:=\ k]$;
use CstShiftEq$[n\ :=\ n,\ c\ :=\ c,\ k\ :=\ k]$;
prove;
use CstComp$[F\ :=\ X,\ m\ :=\ c\ +\ k\ *\ n]$;
prove;
■

The descent, with $n = n_1 + 1$ and $c = c_1 + 1$.

theorem EDesc
$\forall n1, c1 : \mathbb{N} \bullet funE\,((1 + c1) + ((1 + n1) * (1 + c1) * (1 + c1) + 2 * (1 + c1)) * (1 + n1)) = c1$

proof
prove;
use MulNonneg$[a\ :=\ n1,\ b\ :=\ c1]$;
use EAt$[y\ :=\ (1\ +\ c1)\ +\ ((1\ +\ n1)\ *\ (1\ +\ c1)\ *\ (1\ +\ c1)\ +\ 2\ *\ (1\ +\ c1))\ *\ (1\ +\ n1),\ m$
$:=\ (1\ +\ n1)\ *\ (1\ +\ c1)\ +\ 1,\ k\ :=\ c1]$;
prove;
■

theorem KDesc
$\forall n1, c1 : \mathbb{N} \bullet funK\,((1 + c1) + (2 * (1 + n1) * (1 + c1) * (1 + c1) + 3 * (1 + c1)) * (1 + n1)) = c1$

proof
prove;
use MulNonneg$[a\ :=\ n1,\ b\ :=\ c1]$;
use TwoA$[x\ :=\ 2\ *\ (1\ +\ n1)\ *\ (1\ +\ c1)\ +\ 1]$;
use Halve$[a\ :=\ funA\,(2\ *\ (1\ +\ n1)\ *\ (1\ +\ c1)\ +\ 1),\ b\ :=\ (2\ *\ (1\ +\ n1)\ *\ (1\ +\ c1)\ +\ 1)\ *$
$((1\ +\ n1)\ *\ (1\ +\ c1)\ +\ 1)]$;
use KAt$[m\ :=\ (1\ +\ c1)\ +\ (2\ *\ (1\ +\ n1)\ *\ (1\ +\ c1)\ *\ (1\ +\ c1)\ +\ 3\ *\ (1\ +\ c1))\ *\ (1\ +$
$n1),\ t\ :=\ 2\ *\ (1\ +\ n1)\ *\ (1\ +\ c1)\ +\ 1,\ k\ :=\ c1]$;
prove;
■

theorem LDescOne
$\forall\, n1, c : \mathbb{N} \bullet \mathit{funL}\,(c + (2 * (1 + n1) * c * c + 2 * c) * (1 + n1)) = n1 * c$

proof
prove;
use $\mathit{MulNonneg}[a := n1,\ b := c]$;
use $\mathit{TwoA}[x := 2 * (1 + n1) * c]$;
use $\mathit{Halve}[a := \mathit{funA}\,(2 * (1 + n1) * c),\ b := 2 * (1 + n1) * (1 + n1) * c * c + (1 + n1) * c]$;
use $\mathit{LAt}[m := c + (2 * (1 + n1) * c * c + 2 * c) * (1 + n1),\ t := 2 * (1 + n1) * c,\ k := (1 + n1) * c + c]$;
prove;
■

theorem LDescTwo
$\forall\, n1, c1 : \mathbb{N} \bullet \mathit{funL}\,(n1 * (1 + c1) + (2 * (1 + n1) * (1 + c1) * (1 + c1)) * (1 + n1)) = c1$

proof
prove;
use $\mathit{MulNonneg}[a := n1,\ b := c1]$;
use $\mathit{TwoA}[x := (2 * n1 + 1) * (1 + c1) + c1]$;
use $\mathit{Halve}[a := \mathit{funA}\,((2 * n1 + 1) * (1 + c1) + c1),\ b := n1 * (1 + c1) + (2 * (1 + n1) * (1 + c1) * (1 + c1)) * (1 + n1) - (2 * n1 + 1) * (1 + c1)]$;
use $\mathit{LAt}[m := n1 * (1 + c1) + (2 * (1 + n1) * (1 + c1) * (1 + c1)) * (1 + n1),\ t := (2 * n1 + 1) * (1 + c1) + c1,\ k := (2 * n1 + 1) * (1 + c1)]$;
prove;
■

theorem StepE
$\forall\, U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ n, c : \mathbb{N} \mid \mathit{opPlusI} \in U \wedge 1 \leq n \wedge 1 \leq c \wedge \mathit{cst}\, c \in \mathit{clos}\,\{\mathit{cst}\, n, \mathit{funE}\}\ U\ \{\mathit{comp}\} \bullet \mathit{cst}\,(c - 1) \in \mathit{clos}\,\{\mathit{cst}\, n, \mathit{funE}\}\ U\ \{\mathit{comp}\}$

proof
prove;
use $\mathit{CstShift}[X := \mathit{funE},\ U := U,\ n := n,\ c := c,\ k := n * c * c + 2 * c]$;
use $\mathit{EDesc}[n1 := n - 1,\ c1 := c - 1]$;
prove;
■

$\mathit{DescESet} == \{j : \mathbb{N} \mid \forall\, U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ n : \mathbb{N} \mid \mathit{opPlusI} \in U \wedge j + 1 \leq n \bullet \mathit{cst}\,(n - j) \in \mathit{clos}\,\{\mathit{cst}\, n, \mathit{funE}\}\ U\ \{\mathit{comp}\}\}$

proof [DescESet$domainCheck]
prove;
■

theorem rule DescEBase
$0 \in \mathit{DescESet}$

proof
invoke $\mathit{DescESet}$;
prove;
■

theorem rule DescEStep
$\forall\, j : \mathit{DescESet} \bullet 1 + j \in \mathit{DescESet}$

proof
invoke DescESet;
prove;
instantiate $U__0 == U,\ n__0 == n$;
use StepE$[U := U,\ n := n,\ c := n - j]$;
prove;
■

theorem DescESub
$\mathbb{N} \subseteq DescESet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq DescESet$;
prove;
invoke DescESet;
prove;
■

theorem OneInE
$\forall\, U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ m : \mathbb{N} \mid opPlusI \in U \bullet cst\, 1 \in clos\, \{cst\, (1 + m), funE\}\ U\ \{comp\}$

proof
prove;
use DescESub;
invoke DescESet;
with enabled (*inPower*) *prove*;
instantiate $e__0 == m,\ U__0 == U,\ n == 1 + m$;
with enabled (*inPower*) *prove*;
■

theorem StepK
$\forall\, U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ n, c : \mathbb{N} \mid opPlusI \in U \wedge 1 \leq n \wedge 1 \leq c \wedge cst\, c \in clos\, \{cst\, n,$
$funK\}\ U\ \{comp\} \bullet cst\, (c - 1) \in clos\, \{cst\, n, funK\}\ U\ \{comp\}$

proof
prove;
use CstShift$[X := funK,\ U := U,\ n := n,\ c := c,\ k := 2 * n * c * c + 3 * c]$;
use KDesc$[n1 := n - 1,\ c1 := c - 1]$;
prove;
■

$DescKSet == \{j : \mathbb{N} \mid \forall\, U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ n : \mathbb{N} \mid opPlusI \in U \wedge j + 1 \leq n \bullet cst\, (n - j) \in$
$clos\, \{cst\, n, funK\}\ U\ \{comp\}\}$

proof [DescKSet$domainCheck]
prove;
■

theorem rule DescKBase
$0 \in DescKSet$

proof
invoke DescKSet;
prove;
■

theorem rule DescKStep
$\forall j : DescKSet \bullet 1 + j \in DescKSet$

proof
invoke DescKSet;
prove;
instantiate $U__0 == U$, $n__0 == n$;
use $StepK[U := U, n := n, c := n - j]$;
prove;
■

theorem DescKSub
$\mathbb{N} \subseteq DescKSet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq DescKSet$;
prove;
invoke DescKSet;
prove;
■

theorem OneInK
$\forall U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})); m : \mathbb{N} \mid opPlusI \in U \bullet cst\, 1 \in clos\, \{cst\, (1 + m), funK\}\, U\, \{comp\}$

proof
prove;
use DescKSub;
invoke DescKSet;
with enabled (*inPower*) *prove*;
instantiate $e__0 == m$, $U__0 == U$, $n == 1 + m$;
with enabled (*inPower*) *prove*;
■

theorem StepL
$\forall U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})); n, c : \mathbb{N} \mid opPlusI \in U \land 1 \leq n \land 1 \leq c \land cst\, c \in clos\, \{cst\, n,$
$funL\}\, U\, \{comp\} \bullet cst\, (c - 1) \in clos\, \{cst\, n, funL\}\, U\, \{comp\}$

proof
prove;
use $CstShift[X := funL, U := U, n := n, c := c, k := 2 * n * c * c + 2 * c]$;
use $LDescOne[n1 := n - 1, c := c]$;
prove;
use $CstShift[X := funL, U := U, n := n, c := (n - 1) * c, k := 2 * n * c * c]$;
use $LDescTwo[n1 := n - 1, c1 := c - 1]$;
prove;
■

$DescLSet == \{j : \mathbb{N} \mid \forall U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})); n : \mathbb{N} \mid opPlusI \in U \land j + 1 \leq n \bullet cst\, (n - j) \in$
$clos\, \{cst\, n, funL\}\, U\, \{comp\}\}$

proof [DescLSet$domainCheck]
prove;
■

theorem rule DescLBase
$0 \in DescLSet$

proof
invoke DescLSet;
prove;
■

theorem rule DescLStep
$\forall\, j : \mathit{DescLSet} \bullet 1 + j \in \mathit{DescLSet}$

proof
invoke DescLSet;
prove;
instantiate $U__0 == U,\ n__0 == n$;
use StepL$[U := U,\ n := n,\ c := n - j]$;
prove;
■

theorem DescLSub
$\mathbb{N} \subseteq \mathit{DescLSet}$

proof
apply natInduction to predicate $\mathbb{N} \subseteq \mathit{DescLSet}$;
prove;
invoke DescLSet;
prove;
■

theorem OneInL
$\forall\, U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ m : \mathbb{N} \mid \mathit{opPlusI} \in U \bullet \mathit{cst}\, 1 \in \mathit{clos}\, \{\mathit{cst}\, (1 + m), \mathit{funL}\}\ U\ \{\mathit{comp}\}$

proof
prove;
use DescLSub;
invoke DescLSet;
with enabled (*inPower*) *prove*;
instantiate $e__0 == m,\ U__0 == U,\ n == 1 + m$;
with enabled (*inPower*) *prove*;
■

With $\bar{1}$ in the closure, the bases with $\bar{n}$ and with $\bar{1}$ are equivalent.

theorem CnUClosed
$\forall\, X : \mathbb{N} \to \mathbb{N};\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ n : \mathbb{N} \bullet \mathit{clos}\, \{\mathit{cst}\, n, X\}\ U\ \{\mathit{comp}\} \in \mathit{UClosed}\ U$

proof
prove;
use ClosInPow$[I := \{\ \mathit{cst}\, n,\ X\ \},\ U := U,\ B := \{\ \mathit{comp}\ \}]$;
with enabled (*UClosedDef*) *prove*;
■

theorem CnBClosed
$\forall\, X : \mathbb{N} \to \mathbb{N};\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ n : \mathbb{N} \bullet \mathit{clos}\, \{\mathit{cst}\, n, X\}\ U\ \{\mathit{comp}\} \in \mathit{BClosed}\, \{\mathit{comp}\}$

proof
prove;
use ClosInPow$[I := \{\ \mathit{cst}\, n,\ X\ \},\ U := U,\ B := \{\ \mathit{comp}\ \}]$;
with enabled (*BClosedDef*) *prove*;
■

theorem CEqCOne
$\forall X : \mathbb{N} \to \mathbb{N};\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ n : \mathbb{N} \mid \mathit{opPlusI} \in U \wedge 1 \leq n \wedge \mathit{cst}\,1 \in \mathit{clos}\,\{\mathit{cst}\,n, X\}\,U\,\{\mathit{comp}\} \bullet \mathit{clos}\,\{\mathit{cst}\,n, X\}\,U\,\{\mathit{comp}\} = \mathit{clos}\,\{\mathit{cst}\,1, X\}\,U\,\{\mathit{comp}\}$

proof
prove;
use *CstShift*$[X := X,\ U := U,\ n := 1,\ c := 1,\ k := n - 1]$;
use *ClosLeast*$[I := \{\ \mathit{cst}\,1,\ X\ \},\ U := U,\ B := \{\ \mathit{comp}\ \},\ Z := \mathit{clos}\,\{\ \mathit{cst}\,n,\ X\ \}\,U\,\{\ \mathit{comp}\ \}]$;
use *ClosLeast*$[I := \{\ \mathit{cst}\,n,\ X\ \},\ U := U,\ B := \{\ \mathit{comp}\ \},\ Z := \mathit{clos}\,\{\ \mathit{cst}\,1,\ X\ \}\,U\,\{\ \mathit{comp}\ \}]$;
use *ClosInPow*$[I := \{\ \mathit{cst}\,n,\ X\ \},\ U := U,\ B := \{\ \mathit{comp}\ \}]$;
use *ClosInPow*$[I := \{\ \mathit{cst}\,1,\ X\ \},\ U := U,\ B := \{\ \mathit{comp}\ \}]$;
use *CnUClosed*$[X := X,\ U := U,\ n := n]$;
use *CnUClosed*$[X := X,\ U := U,\ n := 1]$;
use *CnBClosed*$[X := X,\ U := U,\ n := n]$;
use *CnBClosed*$[X := X,\ U := U,\ n := 1]$;
use *SubAntisym*$[\mathbb{N} \to \mathbb{N}][A := \mathit{clos}\,\{\ \mathit{cst}\,n,\ X\ \}\,U\,\{\ \mathit{comp}\ \},\ B := \mathit{clos}\,\{\ \mathit{cst}\,1,\ X\ \}\,U\,\{\ \mathit{comp}\ \}]$;
prove;
■

theorem CEq
$\forall U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ n : \mathbb{N} \mid \mathit{opPlusI} \in U \wedge 1 \leq n \bullet \mathit{clos}\,\{\mathit{cst}\,n, \mathit{funE}\}\,U\,\{\mathit{comp}\} = \mathit{clos}\,\{\mathit{cst}\,1, \mathit{funE}\}\,U\,\{\mathit{comp}\} \wedge \mathit{clos}\,\{\mathit{cst}\,n, \mathit{funK}\}\,U\,\{\mathit{comp}\} = \mathit{clos}\,\{\mathit{cst}\,1, \mathit{funK}\}\,U\,\{\mathit{comp}\} \wedge \mathit{clos}\,\{\mathit{cst}\,n, \mathit{funL}\}\,U\,\{\mathit{comp}\} = \mathit{clos}\,\{\mathit{cst}\,1, \mathit{funL}\}\,U\,\{\mathit{comp}\}$

proof
prove;
use *OneInE*$[U := U,\ m := n - 1]$;
use *OneInK*$[U := U,\ m := n - 1]$;
use *OneInL*$[U := U,\ m := n - 1]$;
use *CEqCOne*$[X := \mathit{funE},\ U := U,\ n := n]$;
use *CEqCOne*$[X := \mathit{funK},\ U := U,\ n := n]$;
use *CEqCOne*$[X := \mathit{funL},\ U := U,\ n := n]$;
prove;
■

With $\bar{0}$, every generated function fixes 0 (for $U = \{F^+, F^\Box\}$ or $\{F^+, F^{-1}\}$).

theorem FixZeroUClosedBox
$\mathit{FixZero} \in \mathit{UClosed}\,\{\mathit{opPlusI}, \mathit{opBox}\}$

proof
use *FixZeroInPower*;
with enabled (*UClosedDef*) *prove*;
invoke *FixZero*;
prove;
split $u = \mathit{opBox}$;
cases;
prove;
next;
prove;
next;
■

theorem CZeroFix
$\forall X : \mathbb{N} \to \mathbb{N};\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \mid X\,0 = 0 \wedge \mathit{FixZero} \in \mathit{UClosed}\,U \bullet \neg\ \mathit{cst}\,1 \in \mathit{clos}\,\{\mathit{cst}\,0, X\}\,U\,\{\mathit{comp}\}$

proof
prove;
use ClosLeast[$I := \{ cst\,0,\ X \},\ U := U,\ B := \{ comp \},\ Z := FixZero$];
use FixZeroInPower;
use FixZeroBClosed;
invoke FixZero;
prove;
apply inPower to predicate $clos\,(\{ X \} \cup \{ cst\,0 \})\,U\,\{ comp \} \in \mathbb{P}\,\{ F : \mathbb{N} \to \mathbb{N} \mid F\,0 = 0 \}$;
instantiate e $==$ $cst\,1$;
prove;
■

theorem CnBoxNeq
$\forall X : \mathbb{N} \to \mathbb{N} \mid X\,0 = 0 \bullet \neg\ clos\,\{cst\,0, X\}\,\{opPlusI, opBox\}\,\{comp\} = C6$

proof
use CZeroFix[$X := X,\ U := \{ opPlusI,\ opBox \}$];
use FixZeroUClosedBox;
prove;
split $clos\,(\{ X \} \cup \{ cst\,0 \})\,(\{ opPlusI \} \cup \{ opBox \})\,\{ comp \} = C6$;
cases;
equality substitute $clos\,(\{ X \} \cup \{ cst\,0 \})\,(\{ opPlusI \} \cup \{ opBox \})\,\{ comp \}$;
prove;
next;
prove;
next;
■

theorem CnBox
$\forall n : \mathbb{N} \bullet (clos\,\{cst\,n, funE\}\,\{opPlusI, opBox\}\,\{comp\} = C6 \Leftrightarrow 1 \leq n) \wedge (clos\,\{cst\,n, funK\}\,\{opPlusI, opBox\}\,\{comp\} = C6 \Leftrightarrow 1 \leq n) \wedge (clos\,\{cst\,n, funL\}\,\{opPlusI, opBox\}\,\{comp\} = C6 \Leftrightarrow 1 \leq n)$

proof
prove;
split $n = 0$;
cases;
use CnBoxNeq[$X := funE$];
use CnBoxNeq[$X := funK$];
use CnBoxNeq[$X := funL$];
use KZero;
prove;
next;
use CEq[$U := \{ opPlusI,\ opBox \},\ n := n$];
*use Theorem*68;
with enabled ($C6Def,\ C6KDef,\ C6LDef$) *prove*;
next;
■

theorem CnInvNeq
$\forall X : \mathbb{N} \to \mathbb{N} \mid X\,0 = 0 \bullet \neg\ clos\,\{cst\,0, X\}\,\{opPlusI, opInv\}\,\{comp\} = ClosJR$

proof
use $CZeroFix[X := X,\ U := \{\ opPlusI,\ opInv\ \}]$;
use $FixZeroUClosedInv$;
prove;
split $clos\,(\{\ X\ \} \cup \{\ cst\,0\ \})\,(\{\ opPlusI\ \} \cup \{\ opInv\ \})\,\{\ comp\ \} = ClosJR$;
cases;
equality substitute $clos\,(\{\ X\ \} \cup \{\ cst\,0\ \})\,(\{\ opPlusI\ \} \cup \{\ opInv\ \})\,\{\ comp\ \}$;
prove;
next;
prove;
next;
■

theorem CnInv
$\forall\, n : \mathbb{N} \bullet (clos\,\{cst\,n, funE\}\,\{opPlusI, opInv\}\,\{comp\} = ClosJR \Leftrightarrow 1 \leq n) \wedge (clos\,\{cst\,n,$
$funK\}\,\{opPlusI, opInv\}\,\{comp\} = ClosJR \Leftrightarrow 1 \leq n) \wedge (clos\,\{cst\,n, funL\}\,\{opPlusI,$
$opInv\}\,\{comp\} = ClosJR \Leftrightarrow 1 \leq n)$

proof
prove;
split $n = 0$;
cases;
use $CnInvNeq[X := funE]$;
use $CnInvNeq[X := funK]$;
use $CnInvNeq[X := funL]$;
use $KZero$;
prove;
next;
use $CEq[U := \{\ opPlusI,\ opInv\ \},\ n := n]$;
use $GREqJR$;
with enabled $(CGEDef,\ CGKDef,\ CGLDef)$ *prove*;
next;
■